

MATHEMATICS

Differential and Integral Calculus

Solved Exercises
and Problems

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2026 Edition

Mathematics

Book of Differential and Integral Calculus
Solved Exercises

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Dedication

I dedicate this modest work to my dear parents...

Table of Contents

Chapter 1: Applications of Differentiation

Theorems	2-1
Exercise 1.1 (Speed of a rocket)	2
Exercise 1.2 (Position of a moving body)	4-3
Exercise 1.3 (Minimum length of a ladder)	7-4
Exercise 1.4 (Optimal viewing angle)	9-7
Exercise 1.5 (Rectangle with fixed perimeter)	10-9
Exercise 1.6 (Cylindrical tank)	12-10
Exercise 1.7 (Submarine cable)	14-12
Exercise 1.8 (Submarine gas pipeline)	17-14
Exercise 1.9 (Cone with maximum volume)	19-17
Exercise 1.10 (Optimal path)	21-19
Exercise 1.11 (Distance between a point and a parabola)	23-22
Exercise 1.12 (Light path)	24-23
Exercise 1.13 (Beam in a corridor)	27-25
Exercise 1.14 (Distance between two curves)	28-27
Exercise 1.15 (Conical tank)	30-29
Exercise 1.16 (Minimum cost)	33-31
Exercise 1.17 (Box from a square)	35-33
Exercise 1.18 (Cylinder inside a sphere)	37-35
Exercise 1.19 (Cone containing a sphere)	39-37
Exercise 1.20 (Filling a cone with water)	41-39
Exercise 1.21 (Water flow from a cone)	44-42
Exercise 1.22 (Thin lens)	46-45
Exercise 1.23 (Cylinder inside a cone)	48-46

Chapter 2: Properties of Integrals

Theorems	
Exercise 2.1 (Swapping integration limits)	
Exercise 2.2 (Symmetry $f(a + b - x)$)	
Exercise 2.3 (Application to $\int_0^{\pi/2} \sin^2 x \, dx$)	
Exercise 2.4 ($\int_0^{\pi/2} \frac{\sin t}{\sin t + \cos t} \, dt$)	
Exercise 2.5 (Property $\int_0^{\pi} t f(\sin t) \, dt$)	
Exercise 2.6 (Application: $\int_0^{\pi} x \sin x \, dx$)	
Exercise 2.7 ($\int_0^{\pi} (2x + 1) \frac{\sin^3 x}{1 + \cos^2 x} \, dx$)	
Exercise 2.8 ($\int_0^{\pi/4} \ln(1 + \tan x) \, dx$)	
Exercise 2.9 (Generalization of the symmetry property)	
Exercise 2.10 ($\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} \, dx$)	
Exercise 2.11 (Integration doubling property under symmetry)	
Exercise 2.12 (Recurrence relation for $I_n = \int_0^1 \frac{dx}{(1+x^2)^n}$)	
Exercise 2.13 (Symmetry of the Beta function)	

Exercise 2.14 (Special values of Beta)	
Exercise 2.15 (Recurrence relation for Beta)	
Exercise 2.16 (Trigonometric representation of Beta)	
Exercise 2.17 (Beta representation as an integral over $[0, \infty[$)	
Exercise 2.18 ($\int_0^1 \sqrt{\frac{1-x}{x}} dx$ and $\int_0^1 \sqrt{\frac{x}{1-x}} dx$)	
Exercise 2.19 ($\int_0^\infty \frac{\sin(2x^2)}{x^2} e^{-x^2} dx$)	
Exercise 2.20 ($\int_0^\infty \frac{\cos x}{x^2+1} dx$)	
Exercise 2.21 ($\int_0^\infty \frac{dx}{x\sqrt{x+\sqrt{x}}}$)	
Exercise 2.22 ($\int_0^\infty \frac{\sqrt{x}}{(1+x)^2} dx$)	
Exercise 2.23 (General formula $\int_0^\infty \frac{u^{m-1}}{(1+u^n)^\alpha} du$)	
Exercise 2.24 ($\int_0^\infty \frac{dx}{1+x^4}$)	
Exercise 2.25 (Generalization $\int_0^\infty \frac{x^\rho}{(x^\beta+a^\beta)^\gamma} dx$)	
Exercise 2.26 (Parametric integral $I(\alpha) = \int_0^\infty \frac{e^{-\alpha x} - e^{-\beta x}}{x} dx$)	
Exercise 2.27 (Frullani's formula)	
Exercise 2.28 ($\int_0^\infty \frac{\sin(ax)\sin(bx)}{x^2} dx$)	
Exercise 2.29 ($\int_0^{\pi/2} \ln(\sin x) dx$)	

Chapter 3: Feynman's Technique for Computing Integrals

Theorems	
Exercise 3.1 ($\int_0^1 \frac{x^n-1}{\ln x} dx$)	
Exercise 3.2 ($\int_0^1 \frac{x^a-x^b}{\ln x} dx$)	
Exercise 3.3 ($\int_0^1 \frac{x-1}{\ln x} \cdot \frac{1}{\sqrt{-\ln x}} dx$)	
Exercise 3.4 (Gauss's method for $\int_0^\infty e^{-x^2} dx$)	
Exercise 3.5 (Feynman's method for the same integral)	
Exercise 3.6 ($\int_0^\infty \frac{\sin t}{t} dt$)	
Exercise 3.7 ($\int_0^\infty \frac{e^{-ax}-e^{-bx}}{x} dx$)	
Exercise 3.8 ($\int_0^\infty x^{2n} e^{-x^2} dx$)	
Exercise 3.9 ($\int_0^\infty \frac{\cos(ax)}{1+x^2} dx$)	

Chapter 4: Laplace Transform

Theorems	
Exercise 4.1 ($\mathcal{L}\{At^n\}$)	
Exercise 4.2 ($\mathcal{L}\{1\}$, $\mathcal{L}\{2t\}$, $\mathcal{L}\{\sqrt{t}\}$)	
Exercise 4.3 ($\mathcal{L}\{e^{iat}\}$)	
Exercise 4.4 ($\mathcal{L}\{\cos(at)\}$, $\mathcal{L}\{\sin(at)\}$)	
Exercise 4.5 (Various transforms ($t^2 \sin 3t$, $e^{-2t} \cosh 4t$, ...))	
Exercise 4.6 (Using convolution to compute the inverse of $\frac{1}{(s^2+1)^2}$ and Bessel functions)	
Exercise 4.7 (Solving ordinary differential equations using transforms)	
Exercise 4.8 (Transform of a periodic square wave)	
Exercise 4.9 (Heat equation (semi-infinite line))	
Exercise 4.10 (Bilateral Laplace transform)	
Exercise 4.11 (Verification of initial and final value theorems)	
Exercise 4.12 (Solving integro-differential equations)	
Exercise 4.13 (Transforms of special functions (Bessel, error, ...))	
Exercise 4.14 (Heat equation with fixed boundary conditions)	
Exercise 4.15 (Caputo derivative and Laplace transform)	

Chapter 5: Derivative of Positive Real Order

Theorems	
Exercise 5.1 (Inverse transform of $\frac{1}{s^\kappa}$)	
Exercise 5.2 (Laplace transform of $D^\alpha(t^n)$)	

Exercise 5.3 (Proof of $D^\alpha(t^n) = \frac{\Gamma(n+1)}{\Gamma(n-\alpha+1)}t^{n-\alpha}$ by definition)	
Exercise 5.4 (Same proof using Laplace transform)	
Exercise 5.5 (Numerical examples ($D^{1/2}(t)$, $D^{1/2}D^{1/2}(t)$, ...))	

Chapter 6: Fourier Transforms

Definitions and basic properties	
Table of standard transforms	
Discrete Fourier Transform and FFT	
Physical and signal processing applications	
Exercise 6.1 (Transforms of elementary functions)	
Exercise 6.2 (Convolution, Bessel, cosine modulation)	
Exercise 6.3 (Solving heat, wave, and Laplace equations)	
Exercise 6.4 (AM modulation, Shannon, ideal filter, autocorrelation)	
Exercise 6.5 (Discrete Fourier Transform and FFT)	
Exercise 6.6 (Short-time Fourier Transform (STFT) and uncertainty principle)	
Exercise 6.7 (Fourier optics, diffraction, image filtering)	
Exercise 6.8 (Quantum mechanics, Schrödinger, harmonic oscillator)	
Exercise 6.9 (Distributional transforms: Dirac, principal value, signum)	
Exercise 6.10 (2D and 3D transforms, Hankel transform)	

Chapter 7: Vector Analysis

Basic differential operators	
Curvilinear coordinates	
Integral theorems	
Exercise 7.1 (Gradient and directional derivative)	
Exercise 7.2 (Gradient and tangent plane)	
Exercise 7.3 (Divergence, curl, and flux through a sphere)	
Exercise 7.4 (Conservative field, potential, curl)	
Exercise 7.5 (Gravitational field, flux, divergence contradiction)	
Exercise 7.6 (Velocity field of a fluid and streamlines)	
Exercise 7.7 (Electromagnetic waves and Maxwell's equations)	
Exercise 7.8 (Heat conduction and heat equation)	
Exercise 7.9 (Divergence, curl, and flux through a cube)	
Exercise 7.10 (Gradient and Laplacian of $\ln r^2$)	
Exercise 7.11 (Conservative field, potential, curl)	
Exercise 7.12 (Flux through a conical surface)	
Exercise 7.13 (Rotational, non-conservative field)	
Exercise 7.14 (Field with constant divergence and zero curl)	
Exercise 7.15 (Flux through a hemisphere and verification via divergence theorem)	
Exercise 7.16 (Conservative field and curl along a helical path)	
Exercise 7.17 (Curl, flux, and verification of Stokes' theorem)	
Exercise 7.18 (Divergence, flux, and verification of Gauss's theorem)	

Chapter 8: Multiple Integrals

Basic theorems	
Change of variables and coordinate systems	
Physical and geometric applications	
Exercise 8.1 (Double integral over a rectangle)	
Exercise 8.2 (Region between two curves)	
Exercise 8.3 (Change of variables $u = y/x$, $v = x$)	
Exercise 8.4 (Polar integral over a disk)	
Exercise 8.5 (Integral of $x^2 + y^2$ over a disk)	
Exercise 8.6 (Integral of xy over a triangle)	
Exercise 8.7 (Integral of z over a tetrahedron)	
Exercise 8.8 (Change of variables u, v, w for a tetrahedron)	

Exercise 8.9	(Integral of xyz over a cube)	
Exercise 8.10	(Volume under a paraboloid)	
Exercise 8.11	(Volume of a paraboloid above a disk)	
Exercise 8.12	(Volume between a cone and a hemisphere)	
Exercise 8.13	(Volume of a cylinder with a slanted roof)	
Exercise 8.14	(Volume of the intersection of two cylinders)	
Exercise 8.15	(Volume between a paraboloid and a plane)	
Exercise 8.16	(Integral of z over a cylinder)	
Exercise 8.17	(Integral of $x^2 + y^2$ under a paraboloid)	
Exercise 8.18	(Volume of a sphere (two methods))	
Exercise 8.19	(Integral of z over a hemisphere)	
Exercise 8.20	(Volume between two symmetric paraboloids)	
Exercise 8.21	(Integral of $x^2 + y^2 + z^2$ over a sphere)	
Exercise 8.22	(Integral of y over a cylinder)	
Exercise 8.23	(Transformation $u = x + y, v = x - y$ for $\sin \cos$)	
Exercise 8.24	(Area of a rhombus $ x + y = 1$ via transformation)	
Exercise 8.25	(Transformation u, v, w for a tetrahedron)	
Exercise 8.26	(Integral of $\sqrt{x^2 + y^2}$ between two paraboloids)	
Exercise 8.27	(Center of mass of a hemisphere)	
Exercise 8.28	(Volume between a cone and a sphere)	
Exercise 8.29	(Integral of z over a cylinder (iterated))	
Exercise 8.30	(Integral of $x^2 + y^2$ under a paraboloid (iterated))	
Exercise 8.31	(Center of mass of a hemisphere (another method))	
Exercise 8.32	(Volume and integral of z between two paraboloids)	
Exercise 8.33	(Volume of the intersection of two cylinders)	
Exercise 8.34	(Volume between a paraboloid and an inclined plane)	
Exercise 8.35	(Volume under a paraboloid and a horizontal plane)	
Exercise 8.36	(Integral of $x^2 + z^2$ in a cylindrical domain)	
Exercise 8.37	(Volume of a rectangular box via transformation u, v, w)	
Exercise 8.38	(Integral of y over the intersection of two cylinders)	
Exercise 8.39	(Spherical integral with $1/(1 + r^2)^{3/2}$)	

Chapter 9: Complex Integration

Basic theorems	
Residue theorem and applications	
Laurent series and Rouché's theorem	
Exercise 9.1 (Integral of z^2 over a line segment)	
Exercise 9.2 (Integral of $\frac{1}{z^2+1}$ over the circle $ z = 2$)	
Exercise 9.3 (Integral of $\frac{e^z}{z(z-1)^2}$ over $ z = 2$)	
Exercise 9.4 (Real integral $\int_0^{2\pi} \frac{d\theta}{5+4\cos\theta}$)	
Exercise 9.5 (Laurent series of $\frac{\sin z}{z^3}$ and its integral)	
Exercise 9.6 (Integral of $\frac{e^{2z}}{z^2(z-1)}$ using Cauchy's formula)	
Exercise 9.7 (Entire function bounded by $ f(z) \leq z ^2$)	
Exercise 9.8 (Integral of $\frac{z^2}{(z-1)(z-3)}$ over two circles)	
Exercise 9.9 (Integral $\int_{-\infty}^{\infty} \frac{x \sin x}{x^2+4} dx$)	
Exercise 9.10 (Integral of $\frac{1}{z^4+1}$ over the boundary of a rectangle)	
Exercise 9.11 (Application of Rouché's theorem to count roots of $z^5 + 3z^2 + 1$)	
Exercise 9.12 (Integral $\int_0^{\infty} \frac{\cos(ax)}{x^2+1} dx$)	
Exercise 9.13 (Laurent series of $\frac{1}{(z-1)(z-3)}$ in three regions)	
Exercise 9.14 (Integral of $\frac{z^2+1}{z(z^2+4)}$ over $ z = 3$)	
Exercise 9.15 (Inequality for the derivative of an analytic function in the unit disk)	

Applications of Differentiation

Theorem

1. Let f be a function continuous on $[a, b]$ and differentiable on $]a, b[$ satisfying

$$f(a) = f(b).$$

Then there exists $c \in]a, b[$ such that:

$$f'(c) = 0.$$

2. Let f be continuous on $[a, b]$ and differentiable on $]a, b[$. Then there exists $c \in]a, b[$ such that:

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

3. If $|f'(x)| \leq M$ for all $x \in [a, b]$, then:

$$|f(b) - f(a)| \leq M|b - a|.$$

4. Let f be differentiable on an interval I . If $f'(x_0) = 0$ and the sign of f' changes at x_0 , then f has a local extremum at x_0 .

5. Let f be twice differentiable on an interval I .

- If $f''(x_0) > 0$, then x_0 is a local minimum point.
- If $f''(x_0) < 0$, then x_0 is a local maximum point.

6. Let f be twice differentiable on an interval I .

- If $f''(x) \geq 0$, then f is convex on I .
- If $f''(x) \leq 0$, then f is concave on I .

7. If the sign of f'' changes at x_0 , then the point $(x_0, f(x_0))$ is an inflection point.

8. The equation of the tangent line to the curve $y = f(x)$ at the point with abscissa x_0 is:

$$y = f(x_0) + f'(x_0)(x - x_0).$$

9. If f is differentiable at x_0 , then near x_0 :

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + o(x - x_0).$$

10. If f is continuous on a closed bounded interval $[a, b]$, then f attains a minimum and a maximum value on $[a, b]$.

11. If $f'(x) \geq g'(x)$ on I and $f(x_0) = g(x_0)$, then:

$$f(x) \geq g(x) \quad \forall x \in I.$$

CHAPTER 1. APPLICATIONS OF DIFFERENTIATION

Exercise

The speed of a rocket moving vertically is given by:

$$v(t) = 12t - t^2.$$

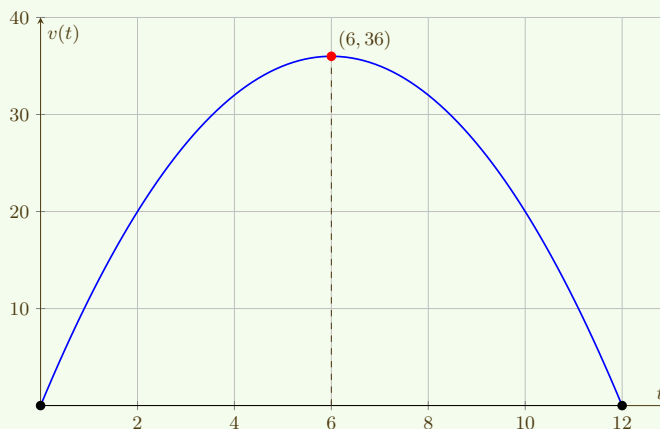
- Find the moment when the speed is maximum.
- Determine the position $y(t)$ given that $y(0) = 0$.

Solution

$$v'(t) = 12 - 2t.$$

The speed is maximum when:

$$12 - 2t = 0 \Rightarrow t = 6.$$



The motion is vertical, so it is along one axis:

$$y(t) = \int (12t - t^2) dt = 6t^2 - \frac{1}{3}t^3 + C.$$

Since $y(0) = 0$, we have $C = 0$.

Exercise

The position of a moving point on a straight line is given as a function of time by:

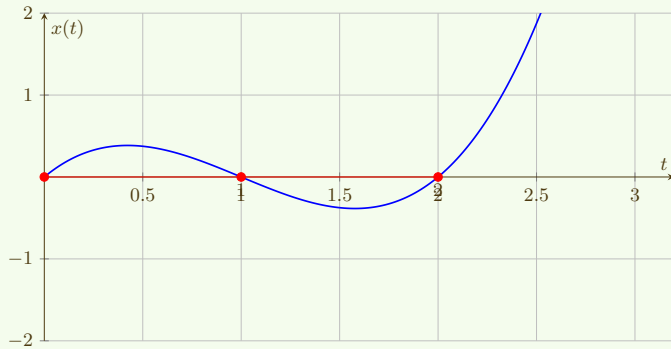
$$x(t) = t^3 - 3t^2 + 2t.$$

Find the velocity and acceleration, determine the stopping times, then calculate the distance traveled between $t = 0$ and $t = 2$.

Solution

Graph of the position as a function of time:

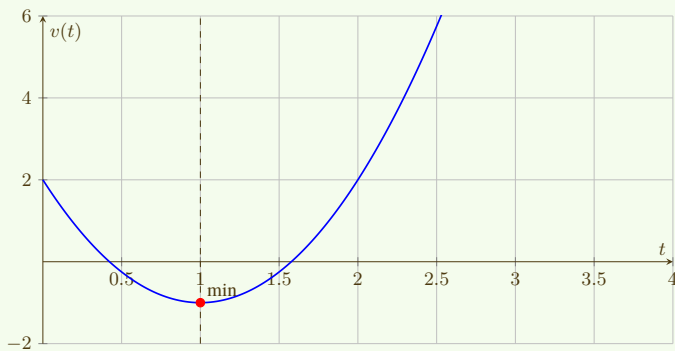
CHAPTER 1. APPLICATIONS OF DIFFERENTIATION



Velocity is the derivative of position:

$$v(t) = x'(t) = 3t^2 - 6t + 2,$$

Graph of velocity:



Acceleration:

$$a(t) = v'(t) = 6t - 6.$$

Stopping times satisfy:

$$v(t) = 0 \iff 3t^2 - 6t + 2 = 0 \iff t = 1 \pm \frac{1}{\sqrt{3}}.$$

Let:

$$t_1 = 1 - \frac{1}{\sqrt{3}}, \quad t_2 = 1 + \frac{1}{\sqrt{3}}.$$

On the interval $[0, 2]$, the sign of velocity changes at t_1 and t_2 , so the distance traveled is:

$$D = \int_0^2 |v(t)| dt = \int_0^{t_1} v(t) dt - \int_{t_1}^{t_2} v(t) dt + \int_{t_2}^2 v(t) dt.$$

Since:

$$\int v(t) dt = x(t),$$

we have:

$$D = [x(t_1) - x(0)] - [x(t_2) - x(t_1)] + [x(2) - x(t_2)].$$

Since:

$$x(0) = 0, \quad x(2) = 0,$$

CHAPTER 1. APPLICATIONS OF DIFFERENTIATION

we obtain:

$$D = 2(x(t_1) - x(t_2)).$$

Substituting:

$$x(t_1) = \frac{2}{3\sqrt{3}}, \quad x(t_2) = -\frac{2}{3\sqrt{3}}.$$

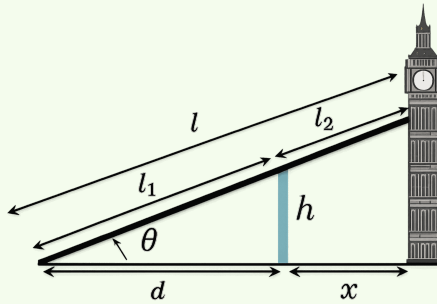
Thus:

$$D = 2 \left(\frac{2}{3\sqrt{3}} + \frac{2}{3\sqrt{3}} \right) = \frac{8}{3\sqrt{3}} = \frac{8\sqrt{3}}{9}.$$

$$D = \frac{8\sqrt{3}}{9}.$$

Exercise

A ladder is used to reach the top of a building. The ladder rests on the horizontal ground and leans against a wall of height h , reaching the wall that is at a distance x from the wall as shown in the figure below.



Let θ be the angle the ladder makes with the ground. Find the minimum length $l_{\min}$ of the ladder.

Solution

From the figure, the ladder length:

$$l = l_1 + l_2.$$

Geometrically:

$$\begin{aligned} \sin \theta &= \frac{h}{l_1} \implies l_1 = \frac{h}{\sin \theta} \\ \cos \theta &= \frac{d}{l_1} = \frac{d+x}{l} = \frac{(d+x) - d}{l - l_1} = \frac{x}{l_2} \implies l_2 = \frac{x}{\cos \theta}. \end{aligned}$$

Thus the function for the ladder length as a function of the angle to find the minimum length is:

$$l(\theta) = \frac{h}{\sin \theta} + \frac{x}{\cos \theta}, \quad 0 < \theta < \frac{\pi}{2}.$$

2. Differentiation

$$l'(\theta) = -\frac{h \cos \theta}{\sin^2 \theta} + \frac{x \sin \theta}{\cos^2 \theta}.$$

Simplifying:

$$l'(\theta) = \frac{x \sin^3 \theta - h \cos^3 \theta}{\sin^2 \theta \cos^2 \theta}.$$

3. Minimum length condition: derivative zero, decreasing then increasing

CHAPTER 1. APPLICATIONS OF DIFFERENTIATION

$$l'(\theta) = 0 \iff x \sin^3 \theta = h \cos^3 \theta \iff \tan^3 \theta = \frac{h}{x}.$$

Thus:

$$\tan \theta_0 = \sqrt[3]{\frac{h}{x}}.$$

The function is decreasing when:

$$l'(\theta) \leq 0 \implies \frac{x \sin^3 \theta - h \cos^3 \theta}{\sin^2 \theta \cos^2 \theta} \leq 0 \implies \tan \theta \leq \sqrt[3]{\frac{h}{x}}$$

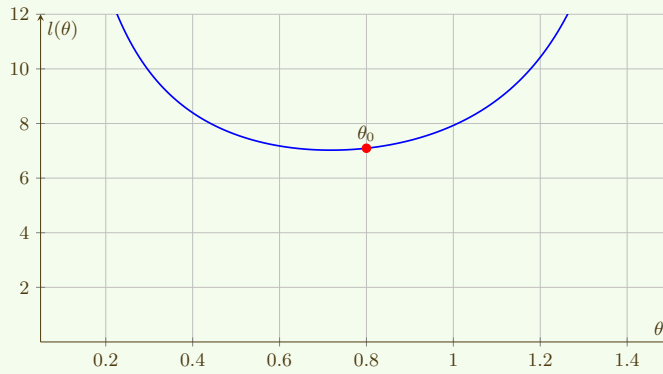
i.e., on the interval:

$$\theta \in \left(0, \arctan \left(\sqrt[3]{\frac{h}{x}} \right) \right]$$

and increasing on:

$$\theta \in \left[\arctan \left(\sqrt[3]{\frac{h}{x}} \right), \frac{\pi}{2} \right)$$

Approximate graph:



4. Minimum length

Let:

$$k = \tan \theta_0 = \sqrt[3]{\frac{h}{x}}.$$

Then:

$$\sin \theta_0 = \frac{k}{\sqrt{1+k^2}}, \quad \cos \theta_0 = \frac{1}{\sqrt{1+k^2}}.$$

Thus:

$$l_{\min} = \frac{h}{\sin \theta_0} + \frac{x}{\cos \theta_0} = \sqrt{1+k^2} \left(\frac{h}{k} + x \right).$$

Substituting $k = \sqrt[3]{h/x}$, and after simplification:

$$l_{\min} = \left(h^{2/3} + x^{2/3} \right)^{3/2}.$$

Result

The minimum length of the ladder is:

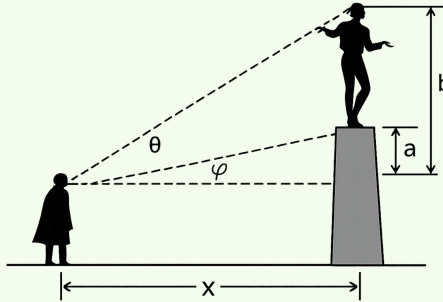
$$l_{\min} = \left(h^{2/3} + x^{2/3} \right)^{3/2},$$

achieved at the angle:

$$\theta_0 = \arctan \left(\sqrt[3]{\frac{h}{x}} \right).$$

Exercise

At what position x should a person stand to view a statue such that the viewing angle θ is maximum?



Solution

Study of the viewing angle function $\theta(x)$

1. Expression of the angle $\theta(x)$

From the geometry:

$$\theta = f(x) = (\theta + \phi) - \phi = \arctan\left(\frac{b}{x}\right) - \arctan\left(\frac{a}{x}\right) \quad ; \quad b > a > 0$$

2. Derivative $\theta'(x)$

$$\frac{d}{dx} \arctan\left(\frac{k}{x}\right) = -\frac{k}{x^2 + k^2}$$

Thus:

$$\theta'(x) = -\frac{b}{x^2 + b^2} + \frac{a}{x^2 + a^2}$$

$$\theta'(x) = \frac{a(x^2 + b^2) - b(x^2 + a^2)}{(x^2 + a^2)(x^2 + b^2)}$$

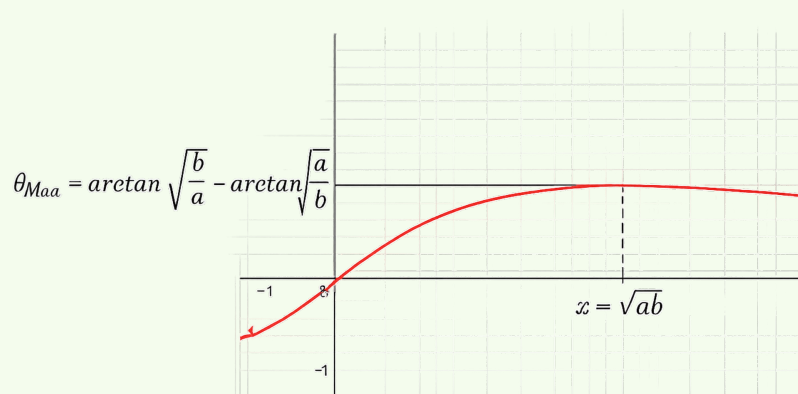
$$\theta'(x) = \frac{(a - b)(x^2 - ab)}{(x^2 + a^2)(x^2 + b^2)}$$

3. Sign analysis

Since $b > a > 0$:

$$\theta'(x) \begin{cases} > 0 & \text{if } 0 < x < \sqrt{ab} \\ = 0 & \text{if } x = \sqrt{ab} \\ < 0 & \text{if } x > \sqrt{ab} \end{cases}$$

4. Table of variations (graphical representation)



5. Maximum angle value

The maximum occurs at:

$$x_0 = \sqrt{ab}$$

Thus:

$$\theta_{\max} = \arctan\left(\frac{b}{\sqrt{ab}}\right) - \arctan\left(\frac{a}{\sqrt{ab}}\right)$$

$$\theta_{\max} = \arctan\left(\sqrt{\frac{b}{a}}\right) - \arctan\left(\sqrt{\frac{a}{b}}\right)$$

After simplification:

$$\theta_{\max} = \arctan\left(\frac{b-a}{2\sqrt{ab}}\right)$$

Conclusion: To maximize the viewing angle of the statue, one should stand at a distance $\sqrt{ab}$ from its base.

Exercise

Let a rectangle have a fixed perimeter $P = 20$ cm.

1. Find its dimensions that maximize its area. 2. Show that among all rectangles with fixed perimeter P , the square has the largest area. 3. Generalize the result for any perimeter P .

Solution

1. For $P = 20$ cm

Let x be the length and y the width.

$$2x + 2y = 20 \Rightarrow y = 10 - x$$

CHAPTER 1. APPLICATIONS OF DIFFERENTIATION

Area:

$$A(x) = x(10 - x) = 10x - x^2, \quad 0 < x < 10$$

Derivative:

$$A'(x) = 10 - 2x$$

$$A'(x) = 0 \Rightarrow x = 5$$

Sign: $A'(x) > 0$ on $(0, 5)$, $A'(x) < 0$ on $(5, 10)$.

Thus the area is maximum at $x = y = 5$: the optimal rectangle is a square.

2. Generalization for perimeter P

$$2x + 2y = P \Rightarrow y = \frac{P}{2} - x$$

$$A(x) = x \left(\frac{P}{2} - x \right) = \frac{P}{2}x - x^2$$

$$A'(x) = \frac{P}{2} - 2x$$

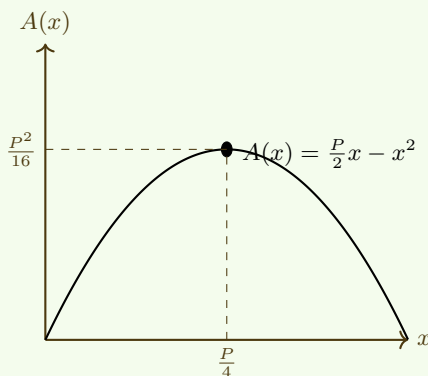
$$A'(x) = 0 \Rightarrow x = \frac{P}{4}, \quad y = \frac{P}{4}$$

Thus the maximum area is achieved for a square.

3. Maximum area

$$A_{\max} = \left(\frac{P}{4} \right)^2 = \frac{P^2}{16}$$

4. Graphical representation

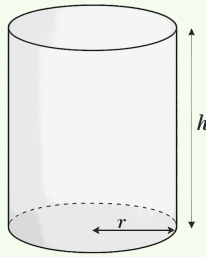


Square side $\frac{P}{4}$, $A_{\max} = \frac{P^2}{16}$

Conclusion: Among all rectangles with fixed perimeter, the square gives the largest possible area.

Exercise

We want to build a closed cylindrical tank with capacity $V = 1 \text{ m}^3$. The cost of the base and lid material is 10 euro/m², and the cost of the side wall material is 8 euro/m².



Find the dimensions r (radius) and h (height) that minimize the total construction cost.

Solution

1. Variables

Volume is:

$$V = \pi r^2 h = 1 \quad \Rightarrow \quad h = \frac{1}{\pi r^2}.$$

2. Areas

Area of base and lid:

$$S_1 = 2\pi r^2.$$

Lateral area:

$$S_2 = 2\pi r h = 2\pi r \cdot \frac{1}{\pi r^2} = \frac{2}{r}.$$

3. Cost function

$$C(r) = 10S_1 + 8S_2 = 10(2\pi r^2) + 8\left(\frac{2}{r}\right)$$

$$C(r) = 20\pi r^2 + \frac{16}{r}.$$

4. Derivative

$$C'(r) = 40\pi r - \frac{16}{r^2}.$$

5. Critical point

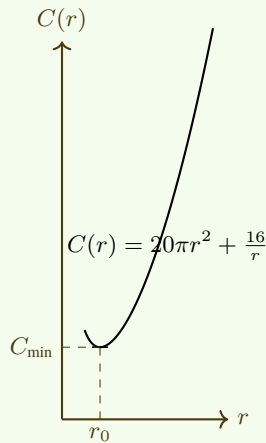
$$C'(r) = 0 \Rightarrow 40\pi r^3 = 16$$

$$r^3 = \frac{2}{5\pi} \quad \Rightarrow \quad r = \sqrt[3]{\frac{2}{5\pi}} \approx 0.503 \text{ m.}$$

6. Optimal height

$$h = \frac{1}{\pi r^2} \approx 1.26 \text{ m.}$$

7. Graphical representation



8. Final result

Radius $r \approx 0.503$ m, Height $h \approx 1.26$ m.

The cost is minimized at these dimensions.

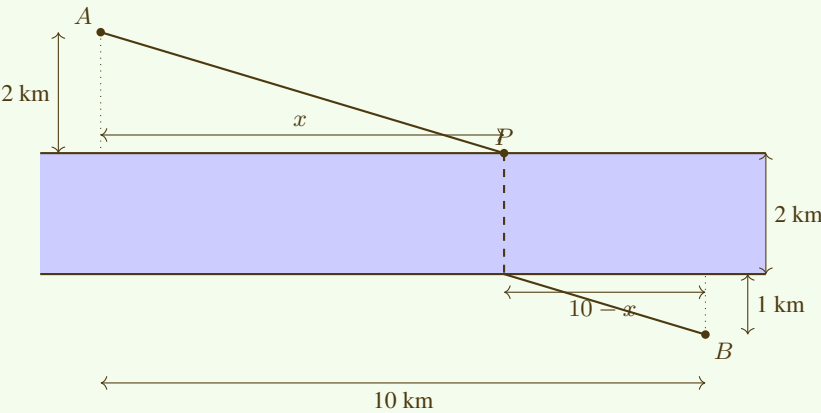
Exercise

An electricity company wants to connect two stations A and B located on opposite sides of a river of width 2 km. Station A is 2 km from the river, and B is 1 km from the river, and the horizontal distance between their projections is 10 km.

Land cost: 50, 000 euro/km, underwater cost: 80, 000 euro/km.

Find the point P on the river where the cable should enter the water so that the total cost is minimized.

Solution



1. Mathematical formulation

Let x be the horizontal distance between A and P , where $0 \leq x \leq 10$.

2. Distances

$$AP = \sqrt{x^2 + 4}, \quad PB = \sqrt{(10 - x)^2 + 1}.$$

The underwater part is constant at 2 km.

3. Function to minimize

The total cost is:

$$C(x) = 50,000(AP + PB) + 80,000 \times 2.$$

Thus it suffices to minimize:

$$f(x) = \sqrt{x^2 + 4} + \sqrt{(10 - x)^2 + 1}.$$

4. Derivative

$$f'(x) = \frac{x}{\sqrt{x^2 + 4}} - \frac{10 - x}{\sqrt{(10 - x)^2 + 1}}.$$

5. Optimality condition

$$f'(x) = 0 \iff \frac{x}{\sqrt{x^2 + 4}} = \frac{10 - x}{\sqrt{(10 - x)^2 + 1}}.$$

After simplification:

$$x^2 = 4(10 - x)^2 \implies x = \frac{20}{3}.$$

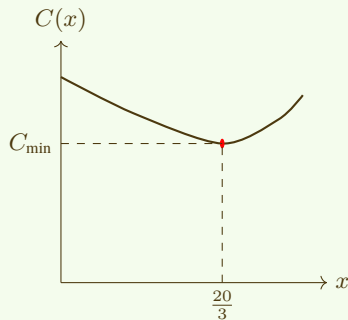
6. Result

$$x = \frac{20}{3} \approx 6.67 \text{ km.}$$

The minimum cost is:

$$C_{\min} \approx 682,750 \text{ euro.}$$

7. Graphical representation

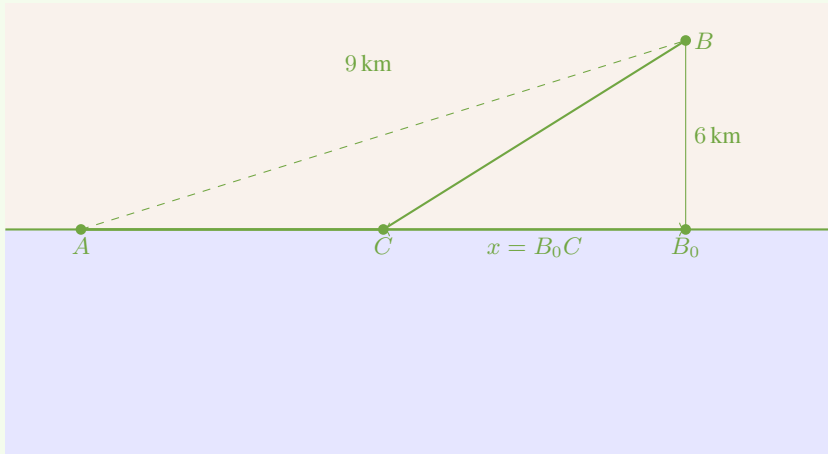


$$x = \frac{20}{3} \text{ km} \approx 6.67 \text{ km}$$

Exercise

A company wants to lay a gas pipeline from point A located on the coast to point B located on an island, such that $AB = 9$ km and $BB_0 = 6$ km.

Pipeline cost is P dollars per km on land, and Q dollars per km underwater. Given: $Q = \sqrt{5}P$.



For a chosen point C on the coast, the pipeline path is adjusted so that the total cost is minimized.

- Express the total cost $K(x)$ in terms of $x = B_0C$.
- Determine the position of C that minimizes this cost.
- Calculate the total pipeline length and the minimum cost.

Solution

1. Geometry of the problem

In right triangle ABB_0 :

$$\begin{aligned} AB^2 &= AB_0^2 + BB_0^2 \\ 9^2 &= AB_0^2 + 6^2 \\ AB_0^2 &= 81 - 36 = 45 \\ AB_0 &= 3\sqrt{5} \end{aligned}$$

Let $x = B_0C$ where $x \geq 0$. Thus:

$$AC = AB_0 - x = 3\sqrt{5} - x$$

Underwater length:

$$CB = \sqrt{x^2 + 6^2} = \sqrt{x^2 + 36}$$

2. Cost function

$$\begin{aligned} K(x) &= P \cdot AC + Q \cdot CB \\ K(x) &= P(3\sqrt{5} - x) + \sqrt{5}P\sqrt{x^2 + 36} \end{aligned}$$

Minimize:

$$f(x) = \frac{K(x)}{P} = 3\sqrt{5} - x + \sqrt{5}\sqrt{x^2 + 36}$$

3. Differentiation

$$f'(x) = -1 + \frac{\sqrt{5}x}{\sqrt{x^2 + 36}}$$

Optimality condition:

$$\begin{aligned}
 f'(x) = 0 &\implies \frac{\sqrt{5}x}{\sqrt{x^2 + 36}} = 1 \\
 \sqrt{5}x &= \sqrt{x^2 + 36} \\
 5x^2 &= x^2 + 36 \\
 4x^2 &= 36 \implies x^2 = 9 \\
 x &= 3 \quad (x > 0)
 \end{aligned}$$

4. Minimum length and cost

$$\begin{aligned}
 AC &= 3\sqrt{5} - 3 \\
 CB &= \sqrt{3^2 + 36} = \sqrt{45} = 3\sqrt{5}
 \end{aligned}$$

Total length:

$$L = AC + CB = (3\sqrt{5} - 3) + 3\sqrt{5} = 6\sqrt{5} - 3$$

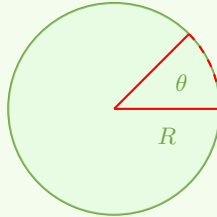
Minimum cost:

$$\begin{aligned}
 K_{\min} &= P(AC + \sqrt{5}CB) \\
 K_{\min} &= P(3\sqrt{5} - 3 + \sqrt{5} \cdot 3\sqrt{5}) \\
 K_{\min} &= P(3\sqrt{5} - 3 + 15) \\
 K_{\min} &= P(12 + 3\sqrt{5})
 \end{aligned}$$

$$x = 3 \text{ km}, \quad L = 6\sqrt{5} - 3 \text{ km}, \quad K_{\min} = P(12 + 3\sqrt{5})$$

Exercise

We want to construct a right circular cone from a cardboard disk of radius $R = 10$ cm by cutting a sector and then bringing the edges together.



Let θ be the angle of the cut sector (in radians), $0 < \theta < 2\pi$.

- Express the base radius r and height h of the cone in terms of θ .
- Express the volume $V(\theta)$ of the cone.
- Find θ that maximizes the volume.
- Calculate the maximum volume.

Solution

1. Geometric relations

The circumference of the cone's base equals the remaining arc length:

$$2\pi r = R(2\pi - \theta) \implies r = \frac{R(2\pi - \theta)}{2\pi}$$

The slant height of the cone is R , thus:

$$h = \sqrt{R^2 - r^2}$$

2. Volume of the cone

$$V(\theta) = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi r^2 \sqrt{R^2 - r^2}$$

where $r = \frac{R(2\pi - \theta)}{2\pi}$.

For $R = 10$:

$$r = \frac{10(2\pi - \theta)}{2\pi} = \frac{10}{\pi} \left(\pi - \frac{\theta}{2} \right)$$

3. Maximizing the volume

Let $t = 2\pi - \theta$, then $t \in (0, 2\pi)$ and $r = \frac{Rt}{2\pi}$.

$$V(t) = \frac{1}{3}\pi \left(\frac{Rt}{2\pi} \right)^2 \sqrt{R^2 - \left(\frac{Rt}{2\pi} \right)^2}$$

$$V(t) = \frac{R^3}{12\pi^2} t^2 \sqrt{4\pi^2 - t^2}$$

Maximize $f(t) = t^2 \sqrt{4\pi^2 - t^2}$.

$$f'(t) = 2t\sqrt{4\pi^2 - t^2} + t^2 \cdot \frac{-t}{\sqrt{4\pi^2 - t^2}}$$

$$= \frac{2t(4\pi^2 - t^2) - t^3}{\sqrt{4\pi^2 - t^2}}$$

$$= \frac{8\pi^2 t - 2t^3 - t^3}{\sqrt{4\pi^2 - t^2}} = \frac{8\pi^2 t - 3t^3}{\sqrt{4\pi^2 - t^2}}$$

$$f'(t) = 0 \Rightarrow 8\pi^2 t - 3t^3 = t(8\pi^2 - 3t^2) = 0$$

Solutions: $t = 0$ (minimum) and $t^2 = \frac{8\pi^2}{3}$.

$$t = \sqrt{\frac{8}{3}}\pi$$

$$\theta = 2\pi - t = 2\pi \left(1 - \sqrt{\frac{2}{3}} \right)$$

4. Maximum volume

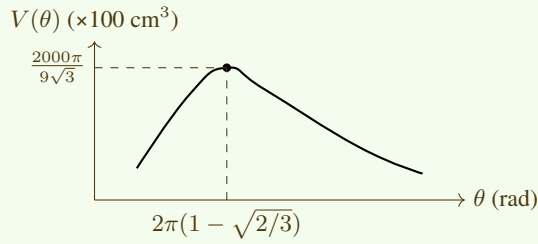
$$r = \frac{10t}{2\pi} = 5\sqrt{\frac{8}{3}} = \frac{10\sqrt{2}}{\sqrt{3}} \text{ cm}$$

$$h = \sqrt{100 - r^2} = \sqrt{100 - \frac{200}{3}} = \sqrt{\frac{100}{3}} = \frac{10}{\sqrt{3}} \text{ cm}$$

$$V_{\max} = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi \cdot \frac{200}{3} \cdot \frac{10}{\sqrt{3}}$$

$$V_{\max} = \frac{2000\pi}{9\sqrt{3}} \text{ cm}^3 \approx 402.6 \text{ cm}^3$$

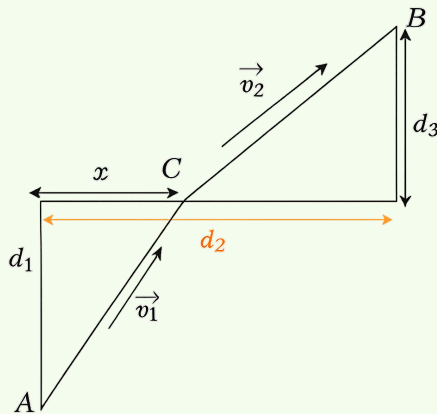
5. Graphical representation



$$\theta = 2\pi \left(1 - \sqrt{\frac{2}{3}} \right) \text{ rad}, \quad V_{\max} = \frac{2000\pi}{9\sqrt{3}} \text{ cm}^3$$

Exercise

A man is at point A on the beach and wants to reach point B in the sea in the shortest possible time. He can run on the beach at speed $v_1 = 5 \text{ m/s}$ and swim at speed $v_2 = 2 \text{ m/s}$.



Given: $d_1 = 100 \text{ m}$, $d_2 = 50 \text{ m}$, $d_3 = 80 \text{ m}$.

Point C lies on the boundary between the beach and the sea. Find the position of C that minimizes the total time.

Solution

1. Variables

Let x be the distance between the point opposite A on the coast and point C (where $0 \leq x \leq d_2$).

Distance on land: $AC = \sqrt{d_1^2 + x^2}$.

Distance in sea: $CB = \sqrt{(d_2 - x)^2 + d_3^2}$.

2. Total time

$$T(x) = \frac{AC}{v_1} + \frac{CB}{v_2}$$

$$T(x) = \frac{\sqrt{100^2 + x^2}}{5} + \frac{\sqrt{(50-x)^2 + 80^2}}{2}$$

3. Derivative

$$T'(x) = \frac{1}{5} \cdot \frac{x}{\sqrt{100^2 + x^2}} - \frac{1}{2} \cdot \frac{50-x}{\sqrt{(50-x)^2 + 80^2}}$$

4. Minimum condition

$$T'(x) = 0 \Rightarrow \frac{x}{5\sqrt{100^2 + x^2}} = \frac{50-x}{2\sqrt{(50-x)^2 + 80^2}}$$

Let $u = \frac{x}{\sqrt{100^2 + x^2}} = \sin \alpha$ where α is the angle between AC and the perpendicular.

We have:

$$\frac{\sin \alpha}{5} = \frac{\sin \beta}{2}$$

where β is the angle between CB and the perpendicular.

This is **Snell-Descartes law**: $\frac{\sin \alpha}{v_2} = \frac{\sin \beta}{v_1}$ with $v_1 > v_2 \rightarrow \sin \alpha < \sin \beta$.

5. Numerical solution

Solve:

$$\frac{x}{5\sqrt{100^2 + x^2}} = \frac{50-x}{2\sqrt{(50-x)^2 + 80^2}}$$

After squaring and simplifying:

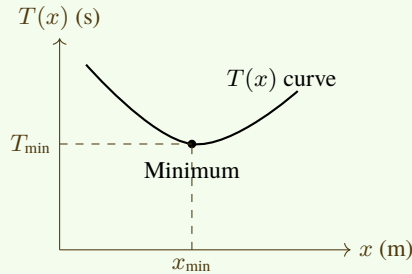
$$4x^2[(50-x)^2 + 6400] = 25(50-x)^2(100^2 + x^2)$$

Numerical solution (by iteration or calculator): $x \approx 31.2$ m.

6. Minimum time

$$\begin{aligned} T_{\min} &\approx \frac{\sqrt{100^2 + 31.2^2}}{5} + \frac{\sqrt{(18.8)^2 + 80^2}}{2} \\ &\approx \frac{104.8}{5} + \frac{82.2}{2} = 20.96 + 41.1 = 62.06 \text{ s} \end{aligned}$$

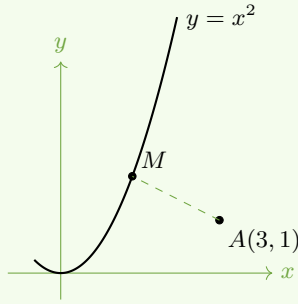
7. Graphical representation



$$x \approx 31.2 \text{ m}, \quad T_{\min} \approx 62.06 \text{ s}$$

Exercise

Consider the parabola $\mathcal{P} : y = x^2$ and the point $A(3, 1)$.



Find the point M on $\mathcal{P}$ closest to A and determine the minimum distance.

Solution

1. Expression of the distance

Let $M(x, x^2)$ be an arbitrary point on the parabola. The squared distance between A and M is:

$$\overrightarrow{AM} = \begin{pmatrix} x-3 \\ y-1 \end{pmatrix}, \quad D(x) = \|\overrightarrow{AM}\| = \sqrt{(x-3)^2 + (y-1)^2}.$$

$$y = x^2, \quad D(x) = \sqrt{(x-3)^2 + (x^2-1)^2}.$$

$$D^2(x) = (x-3)^2 + (x^2-1)^2.$$

2. Study of the function

Let

$$f(x) = (x-3)^2 + (x^2-1)^2.$$

Then

$$f'(x) = 2(x-3) + 2(x^2-1) \cdot 2x = 4x^3 - 2x - 6.$$

3. Finding the minimum point

Solve

$$f'(x) = 0 \iff 4x^3 - 2x - 6 = 0.$$

Numerical solution (using a calculator or Newton's method) gives:

$$x \approx 1.353.$$

Thus,

$$M(1.353, 1.353^2) \approx (1.353, 1.830).$$

4. Minimum distance

$$d_{\min} = \sqrt{(1.353-3)^2 + (1.830-1)^2} \approx \sqrt{3.401} \approx 1.844.$$

$$\boxed{M(1.353, 1.830), \quad d_{\min} \approx 1.844}$$

Exercise

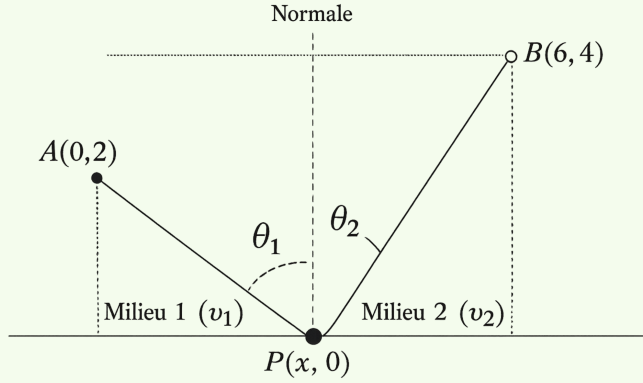
A light ray connects the two points

$$A(0, 2) \quad \text{and} \quad B(6, 4)$$

CHAPTER 1. APPLICATIONS OF DIFFERENTIATION

crossing two media separated by the straight line

$$y = 0.$$



Determine the value of x that minimizes the travel time.

Solution

Assume the speed of light in the first medium is v_1 and in the second medium is v_2 . The crossing point on the interface $y = 0$ is

$$P(x, 0).$$

The distance between A and P is

$$AP = \sqrt{(x-0)^2 + (0-2)^2} = \sqrt{x^2 + 4}.$$

The distance between P and B is

$$PB = \sqrt{(6-x)^2 + (4-0)^2} = \sqrt{(6-x)^2 + 16}.$$

The total travel time is therefore

$$T(x) = \frac{AP}{v_1} + \frac{PB}{v_2} = \frac{\sqrt{x^2 + 4}}{v_1} + \frac{\sqrt{(6-x)^2 + 16}}{v_2}.$$

To minimize this time, we differentiate T with respect to x :

$$T'(x) = \frac{1}{v_1} \cdot \frac{x}{\sqrt{x^2 + 4}} - \frac{1}{v_2} \cdot \frac{6-x}{\sqrt{(6-x)^2 + 16}}.$$

The extremum condition is

$$T'(x) = 0,$$

which gives

$$\frac{x}{v_1 \sqrt{x^2 + 4}} = \frac{6-x}{v_2 \sqrt{(6-x)^2 + 16}}.$$

We now interpret this relation geometrically. If θ_1 and θ_2 are the angles that the rays AP and PB make with the normal to the interface at point P , then

$$\sin \theta_1 = \frac{x}{\sqrt{x^2 + 4}}, \quad \sin \theta_2 = \frac{6-x}{\sqrt{(6-x)^2 + 16}}.$$

CHAPTER 1. APPLICATIONS OF DIFFERENTIATION

Thus the previous relation becomes

$$\frac{\sin \theta_1}{v_1} = \frac{\sin \theta_2}{v_2}.$$

This is Snell-Descartes law of refraction.

Thus the value of x that minimizes the travel time is the solution of the equation

$$\frac{x}{v_1 \sqrt{x^2 + 4}} = \frac{6 - x}{v_2 \sqrt{(6 - x)^2 + 16}}.$$

Without numerical values for v_1 and v_2 we stop at this general formula. If v_1 and v_2 are known, this equation can be solved numerically.

For example, for given values of v_1 and v_2 we obtain approximately

$$x \approx 2.57.$$

Thus,

$$x \text{ is determined by } \frac{x}{v_1 \sqrt{x^2 + 4}} = \frac{6 - x}{v_2 \sqrt{(6 - x)^2 + 16}}.$$

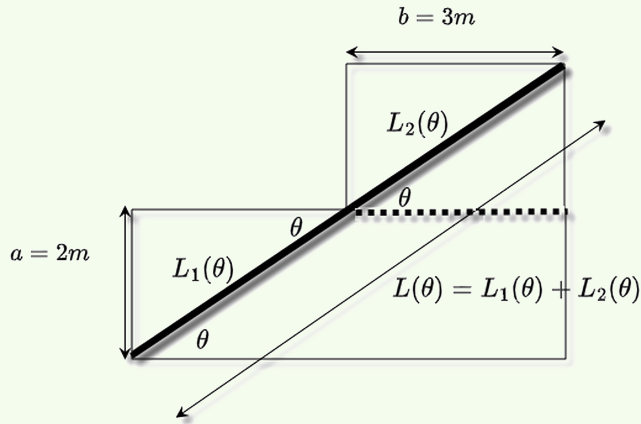
Exercise

A rod must pass through a corridor consisting of two branches of widths

$$a = 2 \text{ m}, \quad b = 3 \text{ m}.$$

Determine the maximum length of a rod that can be transported through this corridor.

Solution



Consider a rod that, in the critical position, touches both outer walls and the inner corner of the corridor. Let θ be the angle the rod makes with the horizontal wall, where

$$0 < \theta < \frac{\pi}{2}.$$

In this position, the rod is divided into two parts: one in the vertical branch, and the other in the horizontal branch.

The part in the corridor of width a has length

$$\frac{a}{\sin \theta},$$

CHAPTER 1. APPLICATIONS OF DIFFERENTIATION

because this segment is the hypotenuse of a right triangle whose side opposite the angle θ equals a . Similarly, the part in the corridor of width b has length

$$\frac{b}{\cos \theta},$$

because this segment is the hypotenuse of a right triangle whose side adjacent to the angle θ equals b .

Thus, the total length of the rod is given by

$$L(\theta) = L_1(\theta) + L_2(\theta) = \frac{a}{\sin \theta} + \frac{b}{\cos \theta}.$$

For a rod to pass, its length must not exceed the smallest possible value of $L(\theta)$. Thus the maximum transportable length is

$$L_{\max} = \min_{0 < \theta < \pi/2} L(\theta).$$

Compute the derivative:

$$L'(\theta) = -a \frac{\cos \theta}{\sin^2 \theta} + b \frac{\sin \theta}{\cos^2 \theta}.$$

Put over a common denominator:

$$L'(\theta) = \frac{-a \cos^3 \theta + b \sin^3 \theta}{\sin^2 \theta \cos^2 \theta}.$$

The extremum condition is

$$L'(\theta) = 0,$$

which gives

$$-a \cos^3 \theta + b \sin^3 \theta = 0.$$

Thus we obtain

$$\tan \theta_0 = \sqrt[3]{\frac{a}{b}}.$$

Substituting this value into the expression for $L(\theta)$,

$$L_{\max} = L_{\min}(\theta) = (a^{2/3} + b^{2/3})^{3/2}.$$

Given

$$a = 2, \quad b = 3.$$

$$L_{\max} = (2^{2/3} + 3^{2/3})^{3/2}.$$

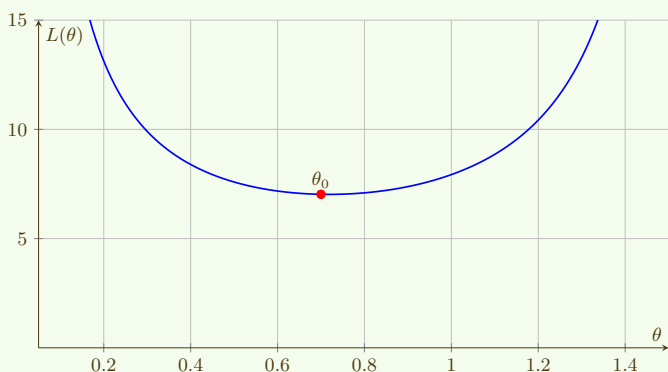
Numerical calculation:

$$2^{2/3} \approx 1.5874, \quad 3^{2/3} \approx 2.0801.$$

$$2^{2/3} + 3^{2/3} \approx 3.6675.$$

$$L_{\max} \approx (3.6675)^{3/2} = 3.6675 \sqrt{3.6675} \approx 3.6675 \times 1.915 \approx 7.02.$$

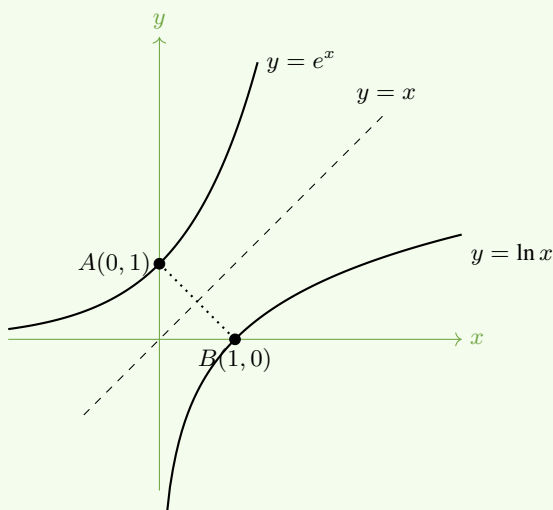
$L_{\max} \approx 7.02 \text{ m.}$



Exercise

Consider the following two curves:

$$C_1 : y = e^x \quad \text{and} \quad C_2 : y = \ln x$$



The two curves are symmetric with respect to the line $y = x$.
Find the minimum distance between a point on C_1 and a point on C_2 .

Solution

1. Using symmetry

The functions $f(x) = e^x$ and $g(x) = \ln x$ are inverse functions. Thus their curves are symmetric with respect to the line $y = x$.

Therefore the minimum distance between the curves is achieved between two symmetric points.

Let

$$A(a, e^a) \in C_1$$

Its reflection across the line $y = x$ is

$$B(e^a, a)$$

and this point belongs to C_2 because $a = \ln(e^a)$.

2. Distance between A and B

$$AB^2 = (a - e^a)^2 + (e^a - a)^2 = 2(a - e^a)^2$$

$$AB = \sqrt{2}|a - e^a|$$

3. Study of the function

Let

$$f(a) = e^a - a > 0$$

(because $e^a > a$ for every real number a)

$$f'(a) = e^a - 1$$

$$f'(a) = 0 \iff e^a = 1 \iff a = 0$$

$$f''(a) = e^a > 0$$

4. Minimum value

For $a = 0$:

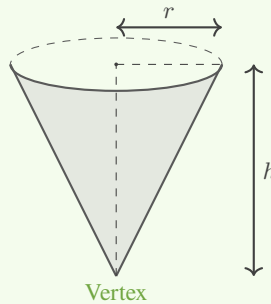
$$A(0, 1), \quad B(1, 0)$$

$$d_{\min} = \sqrt{2}(1 - 0) = \sqrt{2}$$

$$d_{\min} = \sqrt{2}$$

Exercise

A conical tank with its vertex at the bottom must be built to contain a water volume of $V = 100 \text{ m}^3$. The cost of the lateral surface is 30 euro/m², and the cost of the circular base is 20 euro/m².



Find the dimensions r (radius) and h (height) that minimize the total cost.

Solution

1. Volume relation

The volume of the cone is given by:

$$V = \frac{1}{3}\pi r^2 h$$

Given $V = 100$, thus:

$$\frac{1}{3}\pi r^2 h = 100 \implies h = \frac{300}{\pi r^2}$$

2. Areas and cost

- Base area:

$$A_f = \pi r^2$$

Corresponding cost:

$$C_f = 20\pi r^2$$

- Lateral area:

$$A_l = \pi r l \quad \text{where} \quad l = \sqrt{r^2 + h^2}$$

Corresponding cost:

$$C_l = 30\pi r \sqrt{r^2 + h^2}$$

3. Cost function

The total cost is:

$$C(r) = 20\pi r^2 + 30\pi r \sqrt{r^2 + h^2}$$

Substituting $h = \frac{300}{\pi r^2}$:

$$C(r) = 20\pi r^2 + 30\pi r \sqrt{r^2 + \left(\frac{300}{\pi r^2}\right)^2}$$

4. Study of the function

We want to find the minimum value of $C(r)$ for $r > 0$.

Let:

$$f(r) = C(r)/(10\pi)$$

This does not change the position of the minimum:

$$f(r) = 2r^2 + 3r \sqrt{r^2 + \frac{90000}{\pi^2 r^4}}$$

The derivative $f'(r)$ vanishes at a numerical value:

$$r \approx 4.05$$

5. Corresponding value of h

$$h = \frac{300}{\pi r^2} \approx \frac{300}{\pi(4.05)^2} \approx 5.82 \text{ m}$$

6. Minimum cost

$$C_{\min} \approx 20\pi(4.05)^2 + 30\pi(4.05)\sqrt{(4.05)^2 + (5.82)^2}$$

$$C_{\min} \approx 3730 \text{ euro}$$

These dimensions ensure the minimum construction cost.

Exercise

A company produces x units of a product.

The total cost is:

$$C(x) = 0.01x^3 - 0.6x^2 + 14x + 200$$

The selling price per unit is:

$$p(x) = 50 - 0.1x$$

- Express the revenue $R(x)$ and profit $B(x)$.
- Find the quantity x that maximizes the profit.
- Calculate the maximum profit.

Solution

1. Revenue and profit

Revenue is given by:

$$R(x) = x \cdot p(x) = x(50 - 0.1x) = 50x - 0.1x^2$$

Profit is the difference between revenue and cost:

$$B(x) = R(x) - C(x)$$

$$B(x) = (50x - 0.1x^2) - (0.01x^3 - 0.6x^2 + 14x + 200)$$

$$B(x) = -0.01x^3 + 0.5x^2 + 36x - 200$$

2. Maximizing profit

Differentiate the profit function:

$$B'(x) = -0.03x^2 + x + 36$$

Find critical points:

$$B'(x) = 0 \iff -0.03x^2 + x + 36 = 0$$

Multiply by -100 to simplify:

$$3x^2 - 100x - 3600 = 0$$

Discriminant:

$$\Delta = (-100)^2 + 4 \cdot 3 \cdot 3600 = 10000 + 43200 = 53200$$

$$\sqrt{\Delta} \approx 230.65$$

Solutions:

$$x = \frac{100 \pm 230.65}{6}$$

The economically acceptable solution:

$$x_{\max} = \frac{100 + 230.65}{6} \approx 55.1$$

3. Nature of the extremum

Compute the second derivative:

$$B''(x) = -0.06x + 1$$

$$B''(55.1) \approx -3.306 + 1 = -2.306 < 0$$

Thus the profit is **maximum** at $x \approx 55.1$.

4. Maximum profit

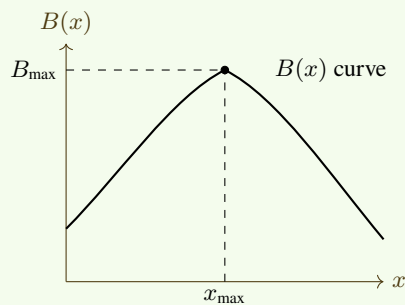
$$B(55.1) \approx -0.01(55.1)^3 + 0.5(55.1)^2 + 36(55.1) - 200$$

$$B_{\max} \approx 1629 \text{ euro}$$

5. Economic interpretation

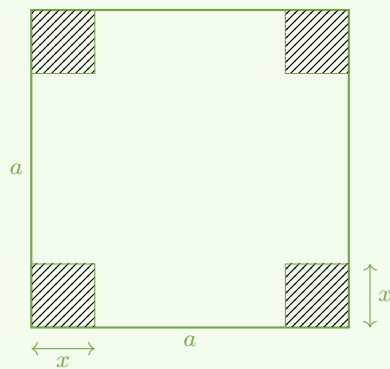
- Optimal production: ≈ 55 units
- Beyond this quantity, marginal cost exceeds marginal revenue
- Before this point, increasing production increases profit

6. Graphical representation



Exercise

We have a square cardboard sheet of side length a . We cut out small squares of side length x from its four corners, then fold the edges to obtain an open box.



The shaded squares are cut out

- Determine the value of x that maximizes the volume of the box.
- Calculate this maximum volume.

Solution

1. Volume expression

After cutting and folding, the resulting box has:

- Base length: $a - 2x$,
- Base width: $a - 2x$,
- Height: x .

The volume of the box is:

$$V(x) = x(a - 2x)^2$$

Domain:

$$0 < x < \frac{a}{2}$$

2. Differentiation

$$V(x) = x(a - 2x)^2$$

$$V'(x) = (a - 2x)^2 + x \cdot 2(a - 2x)(-2)$$

$$V'(x) = (a - 2x)^2 - 4x(a - 2x)$$

$$V'(x) = (a - 2x)(a - 6x)$$

3. Finding the maximum

$$V'(x) = 0 \iff (a - 2x)(a - 6x) = 0$$

Two solutions:

$$x = \frac{a}{2} \quad (\text{boundary of the domain})$$

$$x = \frac{a}{6}$$

Since:

$$V'(x) > 0 \text{ for } 0 < x < \frac{a}{6}$$

$$V'(x) < 0 \text{ for } \frac{a}{6} < x < \frac{a}{2}$$

We obtain a maximum at:

$$x_{\max} = \frac{a}{6}$$

4. Maximum volume

$$V_{\max} = V\left(\frac{a}{6}\right)$$

$$V_{\max} = \frac{a}{6} \left(a - \frac{a}{3}\right)^2$$

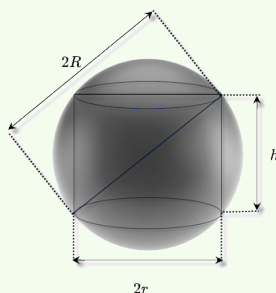
$$V_{\max} = \frac{a}{6} \left(\frac{2a}{3}\right)^2$$

$$V_{\max} = \frac{a}{6} \cdot \frac{4a^2}{9}$$

$$V_{\max} = \frac{2a^3}{27}$$

Exercise

What is the maximum possible volume of a cylinder inscribed in a sphere of fixed radius R ?



Solution

1. Mathematical formulation

Consider a sphere of radius R centered at the origin, and a right cylinder inscribed in it, with axis Oz , base radius r and height h .

A point on the top edge of the cylinder has coordinates:

$$P\left(r, 0, \frac{h}{2}\right)$$

Since this point belongs to the sphere, we have:

$$(2r)^2 + h^2 = (2R)^2$$

Thus the constraint is:

$$r^2 + \frac{h^2}{4} = R^2$$

2. Volume of the cylinder

The volume of the cylinder is:

$$V = \pi r^2 h$$

Using the constraint:

$$r^2 = R^2 - \frac{h^2}{4}$$

Thus:

$$V(h) = \pi \left(R^2 - \frac{h^2}{4} \right) h$$

$$V(h) = \pi R^2 h - \frac{\pi}{4} h^3$$

Domain:

$$0 < h < 2R$$

3. Finding the maximum

$$V'(h) = \pi R^2 - \frac{3\pi}{4} h^2$$

Solve:

$$V'(h) = 0 \iff R^2 - \frac{3}{4} h^2 = 0$$

$$h^2 = \frac{4R^2}{3}$$

$$h = \frac{2R}{\sqrt{3}}$$

4. Optimal radius

$$r^2 = R^2 - \frac{h^2}{4}$$

$$r^2 = R^2 - \frac{1}{4} \cdot \frac{4R^2}{3}$$

$$r^2 = \frac{2R^2}{3}$$

$$r = \frac{R\sqrt{2}}{\sqrt{3}}$$

5. Maximum volume

$$V_{\max} = \pi r^2 h$$

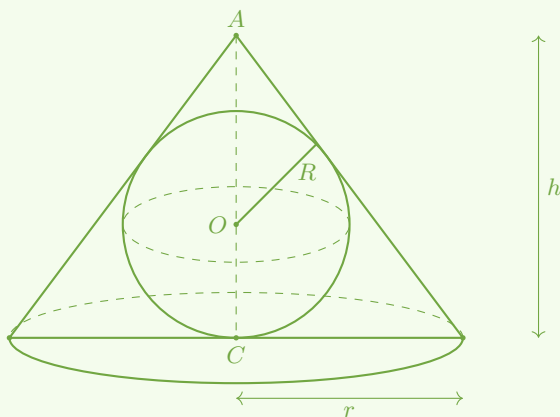
$$V_{\max} = \pi \cdot \frac{2R^2}{3} \cdot \frac{2R}{\sqrt{3}}$$

$$V_{\max} = \frac{4\pi R^3}{3\sqrt{3}}$$

$$V_{\max} = \frac{4\pi\sqrt{3}}{9} R^3$$

Exercise

What is the smallest volume of a right circular cone that can contain a sphere of fixed radius R ?



Right circular cone containing a sphere of radius R
Cone height h and base radius r

Solution

1. Geometric formulation using the angle

Let θ be half of the cone's apex angle, where:

$$\tan \theta = \frac{r}{h} \quad (1)$$

2. Tangency condition

In the axial cross-section, the center of the sphere is at distance R from the base and touches the lateral surface. The distance from the apex A to the center is $h - R$. From the right triangle:

$$\sin \theta = \frac{R}{h - R} \Rightarrow h = R \left(1 + \frac{1}{\sin \theta} \right) \quad (2)$$

3. Volume of the cone in terms of θ

$$V = \frac{1}{3} \pi r^2 h = \frac{1}{3} \pi (h \tan \theta)^2 h = \frac{1}{3} \pi h^3 \tan^2 \theta$$

Substituting from (2):

$$V = \frac{1}{3} \pi R^3 \left(1 + \frac{1}{\sin \theta} \right)^3 \tan^2 \theta$$

4. Simplification using $t = \sin \theta$

Let $t = \sin \theta$, $0 < t < 1$. Then:

$$\begin{aligned} \tan^2 \theta &= \frac{t^2}{1 - t^2}, \quad 1 + \frac{1}{\sin \theta} = \frac{1 + t}{t} \\ V &= \frac{\pi R^3}{3} \cdot \frac{(1 + t)^3}{t^3} \cdot \frac{t^2}{1 - t^2} = \frac{\pi R^3}{3} \cdot \frac{(1 + t)^2}{t(1 - t)} \end{aligned}$$

5. Finding the optimal t (differentiation)

Consider the function $f(t) = \frac{(1+t)^2}{t(1-t)}$. Differentiate $\ln f$:

$$\frac{f'(t)}{f(t)} = \frac{2}{1+t} - \frac{1}{t} + \frac{1}{1-t}$$

Set the derivative to zero:

$$\frac{2}{1+t} - \frac{1}{t} + \frac{1}{1-t} = 0$$

Multiply by $t(1-t)(1+t) = t(1-t^2)$:

$$2t(1-t) - (1-t^2) + t(1+t) = 0$$

$$2t - 2t^2 - 1 + t^2 + t + t^2 = 3t - 1 = 0 \Rightarrow t = \frac{1}{3}$$

Thus:

$$\sin \theta = \frac{1}{3}$$

6. Optimal ratio $\frac{h}{r}$

$$\tan \theta = \frac{\sin \theta}{\sqrt{1 - \sin^2 \theta}} = \frac{1/3}{\sqrt{8/9}} = \frac{1}{2\sqrt{2}}$$

$$\frac{r}{h} = \tan \theta = \frac{1}{2\sqrt{2}} \Rightarrow \boxed{\frac{h}{r} = 2\sqrt{2}}$$

7. Minimum volume in terms of R

From $\sin \theta = \frac{1}{3}$ and (2):

$$h = R(1 + 3) = 4R$$

$$r = h \tan \theta = 4R \cdot \frac{1}{2\sqrt{2}} = \frac{2R}{\sqrt{2}} = R\sqrt{2}$$

$$V_{\min} = \frac{1}{3}\pi(R\sqrt{2})^2(4R) = \frac{1}{3}\pi(2R^2)(4R) = \frac{8\pi}{3}R^3$$

$$\boxed{V_{\min} = \frac{8\pi}{3}R^3}$$

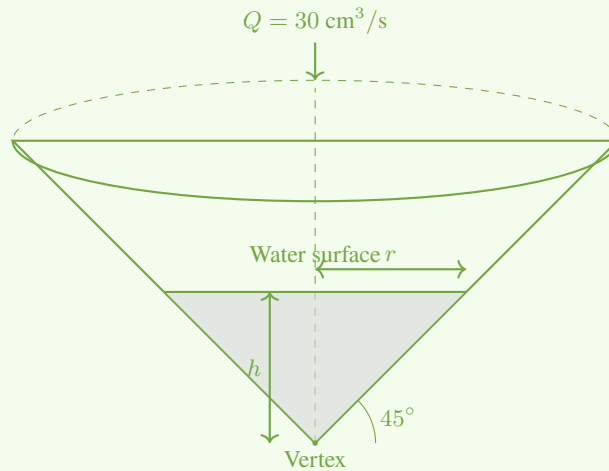
Note that $V_{\min} = 2 V_{\text{sphere}}$.

Exercise

An inverted cone (vertex at the bottom) with half-apex angle 45° is filled with water at a constant rate of $30 \text{ cm}^3/\text{s}$.

When the water height is 60 cm, find the rate at which:

- The water height h changes,
- The radius r of the water surface changes,
- The surface area S of the water changes.



Inverted cone with half-apex angle 45°
Filled at a constant rate

Solution

1. Geometric relation

In a cone with half-apex angle α , we have:

$$\tan \alpha = \frac{r}{h}.$$

Since $\alpha = 45^\circ$, $\tan 45^\circ = 1$, thus:

$$\boxed{r = h}.$$

2. Volume of water

The volume of water in the cone is:

$$V = \frac{1}{3}\pi r^2 h.$$

Using $r = h$, we obtain:

$$V = \frac{1}{3}\pi h^3.$$

3. Rate of change of height

The flow rate is given:

$$\frac{dV}{dt} = 30.$$

Differentiating:

$$\frac{dV}{dt} = \pi h^2 \frac{dh}{dt}.$$

Thus:

$$\frac{dh}{dt} = \frac{30}{\pi h^2}.$$

When $h = 60$:

$$\frac{dh}{dt} = \frac{30}{\pi \times 60^2} = \frac{1}{120\pi}.$$

$$\boxed{\frac{dh}{dt} = \frac{1}{120\pi} \text{ cm/s}}.$$

4. Rate of change of radius

Since $r = h$, we have:

$$\frac{dr}{dt} = \frac{dh}{dt}.$$

$$\boxed{\frac{dr}{dt} = \frac{1}{120\pi} \text{ cm/s}}.$$

5. Rate of change of surface area

The water surface area is:

$$S = \pi r^2.$$

Thus:

$$\frac{dS}{dt} = 2\pi r \frac{dr}{dt}.$$

At the given moment $r = 60$:

$$\frac{dS}{dt} = 2\pi \times 60 \times \frac{1}{120\pi} = 1.$$

$$\boxed{\frac{dS}{dt} = 1 \text{ cm}^2/\text{s}}.$$

Exercise

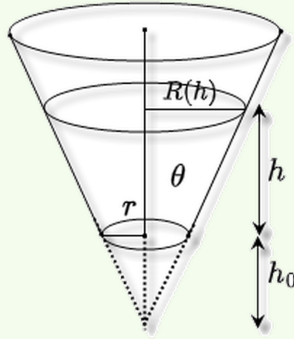
A water container is in the shape of a frustum of a cone with half-apex angle 45° . The bottom circle of the container is a hole of radius r (cm). Water flows out of this hole with speed

$$v = c\sqrt{2gh},$$

CHAPTER 1. APPLICATIONS OF DIFFERENTIATION

where c is a constant, g is the acceleration due to gravity, and h (in cm) is the height of the free water surface above the hole.

- Find the volumetric flow rate in cm^3/s .
- Find the rate at which h changes.



Solution

1. Geometric relation between $R(h)$ and h

Mentally complete the container to form a full cone, with its vertex below the bottom hole. Let h_0 be the distance between this vertex and the level of the hole.

The radius of the cross-section at the hole is r , and the radius of the free water surface is $R(h)$. The height of water above the hole is h . Thus the distance between the vertex of the full cone and the free surface is

$$\tan 45^\circ = \frac{r}{h_0} = \frac{R(h)}{h + h_0} = 1$$

We deduce

$$h_0 = r \quad \text{and} \quad R(h) = h + h_0 = h + r.$$

Thus

$$R(h) = r + h.$$

2. Volume of water

The volume of water is the difference between the volume of the large cone (radius $R(h)$ and height $h + h_0$) and the volume of the small cone (radius r and height h_0). Therefore

$$V(h) = \frac{1}{3}\pi R(h)^2(h + h_0) - \frac{1}{3}\pi r^2 h_0.$$

Using $R(h) = r + h$ and $h_0 = r$, we obtain

$$V(h) = \frac{1}{3}\pi(r + h)^2(h + r) - \frac{1}{3}\pi r^3.$$

Since $h + r = r + h$, we have

$$V(h) = \frac{1}{3}\pi((r + h)^3 - r^3).$$

Expand:

$$(r + h)^3 = r^3 + 3r^2h + 3rh^2 + h^3.$$

Thus

$$(r + h)^3 - r^3 = 3r^2h + 3rh^2 + h^3.$$

Therefore

$$V(h) = \frac{1}{3}\pi(3r^2h + 3rh^2 + h^3).$$

Hence

$$V(h) = \pi r^2 h + \pi r h^2 + \frac{1}{3}\pi h^3.$$

3. Volumetric flow rate

The area of the bottom hole is

$$A = \pi r^2.$$

The flow speed is

$$v = c\sqrt{2gh},$$

so the volumetric flow rate is

$$Q = Av.$$

Thus

$$Q = \pi r^2 c\sqrt{2gh}.$$

That is

$$Q = \pi r^2 c\sqrt{2gh} \text{ cm}^3/\text{s}.$$

4. Rate of change of h

Since water flows out of the container, the volume of water decreases with time. Hence

$$\frac{dV}{dt} = -Q.$$

We have

$$V(h) = \pi r^2 h + \pi r h^2 + \frac{1}{3}\pi h^3.$$

Differentiating V with respect to h :

$$\frac{dV}{dh} = \pi r^2 + 2\pi r h + \pi h^2 = \pi(r + h)^2.$$

By the chain rule:

$$\frac{dV}{dt} = \frac{dV}{dh} \frac{dh}{dt} = \pi(r + h)^2 \frac{dh}{dt}.$$

Since

$$\frac{dV}{dt} = -Q = -\pi r^2 c\sqrt{2gh},$$

we have

$$\pi(r + h)^2 \frac{dh}{dt} = -\pi r^2 c\sqrt{2gh}.$$

Divide by π to obtain

$$(r + h)^2 \frac{dh}{dt} = -r^2 c\sqrt{2gh}.$$

Thus

$$\frac{dh}{dt} = -\frac{r^2 c\sqrt{2gh}}{(r + h)^2}.$$

CHAPTER 1. APPLICATIONS OF DIFFERENTIATION

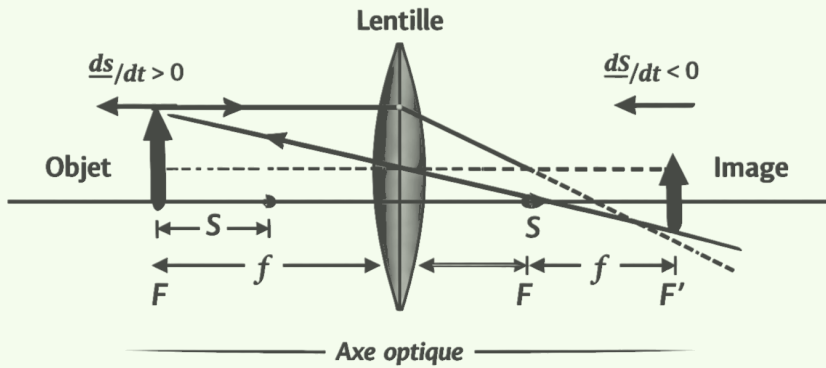
Exercise

The thin lens formula in physics is:

$$\frac{1}{s} + \frac{1}{S} = \frac{1}{f}$$

where s is the distance between the object and the lens, S is the distance between the image and the lens, and f is the focal length (constant).

Assume the object moves away from the lens with speed 3 cm/s, i.e., $\frac{ds}{dt} = 3$. Find the speed $\frac{dS}{dt}$ of the image and determine the direction of its motion.



Solution

Start from the thin lens equation:

$$\frac{1}{s} + \frac{1}{S} = \frac{1}{f}$$

Differentiate with respect to time t :

$$-\frac{1}{s^2} \frac{ds}{dt} - \frac{1}{S^2} \frac{dS}{dt} = 0$$

Isolate $\frac{dS}{dt}$:

$$\frac{dS}{dt} = -\frac{S^2}{s^2} \frac{ds}{dt}$$

From the lens equation:

$$S = \frac{sf}{s - f}$$

Thus:

$$\frac{S}{s} = \frac{f}{s - f}$$

Therefore:

$$\frac{dS}{dt} = -\left(\frac{f}{s - f}\right)^2 \frac{ds}{dt}$$

CHAPTER 1. APPLICATIONS OF DIFFERENTIATION

Since $\frac{ds}{dt} = 3$, we obtain:

$$\frac{dS}{dt} = -\frac{3f^2}{(s-f)^2}$$

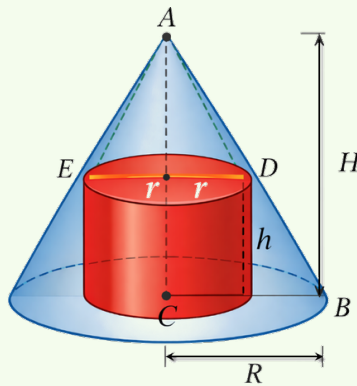
The negative sign indicates that:

- The object moves away from the lens,
- The image moves toward the lens.

Exercise

A right circular cone has base radius R and height H . What is the volume of the largest cylinder that can be inscribed in this cone?

Solution



1. Geometric relation between r , h , R and H

$$\tan \theta = \frac{r}{H-h} = \frac{R}{H} \implies r = R \left(\frac{H-h}{H} \right)$$

$$r = R \left(1 - \frac{h}{H} \right)$$

2. Volume of the cylinder

$$V = \pi r^2 h$$
$$V(h) = \pi \left[R \left(1 - \frac{h}{H} \right) \right]^2 h = \pi R^2 h \left(1 - \frac{h}{H} \right)^2$$

Domain: $h \in [0, H]$ (the cylinder must be contained within the cone).

3. Finding the maximum

Differentiate $V(h)$:

$$\begin{aligned} V'(h) &= \pi R^2 \left[\left(1 - \frac{h}{H}\right)^2 + h \cdot 2 \left(1 - \frac{h}{H}\right) \left(-\frac{1}{H}\right) \right] \\ &= \pi R^2 \left(1 - \frac{h}{H}\right) \left[\left(1 - \frac{h}{H}\right) - \frac{2h}{H} \right] \\ &= \pi R^2 \left(1 - \frac{h}{H}\right) \left(1 - \frac{3h}{H}\right) \end{aligned}$$

The critical points are solutions of $V'(h) = 0$:

$$h = H \implies V = 0, \quad h = \frac{H}{3} \implies V = V_{\max}$$

$$h_0 = \frac{H}{3}$$

Verify that it is a maximum:

- For $0 < h < \frac{H}{3}$: $V'(h) > 0$ (increasing)
- For $\frac{H}{3} < h < H$: $V'(h) < 0$ (decreasing)

4. Corresponding radius

$$r_0 = R \left(1 - \frac{h_0}{H}\right) = R \left(1 - \frac{H/3}{H}\right) = R \left(1 - \frac{1}{3}\right) = \frac{2R}{3}$$

$$r_0 = \frac{2R}{3}$$

5. Maximum volume

$$V_{\max} = \pi(r_0)^2 h_0 = \pi \left(\frac{2R}{3}\right)^2 \cdot \frac{H}{3} = \pi \cdot \frac{4R^2}{9} \cdot \frac{H}{3} = \frac{4\pi R^2 H}{27}$$

$$V_{\max} = \frac{4}{27} \pi R^2 H$$

Properties of Integrals

Theorem

1. Linearity of integration. Let f and g be continuous functions on $[a, b]$, and let $\alpha, \beta \in \mathbb{R}$. Then:

$$\int_a^b (\alpha f(x) + \beta g(x)) dx = \alpha \int_a^b f(x) dx + \beta \int_a^b g(x) dx.$$

2. Direction of integration. For any continuous function f on $[a, b]$, we have:

$$\int_a^b f(x) dx = - \int_b^a f(x) dx.$$

3. Additivity over intervals. If f is continuous on $[a, b]$ and $c \in [a, b]$, then:

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx.$$

4. Positivity. If f is continuous and nonnegative on $[a, b]$, then:

$$\int_a^b f(x) dx \geq 0.$$

5. Comparison. If f and g are continuous on $[a, b]$ and satisfy

$$f(x) \leq g(x) \quad \forall x \in [a, b],$$

then:

$$\int_a^b f(x) dx \leq \int_a^b g(x) dx.$$

6. Triangle inequality. For any continuous function f on $[a, b]$, we have:

$$\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx.$$

Theorem

1. Mean value theorem for integrals. If f is continuous on $[a, b]$, then there exists $c \in [a, b]$ such that:

$$\int_a^b f(x) dx = f(c)(b - a).$$

CHAPTER 2. PROPERTIES OF INTEGRALS

2. Fundamental theorem of calculus. If F is an antiderivative of f on $[a, b]$, then:

$$\int_a^b f(x) dx = F(b) - F(a).$$

3. Symmetry property. If f is continuous on $[a, b]$, then:

$$\int_a^b f(x) dx = \int_a^b f(a + b - x) dx.$$

4. Even and odd functions. If f is continuous on $[-a, a]$, then:

- If f is even, then:

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx;$$

- If f is odd, then:

$$\int_{-a}^a f(x) dx = 0.$$

5. Bounding the integral. If

$$m \leq f(x) \leq M \quad \forall x \in [a, b],$$

then:

$$m(b-a) \leq \int_a^b f(x) dx \leq M(b-a).$$

6. Second mean value theorem. If f is continuous and g is integrable and nonnegative on $[a, b]$, then there exists $c \in [a, b]$ such that:

$$\int_a^b f(x)g(x) dx = f(c) \int_a^b g(x) dx.$$

Theorem

1. General change of variables. Let

$$\varphi : [\alpha, \beta] \rightarrow [a, b]$$

be a C^1 bijective function. Then:

$$\int_a^b f(x) dx = \int_{\varphi^{-1}(a)}^{\varphi^{-1}(b)} f(\varphi(t)) \varphi'(t) dt.$$

2. Affine change of variables. If

$$x = \alpha t + \beta, \quad \alpha \neq 0,$$

then:

$$\int_a^b f(x) dx = \int_{(a-\beta)/\alpha}^{(b-\beta)/\alpha} f(\alpha t + \beta) \alpha dt.$$

3. Integration by parts. If f and g are C^1 on $[a, b]$, then:

$$\int_a^b f(x)g'(x) dx = [f(x)g(x)]_a^b - \int_a^b f'(x)g(x) dx.$$

4. Weighted mean formula. If f is continuous and g is integrable and of constant sign on $[a, b]$, then there exists $c \in [a, b]$ such that:

$$\int_a^b f(x)g(x) dx = f(c) \int_a^b g(x) dx.$$

CHAPTER 2. PROPERTIES OF INTEGRALS

5. Limit under the integral sign. If (f_n) is a sequence of continuous functions converging uniformly to f on $[a, b]$, then:

$$\lim_{n \rightarrow \infty} \int_a^b f_n(x) dx = \int_a^b f(x) dx.$$

Theorem

1. Absolute convergence. If

$$\int_a^b |f(x)| dx$$

converges, then

$$\int_a^b f(x) dx$$

also converges.

2. Comparison test. If

$$0 \leq f(x) \leq g(x)$$

near b^- , and if

$$\int_a^b g(x) dx$$

converges, then

$$\int_a^b f(x) dx$$

converges.

3. Riemann criterion. The integral

$$\int_1^{+\infty} \frac{dx}{x^\alpha}$$

converges if and only if

$$\alpha > 1.$$

4. Reference integrals.

$$\int_0^1 \frac{dx}{x^\alpha} \text{ converges if and only if } \alpha < 1,$$

$$\int_1^{+\infty} \frac{dx}{x^\alpha} \text{ converges if and only if } \alpha > 1,$$

$$\int_0^{+\infty} e^{-ax} dx = \frac{1}{a}, \quad a > 0.$$

5. Fresnel integrals.

$$\int_0^{+\infty} \cos(x^2) dx = \int_0^{+\infty} \sin(x^2) dx = \frac{1}{2} \sqrt{\frac{\pi}{2}}.$$

6. Dirichlet integrals.

$$\int_0^{+\infty} \frac{\sin x}{x} dx = \frac{\pi}{2}, \quad \int_0^{+\infty} \frac{1 - \cos x}{x^2} dx = \frac{\pi}{2}.$$

Theorem

1. Gamma function. For $x > 0$,

$$\Gamma(x) = \int_0^{+\infty} t^{x-1} e^{-t} dt.$$

It satisfies:

$$\Gamma(x+1) = x\Gamma(x), \quad \Gamma(n) = (n-1)!, \quad \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$$

2. Beta function. For $a > 0$ and $b > 0$,

$$B(a, b) = \int_0^1 t^{a-1} (1-t)^{b-1} dt.$$

It satisfies:

$$B(a, b) = B(b, a), \quad B(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}.$$

3. Integral representations of the Beta function.

$$\begin{aligned} B(a, b) &= 2 \int_0^{\pi/2} \sin^{2a-1} \theta \cos^{2b-1} \theta d\theta \\ &= \int_0^{+\infty} \frac{t^{a-1}}{(1+t)^{a+b}} dt. \end{aligned}$$

4. Stirling's formula. As $x \rightarrow +\infty$,

$$\Gamma(x) \sim \sqrt{2\pi} x^{x-\frac{1}{2}} e^{-x}.$$

5. Wallis integrals. For every $n \in \mathbb{N}$,

$$W_n = \int_0^{\pi/2} \sin^n x dx = \int_0^{\pi/2} \cos^n x dx = \begin{cases} \frac{(n-1)!!}{n!!} \cdot \frac{\pi}{2} & \text{if } n \text{ is even,} \\ \frac{(n-1)!!}{n!!} & \text{if } n \text{ is odd.} \end{cases}$$

Basic Properties

Exercise

Show that for any continuous function f on $[a, b]$, we have:

$$\int_a^b f(x) dx = - \int_b^a f(x) dx.$$

Solution

Let F be an antiderivative of f on $[a, b]$. By the fundamental theorem of calculus,

$$\int_a^b f(x) dx = F(b) - F(a).$$

$$\int_b^a f(x) dx = F(a) - F(b).$$

CHAPTER 2. PROPERTIES OF INTEGRALS

Thus,

$$\int_b^a f(x) dx = -(F(b) - F(a)) = -\int_a^b f(x) dx.$$

$$\boxed{\int_a^b f(x) dx = -\int_b^a f(x) dx.}$$

Exercise

Let f be a continuous function on $[a, b]$. Show that:

$$\int_a^b f(x) dx = \int_a^b f(a+b-x) dx.$$

Solution

Make the change of variable

$$t = a + b - x.$$

Then

$$dt = -dx.$$

When $x = a$, $t = b$, and when $x = b$, $t = a$. Thus:

$$\int_a^b f(a+b-x) dx = \int_{t=b}^{t=a} f(t)(-dt) = -\int_b^a f(t) dt.$$

Using the direction of integration property,

$$-\int_b^a f(t) dt = \int_a^b f(t) dt.$$

Renaming the variable t back to x :

$$\boxed{\int_a^b f(a+b-x) dx = \int_a^b f(x) dx.}$$

Exercise

Using the previous property, show that:

$$\int_0^{\pi/2} f(x) dx = \int_0^{\pi/2} f\left(\frac{\pi}{2} - x\right) dx$$

for any continuous function f on $[0, \frac{\pi}{2}]$.

Then deduce the value of:

$$I = \int_0^{\pi/2} \sin^2 t dt.$$

Solution

Apply the property from the previous exercise with

$$a = 0, \quad b = \frac{\pi}{2}.$$

CHAPTER 2. PROPERTIES OF INTEGRALS

Since $a + b = \frac{\pi}{2}$, we directly obtain:

$$\int_0^{\pi/2} f(x) dx = \int_0^{\pi/2} f\left(\frac{\pi}{2} - x\right) dx.$$

To compute

$$I = \int_0^{\pi/2} \sin^2 t dt,$$

apply the previous property to $f(x) = \sin^2 x$:

$$I = \int_0^{\pi/2} \sin^2 x dx = \int_0^{\pi/2} \sin^2\left(\frac{\pi}{2} - x\right) dx.$$

But

$$\sin\left(\frac{\pi}{2} - x\right) = \cos x,$$

so

$$I = \int_0^{\pi/2} \cos^2 x dx.$$

Adding the two expressions for I , we obtain:

$$2I = \int_0^{\pi/2} (\sin^2 x + \cos^2 x) dx = \int_0^{\pi/2} 1 dx = \frac{\pi}{2}.$$

Thus,

$$I = \frac{\pi}{4}.$$

Symmetry Applications

Exercise

Compute:

$$I = \int_0^{\pi/2} \frac{\sin t}{\sin t + \cos t} dt.$$

Solution

Apply the symmetry property on the interval $[0, \frac{\pi}{2}]$:

$$I = \int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x} dx = \int_0^{\pi/2} \frac{\sin\left(\frac{\pi}{2} - x\right)}{\sin\left(\frac{\pi}{2} - x\right) + \cos\left(\frac{\pi}{2} - x\right)} dx.$$

We have:

$$\sin\left(\frac{\pi}{2} - x\right) = \cos x, \quad \cos\left(\frac{\pi}{2} - x\right) = \sin x.$$

Thus:

$$I = \int_0^{\pi/2} \frac{\cos x}{\cos x + \sin x} dx.$$

Adding the two expressions for I , we find:

$$2I = \int_0^{\pi/2} \frac{\sin x + \cos x}{\sin x + \cos x} dx = \int_0^{\pi/2} 1 dx = \frac{\pi}{2}.$$

Therefore

CHAPTER 2. PROPERTIES OF INTEGRALS

$$I = \frac{\pi}{4}.$$

Exercise

Show that for any continuous function f , we have:

$$\int_0^\pi t f(\sin t) dt = \frac{\pi}{2} \int_0^\pi f(\sin t) dt.$$

Solution

Let

$$I = \int_0^\pi x f(\sin x) dx.$$

Make the change of variable

$$t = \pi - x.$$

Then

$$x = \pi - t, \quad dx = -dt.$$

As x goes from 0 to π , t goes from π to 0. Thus:

$$I = \int_0^\pi x f(\sin x) dx = \int_\pi^0 (\pi - t) f(\sin(\pi - t)) dt.$$

But

$$\sin(\pi - t) = \sin t,$$

so we obtain:

$$I = \int_0^\pi (\pi - t) f(\sin t) dt.$$

Adding this relation to the original expression for I , we find:

$$2I = \int_0^\pi [t + (\pi - t)] f(\sin t) dt = \pi \int_0^\pi f(\sin t) dt.$$

Thus:

$$\int_0^\pi t f(\sin t) dt = \frac{\pi}{2} \int_0^\pi f(\sin t) dt.$$

Exercise

Using the property from Exercise 2.5, compute:

$$I = \int_0^\pi x \sin x dx.$$

Solution

By Exercise 2.5, for any continuous function f :

$$\int_0^\pi t f(\sin t) dt = \frac{\pi}{2} \int_0^\pi f(\sin t) dt.$$

Here we choose

$$f(u) = u,$$

so that

$$f(\sin x) = \sin x.$$

CHAPTER 2. PROPERTIES OF INTEGRALS

We obtain:

$$I = \int_0^{\pi} x \sin x \, dx = \frac{\pi}{2} \int_0^{\pi} \sin x \, dx.$$

We have:

$$\int_0^{\pi} \sin x \, dx = [-\cos x]_0^{\pi} = 1 + 1 = 2.$$

Thus:

$$I = \frac{\pi}{2} \cdot 2 = \pi.$$

That is:

$$\boxed{I = \pi.}$$

Exercise

Compute:

$$I = \int_0^{\pi} (2x + 1) \frac{\sin^3 x}{1 + \cos^2 x} \, dx.$$

Solution

Let

$$\varphi(x) = \frac{\sin^3 x}{1 + \cos^2 x}.$$

We verify that

$$\varphi(\pi - x) = \varphi(x),$$

because

$$\sin(\pi - x) = \sin x, \quad \cos(\pi - x) = -\cos x, \quad \cos^2(\pi - x) = \cos^2 x.$$

$$\int_0^{\pi} x \varphi(x) \, dx = \frac{\pi}{2} \int_0^{\pi} \varphi(x) \, dx.$$

Thus:

$$I = 2 \int_0^{\pi} x \varphi(x) \, dx + \int_0^{\pi} \varphi(x) \, dx = (\pi + 1) \int_0^{\pi} \varphi(x) \, dx.$$

Let

$$J = \int_0^{\pi} \varphi(x) \, dx = \int_0^{\pi} \frac{\sin^3 x}{1 + \cos^2 x} \, dx.$$

Write

$$\sin^3 x = \sin x (1 - \cos^2 x),$$

then make the change of variable

$$u = \cos x, \quad du = -\sin x \, dx.$$

When $x = 0$, $u = 1$, and when $x = \pi$, $u = -1$. Thus:

$$J = \int_1^{-1} \frac{1 - u^2}{1 + u^2} (-du) = \int_{-1}^1 \frac{1 - u^2}{1 + u^2} \, du.$$

We have

$$\frac{1 - u^2}{1 + u^2} = \frac{2}{1 + u^2} - 1.$$

Thus:

$$J = 2 \int_{-1}^1 \frac{du}{1 + u^2} - \int_{-1}^1 du.$$

CHAPTER 2. PROPERTIES OF INTEGRALS

Compute:

$$\int_{-1}^1 \frac{du}{1+u^2} = [\arctan u]_{-1}^1 = \frac{\pi}{4} - \left(-\frac{\pi}{4}\right) = \frac{\pi}{2},$$

and

$$\int_{-1}^1 du = 2.$$

Hence

$$J = 2 \cdot \frac{\pi}{2} - 2 = \pi - 2.$$

Thus:

$$I = (\pi + 1)(\pi - 2) = \pi^2 - \pi - 2.$$

That is:

$$I = \pi^2 - \pi - 2.$$

Logarithms and Trigonometric Functions

Exercise

Compute:

$$I = \int_0^{\pi/4} \ln(1 + \tan x) dx.$$

Solution

Apply the symmetry property on the interval $[0, \frac{\pi}{4}]$:

$$I = \int_0^{\pi/4} \ln(1 + \tan x) dx = \int_0^{\pi/4} \ln\left(1 + \tan\left(\frac{\pi}{4} - x\right)\right) dx.$$

We have:

$$\tan\left(\frac{\pi}{4} - x\right) = \frac{1 - \tan x}{1 + \tan x},$$

so:

$$1 + \tan\left(\frac{\pi}{4} - x\right) = 1 + \frac{1 - \tan x}{1 + \tan x} = \frac{2}{1 + \tan x}.$$

Therefore:

$$I = \int_0^{\pi/4} \ln\left(\frac{2}{1 + \tan x}\right) dx = \int_0^{\pi/4} \ln 2 dx - \int_0^{\pi/4} \ln(1 + \tan x) dx.$$

That is

$$I = \frac{\pi}{4} \ln 2 - I.$$

Hence

$$2I = \frac{\pi}{4} \ln 2,$$

so:

$$I = \frac{\pi}{8} \ln 2.$$

Exercise

Let f be a continuous function on $[a, b]$ satisfying:

$$f(a + b - x) = f(x), \quad \forall x \in [a, b].$$

CHAPTER 2. PROPERTIES OF INTEGRALS

Show that:

$$\int_a^b x f(x) dx = \frac{a+b}{2} \int_a^b f(x) dx.$$

Solution

Let

$$I = \int_a^b x f(x) dx.$$

Make the change of variable

$$t = a + b - x.$$

Then

$$x = a + b - t, \quad dt = -dx.$$

When $x = a$, $t = b$, and when $x = b$, $t = a$. Thus:

$$I = \int_a^b x f(x) dx = \int_b^a (a + b - t) f(a + b - t) (-dt) = \int_a^b (a + b - t) f(a + b - t) dt.$$

By hypothesis,

$$f(a + b - t) = f(t).$$

Thus

$$I = \int_a^b (a + b - t) f(t) dt.$$

Renaming t back to x , we obtain:

$$I = \int_a^b (a + b - x) f(x) dx.$$

Adding this relation to the original expression for I , we get:

$$2I = \int_a^b [x + (a + b - x)] f(x) dx = (a + b) \int_a^b f(x) dx.$$

Thus:

$$\int_a^b x f(x) dx = \frac{a+b}{2} \int_a^b f(x) dx.$$

Exercise

Compute:

$$I = \int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx.$$

Solution

Let

$$g(x) = \frac{\sin x}{1 + \cos^2 x}.$$

We verify that

$$g(\pi - x) = g(x),$$

because

$$\sin(\pi - x) = \sin x, \quad \cos(\pi - x) = -\cos x, \quad \cos^2(\pi - x) = \cos^2 x.$$

CHAPTER 2. PROPERTIES OF INTEGRALS

We can apply Exercise 2.9 with $a = 0$ and $b = \pi$:

$$I = \int_0^\pi xg(x) dx = \frac{\pi}{2} \int_0^\pi g(x) dx.$$

Thus

$$I = \frac{\pi}{2} \int_0^\pi \frac{\sin x}{1 + \cos^2 x} dx.$$

Let

$$J = \int_0^\pi \frac{\sin x}{1 + \cos^2 x} dx.$$

Make the change of variable

$$u = \cos x, \quad du = -\sin x dx.$$

When $x = 0$, $u = 1$, and when $x = \pi$, $u = -1$. Thus:

$$J = \int_1^{-1} \frac{-du}{1 + u^2} = \int_{-1}^1 \frac{du}{1 + u^2} = [\arctan u]_{-1}^1 = \frac{\pi}{2}.$$

Therefore:

$$I = \frac{\pi}{2} \cdot \frac{\pi}{2} = \frac{\pi^2}{4}.$$

That is:

$$I = \frac{\pi^2}{4}.$$

Exercise

Show more generally that if

$$f(a + b - x) = f(x), \quad g(a + b - x) = g(x),$$

then:

$$\int_a^b f(x)g(x) dx = 2 \int_a^{\frac{a+b}{2}} f(x)g(x) dx.$$

Solution

Let

$$I = \int_a^b f(x)g(x) dx.$$

Split the integral into two parts:

$$I = \int_a^{\frac{a+b}{2}} f(x)g(x) dx + \int_{\frac{a+b}{2}}^b f(x)g(x) dx.$$

In the second integral, make the change of variable

$$u = a + b - x.$$

Then

$$du = -dx.$$

Note that

$$x = \frac{a+b}{2} \implies u = \frac{a+b}{2}, \quad x = b \implies u = a.$$

Since

$$f(a + b - u) = f(u), \quad g(a + b - u) = g(u),$$

CHAPTER 2. PROPERTIES OF INTEGRALS

we obtain:

$$\int_{\frac{a+b}{2}}^b f(x)g(x) dx = \int_a^a f(u)g(u)(-du) = \int_a^{\frac{a+b}{2}} f(u)g(u) du.$$

Substituting into the expression for I , we find:

$$I = \int_a^{\frac{a+b}{2}} f(x)g(x) dx + \int_a^{\frac{a+b}{2}} f(x)g(x) dx = 2 \int_a^{\frac{a+b}{2}} f(x)g(x) dx.$$

Thus:

$$\int_a^b f(x)g(x) dx = 2 \int_a^{\frac{a+b}{2}} f(x)g(x) dx.$$

Exercise

For each $n \in \mathbb{N}^*$, define:

$$I_n = \int_0^1 \frac{dx}{(1+x^2)^n}.$$

1. Establish a recurrence relation between I_{n+1} and I_n .
2. Compute I_2 and I_3 .

Solution

1. Establishing the recurrence relation

Write:

$$I_n = \int_0^1 \frac{1+x^2-x^2}{(1+x^2)^n} dx = \int_0^1 \frac{dx}{(1+x^2)^{n-1}} - \int_0^1 \frac{x^2}{(1+x^2)^n} dx.$$

In other words,

$$I_n = I_{n-1} - J_n,$$

where

$$J_n = \int_0^1 \frac{x^2}{(1+x^2)^n} dx.$$

To compute J_n , write

$$J_n = \int_0^1 x \cdot \frac{x}{(1+x^2)^n} dx.$$

Integrate by parts with

$$u = x, \quad dv = \frac{x}{(1+x^2)^n} dx.$$

Then

$$du = dx,$$

and

$$v = \int \frac{x}{(1+x^2)^n} dx = -\frac{1}{2(n-1)} \frac{1}{(1+x^2)^{n-1}}.$$

Thus:

$$J_n = \left[-\frac{x}{2(n-1)(1+x^2)^{n-1}} \right]_0^1 + \frac{1}{2(n-1)} \int_0^1 \frac{dx}{(1+x^2)^{n-1}}.$$

We obtain:

$$J_n = -\frac{1}{2^n(n-1)} + \frac{1}{2(n-1)} I_{n-1}.$$

Therefore:

$$I_n = I_{n-1} + \frac{1}{2^n(n-1)} - \frac{1}{2(n-1)} I_{n-1}.$$

CHAPTER 2. PROPERTIES OF INTEGRALS

Simplifying:

$$I_n = \frac{2n-3}{2(n-1)} I_{n-1} + \frac{1}{2^n(n-1)}.$$

Replacing n by $n+1$, we obtain:

$$I_{n+1} = \frac{2n-1}{2n} I_n + \frac{1}{2^{n+1}n}.$$

2. Computing I_2 and I_3

Start with

$$I_1 = \int_0^1 \frac{dx}{1+x^2} = [\arctan x]_0^1 = \frac{\pi}{4}.$$

Using the recurrence relation:

$$I_2 = \frac{1}{2} I_1 + \frac{1}{4} = \frac{\pi}{8} + \frac{1}{4}.$$

Then

$$I_3 = \frac{3}{4} I_2 + \frac{1}{16} = \frac{3}{4} \left(\frac{\pi}{8} + \frac{1}{4} \right) + \frac{1}{16} = \frac{3\pi}{32} + \frac{1}{4}.$$

Thus:

$$I_2 = \frac{\pi}{8} + \frac{1}{4}, \quad I_3 = \frac{3\pi}{32} + \frac{1}{4}.$$

Exercise

Let the Beta function be defined, for $a > 0$ and $b > 0$, by:

$$\beta(a, b) = \int_0^1 x^{a-1} (1-x)^{b-1} dx.$$

Show that:

$$\beta(a, b) = \beta(b, a).$$

Solution

Make the change of variable

$$t = 1 - x.$$

Then

$$x = 1 - t, \quad dx = -dt.$$

As x goes from 0 to 1, t goes from 1 to 0. Thus:

$$\beta(a, b) = \int_0^1 x^{a-1} (1-x)^{b-1} dx = \int_1^0 (1-t)^{a-1} t^{b-1} (-dt).$$

Swapping the limits of integration:

$$\beta(a, b) = \int_0^1 t^{b-1} (1-t)^{a-1} dt.$$

This is exactly $\beta(b, a)$. Hence:

$$\beta(a, b) = \beta(b, a).$$

Exercise

Compute the following values:

- $\beta(1, 1)$

CHAPTER 2. PROPERTIES OF INTEGRALS

- $\beta\left(\frac{1}{2}, \frac{1}{2}\right)$
- $\beta(m, n)$ where $m, n \in \mathbb{N}^*$

Solution

1. Computing $\beta(1, 1)$

By definition:

$$\beta(1, 1) = \int_0^1 x^0(1-x)^0 dx = \int_0^1 dx = 1.$$

2. Computing $\beta\left(\frac{1}{2}, \frac{1}{2}\right)$

Using the relation between the Beta and Gamma functions:

$$\beta\left(\frac{1}{2}, \frac{1}{2}\right) = \frac{\Gamma\left(\frac{1}{2}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma(1)} = \frac{\sqrt{\pi} \cdot \sqrt{\pi}}{1} = \pi.$$

3. The general case $m, n \in \mathbb{N}^*$

Again using the relation

$$\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)},$$

and since

$$\Gamma(m) = (m-1)!, \quad \Gamma(n) = (n-1)!, \quad \Gamma(m+n) = (m+n-1)!,$$

we have:

$$\beta(m, n) = \frac{(m-1)!(n-1)!}{(m+n-1)!}.$$

Thus:

$$\beta(1, 1) = 1, \quad \beta\left(\frac{1}{2}, \frac{1}{2}\right) = \pi, \quad \beta(m, n) = \frac{(m-1)!(n-1)!}{(m+n-1)!}.$$

Exercise

Show that for $a > 1$ and $b > 0$,

$$\beta(a, b) = \frac{a-1}{a+b-1} \beta(a-1, b).$$

Solution

Start from the relation

$$\beta(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}.$$

Similarly,

$$\beta(a-1, b) = \frac{\Gamma(a-1)\Gamma(b)}{\Gamma(a+b-1)}.$$

Dividing the two relations term by term:

$$\frac{\beta(a, b)}{\beta(a-1, b)} = \frac{\Gamma(a)}{\Gamma(a-1)} \cdot \frac{\Gamma(a+b-1)}{\Gamma(a+b)}.$$

We have:

$$\Gamma(a) = (a-1)\Gamma(a-1), \quad \Gamma(a+b) = (a+b-1)\Gamma(a+b-1).$$

Thus:

$$\frac{\Gamma(a)}{\Gamma(a-1)} = a-1, \quad \frac{\Gamma(a+b-1)}{\Gamma(a+b)} = \frac{1}{a+b-1}.$$

CHAPTER 2. PROPERTIES OF INTEGRALS

We obtain:

$$\frac{\beta(a, b)}{\beta(a-1, b)} = \frac{a-1}{a+b-1}.$$

Hence:

$$\beta(a, b) = \frac{a-1}{a+b-1} \beta(a-1, b).$$

Exercise

Show that for $a > 0$ and $b > 0$,

$$\beta(a, b) = 2 \int_0^{\pi/2} \sin^{2a-1} \theta \cos^{2b-1} \theta d\theta.$$

Solution

Start from the definition:

$$\beta(a, b) = \int_0^1 x^{a-1} (1-x)^{b-1} dx.$$

Make the change of variable

$$x = \sin^2 \theta.$$

Then

$$dx = 2 \sin \theta \cos \theta d\theta.$$

Moreover:

$$x^{a-1} = \sin^{2a-2} \theta, \quad (1-x)^{b-1} = \cos^{2b-2} \theta.$$

As x goes from 0 to 1, θ goes from 0 to $\frac{\pi}{2}$. Thus:

$$\beta(a, b) = \int_0^{\pi/2} \sin^{2a-2} \theta \cos^{2b-2} \theta 2 \sin \theta \cos \theta d\theta.$$

Simplify the exponents:

$$\beta(a, b) = 2 \int_0^{\pi/2} \sin^{2a-1} \theta \cos^{2b-1} \theta d\theta.$$

That is:

$$\beta(a, b) = 2 \int_0^{\pi/2} \sin^{2a-1} \theta \cos^{2b-1} \theta d\theta.$$

Exercise

Show that for $a > 0$ and $b > 0$,

$$\beta(a, b) = \int_0^{+\infty} \frac{u^{b-1}}{(1+u)^{a+b}} du = \int_0^{+\infty} \frac{u^{a-1}}{(1+u)^{a+b}} du.$$

Solution

Start from the definition:

$$\beta(a, b) = \int_0^1 x^{a-1} (1-x)^{b-1} dx.$$

Make the change of variable

$$x = \frac{1}{1+u}.$$

CHAPTER 2. PROPERTIES OF INTEGRALS

Then

$$dx = -\frac{du}{(1+u)^2}, \quad 1-x = \frac{u}{1+u}.$$

Also:

$$x^{a-1} = (1+u)^{-(a-1)}, \quad (1-x)^{b-1} = u^{b-1}(1+u)^{-(b-1)}.$$

As x goes from 0 to 1, u goes from $+\infty$ to 0. Thus:

$$\beta(a, b) = \int_{+\infty}^0 (1+u)^{-(a-1)} u^{b-1} (1+u)^{-(b-1)} \left(-\frac{du}{(1+u)^2} \right).$$

Combining the powers of $1+u$ we obtain:

$$\beta(a, b) = \int_0^{+\infty} \frac{u^{b-1}}{(1+u)^{a+b}} du.$$

Using the symmetry $\beta(a, b) = \beta(b, a)$, we also deduce:

$$\beta(a, b) = \int_0^{+\infty} \frac{u^{a-1}}{(1+u)^{a+b}} du.$$

Thus:

$$\beta(a, b) = \int_0^{+\infty} \frac{u^{b-1}}{(1+u)^{a+b}} du = \int_0^{+\infty} \frac{u^{a-1}}{(1+u)^{a+b}} du.$$

Exercise

Compute:

$$I = \int_0^1 \sqrt{\frac{1-x}{x}} dx, \quad J = \int_0^1 \sqrt{\frac{x}{1-x}} dx.$$

Solution

Write

$$I = \int_0^1 x^{-1/2} (1-x)^{1/2} dx.$$

This is a Beta-type integral, with

$$a = \frac{1}{2}, \quad b = \frac{3}{2}.$$

Thus:

$$I = \beta\left(\frac{1}{2}, \frac{3}{2}\right).$$

Using the relation

$$\beta(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)},$$

we find:

$$I = \frac{\Gamma\left(\frac{1}{2}\right)\Gamma\left(\frac{3}{2}\right)}{\Gamma(2)}.$$

We have:

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}, \quad \Gamma\left(\frac{3}{2}\right) = \frac{1}{2}\sqrt{\pi}, \quad \Gamma(2) = 1.$$

Thus:

$$I = \frac{\sqrt{\pi} \cdot \frac{1}{2}\sqrt{\pi}}{1} = \frac{\pi}{2}.$$

Similarly,

$$J = \int_0^1 x^{1/2}(1-x)^{-1/2} dx = \beta\left(\frac{3}{2}, \frac{1}{2}\right).$$

By symmetry of the Beta function,

$$J = \beta\left(\frac{1}{2}, \frac{3}{2}\right) = \frac{\pi}{2}.$$

Thus:

$$I = \frac{\pi}{2}, \quad J = \frac{\pi}{2}.$$

Improper Integrals

Exercise

Compute the improper integral:

$$\omega = \int_0^{+\infty} \frac{\sin(2x^2)}{x^2} e^{-x^2} dx.$$

Solution

Use the identity valid for all $x \neq 0$:

$$\frac{\sin(2x^2)}{x^2} = \int_0^2 \cos(tx^2) dt,$$

because:

$$\int_0^2 \cos(tx^2) dt = \left[\frac{\sin(tx^2)}{x^2} \right]_0^2 = \frac{\sin(2x^2)}{x^2}.$$

Thus:

$$\omega = \int_0^{+\infty} e^{-x^2} \left(\int_0^2 \cos(tx^2) dt \right) dx.$$

Using Fubini's theorem:

$$\omega = \int_0^2 \left(\int_0^{+\infty} e^{-x^2} \cos(tx^2) dx \right) dt.$$

Let

$$J(t) = \int_0^{+\infty} e^{-x^2} \cos(tx^2) dx.$$

Write:

$$J(t) = \Re \left(\int_0^{+\infty} e^{-(1-it)x^2} dx \right).$$

Since $\Re(1-it) = 1 > 0$,

$$\int_0^{+\infty} e^{-(1-it)x^2} dx = \frac{\sqrt{\pi}}{2\sqrt{1-it}}.$$

Thus:

$$J(t) = \frac{\sqrt{\pi}}{2} \Re \left(\frac{1}{\sqrt{1-it}} \right).$$

Therefore:

$$\omega = \frac{\sqrt{\pi}}{2} \Re \left(\int_0^2 \frac{dt}{\sqrt{1-it}} \right).$$

CHAPTER 2. PROPERTIES OF INTEGRALS

An antiderivative is:

$$\int \frac{dt}{\sqrt{1-it}} = 2i\sqrt{1-it}.$$

Thus:

$$\omega = \frac{\sqrt{\pi}}{2} \Re(2i(\sqrt{1-2i}-1)).$$

Write

$$\sqrt{1-2i} = u - iv \quad (u, v > 0).$$

Then:

$$u^2 - v^2 = 1, \quad 2uv = 2 \implies uv = 1.$$

Solving the system gives:

$$u^2 = \frac{1+\sqrt{5}}{2}, \quad v^2 = \frac{\sqrt{5}-1}{2} = \frac{2}{1+\sqrt{5}}.$$

Since:

$$\Re(2i(\sqrt{1-2i}-1)) = 2v,$$

we have:

$$\omega = \sqrt{\pi} v = \sqrt{\pi} \sqrt{\frac{2}{1+\sqrt{5}}}.$$

Finally:

$$\boxed{\omega = \sqrt{\pi} \sqrt{\frac{2}{1+\sqrt{5}}}}.$$

Exercise

Compute the integral:

$$I = \int_0^{+\infty} \frac{\cos x}{x^2 + 1} dx.$$

Solution

More generally, for $a > 0$, define

$$I(a) = \int_0^{+\infty} \frac{\cos(ax)}{1+x^2} dx.$$

Since the function is even,

$$2I(a) = \int_{-\infty}^{+\infty} \frac{\cos(ax)}{1+x^2} dx = \Re\left(\int_{-\infty}^{+\infty} \frac{e^{iax}}{1+x^2} dx\right).$$

Compute

$$F(a) = \int_{-\infty}^{+\infty} \frac{e^{iaz}}{1+z^2} dz, \quad a > 0.$$

Close the contour in the upper half-plane. The only enclosed pole is $z = i$ (simple). Its residue:

$$\text{Res}\left(\frac{e^{iaz}}{1+z^2}, i\right) = \lim_{z \rightarrow i} \frac{e^{iaz}}{z+i} = \frac{e^{-a}}{2i}.$$

By the residue theorem:

$$F(a) = 2\pi i \cdot \frac{e^{-a}}{2i} = \pi e^{-a}.$$

CHAPTER 2. PROPERTIES OF INTEGRALS

Since $F(a)$ is real,

$$2I(a) = \pi e^{-a} \implies I(a) = \frac{\pi}{2} e^{-a}.$$

For $a = 1$:

$$I = \frac{\pi}{2e}.$$

Exercise

Compute:

$$I = \int_0^{+\infty} \frac{dx}{x\sqrt{x} + \sqrt{x}}.$$

Solution

Factor the denominator:

$$x\sqrt{x} + \sqrt{x} = \sqrt{x}(x + 1).$$

Thus:

$$I = \int_0^{+\infty} \frac{dx}{\sqrt{x}(1+x)}.$$

Let

$$t = \sqrt{x}, \quad x = t^2, \quad dx = 2t dt.$$

Then:

$$I = \int_0^{+\infty} \frac{2t dt}{t(1+t^2)} = 2 \int_0^{+\infty} \frac{dt}{1+t^2}.$$

We have:

$$\int_0^{+\infty} \frac{dt}{1+t^2} = [\arctan t]_0^{+\infty} = \frac{\pi}{2}.$$

Hence:

$$I = \pi.$$

Exercise

Compute:

$$I = \int_0^{+\infty} \frac{\sqrt{x} dx}{(1+x)^2}.$$

Solution

Write:

$$I = \int_0^{+\infty} \frac{x^{1/2}}{(1+x)^2} dx.$$

This is a Beta-type integral of the form

$$\beta(p, q) = \int_0^{+\infty} \frac{u^{p-1}}{(1+u)^{p+q}} du.$$

Here:

$$p - 1 = \frac{1}{2} \Rightarrow p = \frac{3}{2}, \quad p + q = 2 \Rightarrow q = \frac{1}{2}.$$

Thus:

$$I = \beta\left(\frac{3}{2}, \frac{1}{2}\right).$$

Using the Beta-Gamma relation:

$$I = \frac{\Gamma\left(\frac{3}{2}\right) \Gamma\left(\frac{1}{2}\right)}{\Gamma(2)}.$$

CHAPTER 2. PROPERTIES OF INTEGRALS

We have:

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}, \quad \Gamma\left(\frac{3}{2}\right) = \frac{1}{2}\sqrt{\pi}, \quad \Gamma(2) = 1.$$

Thus:

$$I = \frac{\frac{1}{2}\sqrt{\pi} \cdot \sqrt{\pi}}{1} = \frac{\pi}{2}.$$

Therefore:

$$I = \frac{\pi}{2}.$$

Exercise

Show that for $m > 0, n > 0$ and $\alpha > \frac{m}{n}$,

$$\int_0^{+\infty} \frac{u^{m-1}}{(1+u^n)^\alpha} du = \frac{\Gamma\left(\frac{m}{n}\right) \Gamma\left(\alpha - \frac{m}{n}\right)}{n \Gamma(\alpha)}.$$

Solution

Make the change of variable

$$t = u^n, \quad u = t^{1/n}, \quad du = \frac{1}{n} t^{\frac{1}{n}-1} dt.$$

As u goes from 0 to $+\infty$, t also goes from 0 to $+\infty$. Substituting into the integral:

$$\int_0^{+\infty} \frac{u^{m-1}}{(1+u^n)^\alpha} du = \int_0^{+\infty} \frac{(t^{1/n})^{m-1}}{(1+t)^\alpha} \cdot \frac{1}{n} t^{\frac{1}{n}-1} dt.$$

Combine the powers of t :

$$\frac{m-1}{n} + \frac{1}{n} - 1 = \frac{m}{n} - 1.$$

Thus:

$$\int_0^{+\infty} \frac{u^{m-1}}{(1+u^n)^\alpha} du = \frac{1}{n} \int_0^{+\infty} \frac{t^{\frac{m}{n}-1}}{(1+t)^\alpha} dt.$$

Recall the Beta function in the form

$$\beta(p, q) = \int_0^{+\infty} \frac{t^{p-1}}{(1+t)^{p+q}} dt.$$

Here:

$$p = \frac{m}{n}, \quad p + q = \alpha \implies q = \alpha - \frac{m}{n}.$$

Thus:

$$\int_0^{+\infty} \frac{u^{m-1}}{(1+u^n)^\alpha} du = \frac{1}{n} \beta\left(\frac{m}{n}, \alpha - \frac{m}{n}\right).$$

Using the relation

$$\beta(p, q) = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)},$$

we obtain:

$$\int_0^{+\infty} \frac{u^{m-1}}{(1+u^n)^\alpha} du = \frac{\Gamma\left(\frac{m}{n}\right) \Gamma\left(\alpha - \frac{m}{n}\right)}{n \Gamma(\alpha)}.$$

CHAPTER 2. PROPERTIES OF INTEGRALS

Exercise

Compute:

$$I = \int_0^\infty \frac{dx}{1+x^4}.$$

Solution

Note that this integral matches the formula from Exercise 2.23 with

$$m = 1, \quad n = 4, \quad \alpha = 1.$$

Then:

$$I = \frac{1}{4} \Gamma\left(\frac{1}{4}\right) \Gamma\left(1 - \frac{1}{4}\right) = \frac{1}{4} \Gamma\left(\frac{1}{4}\right) \Gamma\left(\frac{3}{4}\right).$$

Now use the reflection formula:

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)}.$$

Setting $z = \frac{1}{4}$:

$$\Gamma\left(\frac{1}{4}\right) \Gamma\left(\frac{3}{4}\right) = \frac{\pi}{\sin(\pi/4)} = \frac{\pi}{\sqrt{2}/2} = \pi\sqrt{2}.$$

Thus:

$$I = \frac{\pi\sqrt{2}}{4}.$$

Therefore:

$$I = \frac{\pi\sqrt{2}}{4}.$$

Exercise

Compute, for $\gamma \in \mathbb{N}^*$, $\rho > -1$, $\beta > 0$, $a > 0$ and $\beta\gamma - \rho > 1$,

$$I = \int_0^\infty \frac{x^\rho}{(x^\beta + a^\beta)^\gamma} dx.$$

Solution

Let

$$t = \frac{x^\beta}{a^\beta}, \quad x = a t^{1/\beta}, \quad dx = \frac{a}{\beta} t^{\frac{1}{\beta}-1} dt.$$

Then:

$$x^\beta + a^\beta = a^\beta(1+t), \quad x^\rho = a^\rho t^{\rho/\beta}.$$

Substituting into the integral:

$$I = \int_0^\infty \frac{a^\rho t^{\rho/\beta}}{(a^\beta(1+t))^\gamma} \cdot \frac{a}{\beta} t^{\frac{1}{\beta}-1} dt.$$

Combine the factors:

$$I = \frac{a^{\rho+1-\beta\gamma}}{\beta} \int_0^\infty \frac{t^{\frac{\rho+1}{\beta}-1}}{(1+t)^\gamma} dt.$$

Recall the Beta integral:

$$\beta(p, q) = \int_0^\infty \frac{u^{p-1}}{(1+u)^{p+q}} du.$$

Here:

$$p = \frac{\rho+1}{\beta}, \quad p+q = \gamma \implies q = \gamma - \frac{\rho+1}{\beta}.$$

Thus:

$$I = \frac{a^{\rho+1-\beta\gamma}}{\beta} \beta \left(\frac{\rho+1}{\beta}, \gamma - \frac{\rho+1}{\beta} \right).$$

Using the Beta-Gamma relation:

$$I = \frac{a^{\rho+1-\beta\gamma}}{\beta} \frac{\Gamma\left(\frac{\rho+1}{\beta}\right) \Gamma\left(\gamma - \frac{\rho+1}{\beta}\right)}{\Gamma(\gamma)}.$$

That is:

$$I = \frac{a^{\rho+1-\beta\gamma}}{\beta} \frac{\Gamma\left(\frac{\rho+1}{\beta}\right) \Gamma\left(\gamma - \frac{\rho+1}{\beta}\right)}{\Gamma(\gamma)}.$$

Parametric Integrals

Exercise

For $\alpha > 0$, consider the integral

$$I(\alpha) = \int_0^\infty \frac{e^{-\alpha x} - e^{-\beta x}}{x} dx \quad (\beta > 0).$$

- Show that $I(\alpha)$ is well-defined for $\alpha > 0$.
- Compute $I'(\alpha)$.
- Deduce that $I(\alpha) = \ln\left(\frac{\beta}{\alpha}\right)$.

Solution

1. Convergence of the integral

Near 0, use equivalent limits:

$$e^{-\alpha x} = 1 - \alpha x + o(x), \quad e^{-\beta x} = 1 - \beta x + o(x),$$

then:

$$e^{-\alpha x} - e^{-\beta x} = (\beta - \alpha)x + o(x),$$

thus:

$$\frac{e^{-\alpha x} - e^{-\beta x}}{x} \rightarrow \beta - \alpha \quad (x \rightarrow 0^+).$$

So there is no divergence at 0.

At $+\infty$, the exponential functions decay rapidly, so the integral over $[1, +\infty)$ converges. Hence $I(\alpha)$ is well-defined.

2. Computing $I'(\alpha)$

Differentiate under the integral sign:

$$I'(\alpha) = \int_0^\infty \frac{\partial}{\partial \alpha} \left(\frac{e^{-\alpha x} - e^{-\beta x}}{x} \right) dx.$$

We have:

$$\frac{\partial}{\partial \alpha} \left(\frac{e^{-\alpha x} - e^{-\beta x}}{x} \right) = -e^{-\alpha x}.$$

CHAPTER 2. PROPERTIES OF INTEGRALS

Thus:

$$I'(\alpha) = - \int_0^\infty e^{-\alpha x} dx = -\frac{1}{\alpha}.$$

3. Determining $I(\alpha)$

Since $I'(\alpha) = -\frac{1}{\alpha}$, we have:

$$I(\alpha) = -\ln \alpha + C.$$

To find C , take $\alpha = \beta$. Then:

$$I(\beta) = \int_0^\infty \frac{e^{-\beta x} - e^{-\beta x}}{x} dx = 0.$$

Thus $0 = -\ln \beta + C$, i.e., $C = \ln \beta$. Therefore:

$$I(\alpha) = \ln \beta - \ln \alpha = \ln \left(\frac{\beta}{\alpha} \right).$$

$$I(\alpha) = \ln \left(\frac{\beta}{\alpha} \right).$$

Exercise

Let $f : [0, +\infty) \rightarrow \mathbb{R}$ be a continuous function such that $\lim_{x \rightarrow +\infty} f(x) = L$ exists. Show that for $a, b > 0$,

$$\int_0^\infty \frac{f(ax) - f(bx)}{x} dx = (f(0) - L) \ln \left(\frac{b}{a} \right).$$

Solution

For $\varepsilon > 0$ and $R > \varepsilon$, consider the truncated integral

$$I(\varepsilon, R) = \int_\varepsilon^R \frac{f(ax) - f(bx)}{x} dx.$$

In the first part, let

$$u = ax.$$

Then:

$$dx = \frac{du}{a}, \quad x = \frac{u}{a},$$

and:

$$\int_\varepsilon^R \frac{f(ax)}{x} dx = \int_{a\varepsilon}^{aR} \frac{f(u)}{u} du.$$

Similarly, letting $v = bx$:

$$\int_\varepsilon^R \frac{f(bx)}{x} dx = \int_{b\varepsilon}^{bR} \frac{f(v)}{v} dv.$$

Thus:

$$I(\varepsilon, R) = \int_{a\varepsilon}^{aR} \frac{f(u)}{u} du - \int_{b\varepsilon}^{bR} \frac{f(v)}{v} dv.$$

Rearrange the intervals:

$$I(\varepsilon, R) = \int_{a\varepsilon}^{b\varepsilon} \frac{f(u)}{u} du - \int_{aR}^{bR} \frac{f(v)}{v} dv.$$

Now take the limits $\varepsilon \rightarrow 0^+$ and $R \rightarrow +\infty$.

CHAPTER 2. PROPERTIES OF INTEGRALS

Since f is continuous at 0, $f(u) \rightarrow f(0)$ as $u \rightarrow 0^+$, thus:

$$\int_{a\varepsilon}^{b\varepsilon} \frac{f(u)}{u} du \sim f(0) \int_{a\varepsilon}^{b\varepsilon} \frac{du}{u} = f(0) \ln \left(\frac{b}{a} \right).$$

Since $f(v) \rightarrow L$ as $v \rightarrow +\infty$, we have:

$$\int_{aR}^{bR} \frac{f(v)}{v} dv \sim L \int_{aR}^{bR} \frac{dv}{v} = L \ln \left(\frac{b}{a} \right).$$

Taking the limit:

$$\int_0^\infty \frac{f(ax) - f(bx)}{x} dx = f(0) \ln \left(\frac{b}{a} \right) - L \ln \left(\frac{b}{a} \right) = (f(0) - L) \ln \left(\frac{b}{a} \right).$$

Thus:

$$\boxed{\int_0^\infty \frac{f(ax) - f(bx)}{x} dx = (f(0) - L) \ln \left(\frac{b}{a} \right).}$$

Advanced Trigonometric Integrals

Exercise

For $a, b > 0$, compute:

$$I(a, b) = \int_0^\infty \frac{\sin(ax) \sin(bx)}{x^2} dx.$$

Solution

Use the trigonometric identity:

$$\sin(ax) \sin(bx) = \frac{1}{2} [\cos((a-b)x) - \cos((a+b)x)].$$

Thus:

$$I(a, b) = \frac{1}{2} \int_0^\infty \frac{\cos((a-b)x) - \cos((a+b)x)}{x^2} dx.$$

Use the classical integral:

$$\int_0^\infty \frac{1 - \cos(cx)}{x^2} dx = \frac{\pi}{2} c, \quad c > 0.$$

Write:

$$\cos((a-b)x) - \cos((a+b)x) = [1 - \cos((a+b)x)] - [1 - \cos((a-b)x)].$$

Then:

$$\int_0^\infty \frac{\cos((a-b)x) - \cos((a+b)x)}{x^2} dx = \frac{\pi}{2} (a+b) - \frac{\pi}{2} |a-b|.$$

If $a \geq b$, then $|a-b| = a-b$, so the value becomes:

$$\frac{\pi}{2} (a+b) - \frac{\pi}{2} (a-b) = \pi b.$$

If $b \geq a$, the value becomes πa . Thus:

$$I(a, b) = \frac{1}{2} \pi \min(a, b) = \frac{\pi}{2} \min(a, b).$$

CHAPTER 2. PROPERTIES OF INTEGRALS

Therefore:

$$I(a, b) = \frac{\pi}{2} \min(a, b).$$

Exercise

Show that:

$$\int_0^{\pi/2} \ln(\sin x) dx = -\frac{\pi}{2} \ln 2.$$

Solution

Let

$$I = \int_0^{\pi/2} \ln(\sin x) dx.$$

By the change of variable $x = \frac{\pi}{2} - t$, we obtain:

$$I = \int_0^{\pi/2} \ln(\cos t) dt.$$

Add the two expressions:

$$2I = \int_0^{\pi/2} [\ln(\sin x) + \ln(\cos x)] dx = \int_0^{\pi/2} \ln(\sin x \cos x) dx.$$

We have $\sin x \cos x = \frac{1}{2} \sin(2x)$, hence:

$$2I = \int_0^{\pi/2} \ln\left(\frac{1}{2} \sin(2x)\right) dx = \int_0^{\pi/2} \ln(\sin 2x) dx - \frac{\pi}{2} \ln 2.$$

In the first integral, let $u = 2x$, $dx = du/2$. As x goes from 0 to $\frac{\pi}{2}$, u goes from 0 to π :

$$\int_0^{\pi/2} \ln(\sin 2x) dx = \frac{1}{2} \int_0^{\pi} \ln(\sin u) du.$$

By symmetry of $\sin u$ on $[0, \pi]$,

$$\int_0^{\pi} \ln(\sin u) du = 2 \int_0^{\pi/2} \ln(\sin u) du = 2I.$$

Thus:

$$\int_0^{\pi/2} \ln(\sin 2x) dx = I.$$

Return to the equation:

$$2I = I - \frac{\pi}{2} \ln 2 \implies I = -\frac{\pi}{2} \ln 2.$$

Therefore:

$$\int_0^{\pi/2} \ln(\sin x) dx = -\frac{\pi}{2} \ln 2.$$

Feynman's Technique for Computing Integrals

Theorem

[Basic Principles of Feynman's Technique]

1. Differentiation under the integral sign (bounded domain case). Let

$$F(\alpha) = \int_a^b f(x, \alpha) dx.$$

If $f(x, \alpha)$ and its partial derivative

$$\frac{\partial f}{\partial \alpha}(x, \alpha)$$

are continuous on $[a, b] \times I$, then:

$$F'(\alpha) = \int_a^b \frac{\partial f(x, \alpha)}{\partial \alpha} dx.$$

2. Differentiation under the integral sign (unbounded domain case). Let

$$F(\alpha) = \int_0^{+\infty} f(x, \alpha) dx.$$

If f and $\frac{\partial f}{\partial \alpha}$ are continuous, and if the integrals converge uniformly on every compact subset of I , then:

$$F'(\alpha) = \int_0^{+\infty} \frac{\partial f(x, \alpha)}{\partial \alpha} dx.$$

3. Conditions for validity (Leibniz's theorem). To differentiate under the integral sign, it is practically sufficient that:

- $f(x, \alpha)$ and $\frac{\partial f}{\partial \alpha}(x, \alpha)$ are continuous with respect to (x, α) ;
- The convergence of the integral is uniform with respect to α ;
- The functions are dominated by an integrable function independent of α .

4. General formula. Under sufficient regularity assumptions, we have:

$$\begin{aligned} \frac{d}{d\alpha} \int_{\varphi_1(\alpha)}^{\varphi_2(\alpha)} f(x, \alpha) dx &= \int_{\varphi_1(\alpha)}^{\varphi_2(\alpha)} \frac{\partial f}{\partial \alpha}(x, \alpha) dx \\ &\quad + f(\varphi_2(\alpha), \alpha)\varphi_2'(\alpha) - f(\varphi_1(\alpha), \alpha)\varphi_1'(\alpha). \end{aligned}$$

5. Successive applications. If $f(x, \alpha)$ is n times continuously differentiable with respect to α , then:

$$F^{(n)}(\alpha) = \int_a^b \frac{\partial^n f(x, \alpha)}{\partial \alpha^n} dx.$$

Theorem

[Basic Examples and Applications]

1. Exponential integral. For $a > 0$,

$$\int_0^{+\infty} e^{-ax} dx = \frac{1}{a}.$$

By differentiating with respect to a we obtain:

$$\int_0^{+\infty} x e^{-ax} dx = \frac{1}{a^2}, \quad \int_0^{+\infty} x^2 e^{-ax} dx = \frac{2}{a^3}, \quad \int_0^{+\infty} x^n e^{-ax} dx = \frac{n!}{a^{n+1}}.$$

2. Logarithmic integral. Let

$$F(\alpha) = \int_0^1 \frac{x^\alpha - 1}{\ln x} dx.$$

Then

$$F'(\alpha) = \int_0^1 x^\alpha dx = \frac{1}{\alpha + 1},$$

hence

$$F(\alpha) = \ln(\alpha + 1).$$

3. Gaussian integral. Let

$$I(a) = \int_0^{+\infty} e^{-ax^2} dx.$$

Then

$$I'(a) = - \int_0^{+\infty} x^2 e^{-ax^2} dx,$$

and by direct computation,

$$I(a) = \frac{1}{2} \sqrt{\frac{\pi}{a}}.$$

4. Dirichlet integral. By introducing the parameter

$$J(a) = \int_0^{+\infty} \frac{\sin x}{x} e^{-ax} dx,$$

we obtain

$$J'(a) = - \frac{1}{1 + a^2},$$

hence

$$J(a) = \frac{\pi}{2} - \arctan a,$$

and therefore

$$\int_0^{+\infty} \frac{\sin x}{x} dx = \frac{\pi}{2}.$$

5. Fresnel integrals. Using Feynman's technique, it can be shown that

$$\int_0^{+\infty} \cos(x^2) dx = \int_0^{+\infty} \sin(x^2) dx = \frac{1}{2} \sqrt{\frac{\pi}{2}}.$$

6. Laplace integral. For $a > 0$,

$$\int_0^{+\infty} \frac{\cos(ax)}{1 + x^2} dx = \frac{\pi}{2} e^{-a}.$$

Theorem

[Advanced Strategies and Techniques]

1. Artificial introduction of a parameter. When the integral does not explicitly depend on a

CHAPTER 3. FEYNMAN'S TECHNIQUE FOR COMPUTING INTEGRALS

parameter, one can introduce a parameter in a suitable way, then let it tend to an appropriate value.

2. Using symmetry. For integrals of the form

$$\int_0^1 \frac{x^a - x^b}{\ln x} dx,$$

one can consider the auxiliary function

$$F(t) = \int_0^1 \frac{x^a - x^t}{\ln x} dx.$$

3. Combining with change of variables. Feynman's technique is often used together with an appropriate change of variables to simplify the resulting integral after differentiation.
4. Integration with respect to the parameter. After differentiating under the integral sign, we typically integrate the result with respect to the parameter to recover the original function.
5. Interchanging the order of integration. In double integrals, one can sometimes interchange the order, then apply Feynman's technique to the inner integral.
6. Using series expansions. For some integrals, one can expand the integrand into a series, apply Feynman's technique term by term, then sum the resulting series.
7. Relation with the Laplace transform. Feynman's technique is closely related to the Laplace transform, especially for integrals of the form

$$\int_0^{+\infty} f(x)e^{-ax} dx.$$

8. Applications to special functions. This method is very effective for establishing relations between special functions such as Gamma, Beta, Bessel, etc.

Theorem

[Physical Applications and Classical Cases]

1. In quantum physics. The technique is extensively used in computing path integrals (Feynman integrals).
2. In probability theory. It appears in computing moments of distributions such as the Gaussian or exponential distribution.
3. In electromagnetism. It appears in computing electric and magnetic fields produced by charge distributions.
4. In thermodynamics. It appears in distribution integrals and various thermodynamic functions.
5. Classical case 1.

$$\int_0^{+\infty} \frac{\arctan(ax)}{x(1+x^2)} dx = \frac{\pi}{2} \ln(1+a).$$

6. Classical case 2.

$$\int_0^1 \frac{\ln(1+x)}{1+x^2} dx = \frac{\pi}{8} \ln 2.$$

7. Classical case 3.

$$\int_0^{+\infty} \frac{\ln(1+x^2)}{1+x^2} dx = \pi \ln 2.$$

8. Classical case 4. The correct formula is

$$\int_0^{+\infty} \frac{x^{p-1}}{1+x} dx = \frac{\pi}{\sin(\pi p)}, \quad 0 < p < 1.$$

By differentiating with respect to p we obtain:

$$\int_0^{+\infty} \frac{x^{p-1} \ln x}{1+x} dx = -\frac{\pi^2 \cos(\pi p)}{\sin^2(\pi p)}, \quad 0 < p < 1.$$

Basic Applications

Exercise

Using Feynman's technique, compute

$$I_n = \int_0^1 \frac{x^n - 1}{\ln x} dx, \quad n > 0.$$

Solution

Consider the auxiliary function

$$F(t) = \int_0^1 \frac{x^t - 1}{\ln x} dx, \quad t \geq 0.$$

Then $I_n = F(n)$.

Differentiate F with respect to t :

$$F'(t) = \int_0^1 \frac{\partial}{\partial t} \left(\frac{x^t - 1}{\ln x} \right) dx.$$

We have

$$\frac{\partial}{\partial t} \left(\frac{x^t - 1}{\ln x} \right) = \frac{x^t \ln x}{\ln x} = x^t.$$

Thus

$$F'(t) = \int_0^1 x^t dx.$$

Compute this integral:

$$\int_0^1 x^t dx = \left[\frac{x^{t+1}}{t+1} \right]_0^1 = \frac{1}{t+1}.$$

Hence

$$F'(t) = \frac{1}{t+1}.$$

Integrate with respect to t :

$$F(t) = \ln(t+1) + C.$$

To determine the constant, compute $F(0)$:

$$F(0) = \int_0^1 \frac{1-1}{\ln x} dx = 0.$$

Thus

$$0 = \ln 1 + C \Rightarrow C = 0.$$

Therefore

$$F(t) = \ln(t+1).$$

Setting $t = n$ we obtain

$$\int_0^1 \frac{x^n - 1}{\ln x} dx = \ln(n+1).$$

Exercise

Using Feynman's technique, compute

$$I(a, b) = \int_0^1 \frac{x^a - x^b}{\ln x} dx, \quad a > 0, b > 0.$$

Solution

By linearity of the integral:

$$I(a, b) = \int_0^1 \frac{x^a - 1}{\ln x} dx - \int_0^1 \frac{x^b - 1}{\ln x} dx.$$

From the previous exercise:

$$\int_0^1 \frac{x^a - 1}{\ln x} dx = \ln(a + 1), \quad \int_0^1 \frac{x^b - 1}{\ln x} dx = \ln(b + 1).$$

Thus

$$I(a, b) = \ln(a + 1) - \ln(b + 1) = \ln\left(\frac{a + 1}{b + 1}\right).$$

That is

$$\boxed{\int_0^1 \frac{x^a - x^b}{\ln x} dx = \ln\left(\frac{a + 1}{b + 1}\right)}.$$

Exercise

Compute

$$I = \int_0^1 \frac{x - 1}{\ln x} \cdot \frac{1}{\sqrt{-\ln x}} dx.$$

Solution

Make the change of variable

$$t = -\ln x.$$

Then

$$x = e^{-t}, \quad dx = -e^{-t} dt.$$

As x goes from 0 to 1, t goes from $+\infty$ to 0. Substituting:

$$I = \int_0^{+\infty} \frac{e^{-2t} - e^{-t}}{t^{3/2}} dt.$$

Define

$$F(a) = \int_0^{+\infty} \frac{e^{-at} - e^{-t}}{t^{3/2}} dt, \quad a > 0.$$

Then $I = F(2)$.

Differentiate with respect to a :

$$F'(a) = \int_0^{+\infty} \frac{\partial}{\partial a} \left(\frac{e^{-at} - e^{-t}}{t^{3/2}} \right) dt = - \int_0^{+\infty} e^{-at} t^{-1/2} dt.$$

Recall the Gamma integral:

$$\int_0^{+\infty} e^{-at} t^{-1/2} dt = a^{-1/2} \Gamma\left(\frac{1}{2}\right) = a^{-1/2} \sqrt{\pi}.$$

Thus

$$F'(a) = -\sqrt{\pi} a^{-1/2}.$$

Integrate:

$$F(a) = -2\sqrt{\pi a} + C.$$

We have

$$F(1) = \int_0^{+\infty} \frac{e^{-t} - e^{-t}}{t^{3/2}} dt = 0,$$

so

$$0 = -2\sqrt{\pi} + C \Rightarrow C = 2\sqrt{\pi}.$$

Hence

$$F(a) = 2\sqrt{\pi}(1 - \sqrt{a}).$$

Setting $a = 2$:

$$\int_0^1 \frac{x-1}{\ln x} \cdot \frac{1}{\sqrt{-\ln x}} dx = 2\sqrt{\pi}(1 - \sqrt{2}).$$

Classical Integrals

Exercise

Using Gauss's method, show that

$$\int_0^{+\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}.$$

Solution

Let

$$J = \int_0^{+\infty} e^{-x^2} dx.$$

Then

$$J^2 = \int_0^{+\infty} \int_0^{+\infty} e^{-(x^2+y^2)} dx dy.$$

Switch to polar coordinates:

$$x = r \cos \theta, \quad y = r \sin \theta,$$

with $r \geq 0, 0 \leq \theta \leq \frac{\pi}{2}$. The Jacobian is r , so

$$J^2 = \int_0^{\pi/2} \int_0^{+\infty} e^{-r^2} r dr d\theta.$$

Separate the integrals:

$$J^2 = \left(\int_0^{\pi/2} d\theta \right) \left(\int_0^{+\infty} e^{-r^2} r dr \right) = \frac{\pi}{2} \int_0^{+\infty} e^{-r^2} r dr.$$

Let $u = r^2, du = 2r dr$:

$$\int_0^{+\infty} e^{-r^2} r dr = \frac{1}{2} \int_0^{+\infty} e^{-u} du = \frac{1}{2}.$$

Thus

$$J^2 = \frac{\pi}{4}.$$

Since $J > 0$, we conclude

$$\int_0^{+\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}.$$

Exercise

Compute the Gaussian integral using a parameter for Feynman's technique:

$$I = \int_0^{+\infty} e^{-x^2} dx.$$

Introduce the parameterized function

$$I(t) = \int_0^{+\infty} \frac{e^{-t^2(1+x^2)}}{1+x^2} dx, \quad t \geq 0.$$

Solution

First note that

$$I(0) = \int_0^{+\infty} \frac{dx}{1+x^2} = \frac{\pi}{2},$$

and

$$\lim_{t \rightarrow +\infty} I(t) = 0.$$

Differentiate under the integral sign:

$$I'(t) = \int_0^{+\infty} \frac{\partial}{\partial t} \left(\frac{e^{-t^2(1+x^2)}}{1+x^2} \right) dx = -2t \int_0^{+\infty} e^{-t^2(1+x^2)} dx.$$

In the right-hand integral, let $u = tx$, $dx = du/t$:

$$\int_0^{+\infty} e^{-t^2(1+x^2)} dx = \frac{e^{-t^2}}{t} \int_0^{+\infty} e^{-u^2} du.$$

Denote

$$J = \int_0^{+\infty} e^{-u^2} du.$$

Then

$$I'(t) = -2J e^{-t^2}.$$

Now integrate from 0 to $+\infty$:

$$I(+\infty) - I(0) = -2J \int_0^{+\infty} e^{-t^2} dt.$$

We have $I(+\infty) = 0$, $I(0) = \frac{\pi}{2}$, and $\int_0^{+\infty} e^{-t^2} dt = J$, so:

$$-\frac{\pi}{2} = -2J^2.$$

Thus

$$J^2 = \frac{\pi}{4}, \quad J > 0 \implies J = \frac{\sqrt{\pi}}{2}.$$

This is the desired result:

$$\boxed{\int_0^{+\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}}.$$

Exercise

Using Feynman's technique, compute

$$I = \int_0^{+\infty} \frac{\sin t}{t} dt.$$

Solution

Introduce the parameter

$$J(a) = \int_0^{+\infty} \frac{\sin t}{t} e^{-at} dt, \quad a > 0.$$

Then $I = \lim_{a \rightarrow 0^+} J(a)$.

Differentiate J with respect to a :

$$J'(a) = \int_0^{+\infty} \frac{\partial}{\partial a} \left(\frac{\sin t}{t} e^{-at} \right) dt = - \int_0^{+\infty} e^{-at} \sin t dt.$$

Use the classical Laplace transform:

$$\int_0^{+\infty} e^{-at} \sin t dt = \frac{1}{1+a^2}, \quad a > 0.$$

Thus

$$J'(a) = -\frac{1}{1+a^2}.$$

Integrate:

$$J(a) = -\arctan a + C.$$

To determine the constant, take $a \rightarrow +\infty$. We have

$$\left| \frac{\sin t}{t} e^{-at} \right| \leq e^{-at},$$

so $\lim_{a \rightarrow +\infty} J(a) = 0$. Hence

$$0 = -\frac{\pi}{2} + C \Rightarrow C = \frac{\pi}{2}.$$

Therefore

$$J(a) = \frac{\pi}{2} - \arctan a.$$

Taking the limit as $a \rightarrow 0^+$ we obtain

$$\boxed{\int_0^{+\infty} \frac{\sin t}{t} dt = \frac{\pi}{2}}.$$

Advanced Applications

Exercise

For $a, b > 0$, compute

$$I = \int_0^{+\infty} \frac{e^{-ax} - e^{-bx}}{x} dx.$$

Solution

Define

$$F(t) = \int_0^{+\infty} \frac{e^{-ax} - e^{-tx}}{x} dx, \quad t > 0.$$

Then $I = F(b)$.

Differentiate with respect to t :

$$F'(t) = \int_0^{+\infty} \frac{\partial}{\partial t} \left(\frac{e^{-ax} - e^{-tx}}{x} \right) dx = \int_0^{+\infty} e^{-tx} dx.$$

Compute:

$$\int_0^{+\infty} e^{-tx} dx = \frac{1}{t}.$$

Thus

$$F'(t) = \frac{1}{t}.$$

Integrate:

$$F(t) = \ln t + C.$$

We have

$$F(a) = \int_0^{+\infty} \frac{e^{-ax} - e^{-ax}}{x} dx = 0,$$

so

$$0 = \ln a + C \Rightarrow C = -\ln a.$$

Hence

$$F(t) = \ln \left(\frac{t}{a} \right).$$

Setting $t = b$:

$$\boxed{\int_0^{+\infty} \frac{e^{-ax} - e^{-bx}}{x} dx = \ln \left(\frac{b}{a} \right)}.$$

Exercise

Compute, for $n \in \mathbb{N}$,

$$I_n = \int_0^{+\infty} x^{2n} e^{-x^2} dx.$$

Solution

Consider the function

$$J(a) = \int_0^{+\infty} e^{-ax^2} dx, \quad a > 0.$$

We know that

$$J(a) = \frac{1}{2} \sqrt{\frac{\pi}{a}} = \frac{1}{2} \sqrt{\pi} a^{-1/2}.$$

Differentiate J n times with respect to a . On one hand:

$$J^{(n)}(a) = (-1)^n \int_0^{+\infty} x^{2n} e^{-ax^2} dx.$$

On the other hand:

$$J(a) = \frac{1}{2} \sqrt{\pi} a^{-1/2},$$

so

$$J^{(n)}(a) = \frac{1}{2} \sqrt{\pi} (-1)^n \frac{(2n-1)!!}{2^n} a^{-(2n+1)/2}.$$

Matching, we obtain:

$$\int_0^{+\infty} x^{2n} e^{-ax^2} dx = \frac{1}{2} \sqrt{\pi} \frac{(2n-1)!!}{2^n} a^{-(2n+1)/2}.$$

For the case $a = 1$:

$$I_n = \frac{1}{2} \sqrt{\pi} \frac{(2n-1)!!}{2^n}.$$

Using

$$(2n-1)!! = \frac{(2n)!}{2^n n!},$$

we finally obtain:

$$\boxed{\int_0^{+\infty} x^{2n} e^{-x^2} dx = \frac{\sqrt{\pi}}{2} \cdot \frac{(2n)!}{4^n n!}.$$

Exercise

Compute, for $a > 0$,

$$I(a) = \int_0^{+\infty} \frac{\cos(ax)}{1+x^2} dx.$$

Solution

Consider the complex function

$$f(z) = \frac{e^{iaz}}{1+z^2}, \quad a > 0.$$

Integrate f along the contour consisting of the segment $[-R, R]$ and the upper semicircle of radius R . Since $a > 0$, we have

$$|e^{iaz}| = e^{-a \Im z},$$

and because $a > 0$, the contribution of the semicircle tends to zero as $R \rightarrow +\infty$ (Jordan's lemma). The poles of f are at $z = i$ and $z = -i$. In the upper half-plane, the only pole is $z = i$. Its residue:

$$\text{Res}(f, i) = \lim_{z \rightarrow i} \frac{e^{iaz}}{z+i} = \frac{e^{iai}}{2i} = \frac{e^{-a}}{2i}.$$

By the residue theorem:

$$\int_{-\infty}^{+\infty} \frac{e^{iax}}{1+x^2} dx = 2\pi i \cdot \frac{e^{-a}}{2i} = \pi e^{-a}.$$

Taking the real part:

$$\int_{-\infty}^{+\infty} \frac{\cos(ax)}{1+x^2} dx = \pi e^{-a}.$$

The function $x \mapsto \frac{\cos(ax)}{1+x^2}$ is even, hence

$$2I(a) = \pi e^{-a}.$$

Thus:

$$\boxed{\int_0^{+\infty} \frac{\cos(ax)}{1+x^2} dx = \frac{\pi}{2} e^{-a}, \quad a > 0.}$$

Laplace Transform

Theorem

[Definitions and Basic Properties]

1. Definition. For a function f defined on $[0, +\infty[$ such that the integral converges,

$$\mathcal{L}\{f\}(s) = \int_0^{+\infty} e^{-st} f(t) dt.$$

2. Existence condition. If f is piecewise continuous and of exponential order α , i.e.,

$$|f(t)| \leq M e^{\alpha t} \quad (t \geq 0),$$

then $\mathcal{L}\{f\}(s)$ exists for $\Re(s) > \alpha$.

3. Linearity.

$$\mathcal{L}\{af + bg\} = a\mathcal{L}\{f\} + b\mathcal{L}\{g\}.$$

4. Shift in s . If $\mathcal{L}\{f(t)\} = F(s)$, then

$$\mathcal{L}\{e^{at} f(t)\} = F(s - a).$$

5. Shift in t . If $a > 0$ and u is the Heaviside step function,

$$\mathcal{L}\{f(t - a)u(t - a)\} = e^{-as} F(s).$$

6. Differentiation in t . If f is n times differentiable,

$$\mathcal{L}\{f^{(n)}(t)\} = s^n F(s) - s^{n-1} f(0) - s^{n-2} f'(0) - \dots - f^{(n-1)}(0).$$

7. Differentiation in s .

$$\mathcal{L}\{t^n f(t)\} = (-1)^n F^{(n)}(s), \quad \text{or} \quad \mathcal{L}\{(-t)^n f(t)\} = F^{(n)}(s).$$

8. Integration in t .

$$\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = \frac{1}{s} F(s).$$

9. Division by t . Under the usual assumptions,

$$\mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^{+\infty} F(u) du.$$

10. Convolution. If

$$(f * g)(t) = \int_0^t f(\tau)g(t - \tau) d\tau,$$

CHAPTER 4. LAPLACE TRANSFORM

then

$$\mathcal{L}\{f * g\} = F(s)G(s).$$

11. Scaling. For $a > 0$,

$$\mathcal{L}\{f(at)\}(s) = \frac{1}{a} F\left(\frac{s}{a}\right).$$

12. Initial value theorem. If f has a right-hand limit at 0,

$$\lim_{t \rightarrow 0^+} f(t) = \lim_{s \rightarrow +\infty} sF(s).$$

13. Final value theorem. If all poles of $sF(s)$ lie in the half-plane $\Re(s) < 0$, except possibly a simple pole at 0,

$$\lim_{t \rightarrow +\infty} f(t) = \lim_{s \rightarrow 0} sF(s).$$

14. Periodic function. If $f(t+T) = f(t)$ for $t \geq 0$, then

$$\mathcal{L}\{f(t)\} = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt.$$

Theorem

[Table of Standard Laplace Transforms]

$f(t)$	$\mathcal{L}\{f(t)\}(s)$	Condition
$\delta(t)$	1	—
$1 = u(t)$	$\frac{1}{s}$	$\Re(s) > 0$
$t^n \ (n \in \mathbb{N})$	$\frac{s!}{n!}$	$\Re(s) > 0$
$t^\alpha \ (\alpha > -1)$	$\frac{\Gamma(\alpha+1)}{s^{\alpha+1}}$	$\Re(s) > 0$
e^{at}	$\frac{1}{s-a}$	$\Re(s) > \Re(a)$
$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$	$\Re(s) > 0$
$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$	$\Re(s) > 0$
$\cosh(at)$	$\frac{s}{s^2 - a^2}$	$\Re(s) > a $
$\sinh(at)$	$\frac{a}{s^2 - a^2}$	$\Re(s) > a $
$t^n e^{at}$	$\frac{n!}{(s-a)^{n+1}}$	$\Re(s) > \Re(a)$
$e^{at} \cos(\omega t)$	$\frac{(s-a)\omega}{(s-a)^2 + \omega^2}$	$\Re(s) > \Re(a)$
$e^{at} \sin(\omega t)$	$\frac{\omega}{(s-a)^2 + \omega^2}$	$\Re(s) > \Re(a)$
$J_0(at)$	$\frac{1}{\sqrt{s^2 + a^2}}$	$\Re(s) > 0$
$\operatorname{erf}(\sqrt{at})$	$\frac{\sqrt{a}}{s\sqrt{s+a}}$	$\Re(s) > 0$

Theorem

[Inverse Transform]

1. Bromwich inversion formula.

$$f(t) = \frac{1}{2\pi i} \int_{\gamma-i\infty}^{\gamma+i\infty} e^{st} F(s) ds,$$

where γ is chosen to the right of all singularities of F .

CHAPTER 4. LAPLACE TRANSFORM

2. Residues. If F is meromorphic and the contour can be closed to the left, then

$$f(t) = \sum_k \operatorname{Res}(e^{st}F(s), s_k).$$

3. Partial fractions. If

$$F(s) = \frac{P(s)}{Q(s)}, \quad \deg P < \deg Q,$$

and if the roots of Q are simple $s_1, \dots, s_n$, then

$$\mathcal{L}^{-1}\{F(s)\}(t) = \sum_{k=1}^n \frac{P(s_k)}{Q'(s_k)} e^{s_k t}.$$

Theorem

[Applications]

1. Linear ordinary differential equations. For

$$a_n y^{(n)} + \dots + a_1 y' + a_0 y = g(t),$$

apply the transform, solve algebraically for $Y(s)$, then invert.

2. Systems. For

$$\mathbf{x}' = A\mathbf{x} + \mathbf{g}(t),$$

apply the transform componentwise.

3. Integro-differential equations. If

$$y^{(n)}(t) + \int_0^t K(t-\tau)y(\tau) d\tau = g(t),$$

then $\mathcal{L}\{K * y\} = K(s)Y(s)$, which reduces the equation to an algebraic relation.

4. Partial differential equations. Apply the transform with respect to t , solve the resulting ordinary differential equation in the spatial variable, then invert.

Exercise

Show that for $A \in \mathbb{C}$, $n > -1$ and $s > 0$,

$$\mathcal{L}\{At^n\}(s) = A \frac{\Gamma(n+1)}{s^{n+1}}.$$

Solution

1. By definition,

$$\mathcal{L}\{At^n\}(s) = A \int_0^{+\infty} t^n e^{-st} dt.$$

Set $u = st$. Then $t = u/s$ and $dt = du/s$. Hence

$$\mathcal{L}\{At^n\}(s) = A \int_0^{+\infty} \left(\frac{u}{s}\right)^n e^{-u} \frac{du}{s} = \frac{A}{s^{n+1}} \int_0^{+\infty} u^n e^{-u} du.$$

We have

$$\Gamma(n+1) = \int_0^{+\infty} u^n e^{-u} du.$$

Thus

$$\mathcal{L}\{At^n\}(s) = A \frac{\Gamma(n+1)}{s^{n+1}}.$$

CHAPTER 4. LAPLACE TRANSFORM

Exercise

Compute

$$\mathcal{L}\{1\}, \quad \mathcal{L}\{2t\}, \quad \mathcal{L}\{\sqrt{t}\}.$$

Solution

1. $\mathcal{L}\{1\} = \mathcal{L}\{t^0\} = \frac{\Gamma(1)}{s} = \frac{1}{s}.$
2. $\mathcal{L}\{2t\} = 2\mathcal{L}\{t\} = 2\frac{\Gamma(2)}{s^2} = \frac{2}{s^2}.$
3. Since $\sqrt{t} = t^{1/2},$

$$\mathcal{L}\{\sqrt{t}\} = \frac{\Gamma(\frac{3}{2})}{s^{3/2}} = \frac{\sqrt{\pi}}{2s^{3/2}}.$$

Thus

$$\mathcal{L}\{1\} = \frac{1}{s}, \quad \mathcal{L}\{2t\} = \frac{2}{s^2}, \quad \mathcal{L}\{\sqrt{t}\} = \frac{\sqrt{\pi}}{2s^{3/2}}.$$

Exercise

Compute the Laplace transform of e^{iat} , where $i^2 = -1$ and $a \in \mathbb{R}.$

Solution

1.

$$\mathcal{L}\{e^{iat}\}(s) = \int_0^{+\infty} e^{-(s-ia)t} dt.$$

This integral converges for $\Re(s) > 0.$ We find

$$\mathcal{L}\{e^{iat}\}(s) = \frac{1}{s - ia}.$$

Rationalizing the denominator,

$$\frac{1}{s - ia} = \frac{s + ia}{s^2 + a^2}.$$

Thus

$$\mathcal{L}\{e^{iat}\}(s) = \frac{1}{s - ia} = \frac{s + ia}{s^2 + a^2}, \quad \Re(s) > 0.$$

Exercise

Deduce the Laplace transforms of $\cos(at)$ and $\sin(at).$

Solution

1. Use

$$\cos(at) = \frac{e^{iat} + e^{-iat}}{2}, \quad \sin(at) = \frac{e^{iat} - e^{-iat}}{2i}.$$

Then

$$\mathcal{L}\{\cos(at)\} = \frac{1}{2} \left(\frac{1}{s - ia} + \frac{1}{s + ia} \right) = \frac{s}{s^2 + a^2},$$

and

$$\mathcal{L}\{\sin(at)\} = \frac{1}{2i} \left(\frac{1}{s - ia} - \frac{1}{s + ia} \right) = \frac{a}{s^2 + a^2}.$$

Thus

$$\mathcal{L}\{\cos(at)\} = \frac{s}{s^2 + a^2}, \quad \mathcal{L}\{\sin(at)\} = \frac{a}{s^2 + a^2}.$$

Exercise

Compute:

1. $\mathcal{L}\{t^2 \sin(3t)\}$
2. $\mathcal{L}\{e^{-2t} \cosh(4t)\}$
3. $\mathcal{L}\{te^{-t} \cos(2t)\}$
4. $\mathcal{L}\left\{\frac{\sin t}{t}\right\}$
5. $\mathcal{L}\left\{\int_0^t \frac{\sin \tau}{\tau} d\tau\right\}$

Solution

1. We know that

$$\mathcal{L}\{\sin(3t)\} = \frac{3}{s^2 + 9}.$$

Since $\mathcal{L}\{t^2 f(t)\} = F''(s)$, differentiate twice:

$$F(s) = \frac{3}{s^2 + 9}, \quad F''(s) = \frac{18(s^2 - 3)}{(s^2 + 9)^3}.$$

Thus

$$\mathcal{L}\{t^2 \sin(3t)\} = \frac{18(s^2 - 3)}{(s^2 + 9)^3}.$$

2. Use shift in s . We have

$$\mathcal{L}\{\cosh(4t)\} = \frac{s}{s^2 - 16},$$

substitute s with $s + 2$:

$$\mathcal{L}\{e^{-2t} \cosh(4t)\} = \frac{s + 2}{(s + 2)^2 - 16}.$$

3. First

$$\mathcal{L}\{e^{-t} \cos(2t)\} = \frac{s + 1}{(s + 1)^2 + 4}.$$

Then $\mathcal{L}\{tf(t)\} = -F'(s)$, hence

$$\mathcal{L}\{te^{-t} \cos(2t)\} = \frac{(s + 1)^2 - 4}{((s + 1)^2 + 4)^2}.$$

4. Use

$$\mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^{+\infty} F(u) du, \quad F(u) = \frac{1}{u^2 + 1}.$$

Then

$$\mathcal{L}\left\{\frac{\sin t}{t}\right\} = \int_s^{+\infty} \frac{du}{u^2 + 1} = \frac{\pi}{2} - \arctan s = \arctan\left(\frac{1}{s}\right).$$

Thus

$$\mathcal{L}\left\{\frac{\sin t}{t}\right\} = \arctan\left(\frac{1}{s}\right).$$

5. Finally,

$$\mathcal{L}\left\{\int_0^t \frac{\sin \tau}{\tau} d\tau\right\} = \frac{1}{s} \mathcal{L}\left\{\frac{\sin t}{t}\right\} = \frac{1}{s} \arctan\left(\frac{1}{s}\right).$$

Exercise

1. Compute $\mathcal{L}^{-1} \left\{ \frac{1}{(s^2 + 1)^2} \right\}$ using convolution.
2. Show that $\mathcal{L}\{J_0(t)\} = \frac{1}{\sqrt{s^2 + 1}}$.
3. Compute $\mathcal{L}\{J_0 * J_0\}$.

Solution

1. Write

$$\frac{1}{(s^2 + 1)^2} = \frac{1}{s^2 + 1} \cdot \frac{1}{s^2 + 1}.$$

We have

$$\mathcal{L}^{-1} \left\{ \frac{1}{s^2 + 1} \right\} = \sin t.$$

Thus

$$\mathcal{L}^{-1} \left\{ \frac{1}{(s^2 + 1)^2} \right\} = (\sin * \sin)(t) = \int_0^t \sin \tau \sin(t - \tau) d\tau.$$

Using

$$\sin \tau \sin(t - \tau) = \frac{1}{2} (\cos(2\tau - t) - \cos t),$$

we obtain

$$(\sin * \sin)(t) = \frac{1}{2} \sin t - \frac{t}{2} \cos t.$$

Hence

$$\mathcal{L}^{-1} \left\{ \frac{1}{(s^2 + 1)^2} \right\} = \frac{1}{2} \sin t - \frac{t}{2} \cos t.$$

2. From the series expansion

$$J_0(t) = \sum_{k=0}^{+\infty} \frac{(-1)^k}{(k!)^2} \left(\frac{t}{2} \right)^{2k},$$

transform term by term:

$$\mathcal{L}\{J_0(t)\} = \sum_{k=0}^{+\infty} \frac{(-1)^k}{(k!)^2 2^{2k}} \frac{(2k)!}{s^{2k+1}}.$$

This expansion is known to be the expansion of $(s^2 + 1)^{-1/2}$, hence

$$\mathcal{L}\{J_0(t)\} = \frac{1}{\sqrt{s^2 + 1}}.$$

3. By convolution,

$$\mathcal{L}\{J_0 * J_0\} = \mathcal{L}\{J_0\}^2 = \frac{1}{s^2 + 1}.$$

Thus

$$\mathcal{L}\{J_0 * J_0\} = \frac{1}{s^2 + 1}, \quad (J_0 * J_0)(t) = \sin t.$$

Exercise

Solve using the Laplace transform:

1. $y'' + 4y' + 3y = e^{-t}$, $y(0) = 1$, $y'(0) = 0$
2. $y''' - y = t$, $y(0) = y'(0) = y''(0) = 0$
3. $y'' + y = \sin t$, $y(0) = 0$, $y'(0) = 1$

4.

$$\begin{cases} x' - y = e^t, \\ y' + x = \sin t, \end{cases} \quad x(0) = 1, y(0) = 0.$$

Solution

1. Let $Y(s) = \mathcal{L}\{y\}$,

$$(s^2Y - s) + 4(sY - 1) + 3Y = \frac{1}{s+1}.$$

Thus

$$Y(s) = \frac{s^2 + 5s + 5}{(s+1)^2(s+3)}.$$

Using partial fractions:

$$Y(s) = \frac{5/4}{s+1} + \frac{1/2}{(s+1)^2} - \frac{1/4}{s+3}.$$

Hence

$$y(t) = \frac{5}{4}e^{-t} + \frac{1}{2}te^{-t} - \frac{1}{4}e^{-3t}.$$

2. We have

$$s^3Y(s) - Y(s) = \frac{1}{s^2}, \quad \Rightarrow \quad Y(s) = \frac{1}{s^2(s^3 - 1)}.$$

Since $s^3 - 1 = (s-1)(s^2 + s + 1)$, we obtain

$$Y(s) = -\frac{1}{s^2} + \frac{1}{3(s-1)} - \frac{s-1}{3(s^2 + s + 1)}.$$

Write $s^2 + s + 1 = (s + \frac{1}{2})^2 + \frac{3}{4}$, then invert each term:

$$y(t) = -t + \frac{e^t}{3} - \frac{1}{3}e^{-t/2} \cos\left(\frac{\sqrt{3}}{2}t\right) + \frac{\sqrt{3}}{3}e^{-t/2} \sin\left(\frac{\sqrt{3}}{2}t\right).$$

3. We have

$$(s^2Y - 1) + Y = \frac{1}{s^2 + 1},$$

so

$$Y(s) = \frac{s^2 + 2}{(s^2 + 1)^2} = \frac{1}{s^2 + 1} + \frac{1}{(s^2 + 1)^2}.$$

Using the previous result:

$$y(t) = \sin t + \left(\frac{1}{2} \sin t - \frac{t}{2} \cos t\right) = \frac{3}{2} \sin t - \frac{t}{2} \cos t.$$

4. Let $X(s) = \mathcal{L}\{x\}$, $Y(s) = \mathcal{L}\{y\}$,

$$sX - 1 - Y = \frac{1}{s-1}, \quad sY + X = \frac{1}{s^2 + 1}.$$

Solve the system:

$$X(s) = \frac{1}{2(s-1)} + \frac{s+1}{2(s^2+1)} + \frac{1}{(s^2+1)^2},$$

$$Y(s) = -\frac{1}{2(s-1)} + \frac{s-1}{2(s^2+1)} + \frac{s}{(s^2+1)^2}.$$

Inverting:

$$x(t) = \frac{1}{2}e^t + \frac{1}{2}\cos t + \sin t - \frac{t}{2}\cos t,$$

$$y(t) = -\frac{1}{2}e^t + \frac{1}{2}\cos t - \frac{1}{2}\sin t + \frac{t}{2}\sin t.$$

Exercise

Let f be a periodic function with period $T > 0$, defined on $[0, T[$ by

$$f(t) = \begin{cases} 1, & 0 \leq t < \frac{T}{2}, \\ -1, & \frac{T}{2} \leq t < T. \end{cases}$$

1. Compute $\mathcal{L}\{f(t)\}$.
2. Deduce that

$$\mathcal{L}\{f(t)\} = \frac{1}{s} \tanh\left(\frac{sT}{4}\right).$$

Solution

1. Using the formula for periodic functions,

$$\mathcal{L}\{f(t)\} = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt.$$

We have

$$\int_0^T e^{-st} f(t) dt = \int_0^{T/2} e^{-st} dt - \int_{T/2}^T e^{-st} dt = \frac{(1 - e^{-sT/2})^2}{s}.$$

Thus

$$\mathcal{L}\{f(t)\} = \frac{1}{s} \frac{(1 - e^{-sT/2})^2}{1 - e^{-sT}} = \frac{1}{s} \frac{1 - e^{-sT/2}}{1 + e^{-sT/2}}.$$

2. Since $\tanh z = \frac{1 - e^{-2z}}{1 + e^{-2z}}$, setting $z = sT/4$ we obtain

$$\mathcal{L}\{f(t)\} = \frac{1}{s} \tanh\left(\frac{sT}{4}\right).$$

Exercise

Solve the equation

$$\begin{cases} \frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}, & x > 0, t > 0, \\ u(0, t) = u_0, \\ u(x, 0) = 0, \\ \lim_{x \rightarrow +\infty} u(x, t) = 0. \end{cases}$$

Solution

1. Let $U(x, s) = \mathcal{L}_t\{u(x, t)\}$. Since $u(x, 0) = 0$,

$$sU(x, s) = k U_{xx}(x, s).$$

Solve the ordinary differential equation

$$U_{xx} - \frac{s}{k}U = 0,$$

giving

$$U(x, s) = A(s)e^{\sqrt{s/k}x} + B(s)e^{-\sqrt{s/k}x}.$$

The condition at infinity forces $A(s) = 0$. From $u(0, t) = u_0$,

$$U(0, s) = \frac{u_0}{s} = B(s).$$

Thus

$$U(x, s) = \frac{u_0}{s}e^{-x\sqrt{s/k}}.$$

Using the known formula

$$\mathcal{L}^{-1}\left\{\frac{1}{s}e^{-a\sqrt{s}}\right\} = \operatorname{erfc}\left(\frac{a}{2\sqrt{t}}\right).$$

Setting $a = x/\sqrt{k}$, we find

$$u(x, t) = u_0 \operatorname{erfc}\left(\frac{x}{2\sqrt{kt}}\right).$$

Exercise

The bilateral (two-sided) Laplace transform is defined by

$$\mathcal{L}_b\{f\}(s) = \int_{-\infty}^{+\infty} e^{-st} f(t) dt.$$

1. Show that if $f(t) = 0$ for $t < 0$, then $\mathcal{L}_b\{f\} = \mathcal{L}\{f\}$.
2. Compute $\mathcal{L}_b\{e^{-a|t|}\}$, $a > 0$.
3. Show that $\mathcal{L}_b\{f'\}(s) = s\mathcal{L}_b\{f\}(s)$ if $f(t)e^{-st} \rightarrow 0$ as $t \rightarrow \pm\infty$.
4. Compute $\mathcal{L}_b\{\delta(t - t_0)\}$.

Solution

1. If $f(t) = 0$ for $t < 0$, then

$$\mathcal{L}_b\{f\}(s) = \int_0^{+\infty} e^{-st} f(t) dt = \mathcal{L}\{f\}(s).$$

2. Separate:

$$\mathcal{L}_b\{e^{-a|t|}\}(s) = \int_{-\infty}^0 e^{(a-s)t} dt + \int_0^{+\infty} e^{-(a+s)t} dt.$$

Convergence requires $-a < \Re(s) < a$. We find

$$\int_{-\infty}^0 e^{(a-s)t} dt = \frac{1}{a-s}, \quad \int_0^{+\infty} e^{-(a+s)t} dt = \frac{1}{a+s}.$$

Thus

$$\mathcal{L}_b\{e^{-a|t|}\}(s) = \frac{1}{a-s} + \frac{1}{a+s} = \frac{2a}{a^2 - s^2}, \quad -a < \Re(s) < a.$$

3. Integrating by parts,

$$\mathcal{L}_b\{f'\}(s) = \int_{-\infty}^{+\infty} e^{-st} f'(t) dt = \left[e^{-st} f(t) \right]_{-\infty}^{+\infty} + s \int_{-\infty}^{+\infty} e^{-st} f(t) dt.$$

CHAPTER 4. LAPLACE TRANSFORM

Under the given assumption, the boundary term vanishes, giving

$$\mathcal{L}_b\{f'\}(s) = s\mathcal{L}_b\{f\}(s).$$

4. By the Dirac property,

$$\mathcal{L}_b\{\delta(t - t_0)\}(s) = e^{-st_0}.$$

Exercise

Verify the initial value and final value theorems for:

1. $F(s) = \frac{1}{s(s+1)}$
2. $F(s) = \frac{s+2}{s^2+4}$
3. $F(s) = \frac{e^{-s}}{s+1}$

Solution

- 1.

$$F(s) = \frac{1}{s(s+1)} \implies f(t) = 1 - e^{-t}.$$

We have

$$\lim_{s \rightarrow +\infty} sF(s) = \lim_{s \rightarrow +\infty} \frac{1}{s+1} = 0 = \lim_{t \rightarrow 0^+} f(t),$$

and

$$\lim_{s \rightarrow 0} sF(s) = 1 = \lim_{t \rightarrow +\infty} f(t).$$

- 2.

$$F(s) = \frac{s+2}{s^2+4} = \frac{s}{s^2+4} + \frac{2}{s^2+4},$$

thus

$$f(t) = \cos(2t) + \sin(2t).$$

We have

$$\lim_{s \rightarrow +\infty} sF(s) = 1 = \lim_{t \rightarrow 0^+} f(t).$$

However, $f(t)$ has no limit as $t \rightarrow +\infty$ because it oscillates. The final value theorem does not apply, which is consistent with the presence of purely imaginary poles.

- 3.

$$F(s) = \frac{e^{-s}}{s+1} \implies f(t) = e^{-(t-1)}u(t-1).$$

Then

$$\lim_{s \rightarrow +\infty} sF(s) = 0 = \lim_{t \rightarrow 0^+} f(t),$$

and

$$\lim_{s \rightarrow 0} sF(s) = 0 = \lim_{t \rightarrow +\infty} f(t).$$

Exercise

Solve using the Laplace transform:

1. $y'(t) + \int_0^t y(\tau) d\tau = 1, y(0) = 0$
2. $y(t) = t^2 + \int_0^t y(\tau) \sin(t-\tau) d\tau$

Solution

1. Let $Y(s) = \mathcal{L}\{y\}$. Then

$$sY(s) + \frac{Y(s)}{s} = \frac{1}{s}.$$

Thus

$$Y(s) \left(s + \frac{1}{s} \right) = \frac{1}{s}, \quad Y(s) = \frac{1}{s^2 + 1}.$$

Hence

$$\boxed{y(t) = \sin t.}$$

2. Here

$$y = t^2 + y * \sin.$$

Then

$$Y(s) = \frac{2}{s^3} + \frac{Y(s)}{s^2 + 1}.$$

Hence

$$Y(s) \left(1 - \frac{1}{s^2 + 1} \right) = \frac{2}{s^3}, \quad Y(s) = \frac{2(s^2 + 1)}{s^5} = \frac{2}{s^3} + \frac{2}{s^5}.$$

Inverting:

$$\boxed{y(t) = t^2 + \frac{t^4}{12}.$$

Exercise

Compute:

1. $\mathcal{L}\{J_\nu(at)\}$
2. $\mathcal{L}\{\operatorname{erf}(\sqrt{at})\}$
3. $\mathcal{L}\{t^{\nu/2} J_\nu(2\sqrt{at})\}$

Solution

1. The classical formula:

$$\boxed{\mathcal{L}\{J_\nu(at)\} = \frac{a^\nu}{\sqrt{s^2 + a^2} (s + \sqrt{s^2 + a^2})^\nu}, \quad \Re(s) > 0,$$

which for $\nu = 0$ gives

$$\mathcal{L}\{J_0(at)\} = \frac{1}{\sqrt{s^2 + a^2}}.$$

- 2.

$$\boxed{\mathcal{L}\{\operatorname{erf}(\sqrt{at})\} = \frac{\sqrt{a}}{s\sqrt{s+a}}, \quad \Re(s) > 0.$$

- 3.

$$\boxed{\mathcal{L}\{t^{\nu/2} J_\nu(2\sqrt{at})\} = \frac{a^{\nu/2}}{s^{\nu+1}} e^{-a/s}.$$

Exercise

Solve the problem

$$\begin{cases} u_t = u_{xx}, & 0 < x < \ell, t > 0, \\ u(0, t) = T_0, & u(\ell, t) = T_1, \\ u(x, 0) = 0. \end{cases}$$

Solution

1. Define the steady-state solution

$$u_s(x) = T_0 + \frac{T_1 - T_0}{\ell}x,$$

and set

$$v(x, t) = u(x, t) - u_s(x).$$

Then v satisfies

$$v_t = v_{xx}, \quad v(0, t) = v(\ell, t) = 0, \quad v(x, 0) = -u_s(x).$$

Using the sine Fourier series,

$$v(x, t) = \sum_{n=1}^{+\infty} b_n \sin\left(\frac{n\pi x}{\ell}\right) e^{-n^2 \pi^2 t / \ell^2},$$

where

$$b_n = \frac{2}{\ell} \int_0^\ell \left(-T_0 - \frac{T_1 - T_0}{\ell}x \right) \sin\left(\frac{n\pi x}{\ell}\right) dx = \frac{2}{n\pi} (T_1(-1)^n - T_0).$$

Thus

$$u(x, t) = T_0 + \frac{T_1 - T_0}{\ell}x + \sum_{n=1}^{+\infty} \frac{2}{n\pi} (T_1(-1)^n - T_0) \sin\left(\frac{n\pi x}{\ell}\right) e^{-n^2 \pi^2 t / \ell^2}.$$

Exercise

Show that for the Caputo derivative of order $\alpha > 0$,

$$\mathcal{L}\{ {}^C D^\alpha f(t) \} = s^\alpha F(s) - \sum_{k=0}^{n-1} s^{\alpha-1-k} f^{(k)}(0), \quad n = \lceil \alpha \rceil.$$

Solution

1. Write

$${}^C D^\alpha f(t) = \frac{1}{\Gamma(n-\alpha)} \int_0^t (t-\tau)^{n-\alpha-1} f^{(n)}(\tau) d\tau.$$

This is a convolution:

$${}^C D^\alpha f = g * f^{(n)}, \quad g(t) = \frac{t^{n-\alpha-1}}{\Gamma(n-\alpha)}.$$

Thus

$$\mathcal{L}\{ {}^C D^\alpha f \} = \mathcal{L}\{g\} \mathcal{L}\{f^{(n)}\}.$$

We have

$$\mathcal{L}\{g\} = \frac{1}{s^{n-\alpha}},$$

and

$$\mathcal{L}\{f^{(n)}\} = s^n F(s) - \sum_{k=0}^{n-1} s^{n-1-k} f^{(k)}(0).$$

Hence

$$\mathcal{L}\{ {}^C D^\alpha f \} = \frac{1}{s^{n-\alpha}} \left(s^n F(s) - \sum_{k=0}^{n-1} s^{n-1-k} f^{(k)}(0) \right),$$

which gives

$$\mathcal{L}\{{}^C D^\alpha f(t)\} = s^\alpha F(s) - \sum_{k=0}^{n-1} s^{\alpha-1-k} f^{(k)}(0).$$

Derivative of Positive Real Order

Theorem

[Definitions and Properties]

1. Riemann-Liouville derivative. For $\alpha > 0$ and $n = \lceil \alpha \rceil$,

$$D^\alpha f(t) = \frac{1}{\Gamma(n-\alpha)} \frac{d^n}{dt^n} \int_0^t (t-s)^{n-\alpha-1} f(s) ds.$$

2. Caputo derivative. For $\alpha \in (n-1, n)$,

$${}^C D^\alpha f(t) = \frac{1}{\Gamma(n-\alpha)} \int_0^t (t-s)^{n-\alpha-1} f^{(n)}(s) ds.$$

3. Power rule. For $\beta > -1$,

$$D^\alpha(t^\beta) = \frac{\Gamma(\beta+1)}{\Gamma(\beta-\alpha+1)} t^{\beta-\alpha},$$

provided the right-hand side is well-defined.

4. Linearity.

$$D^\alpha(af + bg) = aD^\alpha f + bD^\alpha g.$$

5. Integer case. If $\alpha = n \in \mathbb{N}$,

$$D^n f = \frac{d^n f}{dt^n}.$$

6. Semigroup property. Under sufficient assumptions,

$$D^\alpha(D^\beta f) = D^{\alpha+\beta} f.$$

7. Limit to integer order. When $\alpha \rightarrow n$, we recover the ordinary derivative of order n , for sufficiently regular functions.

Exercise

Compute

$$\mathcal{L}^{-1} \left\{ \frac{1}{s^k} \right\}, \quad k > 0.$$

Solution

1. We know that

$$\mathcal{L}\{t^{k-1}\} = \frac{\Gamma(k)}{s^k}.$$

Thus

$$\frac{1}{s^k} = \mathcal{L} \left\{ \frac{t^{k-1}}{\Gamma(k)} \right\}.$$

Inverting:

$$\mathcal{L}^{-1} \left\{ \frac{1}{s^k} \right\} = \frac{t^{k-1}}{\Gamma(k)}.$$

Exercise

Let $f(t) = t^n, n \geq 0$. Compute $\mathcal{L}\{D^\alpha(t^n)\}$.

Solution

1. Using

$$D^\alpha(t^n) = \frac{\Gamma(n+1)}{\Gamma(n-\alpha+1)} t^{n-\alpha}.$$

Then

$$\mathcal{L}\{D^\alpha(t^n)\} = \frac{\Gamma(n+1)}{\Gamma(n-\alpha+1)} \mathcal{L}\{t^{n-\alpha}\} = \frac{\Gamma(n+1)}{\Gamma(n-\alpha+1)} \frac{\Gamma(n-\alpha+1)}{s^{n-\alpha+1}}.$$

Thus

$$\mathcal{L}\{D^\alpha(t^n)\} = \frac{\Gamma(n+1)}{s^{n-\alpha+1}}.$$

Exercise

Show that

$$D^\alpha(t^n) = \frac{\Gamma(n+1)}{\Gamma(n-\alpha+1)} t^{n-\alpha}.$$

Solution

1. Start from the Riemann-Liouville definition:

$$D^\alpha(t^n) = \frac{1}{\Gamma(m-\alpha)} \frac{d^m}{dt^m} \int_0^t (t-\tau)^{m-\alpha-1} \tau^n d\tau, \quad m = \lceil \alpha \rceil.$$

Set $\tau = tu$. Then

$$\int_0^t (t-\tau)^{m-\alpha-1} \tau^n d\tau = t^{n+m-\alpha} \int_0^1 (1-u)^{m-\alpha-1} u^n du.$$

The integral is the Beta integral:

$$\int_0^1 (1-u)^{m-\alpha-1} u^n du = B(n+1, m-\alpha) = \frac{\Gamma(n+1)\Gamma(m-\alpha)}{\Gamma(n+m-\alpha+1)}.$$

Thus

$$D^\alpha(t^n) = \frac{\Gamma(n+1)}{\Gamma(n+m-\alpha+1)} \frac{d^m}{dt^m} (t^{n+m-\alpha}).$$

We have

$$\frac{d^m}{dt^m} (t^{n+m-\alpha}) = \frac{\Gamma(n+m-\alpha+1)}{\Gamma(n-\alpha+1)} t^{n-\alpha}.$$

Therefore

$$D^\alpha(t^n) = \frac{\Gamma(n+1)}{\Gamma(n-\alpha+1)} t^{n-\alpha}.$$

Exercise

Deduce the same formula using the Laplace transform.

Solution

1. We have

$$\mathcal{L}\{t^n\} = \frac{\Gamma(n+1)}{s^{n+1}}.$$

Using the symbolic rule

$$\mathcal{L}\{D^\alpha f\} = s^\alpha \mathcal{L}\{f\},$$

in the case of the Riemann-Liouville derivative applied to t^n , we obtain

$$\mathcal{L}\{D^\alpha(t^n)\} = s^\alpha \frac{\Gamma(n+1)}{s^{n+1}} = \frac{\Gamma(n+1)}{s^{n-\alpha+1}}.$$

Inverting:

$$D^\alpha(t^n) = \Gamma(n+1) \mathcal{L}^{-1}\left\{\frac{1}{s^{n-\alpha+1}}\right\} = \Gamma(n+1) \frac{t^{n-\alpha}}{\Gamma(n-\alpha+1)}.$$

Thus

$$D^\alpha(t^n) = \frac{\Gamma(n+1)}{\Gamma(n-\alpha+1)} t^{n-\alpha}.$$

Exercise

Compute:

1. $D^{1/2}(t)$
2. $D^{1/2} D^{1/2}(t)$
3. $D^{1/2}(t^2)$
4. $D^{1/2} D^{1/2}(t^2)$
5. $D^{3/2}(t^2)$

Solution

1. Apply

$$D^\alpha(t^n) = \frac{\Gamma(n+1)}{\Gamma(n-\alpha+1)} t^{n-\alpha}.$$

For $n = 1, \alpha = \frac{1}{2}$:

$$D^{1/2}(t) = \frac{\Gamma(2)}{\Gamma(\frac{3}{2})} t^{1/2} = \frac{2}{\sqrt{\pi}} \sqrt{t}.$$

$$D^{1/2}(t) = \frac{2}{\sqrt{\pi}} \sqrt{t}.$$

2.

$$D^{1/2} D^{1/2}(t) = D^{1/2}\left(\frac{2}{\sqrt{\pi}} t^{1/2}\right) = \frac{2}{\sqrt{\pi}} \frac{\Gamma(\frac{3}{2})}{\Gamma(1)} t^0 = 1.$$

$$D^{1/2} D^{1/2}(t) = 1.$$

3. For $n = 2, \alpha = \frac{1}{2}$:

$$D^{1/2}(t^2) = \frac{\Gamma(3)}{\Gamma(\frac{5}{2})} t^{3/2} = \frac{8}{3\sqrt{\pi}} t^{3/2}.$$

$$D^{1/2}(t^2) = \frac{8}{3\sqrt{\pi}} t^{3/2}.$$

CHAPTER 5. DERIVATIVE OF POSITIVE REAL ORDER

4. By the semigroup property:

$$D^{1/2} D^{1/2}(t^2) = D^1(t^2) = 2t.$$

$$D^{1/2} D^{1/2}(t^2) = 2t.$$

5. For $\alpha = \frac{3}{2}$:

$$D^{3/2}(t^2) = \frac{\Gamma(3)}{\Gamma(\frac{3}{2})} t^{1/2} = \frac{4}{\sqrt{\pi}} \sqrt{t}.$$

$$D^{3/2}(t^2) = \frac{4}{\sqrt{\pi}} \sqrt{t}.$$

Fourier Transforms

Theorem

Definitions, Basic Properties and Precise Notes

We adopt the convention

$$\hat{f}(\omega) = \mathcal{F}\{f\}(\omega) = \int_{-\infty}^{+\infty} f(t)e^{-i\omega t} dt, \quad f(t) = \mathcal{F}^{-1}\{\hat{f}\}(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \hat{f}(\omega)e^{i\omega t} d\omega.$$

Here we denote

$$\text{sinc } x = \frac{\sin x}{x}, \quad \text{sinc}(0) = 1.$$

1. **Fourier transform.** If $f \in L^1(\mathbb{R})$, then $\hat{f}$ is continuous and bounded, with

$$|\hat{f}(\omega)| \leq \|f\|_{L^1}.$$

2. **Inverse transform.** If $f, \hat{f} \in L^1(\mathbb{R})$, then for almost every t ,

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \hat{f}(\omega)e^{i\omega t} d\omega.$$

If f is continuous at t , the equality holds at that point.

3. **Linearity.** For $\alpha, \beta \in \mathbb{C}$,

$$\mathcal{F}\{\alpha f + \beta g\} = \alpha \hat{f} + \beta \hat{g}.$$

4. **Time shift.** For $t_0 \in \mathbb{R}$,

$$\mathcal{F}\{f(t - t_0)\}(\omega) = e^{-i\omega t_0} \hat{f}(\omega).$$

5. **Modulation (frequency shift).** For $\omega_0 \in \mathbb{R}$,

$$\mathcal{F}\{e^{i\omega_0 t} f(t)\}(\omega) = \hat{f}(\omega - \omega_0).$$

6. **Scaling.** If $a \in \mathbb{R}^*$,

$$\mathcal{F}\{f(at)\}(\omega) = \frac{1}{|a|} \hat{f}\left(\frac{\omega}{a}\right).$$

7. **Time differentiation.** If $f^{(n)} \in L^1(\mathbb{R})$ and integration by parts is legitimate,

$$\mathcal{F}\{f^{(n)}\}(\omega) = (i\omega)^n \hat{f}(\omega).$$

8. **Multiplication by t^n .** If $t^n f(t) \in L^1(\mathbb{R})$,

$$\mathcal{F}\{t^n f(t)\}(\omega) = i^n \hat{f}^{(n)}(\omega).$$

9. **Time convolution.** For

$$(f * g)(t) = \int_{-\infty}^{+\infty} f(\tau)g(t - \tau) d\tau,$$

we have

$$\mathcal{F}\{f * g\} = \hat{f} \hat{g}.$$

10. **Multiplication in time.**

$$\mathcal{F}\{fg\}(\omega) = \frac{1}{2\pi} (\hat{f} * \hat{g})(\omega).$$

11. **Parseval.** If $f, g \in L^2(\mathbb{R})$,

$$\int_{-\infty}^{+\infty} f(t) \overline{g(t)} dt = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \hat{f}(\omega) \overline{\hat{g}(\omega)} d\omega.$$

12. **Plancherel.** For $f \in L^2(\mathbb{R})$,

$$\|f\|_{L^2}^2 = \frac{1}{2\pi} \|\hat{f}\|_{L^2}^2.$$

13. **Symmetries.**

- If f is real, then $\hat{f}(-\omega) = \overline{\hat{f}(\omega)}$;
- If f is real and even, then $\hat{f}$ is real and even;
- If f is real and odd, then $\hat{f}$ is purely imaginary and odd.

14. **Standard distributions.**

$$\begin{aligned} \mathcal{F}\{\delta\} &= 1, & \mathcal{F}\{1\} &= 2\pi\delta, \\ \mathcal{F}\{u\} &= \pi\delta + \text{PV}\left(\frac{1}{i\omega}\right), & \mathcal{F}\{\text{sgn } t\} &= \text{PV}\left(\frac{2}{i\omega}\right). \end{aligned}$$

15. **Note on distributions.** Formulas involving u , sgn , δ , $\text{PV}(1/t)$, t^n or pure sinusoids must be understood in the sense of tempered distributions.

Theorem

Table of Standard Fourier Transforms

$f(t)$	$\hat{f}(\omega) = \mathcal{F}\{f\}(\omega)$
$e^{-a t } \ (a > 0)$	$\frac{2a}{a^2 + \omega^2}$
$e^{-at^2} \ (a > 0)$	$\sqrt{\frac{\pi}{a}} e^{-\omega^2/(4a)}$
$\frac{1}{a^2 + t^2} \ (a > 0)$	$\frac{\pi}{a} e^{-a \omega }$
$\text{rect}\left(\frac{t}{T}\right)$	$T \text{sinc}\left(\frac{\omega T}{2}\right)$
$\frac{\sin(at)}{t} \ (a > 0)$	$\pi \text{rect}\left(\frac{\omega}{2a}\right)$
$\cos(\omega_0 t)$	$\pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$
$\sin(\omega_0 t)$	$i\pi [\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$
$e^{i\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$
$u(t)$	$\pi\delta(\omega) + \text{PV}\left(\frac{1}{i\omega}\right)$
$\text{sgn}(t)$	$\text{PV}\left(\frac{2}{i\omega}\right)$
$ t ^{-1/2}$	$\sqrt{2\pi} \omega ^{-1/2}$
$J_0(at) \ (a > 0)$	$\frac{2}{\sqrt{a^2 - \omega^2}} \mathbf{1}_{\{ \omega < a\}}$

Theorem
Discrete Fourier Transform and FFT

1. **Discrete Fourier Transform (DFT).** For a sequence $(x[n])_{0 \leq n \leq N-1}$,

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-i2\pi kn/N}, \quad k = 0, \dots, N-1.$$

2. **Inverse transform.**

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{i2\pi kn/N}.$$

3. **Unitary orthogonality.** If $F = (e^{-i2\pi kn/N})_{k,n}$, then

$$F^* F = N I_N,$$

where I_N is the identity matrix. Hence $\frac{1}{\sqrt{N}} F$ is unitary.

4. **Circular convolution.** If

$$(x \otimes y)[n] = \sum_{m=0}^{N-1} x[m] y[(n-m) \bmod N],$$

then

$$\text{DFT}(x \otimes y) = \text{DFT}(x) \text{DFT}(y).$$

5. **Discrete Parseval.**

$$\sum_{n=0}^{N-1} |x[n]|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X[k]|^2.$$

6. **FFT algorithm.** The Cooley-Tukey algorithm uses even-odd decomposition and reduces complexity from

$$O(N^2) \quad \text{to} \quad O(N \log N).$$

Theorem
Physical and Signal Processing Applications

1. **Heat equation.**

$$u_t = k u_{xx} \implies \partial_t \hat{u} = -k \omega^2 \hat{u}.$$

2. **Wave equation.**

$$u_{tt} = c^2 u_{xx} \implies \partial_t^2 \hat{u} = -c^2 \omega^2 \hat{u}.$$

3. **Linear time-invariant filtering.**

$$y = x * h \implies \hat{y} = \hat{x} \hat{h}.$$

4. **Shannon-Nyquist sampling.** If the signal is bandlimited to B Hz, then alias-free reconstruction requires

$$f_s > 2B.$$

5. **Autocorrelation and spectrum.** If

$$R_{xx}(\tau) = \int_{-\infty}^{+\infty} x(t) \overline{x(t+\tau)} dt,$$

then

$$\mathcal{F}\{R_{xx}\} = |\hat{x}|^2.$$

Exercise

Compute the Fourier transforms of the following functions:

1. $f(t) = e^{-a|t|}, a > 0.$
2. $f(t) = \text{rect}\left(\frac{t}{T}\right).$
3. $f(t) = \cos(\omega_0 t)e^{-a|t|}, a > 0.$
4. $f(t) = te^{-at^2}, a > 0.$
5. $f(t) = \frac{\sin(at)}{t}, a > 0.$

Solution

1. Separate the integral:

$$\hat{f}(\omega) = \int_{-\infty}^0 e^{(a-i\omega)t} dt + \int_0^{\infty} e^{-(a+i\omega)t} dt = \frac{1}{a-i\omega} + \frac{1}{a+i\omega}.$$

Thus

$$\mathcal{F}\{e^{-a|t|}\}(\omega) = \frac{2a}{a^2 + \omega^2}.$$

2. If $\text{rect}(t/T) = 1$ for $|t| < T/2$ and 0 otherwise:

$$\hat{f}(\omega) = \int_{-T/2}^{T/2} e^{-i\omega t} dt = \frac{2 \sin(\omega T/2)}{\omega} = T \text{sinc}\left(\frac{\omega T}{2}\right).$$

Thus

$$\mathcal{F}\left\{\text{rect}\left(\frac{t}{T}\right)\right\}(\omega) = T \text{sinc}\left(\frac{\omega T}{2}\right).$$

3. Write

$$\cos(\omega_0 t) = \frac{1}{2}(e^{i\omega_0 t} + e^{-i\omega_0 t}).$$

By modulation:

$$\mathcal{F}\{\cos(\omega_0 t)e^{-a|t|}\} = \frac{1}{2} \left[\frac{2a}{a^2 + (\omega - \omega_0)^2} + \frac{2a}{a^2 + (\omega + \omega_0)^2} \right].$$

That is

$$\mathcal{F}\{\cos(\omega_0 t)e^{-a|t|}\}(\omega) = \frac{a}{a^2 + (\omega - \omega_0)^2} + \frac{a}{a^2 + (\omega + \omega_0)^2}.$$

4. Start from

$$\mathcal{F}\{e^{-at^2}\}(\omega) = \sqrt{\frac{\pi}{a}} e^{-\omega^2/(4a)}.$$

Since $\mathcal{F}\{tf(t)\} = i \frac{d}{d\omega} \hat{f}(\omega),$

$$\mathcal{F}\{te^{-at^2}\}(\omega) = i \frac{d}{d\omega} \left(\sqrt{\frac{\pi}{a}} e^{-\omega^2/(4a)} \right) = -\frac{i\omega}{2a} \sqrt{\frac{\pi}{a}} e^{-\omega^2/(4a)}.$$

Thus

$$\mathcal{F}\{te^{-at^2}\}(\omega) = -\frac{i\omega}{2a} \sqrt{\frac{\pi}{a}} e^{-\omega^2/(4a)}.$$

5. Using the standard table:

$$\mathcal{F}\left\{\frac{\sin(at)}{t}\right\}(\omega) = \pi \text{rect}\left(\frac{\omega}{2a}\right).$$

That is, π for $|\omega| < a$, and 0 for $|\omega| > a$.

Exercise

1. Compute $\mathcal{F}^{-1}\left\{\frac{1}{(a^2 + \omega^2)^2}\right\}$ using convolution.

2. Show that

$$\mathcal{F}\{J_0(at)\}(\omega) = \frac{2}{\sqrt{a^2 - \omega^2}} \mathbf{1}_{\{|\omega| < a\}}.$$

3. Compute $\mathcal{F}\{J_0(at) * J_0(at)\}$.

4. Show that

$$\mathcal{F}\{f(t) \cos(\omega_0 t)\}(\omega) = \frac{1}{2} [\hat{f}(\omega - \omega_0) + \hat{f}(\omega + \omega_0)].$$

Solution

1. We know that

$$\mathcal{F}^{-1}\left\{\frac{1}{a^2 + \omega^2}\right\}(t) = \frac{1}{2a} e^{-a|t|}.$$

Then

$$\mathcal{F}^{-1}\left\{\frac{1}{(a^2 + \omega^2)^2}\right\} = \frac{1}{2\pi} \left(\frac{1}{2a} e^{-a|t|} \right) * \left(\frac{1}{2a} e^{-a|t|} \right).$$

Compute $h(t) = e^{-a|t|} * e^{-a|t|}$. For $t \geq 0$ we obtain

$$h(t) = e^{-at} \left(t + \frac{1}{a} \right),$$

and by symmetry $h(t) = e^{-a|t|} \left(|t| + \frac{1}{a} \right)$. Hence

$$\mathcal{F}^{-1}\left\{\frac{1}{(a^2 + \omega^2)^2}\right\}(t) = \frac{1 + a|t|}{8\pi a^3} e^{-a|t|}.$$

2. The formula is standard. It follows from the integral representation of Bessel functions.

$$\mathcal{F}\{J_0(at)\}(\omega) = \frac{2}{\sqrt{a^2 - \omega^2}} \mathbf{1}_{\{|\omega| < a\}}.$$

3. By convolution:

$$\mathcal{F}\{J_0(at) * J_0(at)\} = (\mathcal{F}\{J_0(at)\})^2.$$

Thus

$$\mathcal{F}\{J_0(at) * J_0(at)\}(\omega) = \frac{4}{a^2 - \omega^2} \mathbf{1}_{\{|\omega| < a\}}.$$

4. Since $\cos(\omega_0 t) = \frac{1}{2}(e^{i\omega_0 t} + e^{-i\omega_0 t})$,

$$f(t) \cos(\omega_0 t) = \frac{1}{2} (e^{i\omega_0 t} f(t) + e^{-i\omega_0 t} f(t)).$$

By modulation: $\mathcal{F}\{e^{i\omega_0 t} f(t)\}(\omega) = \hat{f}(\omega - \omega_0)$, $\mathcal{F}\{e^{-i\omega_0 t} f(t)\}(\omega) = \hat{f}(\omega + \omega_0)$. Hence

$$\mathcal{F}\{f(t) \cos(\omega_0 t)\}(\omega) = \frac{1}{2} [\hat{f}(\omega - \omega_0) + \hat{f}(\omega + \omega_0)].$$

Exercise

Solve using the Fourier transform:

- 1.

$$u_t = k u_{xx}, \quad u(x, 0) = f(x).$$

2.

$$u_{tt} = c^2 u_{xx}, \quad u(x, 0) = f(x), \quad u_t(x, 0) = g(x).$$

3.

$$u_{xx} + u_{yy} = 0, \quad y > 0, \quad u(x, 0) = f(x).$$

Solution

1. **Heat equation.** Let $\hat{u}(\omega, t) = \mathcal{F}_x\{u(\cdot, t)\}(\omega)$. The equation becomes

$$\partial_t \hat{u} = -k\omega^2 \hat{u}.$$

The solution: $\hat{u}(\omega, t) = C(\omega)e^{-k\omega^2 t}$. The initial condition gives $C(\omega) = \hat{f}(\omega)$. Hence

$$\hat{u}(\omega, t) = \hat{f}(\omega)e^{-k\omega^2 t}.$$

Inverting: $u(x, t) = f * \mathcal{F}^{-1}\{e^{-k\omega^2 t}\}$. But $\mathcal{F}^{-1}\{e^{-k\omega^2 t}\}(x) = \frac{1}{\sqrt{4\pi kt}}e^{-x^2/(4kt)}$. Therefore

$$u(x, t) = \frac{1}{\sqrt{4\pi kt}} \int_{-\infty}^{+\infty} f(\xi) e^{-(x-\xi)^2/(4kt)} d\xi.$$

2. **Wave equation.** After transformation:

$$\partial_t^2 \hat{u} + c^2 \omega^2 \hat{u} = 0.$$

The solution: $\hat{u}(\omega, t) = A(\omega) \cos(c\omega t) + B(\omega) \sin(c\omega t)$. Using the two initial conditions: $\hat{u}(\omega, 0) = \hat{f}(\omega)$, $\partial_t \hat{u}(\omega, 0) = \hat{g}(\omega)$. Thus

$$A(\omega) = \hat{f}(\omega), \quad B(\omega) = \frac{\hat{g}(\omega)}{c\omega}.$$

We obtain

$$\hat{u}(\omega, t) = \hat{f}(\omega) \cos(c\omega t) + \frac{\hat{g}(\omega)}{c\omega} \sin(c\omega t).$$

Inverting yields d'Alembert's formula:

$$u(x, t) = \frac{f(x-ct) + f(x+ct)}{2} + \frac{1}{2c} \int_{x-ct}^{x+ct} g(\xi) d\xi.$$

3. **Laplace equation in the half-plane.** Transform in x :

$$-\omega^2 \hat{u} + \partial_y^2 \hat{u} = 0, \quad \hat{u}(\omega, 0) = \hat{f}(\omega).$$

The equation: $\partial_y^2 \hat{u} - \omega^2 \hat{u} = 0$. Its solution: $\hat{u}(\omega, y) = A(\omega)e^{|\omega|y} + B(\omega)e^{-|\omega|y}$. The boundedness condition forces $A = 0$, leaving $\hat{u}(\omega, y) = \hat{f}(\omega)e^{-|\omega|y}$. We know that $\mathcal{F}^{-1}\{e^{-|\omega|y}\}(x) = \frac{y}{\pi(x^2+y^2)}$. Hence

$$u(x, y) = \frac{y}{\pi} \int_{-\infty}^{+\infty} \frac{f(\xi)}{(x-\xi)^2 + y^2} d\xi.$$

This is Poisson's kernel for the half-plane.

Exercise

1. For

$$s(t) = A_c[1 + k_a m(t)] \cos(\omega_c t),$$

compute the spectrum and explain demodulation.

2. State and justify Shannon's reconstruction formula:

$$x(t) = \sum_{n \in \mathbb{Z}} x(nT_s) \operatorname{sinc}\left(\pi \frac{t - nT_s}{T_s}\right), \quad f_s = \frac{1}{T_s} > 2B.$$

3. For

$$H(\omega) = \operatorname{rect}\left(\frac{\omega}{2\omega_c}\right),$$

compute $h(t)$ and explain why the ideal filter is not physically realizable.

4. Show that

$$\mathcal{F}\{R_{xx}\} = |\hat{x}|^2, \quad R_{xx}(\tau) = \int_{-\infty}^{+\infty} x(t) \overline{x(t + \tau)} dt.$$

Solution

1. **Amplitude modulation AM.** Expand:

$$s(t) = A_c \cos(\omega_c t) + A_c k_a m(t) \cos(\omega_c t).$$

The spectrum is

$$\hat{s}(\omega) = A_c \pi [\delta(\omega - \omega_c) + \delta(\omega + \omega_c)] + \frac{A_c k_a}{2} [\hat{m}(\omega - \omega_c) + \hat{m}(\omega + \omega_c)].$$

The spectrum consists of a carrier at $\pm\omega_c$ and two copies of the spectrum of m around $\pm\omega_c$. Coherent demodulation: multiply by $\cos(\omega_c t)$ and pass through a low-pass filter.

2. **Shannon.** Sampling via the Dirac comb $\operatorname{III}_{T_s}(t) = \sum_{n \in \mathbb{Z}} \delta(t - nT_s)$ multiplies the spectrum by a period $\omega_s = 2\pi/T_s$. If $f_s > 2B$, the copies do not overlap, an ideal low-pass filter extracts the central copy, and we recover the signal. Thus

$$x(t) = \sum_{n \in \mathbb{Z}} x(nT_s) \operatorname{sinc}\left(\pi \frac{t - nT_s}{T_s}\right).$$

3. **Ideal low-pass filter.** Compute:

$$h(t) = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{i\omega t} d\omega = \frac{\sin(\omega_c t)}{\pi t}.$$

$$h(t) = \frac{\sin(\omega_c t)}{\pi t}.$$

This filter is not realizable because:

- It is non-causal ($h(t) \neq 0$ for $t < 0$);
- It has infinite support;
- Its abrupt cut-off requires an unbounded impulse response in time.

4. **Autocorrelation.** Let $\tilde{x}(t) = \overline{x(-t)}$. Note that $R_{xx}(\tau) = (x * \tilde{x})(-\tau)$. Directly, the Wiener-Khinchin theorem gives

$$\mathcal{F}\{R_{xx}\}(\omega) = |\hat{x}(\omega)|^2.$$

For real signals, $|\hat{x}|^2 = \hat{x}(\omega) \overline{\hat{x}(\omega)}$.

CHAPTER 6. FOURIER TRANSFORMS

Exercise

1. Let $x[n] = \{1, 2, 3, 4\}$ for $n = 0, 1, 2, 3$. Compute its Discrete Fourier Transform (DFT).
2. Show the orthogonality of the DFT.
3. Show that

$$\text{TFD}(x \circledast y) = X[k]Y[k].$$

4. Explain the principle of the FFT algorithm.

Solution

1. For $N = 4$:

$$X[k] = \sum_{n=0}^3 x[n]e^{-i2\pi kn/4}.$$

Compute:

$$X[0] = 1 + 2 + 3 + 4 = 10,$$

$$X[1] = 1 + 2e^{-i\pi/2} + 3e^{-i\pi} + 4e^{-i3\pi/2} = 1 - 2i - 3 + 4i = -2 + 2i,$$

$$X[2] = 1 - 2 + 3 - 4 = -2,$$

$$X[3] = 1 + 2i - 3 - 4i = -2 - 2i.$$

Thus

$$X = \{10, -2 + 2i, -2, -2 - 2i\}.$$

2. For $0 \leq k, \ell \leq N - 1$:

$$\sum_{n=0}^{N-1} e^{-i2\pi kn/N} e^{i2\pi \ell n/N} = \sum_{n=0}^{N-1} e^{-i2\pi(k-\ell)n/N} = \begin{cases} N & k = \ell, \\ 0 & k \neq \ell. \end{cases}$$

This is the orthogonality condition.

3. By definition:

$$(x \circledast y)[n] = \sum_{m=0}^{N-1} x[m]y[(n-m) \bmod N].$$

Applying the DFT and swapping sums gives:

$$\sum_{n=0}^{N-1} (x \circledast y)[n]e^{-i2\pi kn/N} = X[k]Y[k].$$

Hence

$$\text{TFD}(x \circledast y) = X[k]Y[k].$$

4. **FFT.** When $N = 2^m$, separate even and odd indices:

$$X[k] = \sum_{r=0}^{N/2-1} x[2r]e^{-i2\pi kr/(N/2)} + e^{-i2\pi k/N} \sum_{r=0}^{N/2-1} x[2r+1]e^{-i2\pi kr/(N/2)}.$$

This reduces a DFT of size N to two DFTs of size $N/2$, applied recursively. This achieves $O(N \log N)$ complexity instead of $O(N^2)$.

Exercise

Short-Time Fourier Transform (STFT). Defined by

$$\text{STFT}_x(\tau, \omega) = \int_{-\infty}^{+\infty} x(t)w(t-\tau)e^{-i\omega t} dt.$$

1. Interpret its meaning.

CHAPTER 6. FOURIER TRANSFORMS

2. For $w(t) = e^{-t^2/(2\sigma^2)}$ and $x(t) = e^{i\omega_0 t}$, compute the STFT.
3. Discuss the trade-off between time and frequency resolution.
4. State the Heisenberg uncertainty inequality.

Solution

1. For each fixed τ , $\text{STFT}_x(\tau, \omega) = \mathcal{F}_t\{x(t)w(t-\tau)\}(\omega)$, i.e., the spectrum of the part of the signal selected by a window centered at τ .
2. Compute:

$$\begin{aligned}\text{STFT}_x(\tau, \omega) &= e^{i(\omega_0 - \omega)\tau} \int_{-\infty}^{+\infty} e^{-u^2/(2\sigma^2)} e^{-i(\omega - \omega_0)u} du \\ &= \sigma\sqrt{2\pi} e^{i(\omega_0 - \omega)\tau} e^{-\sigma^2(\omega - \omega_0)^2/2}.\end{aligned}$$

Thus

$$\text{STFT}_x(\tau, \omega) = \sigma\sqrt{2\pi} e^{i(\omega_0 - \omega)\tau} e^{-\sigma^2(\omega - \omega_0)^2/2}.$$

$$|\text{STFT}_x(\tau, \omega)| = \sigma\sqrt{2\pi} e^{-\sigma^2(\omega - \omega_0)^2/2},$$

which is independent of τ .

3. A short window gives good time resolution but poor frequency resolution, and vice versa. The Gaussian window achieves the optimal trade-off.
4. Uncertainty inequality:

$$\Delta t \Delta \omega \geq \frac{1}{2}.$$

The Gaussian window achieves equality.

Exercise

1. Explain the principle of Fourier optics.
2. For a rectangular slit of width a in the x -direction and b in the y -direction, compute the Fraunhofer diffraction pattern.
3. For a diffraction grating with N regularly spaced slits, show the factor

$$\frac{\sin(N\varphi/2)}{\sin(\varphi/2)}.$$

4. Explain image filtering in the spatial domain.

Solution

1. In Fraunhofer diffraction, the wave amplitude in the focal plane is proportional to the two-dimensional Fourier transform of the aperture $a(x, y)$:

$$U(k_x, k_y) \propto \iint a(x, y) e^{-i(k_x x + k_y y)} dx dy.$$

2. The aperture: $a(x, y) = \text{rect}(x/a) \text{rect}(y/b)$. By separation:

$$U(k_x, k_y) \propto a \text{sinc}\left(\frac{ak_x}{2}\right) b \text{sinc}\left(\frac{bk_y}{2}\right).$$

The intensity:

$$I(k_x, k_y) \propto a^2 b^2 \text{sinc}^2\left(\frac{ak_x}{2}\right) \text{sinc}^2\left(\frac{bk_y}{2}\right).$$

3. For N slits spaced by d , the summed amplitude:

$$\sum_{n=0}^{N-1} e^{-inkd} = e^{-i(N-1)kd/2} \frac{\sin(Nkd/2)}{\sin(kd/2)}.$$

Setting $\varphi = kd$ yields the factor

$$\frac{\sin(N\varphi/2)}{\sin(\varphi/2)}.$$

4. A spatial filter is placed in the Fourier plane of the image: multiply the spectrum by a transfer function $H(k_x, k_y)$, then invert the transform. This allows smoothing, edge enhancement, removing certain frequencies, or selecting orientations.

Exercise

1. Interpret $|\hat{\psi}(p)|^2$ if

$$\hat{\psi}(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{+\infty} \psi(x) e^{-ipx/\hbar} dx.$$

2. Show the uncertainty principle

$$\Delta x \Delta p \geq \frac{\hbar}{2}.$$

3. Solve the free Schrödinger equation using the Fourier transform.
4. Give the form of the Fourier transforms of harmonic oscillator states.

Solution

1. With this normalization, the transform from position to momentum is unitary, and

$$\int |\psi(x)|^2 dx = \int |\hat{\psi}(p)|^2 dp = 1.$$

Thus $|\hat{\psi}(p)|^2 dp$ represents the probability of finding the momentum in $[p, p + dp]$.

2. Heisenberg's inequality follows from applying Cauchy-Schwarz to

$$(X - \langle X \rangle)\psi, \quad (P - \langle P \rangle)\psi,$$

with $P = -i\hbar d/dx$ and the commutator $[X, P] = i\hbar$. The formula gives

$$\Delta x \Delta p \geq \frac{\hbar}{2}.$$

3. Free Schrödinger equation: $i\hbar\psi_t = -\frac{\hbar^2}{2m}\psi_{xx}$. Transform in x :

$$i\hbar \partial_t \hat{\psi}(\xi, t) = \frac{\hbar^2 \xi^2}{2m} \hat{\psi}(\xi, t).$$

The solution: $\hat{\psi}(\xi, t) = \hat{\psi}(\xi, 0) \exp\left(-i\frac{\hbar \xi^2}{2m} t\right)$. Inverting:

$$\psi(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \hat{\psi}(\xi, 0) e^{i\xi x - i\hbar \xi^2 t/(2m)} d\xi.$$

4. Harmonic oscillator eigenstates are Hermite-Gauss functions, and their Fourier transforms have the same form (up to a phase). In dimensionless variables,

$$\mathcal{F}\{\psi_n\} = (-i)^n \psi_n.$$

Exercise

1. Compute $\mathcal{F}\{\delta^{(n)}\}$.
2. Show that

$$\mathcal{F}\{t^n\} = 2\pi i^n \delta^{(n)}.$$

3. Compute $\mathcal{F}\{\text{PV}(1/t)\}$.
4. Define and compute $\mathcal{F}\{\text{sgn}(t)\}$.

Solution

All equalities are in the sense of tempered distributions.

5. $\mathcal{F}\{\delta\} = 1$. By differentiation: $\mathcal{F}\{\delta^{(n)}\}(\omega) = (i\omega)^n$. Hence

$$\mathcal{F}\{\delta^{(n)}\} = (i\omega)^n.$$

6. From the multiplication by t^n rule: $\mathcal{F}\{t^n f(t)\} = i^n \partial_\omega^n \hat{f}$. Take $f = 1$, with $\mathcal{F}\{1\} = 2\pi\delta$, we obtain

$$\mathcal{F}\{t^n\} = i^n \partial_\omega^n (2\pi\delta) = 2\pi i^n \delta^{(n)}(\omega).$$

7. The classic Hilbert transform formula gives

$$\mathcal{F}\left\{\text{PV}\left(\frac{1}{t}\right)\right\}(\omega) = -i\pi \text{sgn}(\omega).$$

8. Since $\frac{d}{dt} \text{sgn}(t) = 2\delta(t)$, transform:

$$i\omega \mathcal{F}\{\text{sgn}(t)\} = 2.$$

Thus

$$\mathcal{F}\{\text{sgn}(t)\} = \text{PV}\left(\frac{2}{i\omega}\right).$$

Exercise

1. Define the two-dimensional and three-dimensional Fourier transforms.
2. Show that

$$\mathcal{F}_{2D}\{f(x)g(y)\} = \hat{f}(k_x)\hat{g}(k_y).$$

3. Compute the two-dimensional transform of $e^{-a(x^2+y^2)}$.
4. For a radial function $f(r)$ in two dimensions, show that its transform can be written as a Hankel transform.

Solution

1. **Definitions:**

$$\hat{f}(k_x, k_y) = \iint_{\mathbb{R}^2} f(x, y) e^{-i(k_x x + k_y y)} dx dy,$$

$$\hat{f}(k_x, k_y, k_z) = \iiint_{\mathbb{R}^3} f(x, y, z) e^{-i(k_x x + k_y y + k_z z)} dx dy dz.$$

2. If $F(x, y) = f(x)g(y)$, then:

$$\hat{F}(k_x, k_y) = \left(\int_{-\infty}^{+\infty} f(x) e^{-ik_x x} dx \right) \left(\int_{-\infty}^{+\infty} g(y) e^{-ik_y y} dy \right) = \hat{f}(k_x) \hat{g}(k_y).$$

$$\mathcal{F}_{2D}\{f(x)g(y)\} = \hat{f}(k_x)\hat{g}(k_y).$$

3. By separation:

$$\begin{aligned}\mathcal{F}_{2D}\{e^{-a(x^2+y^2)}\} &= \mathcal{F}\{e^{-ax^2}\}(k_x) \mathcal{F}\{e^{-ay^2}\}(k_y) \\ &= \sqrt{\frac{\pi}{a}} e^{-k_x^2/(4a)} \sqrt{\frac{\pi}{a}} e^{-k_y^2/(4a)}.\end{aligned}$$

Thus

$$\mathcal{F}_{2D}\{e^{-a(x^2+y^2)}\}(k_x, k_y) = \frac{\pi}{a} e^{-(k_x^2+k_y^2)/(4a)}.$$

4. If $f(x, y) = f(r)$ with $r = \sqrt{x^2 + y^2}$, switch to polar coordinates:

$$\hat{f}(\rho) = \int_0^\infty f(r) \left(\int_0^{2\pi} e^{-i\rho r \cos(\theta-\phi)} d\theta \right) r dr.$$

The angular integral equals $2\pi J_0(\rho r)$. Hence

$$\hat{f}(\rho) = 2\pi \int_0^\infty f(r) J_0(\rho r) r dr.$$

This is the Hankel transform of order zero.

Vector Analysis

Theorem

Vector Differential Operators and Fundamental Theorems

1. **Vector differential operator** ∇ (nabla):

$$\nabla = \frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k} \quad \text{and} \quad \nabla = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix}$$

2. **Gradient of a scalar field** $f(x, y, z)$:

$$\nabla f = \nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) = \frac{\partial f}{\partial x} \vec{i} + \frac{\partial f}{\partial y} \vec{j} + \frac{\partial f}{\partial z} \vec{k}$$

Interpretation: The gradient points in the direction of greatest increase of f and its magnitude is the rate of this increase.

3. **Divergence of a vector field** $\vec{F} = (P, Q, R)$:

$$\vec{F} = P\vec{i} + Q\vec{j} + R\vec{k}$$

$$\nabla \cdot \vec{F} = \nabla \cdot \vec{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

Interpretation: Divergence measures the tendency of the field to diverge from (positive source) or converge toward (negative sink) a point.

4. **Curl of a vector field** $\vec{F} = (P, Q, R)$:

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$$

$$\nabla \times \vec{F} = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}, \frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}, \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right)$$

Interpretation: Curl measures the infinitesimal rotation of the field at a point.

5. **Laplacian of a scalar field** f :

$$\nabla^2 f = \nabla^2 f = \nabla \cdot (\nabla f) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$

6. **Algebraic properties:**

• **Linearity:**

$$\begin{aligned}\nabla(\alpha f + \beta g) &= \alpha \nabla f + \beta \nabla g \\ \nabla \cdot (\alpha \vec{F} + \beta \vec{G}) &= \alpha \nabla \cdot \vec{F} + \beta \nabla \cdot \vec{G} \\ \nabla \times (\alpha \vec{F} + \beta \vec{G}) &= \alpha \nabla \times \vec{F} + \beta \nabla \times \vec{G}\end{aligned}$$

• **Important identities:**

$$\begin{aligned}\nabla \times (\nabla f) &= \vec{0} \quad (\text{irrotational}) \\ \nabla \cdot (\nabla \times \vec{F}) &= 0 \quad (\text{solenoidal}) \\ \nabla \cdot (f \vec{F}) &= f \nabla \cdot \vec{F} + \nabla f \cdot \vec{F} \\ \nabla \times (f \vec{F}) &= f \nabla \times \vec{F} + \nabla f \times \vec{F} \\ \nabla \times (\nabla \times \vec{F}) &= \nabla(\nabla \cdot \vec{F}) - \nabla^2 \vec{F}\end{aligned}$$

7. **Curvilinear coordinates:**

• **Cylindrical coordinates** (r, θ, z) :

$$\begin{aligned}\nabla f &= \frac{\partial f}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial f}{\partial \theta} \hat{\theta} + \frac{\partial f}{\partial z} \hat{z} \\ \nabla \cdot \vec{F} &= \frac{1}{r} \frac{\partial}{\partial r}(r F_r) + \frac{1}{r} \frac{\partial F_\theta}{\partial \theta} + \frac{\partial F_z}{\partial z} \\ \nabla \times \vec{F} &= \left(\frac{1}{r} \frac{\partial F_z}{\partial \theta} - \frac{\partial F_\theta}{\partial z} \right) \hat{r} + \left(\frac{\partial F_r}{\partial z} - \frac{\partial F_z}{\partial r} \right) \hat{\theta} \\ &\quad + \frac{1}{r} \left(\frac{\partial}{\partial r}(r F_\theta) - \frac{\partial F_r}{\partial \theta} \right) \hat{z}\end{aligned}$$

• **Spherical coordinates** (ρ, ϕ, θ) :

$$\begin{aligned}\nabla f &= \frac{\partial f}{\partial \rho} \hat{\rho} + \frac{1}{\rho} \frac{\partial f}{\partial \phi} \hat{\phi} + \frac{1}{\rho \sin \phi} \frac{\partial f}{\partial \theta} \hat{\theta} \\ \nabla \cdot \vec{F} &= \frac{1}{\rho^2} \frac{\partial}{\partial \rho}(\rho^2 F_\rho) \\ &\quad + \frac{1}{\rho \sin \phi} \frac{\partial}{\partial \phi}(\sin \phi F_\phi) \\ &\quad + \frac{1}{\rho \sin \phi} \frac{\partial F_\theta}{\partial \theta} \\ \nabla \times \vec{F} &= \frac{1}{\rho \sin \phi} \left[\frac{\partial}{\partial \phi}(\sin \phi F_\theta) - \frac{\partial F_\phi}{\partial \theta} \right] \hat{\rho} \\ &\quad + \frac{1}{\rho} \left[\frac{1}{\sin \phi} \frac{\partial F_\rho}{\partial \theta} - \frac{\partial}{\partial \rho}(\rho F_\theta) \right] \hat{\phi} \\ &\quad + \frac{1}{\rho} \left[\frac{\partial}{\partial \rho}(\rho F_\phi) - \frac{\partial F_\rho}{\partial \phi} \right] \hat{\theta}\end{aligned}$$

8. **Integral theorems:**

• **Gradient theorem:**

$$\int_C \nabla f \cdot d\vec{r} = f(B) - f(A)$$

where C is a curve from A to B .

- **Green's theorem** (in the plane):

$$\oint_C (P dx + Q dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

where D is a planar domain bounded by C oriented positively.

- **Stokes' theorem:**

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot d\vec{S}$$

where S is an oriented surface bounded by C .

- **Divergence theorem** (Gauss-Ostrogradsky):

$$\iiint_V \vec{F} \cdot d\vec{S} = \iiint_V (\nabla \cdot \vec{F}) dV$$

where V is a volume bounded by the surface S .

Exercise

Let $f(x, y) = x^2 + 3xy - y^2$.

- Compute ∇f .
- Evaluate ∇f at the point $(1, 2)$.
- Find the direction of greatest increase of f at $(1, 2)$.
- Compute the directional derivative of f at $(1, 2)$ in the direction of the vector $\vec{v} = (3, 4)$.

Solution

1. Gradient:

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right) = (2x + 3y, 3x - 2y)$$

2. At the point $(1, 2)$:

$$\nabla f(1, 2) = (2 \cdot 1 + 3 \cdot 2, 3 \cdot 1 - 2 \cdot 2) = (2 + 6, 3 - 4) = (8, -1)$$

3. Direction of greatest increase: This is the direction of the gradient $\nabla f(1, 2) = (8, -1)$.

4. Directional derivative: The unit vector in the direction of $\vec{v} = (3, 4)$ is:

$$\vec{u} = \frac{\vec{v}}{\|\vec{v}\|} = \frac{(3, 4)}{\sqrt{3^2 + 4^2}} = \left(\frac{3}{5}, \frac{4}{5} \right)$$

The directional derivative is:

$$D_{\vec{u}}f(1, 2) = \nabla f(1, 2) \cdot \vec{u} = (8, -1) \cdot \left(\frac{3}{5}, \frac{4}{5} \right) = \frac{24}{5} - \frac{4}{5} = \frac{20}{5} = 4$$

$$\boxed{\nabla f = (2x + 3y, 3x - 2y), \quad \nabla f(1, 2) = (8, -1), \quad D_{\vec{u}}f(1, 2) = 4}$$

Exercise

Let $f(x, y, z) = e^{x+y} \sin z + xyz$.

- Compute ∇f .
- Verify that $\nabla \times (\nabla f) = \vec{0}$.
- Find the equation of the tangent plane to the surface $f(x, y, z) = c$ at the point $(0, 0, \frac{\pi}{4})$.

Solution

1. Gradient:

$$\frac{\partial f}{\partial x} = e^{x+y} \sin z + yz, \quad \frac{\partial f}{\partial y} = e^{x+y} \sin z + xz, \quad \frac{\partial f}{\partial z} = e^{x+y} \cos z + xy$$

$$\nabla f = (e^{x+y} \sin z + yz, e^{x+y} \sin z + xz, e^{x+y} \cos z + xy)$$

2. Verification of $\nabla \times (\nabla f) = \vec{0}$: Compute the first component of $\nabla \times (\nabla f)$:

$$\frac{\partial}{\partial y} (e^{x+y} \cos z + xy) - \frac{\partial}{\partial z} (e^{x+y} \sin z + xz) = (e^{x+y} \cos z + x) - (e^{x+y} \cos z + x) = 0$$

Similarly for the other two components, so the curl is zero.

3. Tangent plane to $f(x, y, z) = c$: At the point $(0, 0, \frac{\pi}{4})$ compute f and ∇f :

$$f\left(0, 0, \frac{\pi}{4}\right) = e^0 \sin\left(\frac{\pi}{4}\right) + 0 = \frac{\sqrt{2}}{2}$$

$$\nabla f\left(0, 0, \frac{\pi}{4}\right) = \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$$

The equation of the tangent plane to the level surface $f(x, y, z) = \frac{\sqrt{2}}{2}$ is:

$$\frac{\sqrt{2}}{2}(x - 0) + \frac{\sqrt{2}}{2}(y - 0) + \frac{\sqrt{2}}{2}\left(z - \frac{\pi}{4}\right) = 0$$

$$x + y + z - \frac{\pi}{4} = 0$$

$$\nabla f = (e^{x+y} \sin z + yz, e^{x+y} \sin z + xz, e^{x+y} \cos z + xy), \text{ Tangent plane: } x + y + z = \frac{\pi}{4}$$

Exercise

Let $\vec{F}(x, y, z) = (x^2y, y^2z, z^2x)$ be a vector field in $\mathbb{R}^3$.

- Compute the divergence $\nabla \cdot \vec{F}$.
- Compute the curl $\nabla \times \vec{F}$.
- Verify the identity $\nabla \cdot (\nabla \times \vec{F}) = 0$.

- Let S be the surface of the unit sphere $x^2 + y^2 + z^2 = 1$. Compute the flux of $\vec{F}$ through S using the divergence theorem.

Solution

1. Divergence:

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(x^2y) + \frac{\partial}{\partial y}(y^2z) + \frac{\partial}{\partial z}(z^2x) = 2xy + 2yz + 2zx = 2(xy + yz + zx)$$

2. Curl:

$$\begin{aligned} \nabla \times \vec{F} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2y & y^2z & z^2x \end{vmatrix} \\ &= \left(\frac{\partial}{\partial y}(z^2x) - \frac{\partial}{\partial z}(y^2z), \frac{\partial}{\partial z}(x^2y) - \frac{\partial}{\partial x}(z^2x), \frac{\partial}{\partial x}(y^2z) - \frac{\partial}{\partial y}(x^2y) \right) \\ &= (0 - y^2, 0 - z^2, 0 - x^2) = (-y^2, -z^2, -x^2) \end{aligned}$$

3. Verification $\nabla \cdot (\nabla \times \vec{F}) = 0$:

$$\nabla \cdot (\nabla \times \vec{F}) = \frac{\partial}{\partial x}(-y^2) + \frac{\partial}{\partial y}(-z^2) + \frac{\partial}{\partial z}(-x^2) = 0$$

4. Flux through the unit sphere (divergence theorem):

$$\iint_S \vec{F} \cdot d\vec{S} = \iiint_{x^2+y^2+z^2 \leq 1} 2(xy + yz + zx) dV$$

By symmetry, xy , yz , zx are odd functions in their respective variables over a symmetric sphere, so their integrals are zero. Therefore:

$$\iint_S \vec{F} \cdot d\vec{S} = 0$$

$$\boxed{\nabla \cdot \vec{F} = 2(xy + yz + zx), \quad \nabla \times \vec{F} = (-y^2, -z^2, -x^2), \quad \iint_S \vec{F} \cdot d\vec{S} = 0}$$

Exercise

Let $\vec{F}(x, y, z) = (3x^2y, x^3 + z^2, 2yz)$ be a vector field.

- Verify whether $\vec{F}$ is conservative (irrotational).
- If $\vec{F}$ is conservative, find a scalar potential f such that $\vec{F} = \nabla f$.
- Compute the circulation of $\vec{F}$ along the curve C given by $\vec{r}(t) = (t, t^2, t^3)$ for $0 \leq t \leq 1$ in two different ways.

Solution

1. Verifying if the field is conservative: Compute the curl:

$$\begin{aligned}\nabla \times \vec{F} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial_x & \partial_y & \partial_z \\ 3x^2y & x^3 + z^2 & 2yz \end{vmatrix} \\ &= \left(\frac{\partial}{\partial y}(2yz) - \frac{\partial}{\partial z}(x^3 + z^2), \frac{\partial}{\partial z}(3x^2y) - \frac{\partial}{\partial x}(2yz), \frac{\partial}{\partial x}(x^3 + z^2) - \frac{\partial}{\partial y}(3x^2y) \right) \\ &= (2z - 2z, 0 - 0, 3x^2 - 3x^2) = (0, 0, 0)\end{aligned}$$

Thus $\vec{F}$ is conservative.

2. Finding the potential: We want f such that $\nabla f = \vec{F}$:

$$\begin{aligned}\frac{\partial f}{\partial x} &= 3x^2y \Rightarrow f = x^3y + g(y, z) \\ \frac{\partial f}{\partial y} &= x^3 + \frac{\partial g}{\partial y} = x^3 + z^2 \Rightarrow \frac{\partial g}{\partial y} = z^2 \Rightarrow g = yz^2 + h(z) \\ f &= x^3y + yz^2 + h(z) \\ \frac{\partial f}{\partial z} &= 2yz + h'(z) = 2yz \Rightarrow h'(z) = 0 \Rightarrow h(z) = C\end{aligned}$$

Take $C = 0$, thus $f(x, y, z) = x^3y + yz^2$.

3. Circulation along C : *First method (using the potential):*

$$\int_C \vec{F} \cdot d\vec{r} = f(1, 1, 1) - f(0, 0, 0) = (1 + 1) - 0 = 2$$

Second method (direct computation):

$$\begin{aligned}\vec{r}(t) &= (t, t^2, t^3), \quad \vec{F}(\vec{r}(t)) = (3t^4, t^3 + t^6, 2t^5), \quad d\vec{r} = (1, 2t, 3t^2)dt \\ \vec{F} \cdot d\vec{r} &= 3t^4 + 2t(t^3 + t^6) + 3t^2(2t^5) = 5t^4 + 8t^7 \\ \int_0^1 (5t^4 + 8t^7)dt &= 1 + 1 = 2\end{aligned}$$

$\vec{F} \text{ conservative, } f(x, y, z) = x^3y + yz^2, \quad \int_C \vec{F} \cdot d\vec{r} = 2$

Exercise

Let $\vec{F}(x, y, z) = \left(\frac{x}{(x^2+y^2+z^2)^{3/2}}, \frac{y}{(x^2+y^2+z^2)^{3/2}}, \frac{z}{(x^2+y^2+z^2)^{3/2}} \right)$ be a vector field.

- Show that $\vec{F} = \nabla f$ where $f(x, y, z) = -\frac{1}{\sqrt{x^2+y^2+z^2}}$.
- Compute $\nabla \cdot \vec{F}$.
- For $R > 0$, let S_R be a sphere of radius R centered at the origin. Compute the flux of $\vec{F}$ through S_R .

- Explain why the flux result does not contradict the divergence theorem even though $\nabla \cdot \vec{F} = 0$ for $(x, y, z) \neq (0, 0, 0)$.

Solution

1. Verification $\vec{F} = \nabla f$:

$$\frac{\partial f}{\partial x} = -\frac{1}{2}(x^2 + y^2 + z^2)^{-3/2} \cdot 2x = \frac{x}{(x^2 + y^2 + z^2)^{3/2}}$$

Similarly for the other components.

2. Computing the divergence: We know that $\nabla \cdot \vec{F} = 0$ for all $(x, y, z) \neq (0, 0, 0)$ (can be computed directly or recall that $\nabla^2(1/r) = 0$).

3. Flux through the sphere S_R : On the sphere S_R : $\hat{n} = \frac{(x, y, z)}{R}$, and $\vec{F} = \frac{(x, y, z)}{R^3}$, so $\vec{F} \cdot \hat{n} = \frac{1}{R^2}$.

$$\iint_{S_R} \vec{F} \cdot d\vec{S} = \iint_{S_R} \frac{1}{R^2} dS = \frac{1}{R^2} \cdot 4\pi R^2 = 4\pi$$

4. Apparent contradiction: The divergence theorem requires the field to be regular throughout the volume. Here the field is undefined at the origin, so the theorem does not apply if the volume contains the origin. The non-zero flux despite zero divergence everywhere except the origin is due to the fact that $\nabla \cdot \vec{F}$ represents a Dirac distribution at the origin: $\nabla \cdot \vec{F} = 4\pi\delta(\vec{r})$.

$$\vec{F} = \nabla \left(-\frac{1}{r} \right), \quad \nabla \cdot \vec{F} = 0 \ (r > 0), \quad \iint_{S_R} \vec{F} \cdot d\vec{S} = 4\pi$$

Exercise

Let the velocity field of an incompressible fluid be:

$$\vec{v}(x, y, z) = (2x - y, x + 2y, -4z)$$

- Verify that the fluid is incompressible.
- Compute the curl of the velocity field.
- Determine whether the flow is irrotational.
- Compute the mass flux through the surface of the cube defined by $0 \leq x \leq 1$, $0 \leq y \leq 1$, $0 \leq z \leq 1$.
- Find the streamlines of the velocity field.
- Compute the acceleration of a fluid particle using the material derivative.

Solution

1. Incompressibility:

$$\nabla \cdot \vec{v} = \frac{\partial}{\partial x}(2x - y) + \frac{\partial}{\partial y}(x + 2y) + \frac{\partial}{\partial z}(-4z) = 2 + 2 - 4 = 0$$

Thus the fluid is incompressible.

2. Curl:

$$\nabla \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial_x & \partial_y & \partial_z \\ 2x-y & x+2y & -4z \end{vmatrix} = (0, 0, 2)$$

3. Irrotational flow: Since $\nabla \times \vec{v} \neq \vec{0}$, the flow is rotational.

4. Mass flux through the cube: Using the divergence theorem and since the divergence is zero, the total flux is zero.

5. Streamlines: Solve the differential equations:

$$\frac{dx}{dt} = 2x - y, \quad \frac{dy}{dt} = x + 2y, \quad \frac{dz}{dt} = -4z$$

From the z equation: $z = z_0 e^{-4t}$. The system in x, y has matrix $A = \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}$ with eigenvalues $2 \pm i$, and the solution:

$$\begin{pmatrix} x \\ y \end{pmatrix} = e^{2t} \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix} \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$$

6. Acceleration (material derivative):

$$\frac{D\vec{v}}{Dt} = (\vec{v} \cdot \nabla) \vec{v}$$

Compute:

$$(\vec{v} \cdot \nabla) v_x = (2x-y) \frac{\partial}{\partial x} (2x-y) + (x+2y) \frac{\partial}{\partial y} (2x-y) = (2x-y) \cdot 2 + (x+2y) \cdot (-1) = 3x-4y$$

$$(\vec{v} \cdot \nabla) v_y = (2x-y) \frac{\partial}{\partial x} (x+2y) + (x+2y) \frac{\partial}{\partial y} (x+2y) = (2x-y) \cdot 1 + (x+2y) \cdot 2 = 4x+3y$$

$$(\vec{v} \cdot \nabla) v_z = (-4z) \frac{\partial}{\partial z} (-4z) = 16z$$

Thus $\vec{a} = (3x - 4y, 4x + 3y, 16z)$.

$$\nabla \cdot \vec{v} = 0, \quad \nabla \times \vec{v} = (0, 0, 2), \quad \iint_S \vec{v} \cdot d\vec{S} = 0, \quad \vec{a} = (3x - 4y, 4x + 3y, 16z)$$

Exercise

Consider the electromagnetic field defined by:

$$\vec{E}(x, y, z, t) = (E_0 \sin(kz - \omega t), 0, 0), \quad \vec{B}(x, y, z, t) = (0, \frac{E_0}{c} \sin(kz - \omega t), 0)$$

where E_0 is the amplitude, k the wave number, ω the angular frequency, and $c = \frac{\omega}{k}$ the speed of light.

- Verify that these fields satisfy Maxwell's equations in vacuum.
- Show that these fields represent a plane electromagnetic wave propagating in the z direction.

- Compute the Poynting vector $\vec{S} = \frac{1}{\mu_0}(\vec{E} \times \vec{B})$.
- Compute the electromagnetic energy density $u = \frac{1}{2} \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right)$.
- Verify the energy continuity equation: $\frac{\partial u}{\partial t} + \nabla \cdot \vec{S} = 0$.
- Compute the average power transmitted through a surface perpendicular to the direction of propagation.

Solution

1. Verification of Maxwell's equations.

Since $\vec{E} = (E_0 \sin(kz - \omega t), 0, 0)$ and $\vec{B} = (0, \frac{E_0}{c} \sin(kz - \omega t), 0)$, we obtain:

$$\nabla \cdot \vec{E} = 0, \quad \nabla \cdot \vec{B} = 0.$$

$$\nabla \times \vec{E} = \left(0, \frac{\partial}{\partial z} E_0 \sin(kz - \omega t), 0 \right) = (0, E_0 k \cos(kz - \omega t), 0).$$

$$-\frac{\partial \vec{B}}{\partial t} = - \left(0, \frac{E_0}{c} (-\omega) \cos(kz - \omega t), 0 \right) = (0, \frac{E_0 \omega}{c} \cos(kz - \omega t), 0).$$

Since $\frac{\omega}{c} = k$, we find $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$.

$$\nabla \times \vec{B} = \left(-\frac{\partial B_y}{\partial z}, 0, 0 \right) = \left(-\frac{E_0 k}{c} \cos(kz - \omega t), 0, 0 \right).$$

$$\mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} = \mu_0 \epsilon_0 (-E_0 \omega \cos(kz - \omega t), 0, 0) = (-\mu_0 \epsilon_0 E_0 \omega \cos(kz - \omega t), 0, 0).$$

Using $\mu_0 \epsilon_0 = 1/c^2$ and $k = \omega/c$, we get $\mu_0 \epsilon_0 E_0 \omega = E_0 k/c$, thus $\nabla \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$. All equations are satisfied.

2. Plane wave: The fields depend only on z and t , $\vec{E} \parallel \hat{x}$, $\vec{B} \parallel \hat{y}$, both perpendicular to the propagation direction $\hat{z}$, and $\vec{E} \perp \vec{B}$.

3. Poynting vector:

$$\vec{E} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ E_0 \sin(kz - \omega t) & 0 & 0 \\ 0 & \frac{E_0}{c} \sin(kz - \omega t) & 0 \end{vmatrix} = \left(0, 0, \frac{E_0^2}{c} \sin^2(kz - \omega t) \right),$$

$$\vec{S} = \frac{1}{\mu_0}(\vec{E} \times \vec{B}) = \left(0, 0, \frac{E_0^2}{\mu_0 c} \sin^2(kz - \omega t) \right).$$

4. Energy density:

$$E^2 = E_0^2 \sin^2(kz - \omega t), \quad B^2 = \frac{E_0^2}{c^2} \sin^2(kz - \omega t),$$

$$u = \frac{1}{2} \epsilon_0 E^2 + \frac{1}{2\mu_0} B^2 = \frac{1}{2} \epsilon_0 E_0^2 \sin^2 + \frac{1}{2\mu_0} \frac{E_0^2}{c^2} \sin^2.$$

Since $\frac{1}{\mu_0 c^2} = \epsilon_0$, we get $u = \epsilon_0 E_0^2 \sin^2(kz - \omega t)$.

5. Energy continuity equation: Let $\theta = kz - \omega t$, then $\partial\theta/\partial t = -\omega$, $\partial\theta/\partial z = k$.

$$\frac{\partial u}{\partial t} = \epsilon_0 E_0^2 \cdot 2 \sin \theta \cos \theta \cdot (-\omega) = -2\epsilon_0 E_0^2 \omega \sin \theta \cos \theta.$$

$$\nabla \cdot \vec{S} = \frac{\partial S_z}{\partial z} = \frac{E_0^2}{\mu_0 c} \cdot 2 \sin \theta \cos \theta \cdot k = \frac{2E_0^2 k}{\mu_0 c} \sin \theta \cos \theta.$$

Using $k = \omega/c$ and $\frac{1}{\mu_0 c} = \epsilon_0 c$, we get $\frac{2E_0^2 k}{\mu_0 c} = 2\epsilon_0 E_0^2 \omega$, so $\nabla \cdot \vec{S} = 2\epsilon_0 E_0^2 \omega \sin \theta \cos \theta$.

$$\frac{\partial u}{\partial t} + \nabla \cdot \vec{S} = 0.$$

6. Average transmitted power: The average of $\sin^2 \theta$ over a period is $1/2$, thus:

$$\langle P \rangle = \langle S_z \rangle \cdot A = \frac{E_0^2}{2\mu_0 c} A = \frac{1}{2} \epsilon_0 c E_0^2 A,$$

where A is the cross-sectional area perpendicular to z .

$$\vec{S} = \frac{E_0^2}{\mu_0 c} \sin^2(kz - \omega t) \hat{k}, \quad u = \epsilon_0 E_0^2 \sin^2(kz - \omega t), \quad \langle P \rangle = \frac{1}{2} \epsilon_0 c E_0^2 A$$

Exercise

Consider heat conduction in a solid body. The temperature $T(x, y, z, t)$ satisfies the heat equation:

$$\frac{\partial T}{\partial t} = \alpha \nabla^2 T$$

where α is the thermal diffusivity. The heat flux is given by Fourier's law:

$$\vec{q} = -k \nabla T$$

where k is the thermal conductivity.

- In steady state ($\frac{\partial T}{\partial t} = 0$), show that T satisfies Laplace's equation $\nabla^2 T = 0$.
- Consider a metallic rod of length L with boundary conditions: $T(0) = T_0$, $T(L) = T_L$. Find the steady-state temperature distribution.
- Compute the total heat flux through a cross-section of the rod.
- For a one-dimensional unsteady system $T(x, t)$ with conditions $T(0, t) = T_0$, $T(L, t) = T_L$ and initial condition $T(x, 0) = f(x)$.
- Compute the rate of change of thermal energy in a volume V in terms of the heat flux through its surface S .
- Derive the thermal energy conservation equation.
- Numerical application: For a copper rod ($k = 400 \text{ W/m}\cdot\text{K}$, $\alpha = 1.1 \times 10^{-4} \text{ m}^2/\text{s}$) of length 1 m with $T_0 = 100^\circ\text{C}$, $T_L = 0^\circ\text{C}$, compute the heat flux and the characteristic time to reach steady state.

Solution

1. Steady state: $\frac{\partial T}{\partial t} = 0$ gives $\alpha \nabla^2 T = 0$, i.e., $\nabla^2 T = 0$.

2. Temperature distribution in a one-dimensional rod:

$$\frac{d^2 T}{dx^2} = 0 \Rightarrow T(x) = C_1 x + C_2$$

Using the boundary conditions: $T(0) = T_0 \Rightarrow C_2 = T_0$, $T(L) = T_L \Rightarrow C_1 L + T_0 = T_L \Rightarrow C_1 = \frac{T_L - T_0}{L}$

$$T(x) = T_0 + \frac{T_L - T_0}{L} x$$

3. Total heat flux:

$$\vec{q} = -k \nabla T = \left(-k \frac{T_L - T_0}{L}, 0, 0 \right)$$

The flux through a cross-section of area A is:

$$Q = -kA \frac{T_L - T_0}{L}$$

4. Unsteady system: Solving the one-dimensional heat equation by standard methods (separation of variables) gives:

$$T(x, t) = T_s(x) + \sum_{n=1}^{\infty} b_n e^{-n^2 \pi^2 \alpha t / L^2} \sin \frac{n \pi x}{L}$$

where T_s is the steady-state solution, and b_n are determined from the initial condition.

5. Rate of change of thermal energy: Thermal energy in volume V : $U = \iiint_V \rho c T dV$, where ρ is density, c is specific heat capacity.

$$\frac{dU}{dt} = \iiint_V \rho c \frac{\partial T}{\partial t} dV = \iiint_V \rho c (\alpha \nabla^2 T) dV = \iiint_V k \nabla^2 T dV$$

Using the divergence theorem: $\iiint_V \nabla^2 T dV = \iint_S (\nabla T) \cdot d\vec{S}$, thus:

$$\frac{dU}{dt} = k \iint_S (\nabla T) \cdot d\vec{S} = - \iint_S \vec{q} \cdot d\vec{S}$$

6. Energy conservation equation:

$$\frac{d}{dt} \iiint_V \rho c T dV + \iint_S \vec{q} \cdot d\vec{S} = 0$$

In differential form: $\rho c \frac{\partial T}{\partial t} + \nabla \cdot \vec{q} = 0$.

7. Numerical application:

Flux: $Q = -400 \times 10^{-4} \times \frac{0-100}{1} = 4 \text{ W}$ (assuming a cross-sectional area of 10^{-4} m^2).

Characteristic time: $\tau \approx L^2 / \alpha = 1 / (1.1 \times 10^{-4}) \approx 9091 \text{ s}$.

This is approximately 2.5 hours.

$$T(x) = T_0 + \frac{T_L - T_0}{L} x, Q = -kA \frac{T_L - T_0}{L}, \frac{dU}{dt} = - \iint_S \vec{q} \cdot d\vec{S}, Q \approx 4 \text{ W}, \tau \approx 2.5 \text{ h}$$

Exercise

Let $\vec{F}(x, y, z) = (xy^2, yz^2, zx^2)$ be a vector field in $\mathbb{R}^3$.

- Compute $\nabla \cdot \vec{F}$.
- Compute $\nabla \times \vec{F}$.
- Let S be the surface of the cube defined by $0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1$. Compute the flux of $\vec{F}$ through the closed surface S using the divergence theorem.
- Compute directly the flux of $\vec{F}$ through the face $z = 1$ of the cube (outward orientation).

Solution

1. Divergence:

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(xy^2) + \frac{\partial}{\partial y}(yz^2) + \frac{\partial}{\partial z}(zx^2) = y^2 + z^2 + x^2$$

2. Curl:

$$\begin{aligned} \nabla \times \vec{F} &= \left(\frac{\partial}{\partial y}(zx^2) - \frac{\partial}{\partial z}(yz^2), \frac{\partial}{\partial z}(xy^2) - \frac{\partial}{\partial x}(zx^2), \frac{\partial}{\partial x}(yz^2) - \frac{\partial}{\partial y}(xy^2) \right) \\ &= (-2yz, -2xz, -2xy) \end{aligned}$$

3. Flux through the cube surface (divergence theorem):

$$\iint_S \vec{F} \cdot d\vec{S} = \iiint_V (x^2 + y^2 + z^2) dV = \int_0^1 \int_0^1 \int_0^1 (x^2 + y^2 + z^2) dx dy dz = 3 \times \frac{1}{3} = 1$$

4. Flux through the face $z = 1$: On this face: $d\vec{S} = (0, 0, 1) dx dy$, and $\vec{F} = (xy^2, y, x^2)$ (since $z = 1$). Thus:

$$\begin{aligned} \vec{F} \cdot d\vec{S} &= x^2 dx dy \\ \iint_{z=1} x^2 dx dy &= \int_0^1 \int_0^1 x^2 dx dy = \frac{1}{3} \end{aligned}$$

$$\boxed{\nabla \cdot \vec{F} = x^2 + y^2 + z^2, \nabla \times \vec{F} = (-2yz, -2xz, -2xy), \iint_S \vec{F} \cdot d\vec{S} = 1, \iint_{z=1} \vec{F} \cdot d\vec{S} = \frac{1}{3}}$$

Exercise

Let $f(x, y, z) = \ln(x^2 + y^2 + z^2)$.

- Compute ∇f .
- Compute $\nabla^2 f$.
- Verify that $\nabla^2 f = \frac{2}{x^2 + y^2 + z^2}$ for $(x, y, z) \neq (0, 0, 0)$.
- Let S_R be a sphere of radius R centered at the origin. Compute the flux of ∇f through S_R .

Solution

1. Gradient:

$$\nabla f = \left(\frac{2x}{x^2 + y^2 + z^2}, \frac{2y}{x^2 + y^2 + z^2}, \frac{2z}{x^2 + y^2 + z^2} \right)$$

2. Laplacian: Compute, for example:

$$\frac{\partial}{\partial x} \left(\frac{2x}{r^2} \right) = \frac{2(r^2) - 2x \cdot 2x}{r^4} = \frac{2(y^2 + z^2 - x^2)}{r^4}$$

$$\nabla^2 f = \frac{2(y^2 + z^2 - x^2) + 2(x^2 + z^2 - y^2) + 2(x^2 + y^2 - z^2)}{r^4} = \frac{2(x^2 + y^2 + z^2)}{r^4} = \frac{2}{r^2}$$

3. Verification: Done above.

4. Flux through the sphere S_R : On S_R : $\vec{n} = \frac{(x,y,z)}{R}$, $\nabla f = \frac{2}{R^2}(x,y,z)$, so $\nabla f \cdot \vec{n} = \frac{2}{R}$.

$$\iint_{S_R} \nabla f \cdot d\vec{S} = \iint_{S_R} \frac{2}{R} dS = \frac{2}{R} \cdot 4\pi R^2 = 8\pi R$$

$$\nabla f = \frac{2}{r^2} \vec{r}, \quad \nabla^2 f = \frac{2}{r^2}, \quad \iint_{S_R} \nabla f \cdot d\vec{S} = 8\pi R$$

Exercise

Let $\vec{F}(x,y,z) = (e^y \cos z, xe^y \cos z, -xe^y \sin z)$.

- Verify that $\nabla \times \vec{F} = \vec{0}$.
- Find a scalar potential f such that $\vec{F} = \nabla f$.
- Compute the circulation of $\vec{F}$ along the curve C given by $\vec{r}(t) = (t, t^2, t^3)$ for $0 \leq t \leq 1$.

Solution

1. Curl of $\vec{F}$:

$$\begin{aligned} \nabla \times \vec{F} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ e^y \cos z & xe^y \cos z & -xe^y \sin z \end{vmatrix} \\ &= (0, 0, e^y \cos z - e^y \cos z) = \vec{0}. \end{aligned}$$

$$\boxed{\nabla \times \vec{F} = \vec{0}}$$

2. Scalar potential f : Integrate $\frac{\partial f}{\partial x} = e^y \cos z$:

$$f(x,y,z) = xe^y \cos z + g(y,z).$$

Then $\frac{\partial f}{\partial y} = xe^y \cos z + \frac{\partial g}{\partial y} = xe^y \cos z \Rightarrow \frac{\partial g}{\partial y} = 0 \Rightarrow g(y,z) = h(z)$. Finally $\frac{\partial f}{\partial z} = -xe^y \sin z + h'(z) = -xe^y \sin z \Rightarrow h'(z) = 0 \Rightarrow h(z) = C$.

$$\boxed{f(x,y,z) = xe^y \cos z + C}$$

3. Circulation along C :

$$\oint_C \vec{F} \cdot d\vec{r} = f(1, 1, 1) - f(0, 0, 0) = (1 \cdot e^1 \cos 1) - (0) = e \cos 1.$$

$$\oint_C \vec{F} \cdot d\vec{r} = e \cos 1$$

Exercise

Let S be the surface parameterized by $\vec{r}(u, v) = (u \cos v, u \sin v, u)$ for $0 \leq u \leq 1$, $0 \leq v \leq 2\pi$. Let $\vec{F}(x, y, z) = (x, y, z)$.

- Compute the tangent vectors $\vec{r}_u$ and $\vec{r}_v$.
- Compute the normal vector $\vec{r}_u \times \vec{r}_v$.
- Compute the area element $dS = \|\vec{r}_u \times \vec{r}_v\| du dv$.
- Compute the flux of $\vec{F}$ through S .

Solution

1. Tangent vectors:

$$\vec{r}_u = (\cos v, \sin v, 1), \quad \vec{r}_v = (-u \sin v, u \cos v, 0).$$

$$\vec{r}_u = (\cos v, \sin v, 1), \quad \vec{r}_v = (-u \sin v, u \cos v, 0)$$

2. Normal vector:

$$\vec{r}_u \times \vec{r}_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos v & \sin v & 1 \\ -u \sin v & u \cos v & 0 \end{vmatrix} = (-u \cos v, -u \sin v, u).$$

$$\vec{r}_u \times \vec{r}_v = (-u \cos v, -u \sin v, u)$$

3. Area element:

$$\|\vec{r}_u \times \vec{r}_v\| = \sqrt{u^2 \cos^2 v + u^2 \sin^2 v + u^2} = \sqrt{2u^2} = \sqrt{2} u.$$

$$dS = \sqrt{2} u du dv$$

4. Flux of $\vec{F}$:

$$\vec{F}(\vec{r}(u, v)) = (u \cos v, u \sin v, u).$$

$$\vec{F} \cdot (\vec{r}_u \times \vec{r}_v) = -u^2 \cos^2 v - u^2 \sin^2 v + u^2 = 0.$$

$$\iint_S \vec{F} \cdot d\vec{S} = \int_0^{2\pi} \int_0^1 0 du dv = 0.$$

$$\text{Flux} = 0$$

Exercise

Let $\vec{F}(x, y) = \left(\frac{-y}{x^2+y^2}, \frac{x}{x^2+y^2} \right)$ be a vector field in the plane (defined for $(x, y) \neq (0, 0)$).

- Compute $\nabla \times \vec{F}$ (in $\mathbb{R}^3$, with $F_z = 0$).
- Let C be the unit circle $x^2 + y^2 = 1$ traversed counterclockwise. Compute the circulation $\oint_C \vec{F} \cdot d\vec{r}$.
- Is the field $\vec{F}$ conservative on $\mathbb{R}^2 \setminus \{(0, 0)\}$? Justify.
- Compute the flux of $\vec{F}$ through the unit circle (outward orientation).

Solution

1. Curl of $\vec{F}$:

$$\begin{aligned} \nabla \times \vec{F} &= \left(0, 0, \frac{\partial}{\partial x} \left(\frac{x}{x^2+y^2} \right) - \frac{\partial}{\partial y} \left(\frac{-y}{x^2+y^2} \right) \right) \\ &= \left(0, 0, \frac{y^2 - x^2}{(x^2+y^2)^2} + \frac{x^2 - y^2}{(x^2+y^2)^2} \right) = (0, 0, 0). \end{aligned}$$

$$\boxed{\nabla \times \vec{F} = \vec{0}}$$

2. Circulation on the unit circle: Parameterization: $\vec{r}(t) = (\cos t, \sin t)$, $0 \leq t \leq 2\pi$.

$$\vec{F} \cdot d\vec{r} = (-\sin t, \cos t) \cdot (-\sin t, \cos t) dt = (\sin^2 t + \cos^2 t) dt = dt.$$

$$\oint_C \vec{F} \cdot d\vec{r} = \int_0^{2\pi} dt = 2\pi.$$

$$\boxed{\oint_C \vec{F} \cdot d\vec{r} = 2\pi}$$

3. Is the field conservative? No, because although $\nabla \times \vec{F} = \vec{0}$ everywhere except the origin, the circulation around a closed curve enclosing the origin is not zero. The domain is not simply connected.

$$\boxed{\text{Not conservative}}$$

4. Flux through the unit circle:

$$\vec{F} \cdot \hat{n} = \left(\frac{-y}{x^2+y^2}, \frac{x}{x^2+y^2} \right) \cdot (x, y) = \frac{-xy + xy}{x^2+y^2} = 0.$$

$$\boxed{\text{Flux} = 0}$$

Exercise

Let $\vec{F}(x, y, z) = (2x + yz, 2y + xz, 2z + xy)$.

- Compute $\nabla \cdot \vec{F}$ and $\nabla \times \vec{F}$.
- Show that $\vec{F}$ is a gradient field (conservative) and find a potential f .

- Let C be the curve of intersection of the sphere $x^2 + y^2 + z^2 = 1$ and the plane $x + y + z = 0$. Compute the circulation of $\vec{F}$ along C .

Solution

1. Divergence and curl:

$$\nabla \cdot \vec{F} = 2 + 2 + 2 = 6.$$

$$\nabla \times \vec{F} = (0, 0, 0).$$

$$\nabla \cdot \vec{F} = 6, \quad \nabla \times \vec{F} = \vec{0}$$

2. Potential f :

$$f(x, y, z) = x^2 + y^2 + z^2 + xyz + C.$$

$$f(x, y, z) = x^2 + y^2 + z^2 + xyz$$

3. Circulation along C : Since $\vec{F}$ is conservative and C is a closed curve:

$$\oint_C \vec{F} \cdot d\vec{r} = 0.$$

$$\oint_C \vec{F} \cdot d\vec{r} = 0$$

Exercise

Let S be the surface of the upper hemisphere $z = \sqrt{1 - x^2 - y^2}$, oriented upward. Let $\vec{F}(x, y, z) = (x^3, y^3, z^3)$.

- Parameterize S using spherical coordinates.
- Compute the unit normal vector associated with this parameterization.
- Compute the flux of $\vec{F}$ through S .
- Verify the result using the divergence theorem (considering the closed surface consisting of the hemisphere and the disk $z = 0, x^2 + y^2 \leq 1$).

Solution

1. Parameterization:

$$\vec{r}(\theta, \phi) = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta), \quad 0 \leq \theta \leq \frac{\pi}{2}, \quad 0 \leq \phi \leq 2\pi.$$

$$\vec{r}(\theta, \phi) = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$$

2. Unit normal vector:

$$\vec{r}_\theta = (\cos \theta \cos \phi, \cos \theta \sin \phi, -\sin \theta),$$

$$\vec{r}_\phi = (-\sin \theta \sin \phi, \sin \theta \cos \phi, 0).$$

$$\vec{r}_\theta \times \vec{r}_\phi = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos \theta \cos \phi & \cos \theta \sin \phi & -\sin \theta \\ -\sin \theta \sin \phi & \sin \theta \cos \phi & 0 \end{vmatrix}$$

$$= (\sin^2 \theta \cos \phi, \sin^2 \theta \sin \phi, \sin \theta \cos \theta).$$

$$\vec{r}_\theta \times \vec{r}_\phi = (\sin^2 \theta \cos \phi, \sin^2 \theta \sin \phi, \sin \theta \cos \theta)$$

3. Flux through S :

$$\vec{F}(\vec{r}(\theta, \phi)) = (\sin^3 \theta \cos^3 \phi, \sin^3 \theta \sin^3 \phi, \cos^3 \theta).$$

$$\vec{F} \cdot (\vec{r}_\theta \times \vec{r}_\phi) = \sin^5 \theta \cos^4 \phi + \sin^5 \theta \sin^4 \phi + \cos^4 \theta \sin \theta.$$

$$\int_0^{2\pi} (\cos^4 \phi + \sin^4 \phi) d\phi = \frac{3\pi}{2}, \quad \int_0^{\pi/2} \sin^5 \theta d\theta = \frac{8}{15},$$

$$\int_0^{\pi/2} \cos^4 \theta \sin \theta d\theta = \frac{1}{5}.$$

$$\Phi_S = \frac{3\pi}{2} \cdot \frac{8}{15} + 2\pi \cdot \frac{1}{5} = \frac{4\pi}{5} + \frac{2\pi}{5} = \frac{6\pi}{5}.$$

$$\text{Flux} = \frac{6\pi}{5}$$

4. Verification using the divergence theorem:

$$\nabla \cdot \vec{F} = 3x^2 + 3y^2 + 3z^2 = 3(x^2 + y^2 + z^2).$$

In spherical coordinates:

$$\begin{aligned} \iiint_V \nabla \cdot \vec{F} dV &= \int_0^{2\pi} \int_0^{\pi/2} \int_0^1 3r^2 \cdot r^2 \sin \theta dr d\theta d\phi \\ &= 3 \cdot 2\pi \cdot 1 \cdot \frac{1}{5} = \frac{6\pi}{5}. \end{aligned}$$

The flux through the disk $z = 0$ is zero because $\vec{F} \cdot \hat{n} = (x^3, y^3, 0) \cdot (0, 0, -1) = 0$. Thus the total flux through the closed surface is $\frac{6\pi}{5}$, which is the same as the flux through the hemisphere.

Verified

Exercise

Let the vector field

$$\vec{V}(x, y, z) = (2x + yz, 2y + xz, 2z + xy).$$

- Compute the divergence of $\vec{V}$.
- Compute the curl of $\vec{V}$.
- Is the field $\vec{V}$ conservative? Justify.
- Find a scalar potential $f(x, y, z)$ such that $\vec{V} = \nabla f$, if it exists.
- Compute the circulation of $\vec{V}$ along the curve $C : \vec{r}(t) = (\cos t, \sin t, t)$, $0 \leq t \leq 2\pi$.

Solution

1. Divergence:

$$\begin{aligned}\nabla \cdot \vec{V} &= \frac{\partial}{\partial x}(2x + yz) + \frac{\partial}{\partial y}(2y + xz) + \frac{\partial}{\partial z}(2z + xy) \\ &= 2 + 2 + 2 = 6.\end{aligned}$$

$$\boxed{\nabla \cdot \vec{V} = 6}$$

2. Curl:

$$\nabla \times \vec{V} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial_x & \partial_y & \partial_z \\ 2x + yz & 2y + xz & 2z + xy \end{vmatrix} = (0, 0, 0).$$

$$\boxed{\nabla \times \vec{V} = \vec{0}}$$

3. Is the field conservative? Yes, because $\nabla \times \vec{V} = \vec{0}$ on $\mathbb{R}^3$ (which is simply connected), so it is a gradient field.

4. Potential f : We look for f such that:

$$\begin{aligned}\frac{\partial f}{\partial x} &= 2x + yz \Rightarrow f = x^2 + xyz + g(y, z), \\ \frac{\partial f}{\partial y} &= 2y + xz \Rightarrow xz + g_y = 2y + xz \Rightarrow g_y = 2y, \\ &\Rightarrow g = y^2 + h(z), \\ \frac{\partial f}{\partial z} &= 2z + xy \Rightarrow xy + h'(z) = 2z + xy \Rightarrow h'(z) = 2z, \\ &\Rightarrow h(z) = z^2 + C.\end{aligned}$$

Thus:

$$f(x, y, z) = x^2 + y^2 + z^2 + xyz + C.$$

$$\boxed{f(x, y, z) = x^2 + y^2 + z^2 + xyz + C}$$

5. Circulation along C : The curve is closed because $\vec{r}(0) = \vec{r}(2\pi)$. Since $\vec{V}$ is conservative:

$$\oint_C \vec{V} \cdot d\vec{r} = 0.$$

$$\boxed{\oint_C \vec{V} \cdot d\vec{r} = 0}$$

Exercise

Let the vector field

$$\vec{F}(x, y, z) = (z^2, x^2, y^2).$$

- Compute $\nabla \times \vec{F}$.
- Compute the flux of $\nabla \times \vec{F}$ through the surface $S : z = 1 - x^2 - y^2, z \geq 0$, oriented upward.

- Verify Stokes' theorem.

Solution

1. Curl:

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial_x & \partial_y & \partial_z \\ z^2 & x^2 & y^2 \end{vmatrix} = (2y, 2z, 2x).$$

$$\boxed{\nabla \times \vec{F} = (2y, 2z, 2x)}$$

2. Flux through S : Parameterization: $\vec{r}(u, v) = (u, v, 1 - u^2 - v^2), u^2 + v^2 \leq 1$.

$$\vec{r}_u = (1, 0, -2u), \quad \vec{r}_v = (0, 1, -2v).$$

$$\vec{r}_u \times \vec{r}_v = (2u, 2v, 1).$$

$$\begin{aligned} \iint_S (\nabla \times \vec{F}) \cdot d\vec{S} &= \iint_D (2v, 2(1 - u^2 - v^2), 2u) \cdot (2u, 2v, 1) \, du \, dv \\ &= \iint_D (4uv + 4v(1 - u^2 - v^2) + 2u) \, du \, dv. \end{aligned}$$

The terms $4uv$ and $2u$ vanish by symmetry. The term $4v(1 - u^2 - v^2)$ is odd in v over a symmetric domain, so its integral is zero.

$$\iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = 0.$$

$$\boxed{\text{Flux} = 0}$$

3. Verification of Stokes' theorem: The boundary of S is the circle $x^2 + y^2 = 1, z = 0$.

$$\vec{r}(t) = (\cos t, \sin t, 0), \quad \vec{r}'(t) = (-\sin t, \cos t, 0).$$

$$\vec{F}(\vec{r}(t)) = (0, \cos^2 t, \sin^2 t).$$

$$\oint_{\partial S} \vec{F} \cdot d\vec{r} = \int_0^{2\pi} \cos^3 t \, dt = 0.$$

Thus Stokes' theorem is verified.

$$\boxed{\oint_{\partial S} \vec{F} \cdot d\vec{r} = 0}$$

Exercise

Let the vector field

$$\vec{H}(x, y, z) = (x^3, y^3, z^3).$$

- Compute $\nabla \cdot \vec{H}$.
- Compute the flux through the sphere $x^2 + y^2 + z^2 = R^2$.
- Verify the divergence theorem.

Solution**1. Divergence:**

$$\nabla \cdot \vec{H} = 3x^2 + 3y^2 + 3z^2 = 3(x^2 + y^2 + z^2).$$

$$\boxed{\nabla \cdot \vec{H} = 3(x^2 + y^2 + z^2)}$$

2. Flux through the sphere: By the divergence theorem:

$$\iint_S \vec{H} \cdot d\vec{S} = \iiint_V 3(x^2 + y^2 + z^2) dV.$$

In spherical coordinates:

$$= 3 \int_0^{2\pi} \int_0^\pi \int_0^R \rho^2 \cdot \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$

$$= 3 \cdot 2\pi \cdot 2 \cdot \frac{R^5}{5} = \frac{12\pi R^5}{5}.$$

$$\boxed{\text{Flux} = \frac{12\pi R^5}{5}}$$

3. Verification: Direct calculation of the flux on the sphere gives the same value, so the divergence theorem is verified.

Multiple Integrals

Theorem

Multiple Integral Calculation – Complete Theorems and Applications

1. **Double integral over a rectangle (Fubini's theorem):** Let f be a continuous function on the rectangle $R = [a, b] \times [c, d]$. Then:

$$\begin{aligned}\iint_R f(x, y) \, dA &= \int_a^b \left(\int_c^d f(x, y) \, dy \right) dx \\ &= \int_c^d \left(\int_a^b f(x, y) \, dx \right) dy\end{aligned}$$

The order of integration can be interchanged.

2. **Double integral over a general domain:** Let D be a bounded domain in the plane. If f is continuous on D , then:

- If $D = \{(x, y) \mid a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$, then:

$$\iint_D f(x, y) \, dA = \int_a^b \left(\int_{g_1(x)}^{g_2(x)} f(x, y) \, dy \right) dx$$

- If $D = \{(x, y) \mid c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\}$, then:

$$\iint_D f(x, y) \, dA = \int_c^d \left(\int_{h_1(y)}^{h_2(y)} f(x, y) \, dx \right) dy$$

3. **Triple integral over a general domain:** Let E be a bounded solid in $\mathbb{R}^3$. If f is continuous on E , then:

- If $E = \{(x, y, z) \mid (x, y) \in D, u_1(x, y) \leq z \leq u_2(x, y)\}$, then:

$$\iiint_E f(x, y, z) \, dV = \iint_D \left(\int_{u_1(x, y)}^{u_2(x, y)} f(x, y, z) \, dz \right) dA$$

- Volume of the solid: $V(E) = \iiint_E dV$

4. **Change of variables in the plane:** Let $T : (u, v) \mapsto (x(u, v), y(u, v))$ be a C^1 change of variables, bijective from D' to D , and let f be continuous on D . Then:

$$\iint_D f(x, y) dx dy = \iint_{D'} f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv$$

where the Jacobian determinant is given by:

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \det \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix}$$

5. **Change of variables in three dimensions:**

$$T : (u, v, w) \longmapsto (x(u, v, w), y(u, v, w), z(u, v, w))$$

$$\iiint_E f(x, y, z) dV = \iiint_{E'} f(x(u, v, w), y(u, v, w), z(u, v, w)) \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| du dv dw$$

where:

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = \det \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{pmatrix}$$

6. **Polar coordinates:** Polar coordinates are defined by $x = r \cos \theta$, $y = r \sin \theta$. The Jacobian determinant is $|J| = r$, thus:

$$\iint_D f(x, y) dx dy = \iint_{D'} f(r \cos \theta, r \sin \theta) r dr d\theta$$

with $r \geq 0$ and $0 \leq \theta \leq 2\pi$ (or an appropriate domain).

7. **Cylindrical coordinates:** Cylindrical coordinates are defined by $x = r \cos \theta$, $y = r \sin \theta$, $z = z$. The Jacobian determinant is $|J| = r$, thus:

$$\iiint_E f(x, y, z) dV = \iiint_{E'} f(r \cos \theta, r \sin \theta, z) r dr d\theta dz$$

8. **Spherical coordinates:** Spherical coordinates are defined by:

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi$$

with $\rho \geq 0$, $0 \leq \phi \leq \pi$, $0 \leq \theta \leq 2\pi$. The Jacobian determinant is $|J| = \rho^2 \sin \phi$, thus:

$$\iiint_E f(x, y, z) dV = \iiint_{E'} f(\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \phi) \rho^2 \sin \phi d\rho d\phi d\theta$$

9. **General curvilinear coordinates:** For a coordinate system (q_1, q_2, q_3) , the scale factors are:

$$h_i = \left\| \frac{\partial \vec{r}}{\partial q_i} \right\|, \quad i = 1, 2, 3$$

Volume element: $dV = h_1 h_2 h_3 dq_1 dq_2 dq_3$

- In cylindrical coordinates: $h_r = 1$, $h_\theta = r$, $h_z = 1$

- In spherical coordinates: $h_\rho = 1$, $h_\phi = \rho$, $h_\theta = \rho \sin \phi$

10. **Mass, center of mass, and moments:** For a solid E with density $\delta(x, y, z)$:

- **Mass:**

$$M = \iiint_E \delta(x, y, z) dV$$

- **Center of mass (centroid):** $(\bar{x}, \bar{y}, \bar{z})$ with

$$\bar{x} = \frac{1}{M} \iiint_E x \delta(x, y, z) dV,$$

$$\bar{y} = \frac{1}{M} \iiint_E y \delta(x, y, z) dV,$$

$$\bar{z} = \frac{1}{M} \iiint_E z \delta(x, y, z) dV$$

- **Static moments (first moments):**

$$M_x = \iiint_E x \delta(x, y, z) dV,$$

$$M_y = \iiint_E y \delta(x, y, z) dV,$$

$$M_z = \iiint_E z \delta(x, y, z) dV$$

- **Moments of inertia (second moments) about the axes:**

$$I_x = \iiint_E (y^2 + z^2) \delta(x, y, z) dV \quad \text{about the } x\text{-axis}$$

$$I_y = \iiint_E (x^2 + z^2) \delta(x, y, z) dV \quad \text{about the } y\text{-axis}$$

$$I_z = \iiint_E (x^2 + y^2) \delta(x, y, z) dV \quad \text{about the } z\text{-axis}$$

- **Moment of inertia about the origin:**

$$I_O = \iiint_E (x^2 + y^2 + z^2) \delta(x, y, z) dV = \frac{1}{2}(I_x + I_y + I_z)$$

- **Moment of inertia about an axis L :** If $d(x, y, z)$ is the distance from the point (x, y, z) to the axis L , then:

$$I_L = \iiint_E d^2(x, y, z) \delta(x, y, z) dV$$

- **Parallel axis theorem (Huygens-Steiner):** If I_G is the moment of inertia about an axis passing through the center of mass G , and I is the moment of inertia about a parallel axis at a distance d from it, then:

$$I = I_G + M d^2$$

- **Radius of gyration:** The radius of gyration k about an axis is defined by:

$$k = \sqrt{\frac{I}{M}}$$

11. **Surface integrals (parametric):** Let S be a surface parameterized by $\vec{r}(u, v) = (x(u, v), y(u, v), z(u, v))$ for $(u, v) \in D$. Then:

- Tangent vectors: $\vec{r}_u = \frac{\partial \vec{r}}{\partial u}$, $\vec{r}_v = \frac{\partial \vec{r}}{\partial v}$
- Normal vector: $\vec{n} = \vec{r}_u \times \vec{r}_v$
- Area element: $dS = \|\vec{r}_u \times \vec{r}_v\| du dv$
- Surface area: $A(S) = \iint_D \|\vec{r}_u \times \vec{r}_v\| du dv$
- Integral of a function f over S :

$$\iint_S f(x, y, z) dS = \iint_D f(\vec{r}(u, v)) \|\vec{r}_u \times \vec{r}_v\| du dv$$

12. **Surface integrals (explicit form):** If S is given by $z = g(x, y)$ for $(x, y) \in D$, then:

- Normal vector: $\vec{n} = \left(-\frac{\partial g}{\partial x}, -\frac{\partial g}{\partial y}, 1\right)$
- Its magnitude: $\|\vec{n}\| = \sqrt{1 + \left(\frac{\partial g}{\partial x}\right)^2 + \left(\frac{\partial g}{\partial y}\right)^2}$
- Area element: $dS = \sqrt{1 + \left(\frac{\partial g}{\partial x}\right)^2 + \left(\frac{\partial g}{\partial y}\right)^2} dx dy$
- Surface area: $A(S) = \iint_D \sqrt{1 + \left(\frac{\partial g}{\partial x}\right)^2 + \left(\frac{\partial g}{\partial y}\right)^2} dx dy$
- Integral of a function f over S :

$$\iint_S f(x, y, z) dS = \iint_D f(x, y, g(x, y)) \sqrt{1 + \left(\frac{\partial g}{\partial x}\right)^2 + \left(\frac{\partial g}{\partial y}\right)^2} dx dy$$

13. **Flux of a vector field through a surface:** Let $\vec{F} = (P, Q, R)$ be a vector field, and S an oriented surface with unit normal vector $\hat{n}$. The flux of $\vec{F}$ through S is:

$$\Phi = \iint_S \vec{F} \cdot d\vec{S} = \iint_S \vec{F} \cdot \hat{n} dS$$

- If S is parameterized by $\vec{r}(u, v)$: $d\vec{S} = (\vec{r}_u \times \vec{r}_v) du dv$
- If S is given by $z = g(x, y)$ with upward orientation:

$$d\vec{S} = \left(-\frac{\partial g}{\partial x}, -\frac{\partial g}{\partial y}, 1\right) dx dy$$

14. **Divergence theorem (Gauss-Ostrogradsky):** Let E be a solid with boundary S oriented outward, and let $\vec{F} = (P, Q, R)$ be a C^1 vector field on E . Then:

$$\iiint_E (\nabla \cdot \vec{F}) dV = \iint_S \vec{F} \cdot d\vec{S}$$

where $\nabla \cdot \vec{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$.

15. **Stokes' theorem:** Let S be an oriented surface with boundary curve C oriented positively (right-hand rule), and let $\vec{F}$ be a C^1 vector field on a domain containing S . Then:

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot d\vec{S}$$

where $\nabla \times \vec{F} = \nabla \times \vec{F}$.

16. **Green's theorem in the plane:** Let C be a positively oriented simple closed curve, and let D be the interior domain of C . Let $\vec{F} = (P, Q)$ be a C^1 vector field on D . Then:

$$\oint_C \vec{F} \cdot d\vec{r} = \oint_C P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

17. **Gradient theorem:** Let f be a C^1 scalar function, and C a curve from A to B . Then:

$$\int_C \nabla f \cdot d\vec{r} = f(B) - f(A)$$

Consequence: If C is closed, then $\oint_C \nabla f \cdot d\vec{r} = 0$.

18. **Green-Riemann formula (area of a domain):** The area of a domain D with positively oriented boundary C is given by:

$$A(D) = \frac{1}{2} \oint_C (x dy - y dx) = \oint_C x dy = - \oint_C y dx$$

19. **Geometric applications:**

- **Length of a plane curve** $y = f(x)$ on $[a, b]$:

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} dx$$

- **Surface area of revolution:**

- About the x -axis: $A = 2\pi \int_a^b y \sqrt{1 + (y')^2} dx$

- About the y -axis: $A = 2\pi \int_c^d x \sqrt{1 + (x')^2} dy$

- **Volume of revolution:**

- Disk method: $V = \pi \int_a^b [f(x)]^2 dx$

- Cylindrical shell method: $V = 2\pi \int_a^b x f(x) dx$

- **Lateral surface area of a frustum** with radii R_1, R_2 and height h :

$$A = \pi(R_1 + R_2) \sqrt{(R_1 - R_2)^2 + h^2}$$

20. **Improper integrals in several variables:**

- **Unbounded domain:** If f is continuous on $\mathbb{R}^2$, then:

$$\iint_{\mathbb{R}^2} f(x, y) dA = \lim_{R \rightarrow \infty} \iint_{D_R} f(x, y) dA$$

where D_R is the disk of radius R centered at the origin.

- **Unbounded function:** If f has a singularity at (x_0, y_0) , then:

$$\iint_D f(x, y) dA = \lim_{\epsilon \rightarrow 0} \iint_{D_\epsilon} f(x, y) dA$$

where D_ϵ is D with a disk of radius ϵ around (x_0, y_0) removed.

- **Convergence in polar coordinates:** For $f(x, y) = \frac{1}{(x^2 + y^2)^p}$ on $\mathbb{R}^2 \setminus \{0\}$:

$$\iint_{\mathbb{R}^2} \frac{1}{(x^2 + y^2)^p} dA = 2\pi \int_0^\infty \frac{r}{r^{2p}} dr = 2\pi \int_0^\infty r^{1-2p} dr$$

Converges if and only if $1 - 2p < -1$, i.e., $p > 1$.

21. Important physical applications:

- **Gauss's law in electrostatics:**

$$\iint_S \vec{E} \cdot d\vec{S} = \frac{Q_{\text{int}}}{\epsilon_0}$$

- **Continuity equation:**

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \vec{J} = 0$$

- **Heat equation:**

$$\frac{\partial T}{\partial t} = \alpha \nabla^2 T, \quad \alpha = \frac{k}{\rho c_p}$$

- **Poisson's equation:**

$$\nabla^2 \phi = -4\pi\rho \quad (\text{gravitation or electrostatics})$$

- **Hydrostatic pressure:** Force on a surface: $\vec{F} = \iint_S P d\vec{S}$

- **Center of pressure:** Point of application of the resultant pressure force.

Exercise

Compute the double integral $\iint_D (2x + y) dA$ where D is the rectangle defined by:

$$D = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq x \leq 2, 1 \leq y \leq 3\}.$$

Interpret the result geometrically as a volume.

Solution

The double integral is computed as an iterated integral:

$$\begin{aligned}
 \iint_D (2x + y) dA &= \int_{x=0}^2 \int_{y=1}^3 (2x + y) dy dx \\
 &= \int_0^2 \left[2xy + \frac{y^2}{2} \right]_{y=1}^{y=3} dx \\
 &= \int_0^2 \left[\left(6x + \frac{9}{2} \right) - \left(2x + \frac{1}{2} \right) \right] dx \\
 &= \int_0^2 (4x + 4) dx \\
 &= [2x^2 + 4x]_0^2 = (8 + 8) - 0 = 16.
 \end{aligned}$$

This result represents the volume of the solid bounded by the rectangle D in the xy -plane and the surface $z = 2x + y$.

$$\boxed{\iint_D (2x + y) dA = 16}$$

Exercise

Compute the double integral $\iint_D x dA$ where D is the plane region bounded by the curves $y = x^2$ and $y = 2x$.

- Sketch the region D graphically.
- Write the integral as an iterated integral in the order $dy dx$.
- Write the integral as an iterated integral in the order $dx dy$.
- Compute the integral by choosing the most convenient order.

Solution

1. Graphical representation: The curves intersect when $x^2 = 2x$, i.e., $x(x - 2) = 0$, so at the points $(0, 0)$ and $(2, 4)$. The region D is bounded between the curves for $x \in [0, 2]$.

2. In the order $dy dx$: Describe D as:

$$D = \{(x, y) \mid 0 \leq x \leq 2, x^2 \leq y \leq 2x\}.$$

Thus:

$$\iint_D x dA = \int_0^2 \int_{x^2}^{2x} x dy dx.$$

3. In the order $dx dy$: Split D into two parts. For each $0 \leq y \leq 4$, we have:

- From the parabola: $x = \sqrt{y}$
- From the line: $x = \frac{y}{2}$

Thus:

$$\iint_D x dA = \int_0^4 \int_{y/2}^{\sqrt{y}} x dx dy.$$

4. Computation: Use the easier order $dy \, dx$:

$$\begin{aligned}
 \iint_D x \, dA &= \int_0^2 \int_{x^2}^{2x} x \, dy \, dx \\
 &= \int_0^2 x [y]_{y=x^2}^{y=2x} \, dx \\
 &= \int_0^2 x(2x - x^2) \, dx \\
 &= \int_0^2 (2x^2 - x^3) \, dx \\
 &= \left[\frac{2x^3}{3} - \frac{x^4}{4} \right]_0^2 \\
 &= \left(\frac{16}{3} - \frac{16}{4} \right) - 0 = \frac{16}{3} - 4 = \frac{4}{3}.
 \end{aligned}$$

$$\boxed{\iint_D x \, dA = \frac{4}{3}}$$

Exercise

Let D be the plane region bounded by the lines $y = x$, $y = 2x$, $x = 1$ and $x = 2$. Compute the double integral:

$$\iint_D (x + y) \, dx \, dy$$

by making the change of variables:

$$u = \frac{y}{x}, \quad v = x.$$

- Determine the new region D' in the (u, v) -plane.
- Compute the Jacobian determinant $\frac{\partial(x, y)}{\partial(u, v)}$.
- Compute the transformed integral.

Solution

1. Description of the region D : D is defined by: $1 \leq x \leq 2$ and $x \leq y \leq 2x$. Using the change $u = y/x$ and $v = x$, we obtain:

$$x = v, \quad y = uv.$$

The boundaries:

- x : $1 \leq v \leq 2$.
- y : $x \leq y \leq 2x \Rightarrow v \leq uv \leq 2v \Rightarrow 1 \leq u \leq 2$ (since $v > 0$). Thus $D' = [1, 2] \times [1, 2]$ (a rectangle in the (u, v) -plane).

2. Jacobian determinant:

$$J = \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ v & u \end{pmatrix}, \quad \left| \frac{\partial(x, y)}{\partial(u, v)} \right| = |\det(J)| = |-v| = v.$$

3. Transformed integral: The function $x + y = v + uv = v(1 + u)$. Thus:

$$\iint_D (x + y) dx dy = \iint_{D'} v(1 + u) \cdot v du dv = \int_{v=1}^2 \int_{u=1}^2 v^2(1 + u) du dv.$$

Compute the integral first with respect to u :

$$\int_1^2 (1 + u) du = \left[u + \frac{u^2}{2} \right]_1^2 = (2 + 2) - (1 + 0.5) = 4 - 1.5 = 2.5.$$

Then with respect to v :

$$\int_1^2 v^2 dv = \left[\frac{v^3}{3} \right]_1^2 = \frac{8}{3} - \frac{1}{3} = \frac{7}{3}.$$

The result:

$$\iint_D (x + y) dx dy = \frac{7}{3} \times \frac{5}{2} = \frac{35}{6}.$$

$$\boxed{\iint_D (x + y) dx dy = \frac{35}{6}}$$

Exercise

Compute the double integral $\iint_D \sqrt{x^2 + y^2} dx dy$, where D is the plane region bounded by the circle centered at $(0, 0)$ with radius $R > 0$.

Solution

The function and the domain strongly suggest using polar coordinates:

$$x = r \cos \theta, \quad y = r \sin \theta, \quad \text{with } r \geq 0, \theta \in [0, 2\pi].$$

The Jacobian determinant: $|J| = r$. The function becomes: $\sqrt{x^2 + y^2} = r$. The domain D in polar coordinates: $0 \leq r \leq R, 0 \leq \theta \leq 2\pi$.

$$\iint_D \sqrt{x^2 + y^2} dx dy = \int_{\theta=0}^{2\pi} \int_{r=0}^R r \cdot r dr d\theta = \int_0^{2\pi} \int_0^R r^2 dr d\theta.$$

$$\int_0^R r^2 dr = \frac{R^3}{3}, \quad \int_0^{2\pi} d\theta = 2\pi \quad \Rightarrow \quad \frac{2\pi R^3}{3}.$$

$$\boxed{\iint_D \sqrt{x^2 + y^2} dx dy = \frac{2\pi R^3}{3}}$$

Exercise

Compute the double integral:

$$\iint_D (x^2 + y^2) dA$$

where D is the closed disk $x^2 + y^2 \leq 4$.

Solution

Using polar coordinates: $0 \leq r \leq 2$, $0 \leq \theta \leq 2\pi$, $dA = r dr d\theta$.

$$\iint_D (x^2 + y^2) dA = \int_{\theta=0}^{2\pi} \int_{r=0}^2 r^2 \cdot r dr d\theta = \int_0^{2\pi} d\theta \int_0^2 r^3 dr = 2\pi \cdot \left[\frac{r^4}{4} \right]_0^2 = 2\pi \cdot 4 = 8\pi.$$

$$\boxed{\iint_D (x^2 + y^2) dA = 8\pi}$$

Exercise

Compute the double integral:

$$\iint_D xy dA$$

where D is the region bounded by the triangle with vertices $(0, 0)$, $(1, 0)$, $(1, 1)$.

Solution

The triangle D is described by: $0 \leq x \leq 1$, $0 \leq y \leq x$.

$$\begin{aligned} \iint_D xy dA &= \int_{x=0}^1 \int_{y=0}^x xy dy dx \\ &= \int_0^1 x \left[\frac{y^2}{2} \right]_0^x dx = \int_0^1 x \cdot \frac{x^2}{2} dx = \frac{1}{2} \int_0^1 x^3 dx = \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}. \end{aligned}$$

$$\boxed{\iint_D xy dA = \frac{1}{8}}$$

Exercise

Compute the triple integral $\iiint_E z dV$, where E is the solid tetrahedron bounded by the planes $x = 0$, $y = 0$, $z = 0$ and $x + y + z = 1$.

Solution

The tetrahedron E is described by:

$$E = \{(x, y, z) \mid 0 \leq x \leq 1, 0 \leq y \leq 1 - x, 0 \leq z \leq 1 - x - y\}.$$

$$\begin{aligned}
 \iiint_E z \, dV &= \int_{x=0}^1 \int_{y=0}^{1-x} \int_{z=0}^{1-x-y} z \, dz \, dy \, dx \\
 &= \int_0^1 \int_0^{1-x} \left[\frac{z^2}{2} \right]_{z=0}^{1-x-y} dy \, dx \\
 &= \int_0^1 \int_0^{1-x} \frac{(1-x-y)^2}{2} dy \, dx.
 \end{aligned}$$

Compute the inner integral: substitute $u = 1 - x - y$, $du = -dy$, limits $y = 0 \rightarrow u = 1 - x$, $y = 1 - x \rightarrow u = 0$:

$$\int_0^{1-x} (1-x-y)^2 dy = \int_{u=1-x}^0 u^2 (-du) = \int_0^{1-x} u^2 du = \left[\frac{u^3}{3} \right]_0^{1-x} = \frac{(1-x)^3}{3}.$$

Thus:

$$\iiint_E z \, dV = \int_0^1 \frac{1}{2} \cdot \frac{(1-x)^3}{3} dx = \frac{1}{6} \int_0^1 (1-x)^3 dx.$$

Substitute $t = 1 - x$, $dt = -dx$: $\int_0^1 (1-x)^3 dx = \int_0^1 t^3 dt = \frac{1}{4}$. Thus the result = $\frac{1}{6} \cdot \frac{1}{4} = \frac{1}{24}$.

$$\boxed{\iiint_E z \, dV = \frac{1}{24}}$$

Exercise

Let E be the solid defined by: $x \geq 0$, $y \geq 0$, $z \geq 0$, $x + y + z \leq 1$. Compute the volume of E using the change of variables:

$$u = x, \quad v = x + y, \quad w = x + y + z.$$

- Express (x, y, z) in terms of (u, v, w) .
- Compute the Jacobian determinant $\frac{\partial(x, y, z)}{\partial(u, v, w)}$.
- Determine the limits of integration for (u, v, w) and compute the triple integral for the volume.

Solution

1. Expression:

$$x = u, \quad y = v - u, \quad z = w - v.$$

2. Jacobian determinant: The Jacobian matrix:

$$J = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}, \quad \det(J) = 1.$$

Absolute value: $|J| = 1$.

3. Limits of integration: From the conditions $x \geq 0$, $y \geq 0$, $z \geq 0$, $x + y + z \leq 1$ we obtain:

$$u \geq 0, \quad v - u \geq 0 \Rightarrow v \geq u, \quad w - v \geq 0 \Rightarrow w \geq v, \quad w \leq 1.$$

Thus $0 \leq u \leq v \leq w \leq 1$. The domain E' is:

$$E' = \{(u, v, w) \mid 0 \leq u \leq v \leq w \leq 1\}.$$

The volume of E is:

$$V = \iiint_E dV = \iiint_{E'} |J| du dv dw = \int_{w=0}^1 \int_{v=0}^w \int_{u=0}^v 1 du dv dw.$$

Compute:

$$\int_{u=0}^v 1 du = v, \quad \int_{v=0}^w v dv = \frac{w^2}{2}, \quad \int_{w=0}^1 \frac{w^2}{2} dw = \frac{1}{6}.$$

$$\boxed{V = \frac{1}{6}}$$

This is the volume of the unit tetrahedron.

Exercise

Compute $\iiint_E xyz dV$ where E is the cube: $0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1$.

Solution

$$\begin{aligned} \iiint_E xyz dV &= \int_0^1 \int_0^1 \int_0^1 xyz dz dy dx \\ &= \left(\int_0^1 x dx \right) \left(\int_0^1 y dy \right) \left(\int_0^1 z dz \right) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}. \end{aligned}$$

$$\boxed{\iiint_E xyz dV = \frac{1}{8}}$$

Exercise

Compute the volume of the solid lying under the elliptic paraboloid $z = 4 - x^2 - y^2$ and above the rectangle $R = [-1, 1] \times [-2, 2]$.

Solution

The volume is given by the double integral:

$$V = \iint_R (4 - x^2 - y^2) dA = \int_{x=-1}^1 \int_{y=-2}^2 (4 - x^2 - y^2) dy dx.$$

First compute the integral with respect to y :

$$\int_{-2}^2 (4 - x^2 - y^2) dy = \left[(4 - x^2)y - \frac{y^3}{3} \right]_{-2}^2 = 2(4 - x^2) \cdot 2 - \frac{16}{3} = 8(4 - x^2) - \frac{16}{3} = \frac{32}{3} - 4x^2.$$

Then integrate with respect to x :

$$V = \int_{-1}^1 \left(\frac{32}{3} - 4x^2 \right) dx = \left[\frac{32}{3}x - \frac{4x^3}{3} \right]_{-1}^1 = \frac{56}{3}.$$

$$V = \frac{56}{3}$$

Exercise

Compute the volume of the solid bounded by the surfaces:

$$z = 0, \quad z = 4 - x^2 - y^2, \quad x^2 + y^2 = 1.$$

Solution

The solid is a “paraboloidal cap” above the unit disk $D = \{(x, y) \mid x^2 + y^2 \leq 1\}$. The volume is:

$$V = \iint_D (4 - x^2 - y^2) dA.$$

Use polar coordinates: $x = r \cos \theta$, $y = r \sin \theta$, with $0 \leq r \leq 1$, $0 \leq \theta \leq 2\pi$. The Jacobian is r .

$$\begin{aligned} V &= \int_{\theta=0}^{2\pi} \int_{r=0}^1 (4 - r^2) \cdot r dr d\theta \\ &= \int_0^{2\pi} \left[2r^2 - \frac{r^4}{4} \right]_0^1 d\theta \\ &= \int_0^{2\pi} \frac{7}{4} d\theta = \frac{7\pi}{2}. \end{aligned}$$

$$V = \frac{7\pi}{2}$$

Exercise

Compute the volume of the solid bounded by:

$$z = \sqrt{x^2 + y^2} \quad (\text{cone}) \quad \text{and} \quad z = \sqrt{1 - x^2 - y^2} \quad (\text{upper hemisphere}).$$

- First method: Use cylindrical coordinates.
- Second method: Use spherical coordinates.

Solution

First method: Cylindrical coordinates

The surfaces intersect when:

$$\sqrt{x^2 + y^2} = \sqrt{1 - x^2 - y^2} \Rightarrow x^2 + y^2 = \frac{1}{2}.$$

In the z coordinate, $z = \sqrt{1/2} = \frac{\sqrt{2}}{2}$.

The volume is the difference between the volume under the hemisphere and the volume under the cone, over the disk $D : x^2 + y^2 \leq \frac{1}{2}$:

$$V = \iint_D \left(\sqrt{1 - x^2 - y^2} - \sqrt{x^2 + y^2} \right) dA.$$

In cylindrical coordinates: $x = r \cos \theta$, $y = r \sin \theta$, with $0 \leq r \leq \frac{\sqrt{2}}{2}$, $0 \leq \theta \leq 2\pi$.

$$\begin{aligned} V &= \int_{\theta=0}^{2\pi} \int_{r=0}^{\frac{\sqrt{2}}{2}} \left(\sqrt{1-r^2} - r \right) r \, dr \, d\theta \\ &= 2\pi \int_0^{\frac{\sqrt{2}}{2}} \left(r\sqrt{1-r^2} - r^2 \right) dr \\ &= 2\pi \left[-\frac{1}{3}(1-r^2)^{3/2} - \frac{r^3}{3} \right]_0^{\frac{\sqrt{2}}{2}} = \frac{\pi(2-\sqrt{2})}{3}. \end{aligned}$$

Second method: Spherical coordinates

In spherical coordinates: $x = \rho \sin \phi \cos \theta$, $y = \rho \sin \phi \sin \theta$, $z = \rho \cos \phi$.

The cone corresponds to $\phi = \pi/4$. The volume above the cone (within the upper hemisphere) is:

$$V_{\text{above}} = \int_{\theta=0}^{2\pi} \int_{\phi=0}^{\pi/4} \int_{\rho=0}^1 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \frac{2\pi}{3} \left(1 - \frac{\sqrt{2}}{2} \right).$$

The volume of the entire hemisphere is $\frac{2\pi}{3}$. Thus the volume outside the cone and inside the hemisphere (which is the desired volume) is:

$$V = \frac{2\pi}{3} - V_{\text{above}} = \frac{2\pi}{3} - \frac{2\pi}{3} \left(1 - \frac{\sqrt{2}}{2} \right) = \frac{\sqrt{2}\pi}{3}.$$

$$V = \frac{\pi(2-\sqrt{2})}{3} \quad \text{First method}$$

$$V = \frac{\sqrt{2}\pi}{3} \quad \text{Second method}$$

Note that $\frac{\pi(2-\sqrt{2})}{3} = \frac{\sqrt{2}\pi}{3}$, so the two results are equal.

Exercise

Compute the volume of the solid bounded by the cylinder $x^2 + y^2 = 4$ and the planes $z = 0$ and $z = 4 - y$.

Solution

The solid is a right cylinder with base the disk $D : x^2 + y^2 \leq 4$, and variable height $4 - y$. The volume:

$$V = \iint_D (4 - y) \, dA.$$

In polar coordinates: $x = r \cos \theta$, $y = r \sin \theta$, with $0 \leq r \leq 2$, $0 \leq \theta \leq 2\pi$.

$$\begin{aligned} V &= \int_{\theta=0}^{2\pi} \int_{r=0}^2 (4 - r \sin \theta) r \, dr \, d\theta \\ &= \int_0^{2\pi} \left[2r^2 - \frac{r^3}{3} \sin \theta \right]_{r=0}^2 d\theta \\ &= \int_0^{2\pi} \left(8 - \frac{8}{3} \sin \theta \right) d\theta = 16\pi - 0 = 16\pi. \end{aligned}$$

$$\boxed{V = 16\pi}$$

Exercise

Compute the volume common to two cylinders of radius R with perpendicular intersecting axes:

$$x^2 + z^2 \leq R^2 \quad \text{and} \quad y^2 + z^2 \leq R^2.$$

Solution

Look at horizontal cross-sections. For a fixed z , the horizontal cross-section is a square of side length $2\sqrt{R^2 - z^2}$. The area of the cross-section is:

$$A(z) = \left(2\sqrt{R^2 - z^2} \right)^2 = 4(R^2 - z^2).$$

Thus the total volume:

$$V = \int_{z=-R}^R A(z) \, dz = 2 \int_0^R 4(R^2 - z^2) \, dz = 8 \left[R^2 z - \frac{z^3}{3} \right]_0^R = \frac{16R^3}{3}.$$

$$\boxed{V = \frac{16R^3}{3}}$$

This solid is known as the “Steinmetz solid” or the intersection of two cylinders.

Exercise

Compute the volume of the solid bounded by the two surfaces:

$$z = x^2 + y^2 \quad \text{and} \quad z = 2x + 2y + 10.$$

Solution

The surfaces intersect when:

$$x^2 + y^2 = 2x + 2y + 10 \Rightarrow (x - 1)^2 + (y - 1)^2 = 12,$$

which is a circle centered at $(1, 1)$ with radius $\sqrt{12} = 2\sqrt{3}$.

After simplification, the integrand becomes $12 - (x - 1)^2 - (y - 1)^2$. Using polar coordinates centered at $(1, 1)$:

$$V = \iint_{u^2 + v^2 \leq 12} (12 - u^2 - v^2) \, du \, dv = \int_{\theta=0}^{2\pi} \int_{r=0}^{2\sqrt{3}} (12 - r^2) r \, dr \, d\theta = 72\pi.$$

$$V = 72\pi$$

Exercise

Compute $\iiint_E z \, dV$ where E is the solid bounded by the cylinder $x^2 + y^2 = 1$ and the planes $z = 0$ and $z = 2$.

Solution

In cylindrical coordinates: $x = r \cos \theta$, $y = r \sin \theta$, $z = z$, with $0 \leq r \leq 1$, $0 \leq \theta \leq 2\pi$, $0 \leq z \leq 2$.

$$\begin{aligned} \iiint_E z \, dV &= \int_{\theta=0}^{2\pi} \int_{r=0}^1 \int_{z=0}^2 z \cdot r \, dz \, dr \, d\theta \\ &= \int_0^{2\pi} d\theta \int_0^1 r \, dr \int_0^2 z \, dz \\ &= (2\pi) \cdot \frac{1}{2} \cdot 2 = 2\pi. \end{aligned}$$

$$\iiint_E z \, dV = 2\pi$$

Exercise

Compute $\iiint_E (x^2 + y^2) \, dV$ where E is the solid bounded by the paraboloid $z = x^2 + y^2$ and the plane $z = 4$.

Solution

In cylindrical coordinates: $x = r \cos \theta$, $y = r \sin \theta$, $z = z$, with $0 \leq r \leq 2$, $0 \leq \theta \leq 2\pi$, $r^2 \leq z \leq 4$.

$$\begin{aligned} \iiint_E (x^2 + y^2) \, dV &= \int_{\theta=0}^{2\pi} \int_{r=0}^2 \int_{z=r^2}^4 r^2 \cdot r \, dz \, dr \, d\theta \\ &= \int_0^{2\pi} d\theta \int_0^2 r^3 (4 - r^2) \, dr = 2\pi \int_0^2 (4r^3 - r^5) \, dr \\ &= 2\pi \left[r^4 - \frac{r^6}{6} \right]_0^2 = 2\pi \left(16 - \frac{64}{6} \right) = 2\pi \cdot \frac{16}{3} = \frac{32\pi}{3}. \end{aligned}$$

$$\iiint_E (x^2 + y^2) \, dV = \frac{32\pi}{3}$$

Exercise

Compute the volume of a sphere of radius R centered at the origin.

- First method: Use spherical coordinates.
- Second method: Use the disk method (slicing method).

Solution

First method: Spherical coordinates

Sphere equation: $x^2 + y^2 + z^2 \leq R^2$. Spherical coordinates:

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi, \quad dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta.$$

Domains: $0 \leq \rho \leq R, 0 \leq \phi \leq \pi, 0 \leq \theta \leq 2\pi$.

$$\begin{aligned} V &= \iiint_E dV = \int_{\theta=0}^{2\pi} \int_{\phi=0}^{\pi} \int_{\rho=0}^R \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta \\ &= \int_0^{2\pi} d\theta \int_0^{\pi} \sin \phi \, d\phi \int_0^R \rho^2 \, d\rho \\ &= (2\pi) \cdot 2 \cdot \frac{R^3}{3} = \frac{4\pi R^3}{3}. \end{aligned}$$

Second method: Disk method

Consider the sphere as a stack of horizontal disks. For a fixed z between $-R$ and R , the radius of the horizontal disk is $r(z) = \sqrt{R^2 - z^2}$, and its area is $\pi r(z)^2 = \pi(R^2 - z^2)$.

$$V = \int_{z=-R}^R \pi(R^2 - z^2) \, dz = 2\pi \int_0^R (R^2 - z^2) \, dz = 2\pi \left[R^2 z - \frac{z^3}{3} \right]_0^R = \frac{4\pi R^3}{3}.$$

$$V = \frac{4\pi R^3}{3}$$

Exercise

Compute $\iiint_E z \, dV$ where E is the upper hemisphere: $x^2 + y^2 + z^2 \leq R^2, z \geq 0$.

Solution

In spherical coordinates: $0 \leq \rho \leq R, 0 \leq \phi \leq \pi/2, 0 \leq \theta \leq 2\pi$.

$$z = \rho \cos \phi, \quad dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta.$$

$$\begin{aligned} \iiint_E z \, dV &= \int_{\theta=0}^{2\pi} \int_{\phi=0}^{\pi/2} \int_{\rho=0}^R (\rho \cos \phi) \cdot \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta \\ &= \int_0^{2\pi} d\theta \int_0^{\pi/2} \cos \phi \sin \phi \, d\phi \int_0^R \rho^3 \, d\rho \\ &= (2\pi) \cdot \frac{1}{2} \cdot \frac{R^4}{4} = \frac{\pi R^4}{4}. \end{aligned}$$

$$\iiint_E z \, dV = \frac{\pi R^4}{4}$$

Exercise

Let E be the solid bounded by the surfaces $z = x^2 + y^2$ and $z = 2 - x^2 - y^2$.

1. Determine the curve of intersection of the two surfaces.

2. Compute the volume of E .
3. Compute $\iiint_E z \, dV$.

Solution

1. Intersection:

$$x^2 + y^2 = 2 - x^2 - y^2 \implies 2(x^2 + y^2) = 2 \implies x^2 + y^2 = 1.$$

Thus the intersection is a circle of radius 1 in the plane $z = 1$.

2. Volume: In cylindrical coordinates: $x = r \cos \theta$, $y = r \sin \theta$, $z = z$, with $0 \leq r \leq 1$, $0 \leq \theta \leq 2\pi$, $r^2 \leq z \leq 2 - r^2$.

$$\begin{aligned} V &= \iiint_E dV = \int_{\theta=0}^{2\pi} \int_{r=0}^1 \int_{z=r^2}^{2-r^2} r \, dz \, dr \, d\theta \\ &= \int_0^{2\pi} d\theta \int_0^1 r(2 - 2r^2) \, dr = 2\pi \int_0^1 (2r - 2r^3) \, dr \\ &= 2\pi \left[r^2 - \frac{r^4}{2} \right]_0^1 = 2\pi \cdot \frac{1}{2} = \pi. \end{aligned}$$

3. $\iiint_E z \, dV$:

$$\begin{aligned} \iiint_E z \, dV &= \int_0^{2\pi} \int_0^1 \int_{r^2}^{2-r^2} z \cdot r \, dz \, dr \, d\theta \\ &= \int_0^{2\pi} d\theta \int_0^1 r \left[\frac{z^2}{2} \right]_{z=r^2}^{z=2-r^2} dr = 2\pi \int_0^1 \frac{r}{2} ((2 - r^2)^2 - r^4) \, dr \\ &= \pi \int_0^1 r (4 - 4r^2 + r^4 - r^4) \, dr = \pi \int_0^1 (4r - 4r^3) \, dr \\ &= \pi [2r^2 - r^4]_0^1 = \pi. \end{aligned}$$

$$\boxed{V = \pi, \quad \iiint_E z \, dV = \pi}$$

Exercise

Compute $\iiint_E (x^2 + y^2 + z^2) \, dV$ where E is the solid bounded by the sphere $x^2 + y^2 + z^2 \leq 1$.

Solution

In spherical coordinates: $0 \leq \rho \leq 1$, $0 \leq \phi \leq \pi$, $0 \leq \theta \leq 2\pi$.

$$x^2 + y^2 + z^2 = \rho^2, \quad dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta.$$

$$\begin{aligned}
 \iiint_E (x^2 + y^2 + z^2) dV &= \int_0^{2\pi} \int_0^\pi \int_0^1 \rho^2 \cdot \rho^2 \sin \phi d\rho d\phi d\theta \\
 &= \int_0^{2\pi} d\theta \int_0^\pi \sin \phi d\phi \int_0^1 \rho^4 d\rho \\
 &= (2\pi) \cdot 2 \cdot \frac{1}{5} = \frac{4\pi}{5}.
 \end{aligned}$$

$$\boxed{\iiint_E (x^2 + y^2 + z^2) dV = \frac{4\pi}{5}}$$

Exercise

Compute $\iiint_E y dV$ where E is the solid bounded by the cylinder $x^2 + y^2 = 1$ and the planes $z = 0$ and $z = x + 2$.

Solution

In cylindrical coordinates: $x = r \cos \theta$, $y = r \sin \theta$, $z = z$, with $0 \leq r \leq 1$, $0 \leq \theta \leq 2\pi$, $0 \leq z \leq r \cos \theta + 2$.

$$\begin{aligned}
 \iiint_E y dV &= \int_{\theta=0}^{2\pi} \int_{r=0}^1 \int_{z=0}^{r \cos \theta + 2} (r \sin \theta) \cdot r dz dr d\theta \\
 &= \int_0^{2\pi} \int_0^1 r^2 \sin \theta (r \cos \theta + 2) dr d\theta \\
 &= \int_0^{2\pi} \sin \theta \left[\int_0^1 (r^3 \cos \theta + 2r^2) dr \right] d\theta \\
 &= \int_0^{2\pi} \sin \theta \left(\frac{\cos \theta}{4} + \frac{2}{3} \right) d\theta = \frac{1}{4} \int_0^{2\pi} \cos \theta \sin \theta d\theta + \frac{2}{3} \int_0^{2\pi} \sin \theta d\theta = 0.
 \end{aligned}$$

$$\boxed{\iiint_E y dV = 0}$$

Exercise

Use the transformation $u = x + y$, $v = x - y$ to compute the integral:

$$\iint_R \sin(x + y) \cos(x - y) dA$$

where R is the square with vertices $(0, 0)$, $(\pi, 0)$, (π, π) , $(0, \pi)$.

Solution

The transformation: $u = x + y$, $v = x - y$.

Inverse: $x = \frac{u+v}{2}$, $y = \frac{u-v}{2}$.

Jacobian determinant:

$$J = \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{vmatrix} = -\frac{1}{2}, \quad |J| = \frac{1}{2}, \quad dA = \frac{1}{2} du dv.$$

The domain R in the uv -plane: we find that $0 \leq u \leq 2\pi$, $-u \leq v \leq u$ for $0 \leq u \leq \pi$,

and $u - 2\pi \leq v \leq 2\pi - u$ for $\pi \leq u \leq 2\pi$. By symmetry:

$$I = \frac{1}{2} \int_{u=0}^{2\pi} \sin u \left(\int_{v=-u}^u \cos v \, dv \right) du = \frac{1}{2} \int_0^\pi \sin u [\sin v]_{-u}^u du = \int_0^\pi \sin^2 u \, du = \frac{\pi}{2}.$$

$$\boxed{\iint_R \sin(x+y) \cos(x-y) \, dA = \frac{\pi}{2}}$$

Exercise

Use the transformation $x = u + v$, $y = u - v$ to compute the area of the region bounded by the curve $|x| + |y| = 1$.

Solution

The transformation: $u = \frac{x+y}{2}$, $v = \frac{x-y}{2}$.

Inverse: $x = u + v$, $y = u - v$.

Jacobian determinant:

$$J = \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = -2, \quad |J| = 2, \quad dA = 2 \, du \, dv.$$

In the uv -coordinates, the equation $|x| + |y| = 1$ becomes $|u + v| + |u - v| = 1$, which represents a square with vertices $(\pm\frac{1}{2}, 0)$ and $(0, \pm\frac{1}{2})$, i.e., exactly the region $|u| + |v| \leq \frac{1}{2}$.

The area of this square in uv is $\left(\frac{\sqrt{2}}{2}\right)^2 = \frac{1}{2}$.

Thus the original area:

$$A = \iint_R dA = \iint_{R'} 2 \, du \, dv = 2 \times \frac{1}{2} = 1.$$

$$\boxed{A = 1}$$

Exercise

Compute $\iiint_E e^{(x+y+z)/(x+y-z)} \, dV$ using the transformation:

$$u = x + y + z, \quad v = x + y - z, \quad w = x - y + z$$

where E is the tetrahedron bounded by the planes $x = 0$, $y = 0$, $z = 0$, $x + y + z = 1$.

Solution

The transformation is linear. Express x, y, z in terms of u, v, w :

$$\begin{cases} u = x + y + z \\ v = x + y - z \\ w = x - y + z \end{cases} \Rightarrow \begin{cases} x = \frac{1}{4}(u + v + 2w) \\ y = \frac{1}{4}(u + v - 2w) \\ z = \frac{1}{4}(u - 3v + 2w) \end{cases}$$

The Jacobian determinant of this transformation:

$$J = \begin{vmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{4} & -\frac{1}{2} \\ \frac{1}{4} & -\frac{3}{4} & \frac{1}{2} \end{vmatrix} = -2, \quad |J| = 2, \quad dV = 2 \, du \, dv \, dw.$$

The integrand $e^{(x+y+z)/(x+y-z)} = e^{u/v}$.

The tetrahedron E maps to a tetrahedron E' in uvw . The transformed integral becomes:

$$\iiint_E e^{(x+y+z)/(x+y-z)} dV = 2 \iiint_{E'} e^{u/v} du \, dv \, dw.$$

$$\boxed{2 \iiint_{E'} e^{u/v} du \, dv \, dw}$$

Exercise

Compute $\iiint_E \sqrt{x^2 + y^2} \, dV$ where E is the solid bounded by the surfaces:

$$z = x^2 + y^2, \quad z = 4(x^2 + y^2), \quad y \geq 0.$$

Solution

In cylindrical coordinates: $x = r \cos \theta$, $y = r \sin \theta$, $z = z$, $dV = r \, dr \, d\theta \, dz$.

$$\sqrt{x^2 + y^2} = r.$$

Domains: $0 \leq \theta \leq \pi$ (since $y \geq 0$), r is bounded, but z ranges from r^2 to $4r^2$. Without an upper bound for r , the integral is improper. Assume there is a maximum radius R (e.g., R is the radius of the intersection region with some plane). Then:

$$\iiint_E \sqrt{x^2 + y^2} \, dV = \int_{\theta=0}^{\pi} \int_{r=0}^R \int_{z=r^2}^{4r^2} r \cdot r \, dz \, dr \, d\theta = \pi \int_0^R 3r^4 \, dr = \frac{3\pi R^5}{5}.$$

$$\boxed{\frac{3\pi R^5}{5} \text{ where } R \text{ is the maximum radius of the region}}$$

Exercise

Compute the center of mass of a solid hemisphere of radius R with constant density δ_0 .

Solution

By symmetry $\bar{x} = \bar{y} = 0$. Compute $\bar{z}$.

Mass: Volume of hemisphere $= \frac{1}{2} \cdot \frac{4\pi R^3}{3} = \frac{2\pi R^3}{3}$, so $M = \delta_0 \cdot \frac{2\pi R^3}{3}$.

Moment M_{xy} :

$$M_{xy} = \iiint_E z \delta_0 \, dV = \delta_0 \iiint_E z \, dV.$$

In spherical coordinates for the upper hemisphere: $0 \leq \rho \leq R, 0 \leq \phi \leq \pi/2, 0 \leq \theta \leq 2\pi$.

$$\iiint_E z \, dV = \int_0^{2\pi} \int_0^{\pi/2} \int_0^R \rho \cos \phi \cdot \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \frac{\pi R^4}{4}.$$

Thus $M_{xy} = \delta_0 \cdot \frac{\pi R^4}{4}$.

Center of mass:

$$\bar{z} = \frac{M_{xy}}{M} = \frac{\frac{\pi R^4}{4}}{\frac{2\pi R^3}{3}} = \frac{3R}{8}.$$

$$(\bar{x}, \bar{y}, \bar{z}) = \left(0, 0, \frac{3R}{8}\right)$$

Exercise

Compute the volume of the solid bounded by the cone $z^2 = x^2 + y^2$ and the sphere $x^2 + y^2 + z^2 = 1$.

Solution

Spherical coordinates:

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi,$$

$$dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta, \quad 0 \leq \rho \leq 1, \quad 0 \leq \phi \leq \pi, \quad 0 \leq \theta \leq 2\pi.$$

Cone equation in spherical coordinates:

$$z^2 = x^2 + y^2 \implies \rho^2 \cos^2 \phi = \rho^2 \sin^2 \phi \implies \tan^2 \phi = 1 \implies \phi = \frac{\pi}{4} \text{ or } \frac{3\pi}{4}.$$

Volume inside the cone: The region inside the cone in the upper half is $0 \leq \phi \leq \pi/4$, and in the lower half $3\pi/4 \leq \phi \leq \pi$. By symmetry, compute the volume in the upper half and double it:

$$\begin{aligned} V_{\text{inside}} &= 2 \int_0^{2\pi} \int_0^{\pi/4} \int_0^1 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta \\ &= 2 \left(\int_0^{2\pi} d\theta \right) \left(\int_0^{\pi/4} \sin \phi \, d\phi \right) \left(\int_0^1 \rho^2 \, d\rho \right) \\ &= 2 \times (2\pi) \times [-\cos \phi]_0^{\pi/4} \times \frac{1}{3} \\ &= 4\pi \times \left(1 - \frac{\sqrt{2}}{2}\right) \times \frac{1}{3} = \frac{4\pi}{3} \left(1 - \frac{\sqrt{2}}{2}\right). \end{aligned}$$

Total sphere volume:

$$V_{\text{sphere}} = \frac{4\pi}{3}.$$

Desired volume (outside the cone but inside the sphere):

$$V = V_{\text{sphere}} - V_{\text{inside}} = \frac{4\pi}{3} - \frac{4\pi}{3} \left(1 - \frac{\sqrt{2}}{2}\right) = \frac{4\pi}{3} \cdot \frac{\sqrt{2}}{2} = \frac{2\sqrt{2}\pi}{3}.$$

$$V = \frac{2\sqrt{2}\pi}{3}$$

Exercise

Compute $\iiint_E z \, dV$ where E is the solid bounded by the cylinder $x^2 + y^2 = 1$ and the planes $z = 0$ and $z = 2$.

Solution

Transformation to cylindrical coordinates:

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z.$$

Jacobian determinant:

$$\frac{\partial(x, y, z)}{\partial(r, \theta, z)} = \det \begin{pmatrix} \cos \theta & -r \sin \theta & 0 \\ \sin \theta & r \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} = \cos \theta \cdot (r \cos \theta) - (-r \sin \theta) \cdot (\sin \theta) = r.$$

Thus $|J| = r$, and the volume element $dV = r \, dr \, d\theta \, dz$.

Integration limits:

$$0 \leq r \leq 1, \quad 0 \leq \theta \leq 2\pi, \quad 0 \leq z \leq 2.$$

Compute the integral:

$$\begin{aligned} \iiint_E z \, dV &= \int_{\theta=0}^{2\pi} \int_{r=0}^1 \int_{z=0}^2 z \cdot r \, dz \, dr \, d\theta \\ &= \left(\int_0^{2\pi} d\theta \right) \left(\int_0^1 r \, dr \right) \left(\int_0^2 z \, dz \right) \\ &= (2\pi) \cdot \left(\frac{1}{2} \right) \cdot (2) = 2\pi. \end{aligned}$$

$$\iiint_E z \, dV = 2\pi$$

Exercise

Compute $\iiint_E (x^2 + y^2) \, dV$ where E is the solid bounded by the paraboloid $z = x^2 + y^2$ and the plane $z = 4$.

Solution

In cylindrical coordinates: $x = r \cos \theta$, $y = r \sin \theta$, $z = z$, with $0 \leq r \leq 2$, $0 \leq \theta \leq 2\pi$,

$$r^2 \leq z \leq 4.$$

$$\begin{aligned} \iiint_E (x^2 + y^2) dV &= \int_0^{2\pi} \int_0^2 \int_{r^2}^4 r^2 \cdot r dz dr d\theta \\ &= 2\pi \int_0^2 r^3 (4 - r^2) dr = 2\pi \left[r^4 - \frac{r^6}{6} \right]_0^2 = \frac{32\pi}{3}. \end{aligned}$$

$$\boxed{\iiint_E (x^2 + y^2) dV = \frac{32\pi}{3}}$$

Exercise

Compute the center of mass of a solid hemisphere of radius R with constant density δ_0 .

Solution

General relations:

$$M = \iiint_E \delta dV, \quad M_{xy} = \iiint_E z \delta dV, \quad \bar{z} = \frac{M_{xy}}{M}.$$

By symmetry $\bar{x} = \bar{y} = 0$.

Application: Density $\delta = \delta_0$ is constant.

$$M = \delta_0 \cdot \frac{2\pi R^3}{3}, \quad M_{xy} = \delta_0 \iiint_E z dV.$$

In spherical coordinates for the upper hemisphere:

$$\iiint_E z dV = \int_0^{2\pi} \int_0^{\pi/2} \int_0^R \rho \cos \phi \cdot \rho^2 \sin \phi d\rho d\phi d\theta = \frac{\pi R^4}{4}.$$

$$\bar{z} = \frac{\delta_0 \cdot \pi R^4 / 4}{\delta_0 \cdot 2\pi R^3 / 3} = \frac{3R}{8}.$$

$$\boxed{(\bar{x}, \bar{y}, \bar{z}) = \left(0, 0, \frac{3R}{8}\right)}$$

Exercise

Let E be the solid bounded by the surfaces $z = x^2 + y^2$ and $z = 2 - x^2 - y^2$.

- Determine the curve of intersection of the two surfaces.
- Compute the volume of E .
- Compute $\iiint_E z dV$.

Solution

1. Intersection:

$$x^2 + y^2 = 2 - x^2 - y^2 \Rightarrow x^2 + y^2 = 1, \quad z = 1.$$

2. Volume: In cylindrical coordinates: $0 \leq r \leq 1, 0 \leq \theta \leq 2\pi, r^2 \leq z \leq 2 - r^2$.

$$V = \int_0^{2\pi} \int_0^1 \int_{r^2}^{2-r^2} r \, dz \, dr \, d\theta = 2\pi \int_0^1 2r(1 - r^2) \, dr = \pi.$$

3. $\iiint_E z \, dV$:

$$\iiint_E z \, dV = \int_0^{2\pi} \int_0^1 \int_{r^2}^{2-r^2} zr \, dz \, dr \, d\theta = \pi \int_0^1 r((2 - r^2)^2 - r^4) \, dr = \pi.$$

$$V = \pi, \quad \iiint_E z \, dV = \pi$$

Exercise

Compute the volume of the intersection of the two cylinders: $x^2 + y^2 \leq 1$ and $x^2 + z^2 \leq 1$.

Solution

By symmetry:

$$V = 8V_{\text{octant}}, \quad V_{\text{octant}} = \int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2}} dz \, dy \, dx.$$

$$\int_0^{\sqrt{1-x^2}} dz = \sqrt{1-x^2}, \quad \int_0^{\sqrt{1-x^2}} \sqrt{1-x^2} \, dy = (1-x^2).$$

$$V_{\text{octant}} = \int_0^1 (1-x^2) \, dx = \left[x - \frac{x^3}{3} \right]_0^1 = \frac{2}{3}.$$

$$V = 8 \cdot \frac{2}{3} = \frac{16}{3}$$

Exercise

Compute the volume of the solid bounded by the surfaces $z = x^2 + y^2$ and $z = 2x + 2y + 10$.

Solution

We have the two surfaces:

$$z_1(x, y) = x^2 + y^2, \quad z_2(x, y) = 2x + 2y + 10.$$

Determine the intersection region in the xy -plane by solving $z_1 = z_2$:

$$x^2 + y^2 = 2x + 2y + 10 \implies x^2 - 2x + y^2 - 2y = 10.$$

Complete the squares:

$$(x-1)^2 - 1 + (y-1)^2 - 1 = 10 \implies (x-1)^2 + (y-1)^2 = 12.$$

Thus the projection D onto the xy -plane is a disk centered at $(1, 1)$ with radius $R = \sqrt{12} = 2\sqrt{3}$. In this region, $z_2 - z_1 = (2x + 2y + 10) - (x^2 + y^2) = 12 - (x-1)^2 - (y-1)^2 \geq 0$,

CHAPTER 8. MULTIPLE INTEGRALS

so z_2 is above z_1 , and the volume is:

$$V = \iint_D (z_2(x, y) - z_1(x, y)) \, dA = \iint_D (12 - (x - 1)^2 - (y - 1)^2) \, dA.$$

Switch to polar coordinates centered at $(1, 1)$:

$$x - 1 = r \cos \theta, \quad y - 1 = r \sin \theta, \quad dA = r \, dr \, d\theta,$$

with $0 \leq r \leq 2\sqrt{3}$, $0 \leq \theta \leq 2\pi$. The integrand becomes $12 - r^2$, thus:

$$\begin{aligned} V &= \int_{\theta=0}^{2\pi} \int_{r=0}^{2\sqrt{3}} (12 - r^2) r \, dr \, d\theta \\ &= \int_0^{2\pi} d\theta \int_0^{2\sqrt{3}} (12r - r^3) \, dr \\ &= 2\pi \left[6r^2 - \frac{r^4}{4} \right]_0^{2\sqrt{3}}. \end{aligned}$$

Compute the bracket:

$$r^2 = (2\sqrt{3})^2 = 12, \quad r^4 = (12)^2 = 144.$$

$$6r^2 - \frac{r^4}{4} = 6 \cdot 12 - \frac{144}{4} = 72 - 36 = 36.$$

Thus:

$$V = 2\pi \times 36 = 72\pi.$$

$$\boxed{V = 72\pi}$$

Exercise

Compute the volume of the solid bounded by the paraboloid $z = x^2 + y^2$ and the plane $z = 4$.

Solution

The solid E is the region below the plane $z = 4$ and above the paraboloid $z = x^2 + y^2$. To determine the projection onto the xy -plane, find the intersection of the two surfaces:

$$x^2 + y^2 = 4 \implies r = 2 \text{ (in polar coordinates).}$$

Thus the base is the disk $D: x^2 + y^2 \leq 4$.

Using cylindrical coordinates:

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z, \quad dV = r \, dr \, d\theta \, dz.$$

Integration limits:

$$0 \leq r \leq 2, \quad 0 \leq \theta \leq 2\pi, \quad r^2 \leq z \leq 4.$$

The volume is:

$$V = \iiint_E dV = \int_{\theta=0}^{2\pi} \int_{r=0}^2 \int_{z=r^2}^4 r \, dz \, dr \, d\theta.$$

CHAPTER 8. MULTIPLE INTEGRALS

Compute the integral first with respect to z :

$$\int_{z=r^2}^4 r \, dz = r(4 - r^2).$$

Then with respect to r :

$$\int_0^2 r(4 - r^2) \, dr = \int_0^2 (4r - r^3) \, dr = \left[2r^2 - \frac{r^4}{4} \right]_0^2 = 8 - 4 = 4.$$

Finally, with respect to θ :

$$\int_0^{2\pi} 4 \, d\theta = 8\pi.$$

Thus:

$$V = 8\pi.$$

$$\boxed{V = 8\pi}$$

Exercise

Compute $\iiint_E (x^2 + z^2) \, dV$ where E is the region bounded by the cylinder $y = x^2$ and the planes $z = 0$, $y = 1$, $z = y$.

Solution

The domain: $-1 \leq x \leq 1$, $x^2 \leq y \leq 1$, $0 \leq z \leq y$.

$$\begin{aligned} I &= \int_{-1}^1 \int_{x^2}^1 \int_0^y (x^2 + z^2) \, dz \, dy \, dx = \int_{-1}^1 \int_{x^2}^1 \left(x^2 y + \frac{y^3}{3} \right) \, dy \, dx \\ &= \int_{-1}^1 \left[\frac{x^2 y^2}{2} + \frac{y^4}{12} \right]_{x^2}^1 \, dx = \int_{-1}^1 \left(\frac{x^2}{2} + \frac{1}{12} - \frac{x^6}{2} - \frac{x^8}{12} \right) \, dx = \frac{88}{189}. \end{aligned}$$

$$\boxed{\iiint_E (x^2 + z^2) \, dV = \frac{88}{189}}$$

Exercise

Use the transformation $u = x + y$, $v = x - y$, $w = z$ to compute the volume of the solid bounded by the planes:

$$x + y = 1, \quad x - y = 1, \quad x + y = -1, \quad x - y = -1, \quad z = 0, \quad z = 2.$$

Solution

Define the transformation and its inverse:

$$u = x + y, \quad v = x - y, \quad w = z.$$

Solving the system we obtain:

$$x = \frac{u + v}{2}, \quad y = \frac{u - v}{2}, \quad z = w.$$

Compute the Jacobian determinant:

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = \det \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} = \frac{1}{2} \cdot \left(-\frac{1}{2}\right) - \frac{1}{2} \cdot \frac{1}{2} = -\frac{1}{4} - \frac{1}{4} = -\frac{1}{2}.$$

Thus:

$$\left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| = \frac{1}{2}, \quad dV = \frac{1}{2} du dv dw.$$

Determine the new domain E' in (u, v, w) -coordinates: From the given conditions:

$$-1 \leq x + y \leq 1 \implies -1 \leq u \leq 1,$$

$$-1 \leq x - y \leq 1 \implies -1 \leq v \leq 1,$$

$$0 \leq z \leq 2 \implies 0 \leq w \leq 2.$$

Thus:

$$E' = [-1, 1] \times [-1, 1] \times [0, 2].$$

The volume of the original solid is:

$$V = \iiint_E dV = \iiint_{E'} \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| du dv dw = \iiint_{E'} \frac{1}{2} du dv dw.$$

Compute the triple integral:

$$V = \frac{1}{2} \int_{w=0}^2 \int_{v=-1}^1 \int_{u=-1}^1 du dv dw = \frac{1}{2} \left(\int_{-1}^1 du \right) \left(\int_{-1}^1 dv \right) \left(\int_0^2 dw \right).$$

$$\int_{-1}^1 du = 2, \quad \int_{-1}^1 dv = 2, \quad \int_0^2 dw = 2.$$

$$V = \frac{1}{2} \times (2 \times 2 \times 2) = \frac{1}{2} \times 8 = 4.$$

$$\boxed{V = 4}$$

Exercise

Compute $\iiint_E y dV$ where E is the region common to the two cylinders $x^2 + y^2 \leq 1$ and $y^2 + z^2 \leq 1$.

Solution

The region E is defined by:

$$E = \{(x, y, z) \mid x^2 + y^2 \leq 1, y^2 + z^2 \leq 1\}.$$

Note that E is symmetric about the plane $y = 0$, i.e., if $(x, y, z) \in E$ then $(x, -y, z) \in E$. Split E into two parts:

$$E_1 = E \cap \{y \geq 0\}, \quad E_2 = E \cap \{y \leq 0\}.$$

By symmetry, we have:

$$\iiint_{E_2} y \, dV = - \iiint_{E_1} y \, dV.$$

Thus:

$$\iiint_E y \, dV = \iiint_{E_1} y \, dV + \iiint_{E_2} y \, dV = \iiint_{E_1} y \, dV - \iiint_{E_1} y \, dV = 0.$$

$$\boxed{\iiint_E y \, dV = 0}$$

Exercise

Compute $\iiint_E \frac{1}{(1+x^2+y^2+z^2)^{3/2}} \, dV$ where E is the sphere $x^2 + y^2 + z^2 \leq a^2$.

Solution

In spherical coordinates: $\rho \leq a$, $dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$.

$$I = \int_0^{2\pi} \int_0^\pi \int_0^a \frac{\rho^2}{(1+\rho^2)^{3/2}} \sin \phi \, d\rho \, d\phi \, d\theta = 4\pi \int_0^a \frac{\rho^2}{(1+\rho^2)^{3/2}} d\rho.$$

Compute $\int \frac{\rho^2}{(1+\rho^2)^{3/2}} d\rho$ by letting $\rho = \tan u$:

$$\int \frac{\rho^2}{(1+\rho^2)^{3/2}} d\rho = \ln(\sqrt{1+\rho^2} + \rho) - \frac{\rho}{\sqrt{1+\rho^2}}.$$

Thus:

$$I = 4\pi \left[\ln(\sqrt{1+a^2} + a) - \frac{a}{\sqrt{1+a^2}} \right].$$

$$\boxed{I = 4\pi \left(\ln(\sqrt{1+a^2} + a) - \frac{a}{\sqrt{1+a^2}} \right)}$$

Complex Integration

Theorem

Fundamental Theorems of Complex Integration

1. **Complex line integral:** Let $f(z)$ be a continuous complex function on a domain D and $\gamma : [a, b] \rightarrow \mathbb{C}$ a piecewise C^1 path. The integral of f along γ is:

$$\int_{\gamma} f(z) dz = \int_a^b f(\gamma(t)) \gamma'(t) dt$$

Writing $f = u + iv$ and $dz = dx + i dy$, we obtain:

$$\int_{\gamma} f(z) dz = \int_{\gamma} (u dx - v dy) + i \int_{\gamma} (v dx + u dy)$$

2. **Properties of line integrals:**

- Linearity: $\int_{\gamma} (\alpha f(z) + \beta g(z)) dz = \alpha \int_{\gamma} f(z) dz + \beta \int_{\gamma} g(z) dz$
- Additivity with respect to path: $\int_{\gamma_1 + \gamma_2} f(z) dz = \int_{\gamma_1} f(z) dz + \int_{\gamma_2} f(z) dz$
- Reparameterization: The integral is invariant under increasing reparameterization.
- Reverse path: $\int_{-\gamma} f(z) dz = - \int_{\gamma} f(z) dz$
- Estimation: $\left| \int_{\gamma} f(z) dz \right| \leq \int_{\gamma} |f(z)| |dz| \leq ML$ where $M = \max_{z \in \gamma} |f(z)|$ and L is the length of γ .

3. **Cauchy-Goursat theorem (simple form):** Let f be analytic (holomorphic) in a simply connected domain D . For every simple closed curve γ contained in D :

$$\oint_{\gamma} f(z) dz = 0$$

More generally, if f is analytic in D and continuous on $\overline{D}$, then the integral over the boundary of D is zero.

4. **Cauchy-Goursat theorem (extended form):** Let f be analytic in a domain D except for a finite number of isolated singular points. Let γ be a positively oriented simple closed curve in D that avoids the singularities. Then:

$$\oint_{\gamma} f(z) dz = 2\pi i \sum_{k=1}^n \text{Res}(f, z_k)$$

where the sum is over all singularities z_k inside γ .

5. **Cauchy integral formula:** Let f be analytic in a simply connected domain D , and γ a positively oriented simple closed curve contained in D . For every point z_0 inside γ :

$$f(z_0) = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{z - z_0} dz$$

More generally, for the n -th derivative:

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_{\gamma} \frac{f(z)}{(z - z_0)^{n+1}} dz$$

This result shows that any analytic function is infinitely differentiable (expandable as a power series).

6. **Residue theorem:** Let f be a meromorphic function in a domain D , with isolated singularities $z_1, z_2, \dots, z_n$ inside a simple closed curve $\gamma \subset D$. Then:

$$\oint_{\gamma} f(z) dz = 2\pi i \sum_{k=1}^n \text{Res}(f, z_k)$$

where $\text{Res}(f, z_k)$ is the residue of f at z_k .

7. **Computing residues:**

- **Simple pole:** If f has a simple pole at z_0 , then:

$$\text{Res}(f, z_0) = \lim_{z \rightarrow z_0} (z - z_0) f(z)$$

- **Pole of order m :** If f has a pole of order m at z_0 , then:

$$\text{Res}(f, z_0) = \frac{1}{(m-1)!} \lim_{z \rightarrow z_0} \frac{d^{m-1}}{dz^{m-1}} [(z - z_0)^m f(z)]$$

- **Residue at infinity:** The residue of f at infinity is defined by:

$$\text{Res}(f, \infty) = -\frac{1}{2\pi i} \oint_{C^-} f(z) dz$$

where C^- is a large circle traversed in the negative direction.

8. **Argument principle (zeros and poles theorem):** Let f be a meromorphic function in a domain D , and γ a simple closed curve in D that does not pass through any zero or pole of f . Then:

$$\frac{1}{2\pi i} \oint_{\gamma} \frac{f'(z)}{f(z)} dz = N - P$$

where N is the number of zeros (counting multiplicities) and P the number of poles (counting orders) inside γ .

9. **Argument principle (geometric form):** Let f be analytic in a domain D , except for possible poles, and γ a simple closed curve in D that does not pass through any zero or pole of f . Then:

$$\frac{1}{2\pi} \Delta_{\gamma} \arg f(z) = N - P$$

where $\Delta_\gamma \arg f(z)$ is the total change in the argument of $f(z)$ as z traverses γ .

10. **Rouché's theorem:** Let f and g be analytic in a domain D , and γ a simple closed curve in D . If $|g(z)| < |f(z)|$ for all $z \in \gamma$, then f and $f+g$ have the same number of zeros inside γ (counting multiplicities).
11. **Morera's theorem (converse of Cauchy's theorem):** Let f be continuous in a domain D . If $\oint_\gamma f(z) dz = 0$ for every closed curve $\gamma \subset D$, then f is analytic in D .
12. **Liouville's theorem:** Every entire (analytic on $\mathbb{C}$) and bounded function is constant. Consequence: The fundamental theorem of algebra (every non-constant polynomial with complex coefficients has at least one root) follows from Liouville's theorem.
13. **Riemann mapping theorem:** Every simply connected domain in the complex plane (different from $\mathbb{C}$ itself) can be conformally mapped onto the open unit disk.
14. **Poisson integral formula:** Let f be analytic in the disk $|z| < R$ and continuous on $|z| \leq R$. For $0 \leq r < R$ and $0 \leq \theta < 2\pi$:

$$f(re^{i\theta}) = \frac{1}{2\pi} \int_0^{2\pi} \frac{R^2 - r^2}{R^2 - 2Rr \cos(\theta - \phi) + r^2} f(Re^{i\phi}) d\phi$$

For the unit disk ($R = 1$):

$$f(re^{i\theta}) = \frac{1}{2\pi} \int_0^{2\pi} \frac{1 - r^2}{1 - 2r \cos(\theta - \phi) + r^2} f(e^{i\phi}) d\phi$$

15. **Maximum modulus principle:** Let f be analytic in a domain D . If $|f|$ attains a local maximum at a point $z_0 \in D$, then f is constant in D . Consequence: For f analytic in a bounded domain D and continuous on $\overline{D}$, the maximum of $|f|$ is attained on the boundary ∂D .
16. **Schwarz lemma:** Let f be analytic in the unit disk $|z| < 1$, with $f(0) = 0$ and $|f(z)| < 1$ for all $|z| < 1$. Then:
 - $|f(z)| \leq |z|$ for all $|z| < 1$
 - $|f'(0)| \leq 1$
 - If $|f(z_0)| = |z_0|$ for some $z_0 \neq 0$, or if $|f'(0)| = 1$, then $f(z) = e^{i\alpha}z$ for some real constant α .
17. **Real integrals using residues:** To compute real integrals of the form $\int_0^{2\pi} R(\cos \theta, \sin \theta) d\theta$, set $z = e^{i\theta}$, where:

$$\cos \theta = \frac{z + z^{-1}}{2}, \quad \sin \theta = \frac{z - z^{-1}}{2i}, \quad d\theta = \frac{dz}{iz}$$

The integral becomes:

$$\int_0^{2\pi} R(\cos \theta, \sin \theta) d\theta = \oint_{|z|=1} R\left(\frac{z + z^{-1}}{2}, \frac{z - z^{-1}}{2i}\right) \frac{dz}{iz}$$

This integral over the unit circle is evaluated using the residue theorem.

18. **Fourier and Laplace transforms via residues:**

- Inverse Fourier transform: $f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega t} d\omega$

- Inverse Laplace transform: $f(t) = \frac{1}{2\pi i} \int_{\gamma-i\infty}^{\gamma+i\infty} F(s)e^{st} ds$

These integrals are typically computed by closing the contour and using the residue theorem.

19. **Essential singularities:** An essential singularity is a point where the Laurent expansion has an infinite number of non-zero terms with negative exponents.

- Casorati-Weierstrass theorem: In any neighborhood of an essential singularity, f takes values arbitrarily close to any complex number.
- Picard's theorem (Great Picard theorem): In any neighborhood of an essential singularity, f takes every complex value except at most one.

20. **Conformal mappings:** A mapping $f : D \rightarrow \mathbb{C}$ is conformal at z_0 if it preserves angles (both magnitude and direction) at z_0 . An analytic map with $f'(z_0) \neq 0$ is conformal at z_0 .

- Möbius transformation: $f(z) = \frac{az+b}{cz+d}$ where $ad - bc \neq 0$
- Joukowski transformation: $f(z) = z + \frac{1}{z}$
- Exponential map: $f(z) = e^z$
- Schwarz-Christoffel transformation: Maps the upper half-plane to polygons.

21. **Harmonic functions:** A function $u(x, y)$ is called harmonic if $\nabla^2 u = 0$. For any analytic function $f = u + iv$, both the real part u and the imaginary part v are harmonic functions and are harmonic conjugates.

- Poisson formula for harmonic functions:

$$u(re^{i\theta}) = \frac{1}{2\pi} \int_0^{2\pi} \frac{1-r^2}{1-2r\cos(\theta-\phi)+r^2} u(e^{i\phi}) d\phi$$

- Maximum principle for harmonic functions: A non-constant harmonic function in a domain cannot attain its maximum or minimum value except on the boundary.

Exercise

Let γ be the line segment from $z_1 = 0$ to $z_2 = 1 + i$. Compute the integral:

$$\int_{\gamma} z^2 dz$$

Solution

Parameterize the segment: $\gamma(t) = t(1 + i)$ for $0 \leq t \leq 1$. Then $\gamma'(t) = 1 + i$.

$$\begin{aligned} \int_{\gamma} z^2 dz &= \int_0^1 [\gamma(t)]^2 \gamma'(t) dt \\ &= \int_0^1 [t(1+i)]^2 (1+i) dt = (1+i)^3 \int_0^1 t^2 dt = \frac{(1+i)^3}{3} \end{aligned}$$

We have $(1+i)^3 = (1+i)^2(1+i) = (2i)(1+i) = 2i - 2 = -2 + 2i$. Thus:

$$\int_{\gamma} z^2 dz = \frac{-2+2i}{3} = \frac{2}{3}(-1+i)$$

$$\boxed{\int_{\gamma} z^2 dz = \frac{2}{3}(-1+i)}$$

Exercise

Compute $\oint_{|z|=2} \frac{dz}{z^2+1}$ with the circle oriented positively.

Solution

Factor the denominator: $z^2 + 1 = (z-i)(z+i)$. The poles inside the circle $|z| = 2$ are $z = i$ and $z = -i$ (since $|i| = 1 < 2$, $|-i| = 1 < 2$). These are simple poles. Compute the residues:

$$\text{Res}\left(\frac{1}{z^2+1}, i\right) = \lim_{z \rightarrow i} (z-i) \frac{1}{z^2+1} = \lim_{z \rightarrow i} \frac{1}{z+i} = \frac{1}{2i}$$

$$\text{Res}\left(\frac{1}{z^2+1}, -i\right) = \lim_{z \rightarrow -i} (z+i) \frac{1}{z^2+1} = \lim_{z \rightarrow -i} \frac{1}{z-i} = \frac{1}{-2i} = -\frac{1}{2i}$$

By the residue theorem:

$$\oint_{|z|=2} \frac{dz}{z^2+1} = 2\pi i \left(\frac{1}{2i} - \frac{1}{2i} \right) = 0$$

$$\boxed{\oint_{|z|=2} \frac{dz}{z^2+1} = 0}$$

Exercise

Let $f(z) = \frac{e^z}{z(z-1)^2}$. Compute $\oint_{|z|=2} f(z) dz$.

Solution

The singularities inside $|z| = 2$: $z = 0$: simple pole - $z = 1$: double pole
Compute the residues:

- At $z = 0$ (simple pole):

$$\text{Res}(f, 0) = \lim_{z \rightarrow 0} z \cdot \frac{e^z}{z(z-1)^2} = \lim_{z \rightarrow 0} \frac{e^z}{(z-1)^2} = \frac{1}{1} = 1$$

- At $z = 1$ (double pole):

$$\text{Res}(f, 1) = \lim_{z \rightarrow 1} \frac{d}{dz} \left[(z-1)^2 \cdot \frac{e^z}{z(z-1)^2} \right] = \lim_{z \rightarrow 1} \frac{d}{dz} \left(\frac{e^z}{z} \right)$$

$$\frac{d}{dz} \left(\frac{e^z}{z} \right) = \frac{ze^z - e^z}{z^2} = \frac{e^z(z-1)}{z^2}$$

$$\text{Res}(f, 1) = \lim_{z \rightarrow 1} \frac{e^z(z-1)}{z^2} = 0$$

By the residue theorem:

$$\oint_{|z|=2} f(z) dz = 2\pi i(1+0) = 2\pi i$$

$$\oint_{|z|=2} \frac{e^z}{z(z-1)^2} dz = 2\pi i$$

Exercise

Compute the real integral $\int_0^{2\pi} \frac{d\theta}{5+4\cos\theta}$.

Solution

Set $z = e^{i\theta}$, then $d\theta = \frac{dz}{iz}$ and $\cos\theta = \frac{z+z^{-1}}{2}$.

$$\begin{aligned} \int_0^{2\pi} \frac{d\theta}{5+4\cos\theta} &= \oint_{|z|=1} \frac{1}{5+4 \cdot \frac{z+z^{-1}}{2}} \cdot \frac{dz}{iz} \\ &= \oint_{|z|=1} \frac{1}{5+2z+2z^{-1}} \cdot \frac{dz}{iz} \end{aligned}$$

Simplify the denominator:

$$5+2(z+z^{-1}) = \frac{5z+2z^2+2}{z} \Rightarrow \frac{1}{5+2(z+z^{-1})} = \frac{z}{2z^2+5z+2}$$

Thus:

$$\int_0^{2\pi} \frac{d\theta}{5+4\cos\theta} = \oint_{|z|=1} \frac{z}{2z^2+5z+2} \cdot \frac{1}{iz} dz = \frac{1}{i} \oint_{|z|=1} \frac{dz}{2z^2+5z+2}$$

The roots of $2z^2+5z+2=0$ are $z_1 = -\frac{1}{2}$, $z_2 = -2$. Inside $|z|=1$, only $z_1 = -\frac{1}{2}$ is present. Compute the residue at this simple pole:

$$\text{Res} \left(\frac{1}{2z^2+5z+2}, -\frac{1}{2} \right) = \lim_{z \rightarrow -\frac{1}{2}} \left(z + \frac{1}{2} \right) \frac{1}{2z^2+5z+2}$$

Factor the denominator: $2z^2+5z+2 = 2(z+\frac{1}{2})(z+2)$. Thus:

$$\text{Res} = \lim_{z \rightarrow -\frac{1}{2}} \left(z + \frac{1}{2} \right) \frac{1}{2(z+\frac{1}{2})(z+2)} = \frac{1}{2} \lim_{z \rightarrow -\frac{1}{2}} \frac{1}{z+2} = \frac{1}{2} \cdot \frac{1}{3/2} = \frac{1}{3}$$

Therefore:

$$\frac{1}{i} \oint_{|z|=1} \frac{dz}{2z^2+5z+2} = \frac{1}{i} \cdot 2\pi i \cdot \frac{1}{3} = \frac{2\pi}{3}$$

$$\int_0^{2\pi} \frac{d\theta}{5+4\cos\theta} = \frac{2\pi}{3}$$

Exercise

Let $f(z) = \frac{\sin z}{z^3}$. Determine the Laurent expansion of f about $z = 0$ and compute $\oint_{|z|=1} f(z) dz$.

Solution

Expand $\sin z$:

$$\sin z = z - \frac{z^3}{6} + \frac{z^5}{120} - \cdots$$

Divide by z^3 :

$$\frac{\sin z}{z^3} = \frac{1}{z^2} - \frac{1}{6} + \frac{z^2}{120} - \cdots$$

The coefficient of $\frac{1}{z}$ in this expansion is zero. Therefore the residue at 0 is zero. By the residue theorem:

$$\oint_{|z|=1} \frac{\sin z}{z^3} dz = 2\pi i \cdot 0 = 0$$

$$\boxed{\oint_{|z|=1} \frac{\sin z}{z^3} dz = 0}$$

Exercise

Using the Cauchy integral formula, compute $\oint_{|z|=2} \frac{e^{2z}}{z^2(z-1)} dz$.

Solution

The singularities inside $|z| = 2$: $z = 0$ (double pole) and $z = 1$ (simple pole). Decompose into partial fractions:

$$\frac{1}{z^2(z-1)} = \frac{A}{z} + \frac{B}{z^2} + \frac{C}{z-1}$$

Multiply by $z^2(z-1)$:

$$1 = Az(z-1) + B(z-1) + Cz^2$$

Substitute $z = 0$: $1 = -B \Rightarrow B = -1$. $z = 1$: $1 = C \Rightarrow C = 1$. Substitute $z = 2$ for example: $1 = A(2)(1) - 1(1) + 4 = 2A - 1 + 4 = 2A + 3 \Rightarrow 2A = -2 \Rightarrow A = -1$. Thus:

$$\frac{e^{2z}}{z^2(z-1)} = e^{2z} \left(-\frac{1}{z} - \frac{1}{z^2} + \frac{1}{z-1} \right)$$

Using Cauchy's formulas:

$$\oint_{|z|=2} \frac{e^{2z}}{z} dz = 2\pi i e^0 = 2\pi i$$

$$\oint_{|z|=2} \frac{e^{2z}}{z^2} dz = \frac{2\pi i}{1!} \cdot \frac{d}{dz} e^{2z} \Big|_{z=0} = 2\pi i \cdot 2 = 4\pi i$$

$$\oint_{|z|=2} \frac{e^{2z}}{z-1} dz = 2\pi i e^2$$

Therefore:

$$\oint_{|z|=2} \frac{e^{2z}}{z^2(z-1)} dz = -2\pi i - 4\pi i + 2\pi i e^2 = 2\pi i(e^2 - 3)$$

$$\oint_{|z|=2} \frac{e^{2z}}{z^2(z-1)} dz = 2\pi i(e^2 - 3)$$

Exercise

Show that every entire function f satisfying $|f(z)| \leq |z|^2$ for all z is a polynomial of degree at most 2.

Solution

Let f be an entire function. By the inequality, $|f(z)| \leq |z|^2$ for sufficiently large $|z|$. Expand f in a power series about 0: $f(z) = \sum_{n=0}^{\infty} a_n z^n$, and this power series converges on all of $\mathbb{C}$. By Cauchy's inequalities, for any $R > 0$ and all $n \geq 0$:

$$|a_n| \leq \frac{\max_{|z|=R} |f(z)|}{R^n} \leq \frac{R^2}{R^n} = R^{2-n}$$

If $n > 2$, then $\lim_{R \rightarrow \infty} R^{2-n} = 0$, hence $a_n = 0$ for all $n > 2$. Thus $f(z) = a_0 + a_1 z + a_2 z^2$ is a polynomial of degree at most 2.

$$f \text{ is a polynomial of degree } \leq 2$$

Exercise

Let $f(z) = \frac{z^2}{(z-1)(z-3)}$. Compute $\oint_{\gamma} f(z) dz$ for:

- γ the circle $|z| = 2$ oriented positively.
- γ the circle $|z| = 4$ oriented positively.

Solution

The singularities: $z = 1$ and $z = 3$ (simple poles).

1. For $|z| = 2$: Inside the circle, only $z = 1$ is present ($|1| = 1 < 2$, $|3| = 3 > 2$).

$$\text{Res}(f, 1) = \lim_{z \rightarrow 1} (z-1) \frac{z^2}{(z-1)(z-3)} = \lim_{z \rightarrow 1} \frac{z^2}{z-3} = \frac{1}{-2} = -\frac{1}{2}$$

$$\oint_{|z|=2} f(z) dz = 2\pi i \left(-\frac{1}{2} \right) = -\pi i$$

2. For $|z| = 4$: Inside the circle, both $z = 1$ and $z = 3$ are present.

$$\text{Res}(f, 3) = \lim_{z \rightarrow 3} (z-3) \frac{z^2}{(z-1)(z-3)} = \lim_{z \rightarrow 3} \frac{z^2}{z-1} = \frac{9}{2}$$

$$\oint_{|z|=4} f(z) dz = 2\pi i \left(-\frac{1}{2} + \frac{9}{2} \right) = 2\pi i \cdot 4 = 8\pi i$$

$$\oint_{|z|=2} \frac{z^2}{(z-1)(z-3)} dz = -\pi i, \quad \oint_{|z|=4} \frac{z^2}{(z-1)(z-3)} dz = 8\pi i$$

Exercise

Compute $\int_{-\infty}^{\infty} \frac{x \sin x}{x^2+4} dx$.

Solution

Consider $\int_{-\infty}^{\infty} \frac{x e^{ix}}{x^2+4} dx$ (its imaginary part gives the desired integral). Let $f(z) = \frac{z e^{iz}}{z^2+4}$. The poles are at $z = \pm 2i$. Take a contour consisting of the segment $[-R, R]$ and the upper semicircle γ_R (radius R large). Inside the contour (for $R > 2$) lies only the pole $z = 2i$. Compute the residue at $z = 2i$ (simple pole):

$$\text{Res}(f, 2i) = \lim_{z \rightarrow 2i} (z - 2i) \frac{z e^{iz}}{(z - 2i)(z + 2i)} = \frac{2i e^{i(2i)}}{4i} = \frac{2i e^{-2}}{4i} = \frac{e^{-2}}{2}$$

By the residue theorem:

$$\oint_{\Gamma_R} f(z) dz = 2\pi i \cdot \frac{e^{-2}}{2} = \pi i e^{-2}$$

Since $|e^{iz}| = e^{-y} \leq 1$ in the upper half-plane, the integral over γ_R tends to 0 as $R \rightarrow \infty$ (Jordan's lemma). Thus:

$$\int_{-\infty}^{\infty} \frac{x e^{ix}}{x^2+4} dx = \pi i e^{-2}$$

Take the imaginary part:

$$\int_{-\infty}^{\infty} \frac{x \sin x}{x^2+4} dx = \text{Im}(\pi i e^{-2}) = \pi e^{-2}$$

$$\boxed{\int_{-\infty}^{\infty} \frac{x \sin x}{x^2+4} dx = \pi e^{-2}}$$

Exercise

Let $f(z) = \frac{1}{z^4+1}$. Compute $\oint_{\gamma} f(z) dz$ where γ is the boundary of the rectangle with vertices $\pm R$ and $\pm R + iR$, with $R > 1$.

Solution

The poles of f are the fourth roots of -1 : $z_k = e^{i(\pi+2k\pi)/4}$, $k = 0, 1, 2, 3$. Inside the rectangle (for $R > 1$) lies the pole in the first quadrant: $z_0 = e^{i\pi/4} = \frac{\sqrt{2}}{2}(1+i)$. The pole is simple, and the residue:

$$\text{Res}(f, z_0) = \lim_{z \rightarrow z_0} (z - z_0) \frac{1}{z^4+1}$$

Since $z^4+1 = (z - z_0)(z^3 + z_0 z^2 + z_0^2 z + z_0^3)$, we have $g(z_0) = 4z_0^3$ (the derivative of

the denominator). Thus:

$$\operatorname{Res}(f, z_0) = \frac{1}{4z_0^3}$$

where $z_0^3 = e^{i3\pi/4} = -\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}$. By the residue theorem:

$$\oint_{\gamma} f(z) dz = 2\pi i \cdot \frac{1}{4z_0^3}$$

After simplification we obtain $\frac{\pi}{2\sqrt{2}}(1 - i)$.

$$\oint_{\gamma} \frac{dz}{z^4 + 1} = \frac{\pi}{2\sqrt{2}}(1 - i)$$

Exercise

Using Rouché's theorem, show that the polynomial $z^5 + 3z^2 + 1$ has exactly 2 roots in the disk $|z| < 1$.

Solution

Set $f(z) = 3z^2$ and $g(z) = z^5 + 1$. On the circle $|z| = 1$:

$$|g(z)| = |z^5 + 1| \leq |z|^5 + 1 = 1 + 1 = 2, \quad |f(z)| = |3z^2| = 3$$

Thus $|g(z)| < |f(z)|$ on $|z| = 1$. By Rouché's theorem, f and $f + g (= z^5 + 3z^2 + 1)$ have the same number of zeros inside $|z| < 1$. $f(z) = 3z^2$ has a double zero at 0, i.e., 2 zeros (counting multiplicity). Therefore the given polynomial has 2 zeros inside the disk.

The polynomial has 2 roots inside $|z| < 1$

Exercise

Compute $\int_0^{\infty} \frac{\cos(ax)}{x^2+1} dx$ for $a > 0$.

Solution

Consider $\int_{-\infty}^{\infty} \frac{e^{iax}}{x^2+1} dx$ (the real part gives twice the desired integral). Let $f(z) = \frac{e^{iaz}}{z^2+1}$. The poles are at $z = \pm i$. In the upper half-plane, only the pole $z = i$ is present (since $a > 0$). Compute the residue at $z = i$:

$$\operatorname{Res}(f, i) = \lim_{z \rightarrow i} (z - i) \frac{e^{iaz}}{(z - i)(z + i)} = \frac{e^{iai}}{2i} = \frac{e^{-a}}{2i}$$

Using a contour consisting of the segment $[-R, R]$ and the upper semicircle, and taking the limit $R \rightarrow \infty$ (Jordan's lemma):

$$\int_{-\infty}^{\infty} \frac{e^{iax}}{x^2+1} dx = 2\pi i \cdot \frac{e^{-a}}{2i} = \pi e^{-a}$$

Taking the real part and using evenness:

$$\int_0^{\infty} \frac{\cos(ax)}{x^2 + 1} dx = \frac{1}{2} \int_{-\infty}^{\infty} \frac{\cos(ax)}{x^2 + 1} dx = \frac{\pi}{2} e^{-a}$$

$$\boxed{\int_0^{\infty} \frac{\cos(ax)}{x^2 + 1} dx = \frac{\pi}{2} e^{-a}}$$

Exercise

Find the Laurent expansion of $f(z) = \frac{1}{(z-1)(z-3)}$ in the regions:

- $|z| < 1$
- $1 < |z| < 3$
- $|z| > 3$

Solution

Decompose into partial fractions:

$$\frac{1}{(z-1)(z-3)} = \frac{1}{2} \left(\frac{1}{z-3} - \frac{1}{z-1} \right)$$

Region $|z| < 1$:

$$\frac{1}{z-1} = -\frac{1}{1-z} = -\sum_{n=0}^{\infty} z^n$$

$$\frac{1}{z-3} = -\frac{1}{3} \cdot \frac{1}{1-\frac{z}{3}} = -\sum_{n=0}^{\infty} \frac{z^n}{3^{n+1}}$$

Thus:

$$f(z) = \frac{1}{2} \left(-\sum_{n=0}^{\infty} \frac{z^n}{3^{n+1}} + \sum_{n=0}^{\infty} z^n \right) = \frac{1}{2} \sum_{n=0}^{\infty} \left(1 - \frac{1}{3^{n+1}} \right) z^n$$

Region $1 < |z| < 3$:

$$\frac{1}{z-1} = \frac{1}{z} \cdot \frac{1}{1-\frac{1}{z}} = \sum_{n=0}^{\infty} \frac{1}{z^{n+1}}$$

$$\frac{1}{z-3} = -\frac{1}{3} \cdot \frac{1}{1-\frac{z}{3}} = -\sum_{n=0}^{\infty} \frac{z^n}{3^{n+1}}$$

Thus:

$$f(z) = \frac{1}{2} \left(-\sum_{n=0}^{\infty} \frac{z^n}{3^{n+1}} - \sum_{n=0}^{\infty} \frac{1}{z^{n+1}} \right) = -\frac{1}{2} \sum_{n=0}^{\infty} \frac{z^n}{3^{n+1}} - \frac{1}{2} \sum_{n=0}^{\infty} \frac{1}{z^{n+1}}$$

Region $|z| > 3$:

$$\frac{1}{z-1} = \frac{1}{z} \cdot \frac{1}{1-\frac{1}{z}} = \sum_{n=0}^{\infty} \frac{1}{z^{n+1}}$$

$$\frac{1}{z-3} = \frac{1}{z} \cdot \frac{1}{1-\frac{3}{z}} = \sum_{n=0}^{\infty} \frac{3^n}{z^{n+1}}$$

Thus:

$$f(z) = \frac{1}{2} \left(\sum_{n=0}^{\infty} \frac{3^n}{z^{n+1}} - \sum_{n=0}^{\infty} \frac{1}{z^{n+1}} \right) = \frac{1}{2} \sum_{n=0}^{\infty} \frac{3^n - 1}{z^{n+1}}$$

$$|z| < 1 : f(z) = \frac{1}{2} \sum_{n=0}^{\infty} \left(1 - \frac{1}{3^{n+1}} \right) z^n$$

$$1 < |z| < 3 : f(z) = -\frac{1}{2} \sum_{n=0}^{\infty} \frac{z^n}{3^{n+1}} - \frac{1}{2} \sum_{n=0}^{\infty} \frac{1}{z^{n+1}}$$

$$|z| > 3 : f(z) = \frac{1}{2} \sum_{n=0}^{\infty} \frac{3^n - 1}{z^{n+1}}$$

Exercise

Let $f(z) = \frac{z^2+1}{z(z^2+4)}$. Compute $\oint_{|z|=3} f(z) dz$.

Solution

The singularities inside $|z| = 3$ are: - $z = 0$ (simple pole) - $z = 2i$ (simple pole, $|2i| = 2 < 3$) - $z = -2i$ (simple pole, $|-2i| = 2 < 3$)

Compute the residues:

$$\text{Res}(f, 0) = \lim_{z \rightarrow 0} z \cdot \frac{z^2 + 1}{z(z^2 + 4)} = \frac{0^2 + 1}{0^2 + 4} = \frac{1}{4}$$

$$\text{Res}(f, 2i) = \lim_{z \rightarrow 2i} (z - 2i) \frac{z^2 + 1}{z(z - 2i)(z + 2i)} = \frac{(2i)^2 + 1}{2i \cdot 4i} = \frac{-4 + 1}{-8} = \frac{-3}{-8} = \frac{3}{8}$$

$$\text{Res}(f, -2i) = \lim_{z \rightarrow -2i} (z + 2i) \frac{z^2 + 1}{z(z - 2i)(z + 2i)} = \frac{(-2i)^2 + 1}{-2i \cdot (-4i)} = \frac{-4 + 1}{-8} = \frac{-3}{-8} = \frac{3}{8}$$

Sum of residues = $\frac{1}{4} + \frac{3}{8} + \frac{3}{8} = \frac{1}{4} + \frac{3}{4} = 1$. By the residue theorem:

$$\oint_{|z|=3} f(z) dz = 2\pi i \cdot 1 = 2\pi i$$

$$\oint_{|z|=3} \frac{z^2 + 1}{z(z^2 + 4)} dz = 2\pi i$$

Exercise

Show that if f is analytic in the unit disk and $|f(z)| \leq 1$ for $|z| < 1$, then for $|z| < 1$:

$$|f'(z)| \leq \frac{1}{1 - |z|^2}$$

Solution

Use Cauchy's formula for the derivative. Let $|z_0| = r < 1$. Choose R such that $r < R < 1$. On the circle $|z| = R$, we have $|f(z)| \leq 1$. Then:

$$f'(z_0) = \frac{1}{2\pi i} \oint_{|z|=R} \frac{f(z)}{(z - z_0)^2} dz$$

Using the estimation lemma:

$$|f'(z_0)| \leq \frac{1}{2\pi} \oint_{|z|=R} \frac{|f(z)|}{|z - z_0|^2} |dz| \leq \frac{1}{2\pi} \cdot \frac{1}{(R - r)^2} \cdot 2\pi R = \frac{R}{(R - r)^2}$$

Let $R \rightarrow 1^-$:

$$|f'(z_0)| \leq \frac{1}{(1 - r)^2}$$

However, this result is weaker than desired. In fact, the stronger result (Schwarz-Pick lemma) gives:

$$|f'(z)| \leq \frac{1 - |f(z)|^2}{1 - |z|^2} \leq \frac{1}{1 - |z|^2}$$

since $|f(z)| \leq 1$.

$$|f'(z)| \leq \frac{1}{1 - |z|^2}$$

Polar Coordinates

Theorem

1. Let M be a point in the plane. Cartesian coordinates (x, y) and polar coordinates (r, θ) are related by:

$$x = r \cos \theta, \quad y = r \sin \theta,$$

and conversely:

$$r = \sqrt{x^2 + y^2}, \quad \theta = \arctan\left(\frac{y}{x}\right).$$

2. Let a curve be defined by $r = f(\theta)$, continuous on $[\alpha, \beta]$. The area swept by the radius vector is given by:

$$S = \frac{1}{2} \int_{\alpha}^{\beta} r^2(\theta) d\theta.$$

3. The arc length of a polar curve $r = f(\theta)$ between $\theta = a$ and $\theta = b$ is:

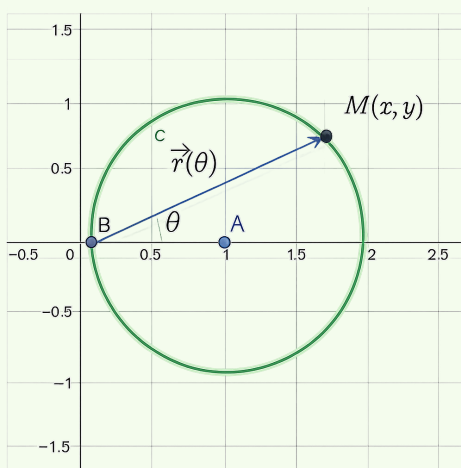
$$L = \int_a^b \sqrt{r^2(\theta) + (r'(\theta))^2} d\theta.$$

Exercise

Plot the polar curve given by the equation:

$$r = 2 \cos \theta.$$

Describe the geometric nature of the curve.

Solution


We have $r = 2 \cos \theta$. Convert to Cartesian coordinates:

$$r^2 = 2r \cos \theta \implies x^2 + y^2 = 2x.$$

Thus:

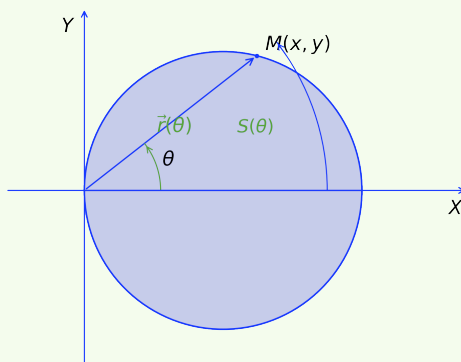
$$(x - 1)^2 + y^2 = 1.$$

This is a circle with center $(1, 0)$ and radius 1.

Exercise

Compute the area of the region bounded by the cardioid with polar equation:

$$r = a(1 + \cos \theta).$$

Solution


Using the area formula:

$$S = \frac{1}{2} \int_0^{2\pi} r^2 d\theta = \frac{1}{2} \int_0^{2\pi} a^2(1 + \cos \theta)^2 d\theta.$$

CHAPTER 10. POLAR COORDINATES

We have:

$$(1 + \cos \theta)^2 = 1 + 2 \cos \theta + \cos^2 \theta, \quad \cos^2 \theta = \frac{1 + \cos 2\theta}{2}.$$

Computing the integral we find:

$$S = \frac{3\pi a^2}{2}.$$

Exercise

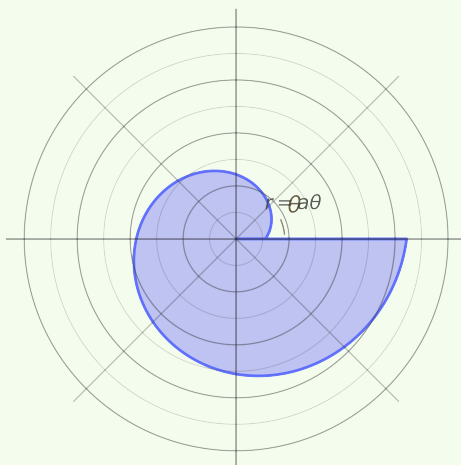
Consider the Archimedean spiral with polar equation:

$$r = a\theta.$$

Show that the area swept by the radius vector between 0 and θ is:

$$S(\theta) = \frac{a^2 \theta^3}{6}.$$

Solution



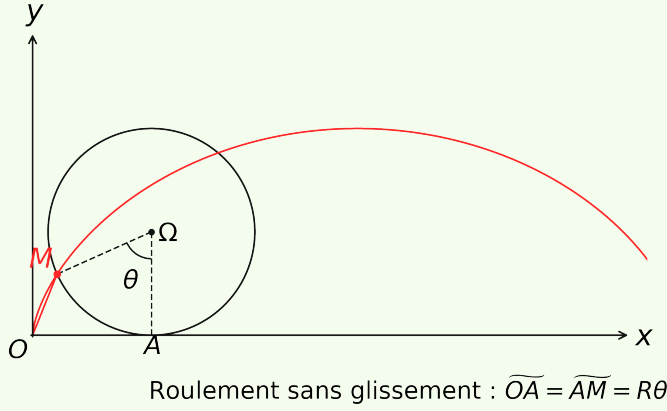
$$S(\theta) = \frac{1}{2} \int_0^\theta r^2 d\theta = \frac{1}{2} \int_0^\theta a^2 \theta^2 d\theta = \frac{a^2}{2} \cdot \frac{\theta^3}{3} = \frac{a^2 \theta^3}{6}.$$

Exercise

A circle of radius R rolls without slipping along the (Ox) axis. A fixed point M on the circumference traces a curve called a cycloid.

1. Find the parametric equations of the cycloid.
2. Find the Cartesian equation of the path.
3. Give the expression in polar coordinates.

Solution



1. Geometric setting

The circle rolls without slipping, so the distance traveled on the axis equals the arc length:

$$OA = \widetilde{AM} = R\theta.$$

2. Parametric equations

We have:

$$\begin{aligned}\overrightarrow{OM} &= \overrightarrow{O\Omega} + \overrightarrow{\Omega M}. \\ \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} R\theta \\ R \end{pmatrix} + \begin{pmatrix} -R \sin \theta \\ -R \cos \theta \end{pmatrix}\end{aligned}$$

Thus we obtain:

$$\begin{cases} x = R(\theta - \sin \theta), \\ y = R(1 - \cos \theta). \end{cases}$$

3. Cartesian equation

We have:

$$\cos \theta = 1 - \frac{y}{R}.$$

Thus:

$$\theta = \arccos \left(1 - \frac{y}{R} \right), \quad \sin \theta = \sqrt{1 - \left(1 - \frac{y}{R} \right)^2}.$$

We obtain:

$$x = R \arccos \left(1 - \frac{y}{R} \right) - \sqrt{2Ry - y^2}$$

for $0 \leq y \leq 2R$.

4. Polar coordinates

We have:

$$\rho = \sqrt{x^2 + y^2}, \quad \varphi = \arctan\left(\frac{y}{x}\right),$$

Substituting the coordinates of the moving point:

$$\begin{cases} \rho(\theta) = R\sqrt{(\theta - \sin \theta)^2 + (1 - \cos \theta)^2}, \\ \varphi(\theta) = \arctan\left(\frac{1 - \cos \theta}{\theta - \sin \theta}\right). \end{cases}$$

Exercise

Consider a circle of radius R rolling without slipping on a line. Let P be a fixed point at distance d from the center of the circle.

1. Find the parametric equations when:

- $d > R$ (prolate cycloid),
- $d < R$ (curtate cycloid).

2. Study the limiting cases:

- $d = R$ (ordinary cycloid),
- $d = 0$.

3. Plot the curves for $R = 2$ and $d = 1, 2, 3$.

Solution

1) General parametric equations

When a circle of radius R rolls without slipping, the abscissa of its center after a rotation by angle θ is $R\theta$, and its ordinate is R . Thus:

$$C(R\theta, R)$$

If point P is at distance d from the center, its relative coordinates in the frame attached to the circle are:

$$(-d \sin \theta, -d \cos \theta)$$

The absolute coordinates of P become:

$$\begin{cases} x(\theta) = R\theta - d \sin \theta, \\ y(\theta) = R - d \cos \theta. \end{cases}$$

These equations describe:

- A prolate cycloid if $d > R$,
- A curtate cycloid if $d < R$.

2) Limiting cases

a) $d = R$: **ordinary cycloid**

$$\begin{cases} x = R(\theta - \sin \theta), \\ y = R(1 - \cos \theta). \end{cases}$$

This is the traditional cycloid with cusps.

b) $d = 0$

$$\begin{cases} x = R\theta, \\ y = R. \end{cases}$$

The point remains fixed on the horizontal line $y = R$: it is the center of the circle.

3) Study of cusps

Compute:

$$\frac{dx}{d\theta} = R - d \cos \theta, \quad \frac{dy}{d\theta} = d \sin \theta.$$

A cusp satisfies:

$$\frac{dx}{d\theta} = 0 \quad \text{and} \quad \frac{dy}{d\theta} = 0.$$

Thus:

$$\sin \theta = 0 \Rightarrow \theta = k\pi,$$

$$R - d \cos \theta = 0 \Rightarrow \cos \theta = \frac{R}{d}.$$

- If $d > R$, then $\frac{R}{d} < 1$: solutions exist $\rightarrow$ loops.
- If $d = R$, we obtain cusps.
- If $d < R$, no solutions $\rightarrow$ smooth curve.

4) Numerical applications

For $R = 2$:

- $d = 1$: curtate cycloid,

$$x = 2\theta - \sin \theta, \quad y = 2 - \cos \theta.$$

- $d = 2$: ordinary cycloid,

$$x = 2(\theta - \sin \theta), \quad y = 2(1 - \cos \theta).$$

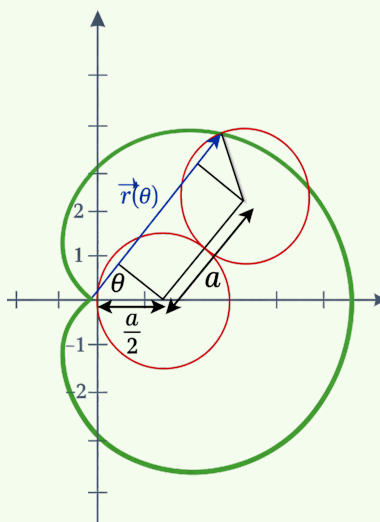
- $d = 3$: prolate cycloid,

$$x = 2\theta - 3 \sin \theta, \quad y = 2 - 3 \cos \theta.$$

Exercise

1. Find the polar equation of the cardioid.
2. Find the parametric equations of the cardioid.
3. Find the Cartesian equation of the path.

Solution



1) Polar equation

From the geometric construction we have:

$$r = a(1 + \cos \theta), \quad \theta \in [0, 2\pi].$$

2) Parametric equations

We have:

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} r \cos \theta \\ r \sin \theta \end{pmatrix} = \begin{pmatrix} a(1 + \cos \theta) \cos \theta \\ a(1 + \cos \theta) \sin \theta \end{pmatrix}$$

3) Cartesian equation

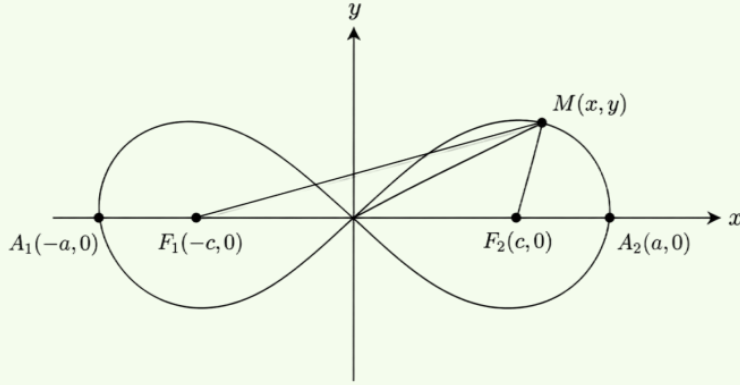
$$(x^2 + y^2 - ax)^2 = a^2(x^2 + y^2).$$

Exercise

The set of points M of the lemniscate is defined by:

$$|MF_1| \cdot |MF_2| = c^2$$

where $F_1(-c, 0)$ and $F_2(c, 0)$ are fixed points.



1. Show that $a^2 = 2c^2$.
2. Then prove that $r^2 = a^2 \cos(2\theta)$.
3. Deduce the polar equation.
4. Find a parametric representation.
5. Find the Cartesian equation of the path.

Solution

1. Proof that $a^2 = 2c^2$.

$$\|\overrightarrow{A_2F_1}\| = a + c \quad \|\overrightarrow{A_2F_2}\| = a - c$$

$$M \equiv A_2 \implies \|\overrightarrow{A_2F_1}\| \cdot \|\overrightarrow{A_2F_2}\| = c^2 = (a + c)(a - c) \implies a^2 = 2c^2$$

2. Proof that $r^2 = a^2 \cos(2\theta)$. We have:

$$\overrightarrow{F_1M} = \overrightarrow{F_1O} + \overrightarrow{r} \implies \|\overrightarrow{F_1M}\|^2 = c^2 + r^2 + 2rc \cos \theta$$

$$\overrightarrow{F_2M} = \overrightarrow{r} - \overrightarrow{OF_2} \implies \|\overrightarrow{F_2M}\|^2 = r^2 + c^2 - 2rc \cos \theta$$

$$\|\overrightarrow{F_1M}\|^2 \times \|\overrightarrow{F_2M}\|^2 = (r^2 + c^2)^2 - 4r^2c^2 \cos^2 \theta = c^4$$

$$r^4 - 2r^2c^2 (2 \cos^2 \theta - 1) = 0 \implies r^2 = 2c^2 \cos(2\theta)$$

3. Deduction of the polar equation.

$$r^2 = 2c^2 \cos(2\theta) = a^2 \cos(2\theta) \implies r = a\sqrt{|\cos(2\theta)|}$$

4. Parametric representation.

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} r \cos \theta \\ r \sin \theta \end{pmatrix} = \begin{pmatrix} a\sqrt{|\cos(2\theta)|} \cos \theta \\ a\sqrt{|\cos(2\theta)|} \sin \theta \end{pmatrix}$$

5. Cartesian equation.

$$r^2 = x^2 + y^2, \quad \cos 2\theta = \frac{x^2 - y^2}{x^2 + y^2}.$$

Thus:

$$x^2 + y^2 = a^2 \frac{x^2 - y^2}{x^2 + y^2}.$$

We obtain:

$$(x^2 + y^2)^2 = a^2(x^2 - y^2)$$

Exercise

The lemniscate can be obtained as the inversion of a rectangular hyperbola.

1. Prove this property.
2. Deduce the relations between the coefficients.
3. Study the properties of this transformation.

Solution

1) Proof by inversion

a) Definition of inversion

Let an inversion circle with center O and radius R . The image of point M is point M' on ray OM such that:

$$OM \cdot OM' = R^2$$

In polar coordinates:

$$r' = \frac{R^2}{r}, \quad \theta' = \theta.$$

b) Rectangular hyperbola

Consider the rectangular hyperbola:

$$xy = k^2$$

In polar coordinates:

$$x = r \cos \theta, \quad y = r \sin \theta$$

We obtain:

$$r^2 \sin \theta \cos \theta = k^2$$

$$\frac{r^2}{2} \sin 2\theta = k^2$$

$$r^2 = \frac{2k^2}{\sin 2\theta}$$

c) Image under inversion

Under inversion:

$$r' = \frac{R^2}{r} \implies r'^2 = \frac{R^4}{r^2}$$

Thus:

$$r'^2 = \frac{R^4}{\frac{2k^2}{\sin 2\theta}} = \frac{R^4}{2k^2} \sin 2\theta$$

Setting:

$$a^2 = \frac{R^4}{2k^2}$$

We obtain:

$$r'^2 = a^2 \sin 2\theta$$

After a rotation by $\pi/4$, we transform $\sin 2\theta$ into $\cos 2\theta$, giving:

$$r^2 = a^2 \cos 2\theta$$

This is exactly the polar equation of the lemniscate.

2) Relations between coefficients

We have:

$$a^2 = \frac{R^4}{2k^2}$$

Thus:

$$a = \frac{R^2}{\sqrt{2}k}$$

and conversely:

$$k = \frac{R^2}{a\sqrt{2}}$$

3) Properties of the transformation

1. Inversion preserves angles (conformal transformation).
2. A hyperbola of degree 2 becomes a curve of degree 4 (the lemniscate).
3. The symmetry of the hyperbola is preserved in the lemniscate.
4. The origin is a double point of the lemniscate (image of the asymptotes).

4) Conclusion

Inversion of a rectangular hyperbola with respect to a circle centered at its center yields exactly Bernoulli's lemniscate.

$$\text{Hyperbola } xy = k^2 \xrightarrow{\text{inversion}} r^2 = a^2 \cos 2\theta$$

Exercise

For each of the following curves:

1. Cycloid: $x = R(\theta - \sin \theta)$, $y = R(1 - \cos \theta)$
2. Cardioid: $r = a(1 + \cos \theta)$
3. Lemniscate: $r^2 = a^2 \cos 2\theta$
1. Compute the arc length.
2. Compute the curvature.
3. Compute the area enclosed.
4. Find the equations of the tangent and normal lines.

Solution

1) Arc length

a) Cycloid

$$x' = R(1 - \cos \theta), \quad y' = R \sin \theta$$

$$ds = \sqrt{x'^2 + y'^2} d\theta = 2R \sin \frac{\theta}{2} d\theta$$

$$L = \int_0^{2\pi} ds = 8R$$

b) Cardioid

$$r = a(1 + \cos \theta)$$

$$L = \int_0^{2\pi} \sqrt{r^2 + (r')^2} d\theta = 8a$$

c) Lemniscate

$$r^2 = a^2 \cos 2\theta$$

$$L = 4 \int_0^{\pi/4} \frac{a}{\sqrt{\cos 2\theta}} d\theta = 4aK(1/\sqrt{2})$$

(where K is the elliptic integral)

2) Curvature

a) Cycloid

$$\kappa = \frac{1}{4R \sin(\theta/2)}$$

b) Cardioid

$$\kappa = \frac{3}{4a} \sec^3 \left(\frac{\theta}{2} \right)$$

c) Lemniscate

$$\kappa = \frac{3}{a} \sqrt{\cos 2\theta}$$

3) Areas

a) Cycloid (one arch)

$$A = 3\pi R^2$$

b) Cardioid

$$A = \frac{3}{2} \pi a^2$$

c) Lemniscate (one loop)

$$A = \frac{a^2}{2}$$

4) Tangents

a) Cycloid

$$\text{Slope} = \frac{dy}{dx} = \cot \frac{\theta}{2}$$

b) Cardioid

$$\frac{dy}{dx} = \frac{\sin \theta (2 + \cos \theta)}{\cos \theta (2 + \cos \theta) - 1}$$

c) Lemniscate

For $F(x, y) = (x^2 + y^2)^2 - a^2(x^2 - y^2)$, the tangent at (x_0, y_0) is:

$$2(x^2 + y^2)(xx_0 + yy_0) = a^2(xx_0 - yy_0)$$

Exercise

1. **Cycloidal pendulum:** Show that for a pendulum whose mass moves on a cycloid, the period of oscillation is independent of amplitude.
2. **Brachistochrone problem:** Prove that the cycloid is the curve of fastest descent between two points.
3. **Cardioid antenna:** Explain why the directional pattern of a cardioid antenna has this shape.
4. **Magnetic field:** Study the shape of the magnetic field lines of a dipole.

Solution

1) Cycloidal pendulum (tautochrone property)

a) Model

Consider a mass m sliding without friction on an inverted cycloid:

$$x = R(\theta + \sin \theta), \quad y = R(1 - \cos \theta)$$

The arc length element is:

$$ds = 2R \cos \frac{\theta}{2} d\theta$$

b) Mechanical energy

By conservation of energy:

$$\frac{1}{2}mv^2 + mgy = \text{constant}$$

Thus:

$$\frac{1}{2}m \left(2R \cos \frac{\theta}{2} \frac{d\theta}{dt} \right)^2 + mgR(1 - \cos \theta) = E$$

After simplification:

$$\left(\frac{d\theta}{dt} \right)^2 = \frac{g}{R} \left(\cos^2 \frac{\theta_0}{2} - \cos^2 \frac{\theta}{2} \right)$$

c) Period

The period is:

$$T = 4\sqrt{\frac{R}{g}} \int_0^{\theta_0} \frac{d\theta}{\sqrt{\cos^2 \frac{\theta_0}{2} - \cos^2 \frac{\theta}{2}}}$$

Using the classical change of variables:

$$T = 4\pi\sqrt{\frac{R}{g}}$$

The period is independent of amplitude: the cycloid is tautochrone.

2) Brachistochrone problem

a) Travel time

A particle slides under gravity:

$$v = \sqrt{2gy}$$

The time is:

$$T = \int \frac{ds}{v} = \int \frac{\sqrt{1+y'^2}}{\sqrt{2gy}} dx$$

The Lagrangian is:

$$L(y, y') = \frac{\sqrt{1+y'^2}}{\sqrt{y}}$$

b) Euler-Lagrange equation

Since L does not depend explicitly on x , we use Beltrami's identity:

$$L - y' \frac{\partial L}{\partial y'} = C$$

This leads to:

$$y(1+y'^2) = C$$

c) Solution

This differential equation is exactly the equation of the cycloid. We verify that:

$$x = R(\theta - \sin \theta), \quad y = R(1 - \cos \theta)$$

satisfy this relation. Thus the curve of fastest descent is the cycloid.

3) Cardioid antenna

a) Radiation pattern

For a cardioid antenna, the angular gain is:

$$G(\theta) = G_0(1 + \cos \theta)$$

This pattern is exactly the polar equation of the cardioid.

b) Physical origin

This pattern is obtained by combining:

- An omnidirectional antenna
- A dipole antenna whose gain is proportional to $\cos \theta$

c) Result

- Maximum value in one direction
- Nearly zero minimum value in the opposite direction
- Often used to reduce back interference

4) Magnetic field lines of a dipole

a) Magnetic field

In polar coordinates:

$$B_r = \frac{2k \cos \theta}{r^3}, \quad B_\theta = \frac{k \sin \theta}{r^3}$$

Field lines satisfy:

$$\frac{dr}{B_r} = \frac{r d\theta}{B_\theta}$$

We obtain:

$$\frac{dr}{d\theta} = 2r \cot \theta$$

b) Integration

$$\begin{aligned} \frac{dr}{r} &= 2 \cot \theta d\theta \\ \ln r &= 2 \ln(\sin \theta) + C \end{aligned}$$

Thus:

$$r = k \sin^2 \theta$$

c) Interpretation

Field lines form closed loops around the dipole, with geometry close to that of the lemniscate in certain planar cross-sections.

Exercise

Let $\vec{T} = (T_x, T_y)$ be a unit tangent vector to the curve Γ of function f at point M_0 . The vector $\vec{T}$ depends on the angle θ .

1. Express $\vec{T}$ in the Cartesian coordinate system.
2. Express the vector $\vec{N} = \frac{d\vec{T}}{d\theta}$.
3. Show that $\vec{N}$ is perpendicular to $\vec{T}$.

Solution

Write:

$$\vec{T} = T_x \vec{i} + T_y \vec{j}, \quad \|\vec{T}\| = 1.$$

By trigonometric definition:

$$T_x = \cos \theta, \quad T_y = \sin \theta.$$

Thus:

$$\vec{T} = \cos \theta \vec{i} + \sin \theta \vec{j}.$$

Differentiate with respect to θ :

$$\vec{N} = \frac{d\vec{T}}{d\theta} = -\sin \theta \vec{i} + \cos \theta \vec{j}.$$

The dot product:

$$\vec{T} \cdot \vec{N} = (\cos \theta, \sin \theta) \cdot (-\sin \theta, \cos \theta) = -\cos \theta \sin \theta + \sin \theta \cos \theta = 0.$$

Thus:

$$\vec{T} \perp \vec{N}.$$

Exercise

Let γ be the curve represented by the function $y = f(x)$ of class $\mathcal{C}^2$.

1. Define the concept of curvature of a curve.
2. Determine the expression for the radius of curvature.
3. Prove the obtained formula.

Solution

1. Definition of curvature

The *curvature* k of a curve at a point M is the rate of change of the angle θ that the tangent makes with the x-axis with respect to the arc length l :

$$k = \left| \frac{d\theta}{dl} \right|.$$

2. Arc length element

We have:

$$dl = \sqrt{dx^2 + dy^2}.$$

Parameterizing by dx , we obtain:

$$dl = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \sqrt{1 + (y')^2} dx.$$

3. Tangent angle

The slope of the tangent satisfies:

$$\tan \theta = y'.$$

Thus:

$$\theta = \arctan(y').$$

Differentiating with respect to x :

$$\frac{d\theta}{dx} = \frac{1}{1 + (y')^2} y''.$$

4. Expression for curvature

By definition:

$$k = \left| \frac{d\theta}{dl} \right| = \left| \frac{d\theta/dx}{dl/dx} \right|.$$

We have:

$$\frac{dl}{dx} = \sqrt{1 + (y')^2}.$$

Thus:

$$k = \left| \frac{\frac{y''}{1 + (y')^2}}{\sqrt{1 + (y')^2}} \right| = \left| \frac{y''}{(1 + (y')^2)^{3/2}} \right|.$$

5. Radius of curvature

The radius of curvature is defined as:

$$\rho = \frac{1}{k}.$$

We obtain:

$$\rho = \left| \frac{(1 + (y')^2)^{3/2}}{y''} \right|.$$

Exercise

Using Newton's second law, prove that the radius of curvature of a path $y = f(x)$ is given by:

$$\rho(x) = \left| \frac{(1 + (y')^2)^{3/2}}{y''} \right|.$$

Solution

1. Kinematics of motion

Let $M(x(t), y(t))$ be a moving point in the plane. Set:

$$\vec{r}(t) = (x(t), y(t)), \quad \vec{v} = \frac{d\vec{r}}{dt}, \quad \vec{a} = \frac{d\vec{v}}{dt}.$$

The velocity can be written as:

$$\vec{v} = v\vec{T}$$

where $\vec{T}$ is the unit tangent vector and $v = \|\vec{v}\|$ is the speed.

Differentiate:

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d}{dt}(v\vec{T}) = \frac{dv}{dt}\vec{T} + v\frac{d\vec{T}}{dt}.$$

In differential geometry:

$$\frac{d\vec{T}}{dt} = \frac{v}{\rho}\vec{N},$$

where $\vec{N}$ is the unit normal vector and ρ is the radius of curvature.

Thus:

$$\vec{a} = \frac{dv}{dt} \vec{T} + \frac{v^2}{\rho} \vec{N}.$$

2. Analytical expressions

We have:

$$v^2 = \left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2, \quad \|\vec{a}\|^2 = \left(\frac{d^2x}{dt^2} \right)^2 + \left(\frac{d^2y}{dt^2} \right)^2.$$

On the other hand:

$$\vec{a} = a_T \vec{T} + a_N \vec{N} \Rightarrow \|\vec{a}\|^2 = a_T^2 + a_N^2$$

with:

$$a_T = \frac{dv}{dt}, \quad a_N = \frac{v^2}{\rho}.$$

Thus:

$$\left(\frac{d^2x}{dt^2} \right)^2 + \left(\frac{d^2y}{dt^2} \right)^2 = \left(\frac{dv}{dt} \right)^2 + \left(\frac{v^2}{\rho} \right)^2.$$

3. Switching to the variable x

Assume x is the parameter of motion:

$$y = y(x), \quad \frac{dy}{dt} = y' \frac{dx}{dt}.$$

After classical calculations (standard in differential mechanics), we obtain:

$$\frac{1}{\rho^2} = \frac{(y'')^2}{(1 + (y')^2)^3}.$$

Thus:

$$\rho(x) = \left| \frac{(1 + (y')^2)^{3/2}}{y''} \right|.$$

Exercise

Compute the radius of curvature of the curve of the function f defined by $f(x) = x^2$. Then plot $f(x)$ and its radius of curvature.

Solution

1. Computing the radius of curvature

Recall that for a curve $y = f(x)$, the radius of curvature is given by:

$$\rho(x) = \left| \frac{(1 + (y')^2)^{3/2}}{y''} \right|.$$

Here, $y = x^2$, thus:

$$y' = 2x, \quad y'' = 2.$$

We obtain:

$$\rho(x) = \frac{(1 + 4x^2)^{3/2}}{2}.$$

Thus:

$$\rho(x) = \frac{1}{2}(1 + 4x^2)^{3/2}.$$

2. Plot of $f(x)$ and $\rho(x)$

(Image will be inserted here)

3. Interpretation

- At the vertex of the parabola ($x = 0$):

$$\rho(0) = \frac{1}{2},$$

which is the smallest radius of curvature.

- As $|x|$ increases, $\rho(x)$ increases rapidly.
- This means the parabola is highly curved near the vertex and becomes flatter as we move away.

Note: The formula is valid as long as $y'' \neq 0$. Here $y'' = 2 > 0$, so it applies on all $\mathbb{R}$.

Exercise

Show that the radius of curvature of a circle of radius R is constant and equals R .

Solution

Consider a circle with center $\Omega(x_0, y_0)$ and radius R with equation:

$$(x - x_0)^2 + (y - y_0)^2 = R^2.$$

Write the function explicitly:

$$y = y_0 \pm \sqrt{R^2 - (x - x_0)^2}.$$

Computing the derivatives:

$$y' = \pm \frac{x - x_0}{\sqrt{R^2 - (x - x_0)^2}}.$$

Thus:

$$1 + (y')^2 = 1 + \frac{(x - x_0)^2}{R^2 - (x - x_0)^2} = \frac{R^2}{R^2 - (x - x_0)^2}.$$

We obtain:

$$(1 + (y')^2)^{3/2} = \frac{R^3}{(R^2 - (x - x_0)^2)^{3/2}}.$$

Computing the second derivative:

$$y'' = \pm \frac{R^2}{(R^2 - (x - x_0)^2)^{3/2}}.$$

Radius of curvature: By definition:

$$\rho(x) = \left| \frac{(1 + (y')^2)^{3/2}}{y''} \right|.$$

We obtain:

$$\rho(x) = \left| \frac{\frac{R^3}{(R^2 - (x - x_0)^2)^{3/2}}}{\pm \frac{R^2}{(R^2 - (x - x_0)^2)^{3/2}}} \right| = R.$$

$$\boxed{\rho(x) = R}$$

The radius of curvature is constant and equals the radius of the circle.

Geometric illustration: (Image will be inserted here)

Exercise

Compute the radius of curvature $\rho(x)$ of an ellipse centered at O with equation:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

Then compute $\rho(0)$ and $\rho(a)$ and study the cases $a > b$, $a = b$ and $a < b$.

Solution

The ellipse can be written explicitly as:

$$y = \pm b \sqrt{1 - \frac{x^2}{a^2}} = \pm \frac{b}{a} \sqrt{a^2 - x^2}.$$

Derivatives:

$$y' = \pm \frac{b}{a} \frac{x}{\sqrt{a^2 - x^2}}.$$

$$1 + (y')^2 = 1 + \frac{b^2 x^2}{a^2 (a^2 - x^2)} = \frac{a^4 + (b^2 - a^2)x^2}{a^2 (a^2 - x^2)}.$$

$$(1 + (y')^2)^{3/2} = \frac{(a^4 + (b^2 - a^2)x^2)^{3/2}}{a^3 (a^2 - x^2)^{3/2}}.$$

The second derivative:

$$y'' = \pm \frac{b}{a} \frac{a^2}{(a^2 - x^2)^{3/2}}.$$

Radius of curvature:

$$\rho(x) = \left| \frac{(1 + (y')^2)^{3/2}}{y''} \right| = \frac{1}{ab} (a^4 + (b^2 - a^2)x^2)^{3/2}.$$

$$\boxed{\rho(x) = \frac{(a^4 + (b^2 - a^2)x^2)^{3/2}}{ab}}.$$

Special values:

$$\rho(0) = \frac{(a^4)^{3/2}}{ab} = \frac{a^2}{b}, \quad \rho(a) = \frac{(a^2b^2)^{3/2}}{ab} = \frac{b^2}{a}.$$

Special cases: If $a = b = R$, the ellipse becomes a circle and $\rho(x) = R$ (constant).

(Images will be inserted here)

Exercise

Consider the cycloid defined by:

$$\begin{cases} x = R(\theta - \sin \theta), \\ y = R(1 - \cos \theta), \end{cases} \quad \theta \in [0, 2\pi].$$

1. Find the velocity vector and its magnitude.
2. Using Newton's second law, find the radius of curvature.
3. What is the maximum value of the radius of curvature?
4. What is the largest area occupied by this radius?
5. Compute the path length.
6. Compute the arc length over one complete cycle.
7. Find the expression for the area under one arch.
8. Give the maximum value of this area.

Solution

1) Velocity vector

$$\vec{r}(\theta) = (R(\theta - \sin \theta), R(1 - \cos \theta))$$

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{d\vec{r}}{d\theta} \frac{d\theta}{dt} = R\omega(1 - \cos \theta, \sin \theta)$$

$$\|\vec{v}\| = R\omega\sqrt{(1 - \cos \theta)^2 + \sin^2 \theta} = R\omega\sqrt{2 - 2\cos \theta} = 2R\omega \sin(\theta/2)$$

2) Radius of curvature Using the formula:

$$\rho = \frac{(x'^2 + y'^2)^{3/2}}{|x'y'' - y'x''|}$$

After calculation we obtain:

$$\rho(\theta) = 4R \sin(\theta/2)$$

3) Maximum value

$$\rho_{\max} = 4R \quad \text{at } \theta = \pi$$

4) Largest area occupied by this radius

$$S_{\max} = \pi(4R)^2 = 16\pi R^2$$

5) Path length

$$ds = \sqrt{x'^2 + y'^2} d\theta = 2R \sin(\theta/2) d\theta$$

$$L = \int_0^{2\pi} 2R \sin(\theta/2) d\theta = 8R$$

6) Arc length over one cycle

$$L_{\text{cycle}} = 8R$$

7) Area under one arch

$$A = \int_0^{2\pi} y \, dx = \int_0^{2\pi} R(1 - \cos \theta) R(1 - \cos \theta) d\theta$$

$$A = R^2 \int_0^{2\pi} (1 - \cos \theta)^2 d\theta = 3\pi R^2$$

8) Maximum area of a cycle

$$A_{\text{max}} = 3\pi R^2$$

(Images will be inserted here)

Exercise

What is the area swept by the position vector $\vec{r} = \overrightarrow{OM}$ as its magnitude changes from $r = 0$ to $r = \|\vec{r}\| = f(\theta)$?

Study the following cases:

1. r independent of θ (deduce the area of a disk of radius R),
2. $r = k\theta$ with $\theta_0 = 0$,
3. $r = k \sin \theta$ with $\theta_0 = 0$.

Solution

In polar coordinates, the area element is:

$$dS = r \, dr \, d\theta.$$

The area swept by the vector between θ_0 and θ is:

$$S = \int_{\theta_0}^{\theta} \left(\int_0^r r \, dr \right) d\theta = \int_{\theta_0}^{\theta} \frac{r^2}{2} d\theta.$$

If $r = f(\theta)$, we obtain the general formula:

$$S(\theta) = \frac{1}{2} \int_{\theta_0}^{\theta} f^2(\theta) d\theta.$$

1) **Case r independent of θ** If $r = R$ (constant), then:

$$S = \frac{1}{2} \int_{\theta_0}^{\theta} R^2 d\theta = \frac{R^2}{2} (\theta - \theta_0).$$

For a full cycle: $\theta - \theta_0 = 2\pi$, thus:

$$S = \pi R^2,$$

which is the area of a disk of radius R .

(Image will be inserted here)

2) Case $r = k\theta$ with $\theta_0 = 0$

$$S(\theta) = \frac{1}{2} \int_0^\theta (k\theta)^2 d\theta = \frac{k^2}{2} \int_0^\theta \theta^2 d\theta = \frac{k^2 \theta^3}{6}.$$

$$S(\theta) = \frac{k^2 \theta^3}{6}.$$

(Image will be inserted here)

3) Case $r = k \sin \theta$ with $\theta_0 = 0$

$$S = \frac{1}{2} \int_0^\theta k^2 \sin^2 \theta d\theta.$$

We have:

$$\sin^2 \theta = \frac{1 - \cos 2\theta}{2}.$$

Thus:

$$S = \frac{k^2}{4} \int_0^\theta (1 - \cos 2\theta) d\theta = \frac{k^2}{4} \left(\theta - \frac{\sin 2\theta}{2} \right).$$

$$S(\theta) = \frac{k^2}{4} \left(\theta - \frac{\sin 2\theta}{2} \right).$$

(Image will be inserted here)

Exercise

Compute the area of Bernoulli's lemniscate defined in polar coordinates by:

$$r^2 = a^2 \cos(2\theta).$$

Solution

We can write:

$$r = a \sqrt{|\cos(2\theta)|}.$$

Note that:

$$r_{\max} = a \quad \text{for } \theta = 0, \quad r_{\min} = 0 \quad \text{for } \theta = \frac{\pi}{4}.$$

In polar coordinates, the area element is:

$$dS = r dr d\theta.$$

The area of a sector is:

$$S = \int_0^\theta \left(\int_0^r r dr \right) d\theta = \int_0^\theta \frac{r^2}{2} d\theta.$$

Thus:

$$S = \frac{a^2}{2} \int_0^\theta |\cos(2\theta)| d\theta.$$

On the interval $0 \leq \theta \leq \frac{\pi}{4}$, we have $\cos(2\theta) \geq 0$, so:

$$S_1 = \frac{a^2}{2} \int_0^{\pi/4} \cos(2\theta) d\theta.$$

We have:

$$\int \cos(2\theta) d\theta = \frac{1}{2} \sin(2\theta).$$

Thus:

$$S_1 = \frac{a^2}{2} \cdot \frac{1}{2} \left[\sin(2\theta) \right]_0^{\pi/4} = \frac{a^2}{4} (1 - 0) = \frac{a^2}{4}.$$

The lemniscate contains four symmetric sectors, therefore:

$$S = 4S_1 = a^2.$$

$$\boxed{S = a^2}.$$

If we set $a^2 = 2c^2$, then:

$$\boxed{S = 2c^2}.$$

(Image will be inserted here)

Exercise

Compute the area of the cardioid defined by the polar equation:

$$r = a(1 + \cos \theta).$$

Solution

Use the area formula in polar coordinates:

$$S = \int_{\theta_0}^{\theta_1} \frac{r^2}{2} d\theta.$$

Here:

$$r = a(1 + \cos \theta).$$

Note that:

$$r_{\min} = r(\pi) = 0, \quad r_{\max} = r(0) = 2a.$$

The cardioid is symmetric about the Ox -axis, so it suffices to integrate over $[0, \pi]$ and multiply by 2.

$$S = 2 \times \frac{1}{2} \int_0^{\pi} r^2 d\theta = \int_0^{\pi} a^2 (1 + \cos \theta)^2 d\theta.$$

$$S = a^2 \int_0^{\pi} (1 + 2 \cos \theta + \cos^2 \theta) d\theta.$$

We have:

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}.$$

Thus:

$$S = a^2 \int_0^{\pi} \left(1 + 2 \cos \theta + \frac{1 + \cos 2\theta}{2} \right) d\theta = a^2 \int_0^{\pi} \left(\frac{3}{2} + 2 \cos \theta + \frac{1}{2} \cos 2\theta \right) d\theta.$$

Integrate:

$$S = a^2 \left[\frac{3}{2}\theta + 2 \sin \theta + \frac{1}{4} \sin 2\theta \right]_0^\pi.$$

Since $\sin \pi = 0$ and $\sin 2\pi = 0$, we obtain:

$$S = a^2 \cdot \frac{3}{2}\pi = \frac{3\pi a^2}{2}.$$

$$S = \frac{3\pi a^2}{2}.$$

(Image will be inserted here)

Exercise

Area of a polar rose with an odd number of petals

Consider the polar curve defined by:

$$r = a \cos((2n + 1)\theta), \quad n \geq 1$$

1. Determine the area of a rose with $(2n + 1)$ petals.
2. Applications:
 - Rose with 3 petals,
 - Rose with 5 petals.

Solution

1) Area of one petal

In polar coordinates, the area is given by:

$$S = \frac{1}{2} \int_{\theta_1}^{\theta_2} r^2(\theta) d\theta$$

Here:

$$r(\theta) = a \cos((2n + 1)\theta)$$

One petal is described when:

$$0 \leq \theta \leq \frac{\pi}{2(2n + 1)}$$

Thus the area of one petal is:

$$S_1 = \frac{1}{2} \int_0^{\frac{\pi}{2(2n+1)}} a^2 \cos^2((2n + 1)\theta) d\theta$$

$$S_1 = \frac{a^2}{2} \int_0^{\frac{\pi}{2(2n+1)}} \cos^2((2n + 1)\theta) d\theta$$

We have:

$$\cos^2 u = \frac{1 + \cos(2u)}{2}$$

Thus:

$$S_1 = \frac{a^2}{4} \int_0^{\frac{\pi}{2(2n+1)}} (1 + \cos(2(2n+1)\theta)) d\theta$$

$$S_1 = \frac{a^2}{4} \left[\theta + \frac{\sin(2(2n+1)\theta)}{2(2n+1)} \right]_0^{\frac{\pi}{2(2n+1)}}$$

The sine term vanishes at both limits, so:

$$S_1 = \frac{a^2}{4} \cdot \frac{\pi}{2(2n+1)}$$

2) Total area of the rose

The rose contains $(2n+1)$ petals, therefore:

$$S = (2n+1)S_1 = (2n+1) \frac{a^2\pi}{8(2n+1)}$$

$$\boxed{S = \frac{\pi a^2}{8}}$$

Remarkable result: The area of the rose is independent of the number of petals.

3) Special cases

Rose with 3 petals: $r = a \cos(3\theta)$

Rose with 5 petals: $r = a \cos(5\theta)$

Rose with 7 petals: $r = a \cos(7\theta)$

In all cases:

$$S = \frac{\pi a^2}{8}$$

Exercise

Area of a polar rose with an even number of petals

Consider the polar curve defined by:

$$r = a \cos\left(\frac{k\theta}{2}\right), \quad k \geq 2$$

1. Determine the area of a rose with $k = 2n$ and $k = 4n$ petals.
2. Applications:
 - Rose with 2 petals,
 - Rose with 4 petals.

Solution

Recall: In polar coordinates, the area is given by:

$$S = \int \frac{r^2}{2} d\theta.$$

1) Rose with $k = 2n$ petals: $r = a \cos(n\theta)$

One petal is described when:

$$r = 0 \iff \cos(n\theta) = 0 \iff n\theta = \frac{\pi}{2} \iff \theta = \frac{\pi}{2n}.$$

Thus for one petal:

$$0 \leq \theta \leq \frac{\pi}{2n}.$$

Area of one petal:

$$S_1 = \int_0^{\pi/2n} \frac{a^2 \cos^2(n\theta)}{2} d\theta.$$

Using:

$$\cos^2 x = \frac{1 + \cos(2x)}{2},$$

we obtain:

$$S_1 = \frac{a^2}{4} \int_0^{\pi/2n} (1 + \cos(2n\theta)) d\theta.$$

Thus:

$$S_1 = \frac{a^2}{4} \left[\theta + \frac{\sin(2n\theta)}{2n} \right]_0^{\pi/2n} = \frac{a^2}{4} \cdot \frac{\pi}{2n}.$$

$$S_1 = \frac{\pi a^2}{8n}.$$

The rose has $2n$ petals, so its total area is:

$$S = 2n \times S_1 = 2n \times \frac{\pi a^2}{8n} = \frac{\pi a^2}{4}.$$

2) Rose with $k = 4n$ petals: $r = a \cos(2n\theta)$

One petal is described when:

$$r = 0 \iff \cos(2n\theta) = 0 \iff 2n\theta = \frac{\pi}{2} \iff \theta = \frac{\pi}{4n}.$$

Thus for one petal:

$$0 \leq \theta \leq \frac{\pi}{4n}.$$

Area of one petal:

$$S_1 = \int_0^{\pi/4n} \frac{a^2 \cos^2(2n\theta)}{2} d\theta.$$

Using:

$$\cos^2 x = \frac{1 + \cos(2x)}{2},$$

we obtain:

$$S_1 = \frac{a^2}{4} \int_0^{\pi/4n} (1 + \cos(4n\theta)) d\theta.$$

Thus:

$$S_1 = \frac{a^2}{4} \left[\theta + \frac{\sin(4n\theta)}{4n} \right]_0^{\pi/4n} = \frac{a^2}{4} \cdot \frac{\pi}{4n}.$$

$$S_1 = \frac{\pi a^2}{16n}.$$

The rose has $4n$ petals, so its total area is:

$$S = 4n \times S_1 = 4n \times \frac{\pi a^2}{16n} = \frac{\pi a^2}{4}.$$

Conclusion

In both cases:

$$S = \frac{\pi a^2}{4}.$$

The area of the rose is independent of the number of petals.

Special cases

- 2 petals: $r = a \cos \theta \Rightarrow S = \frac{\pi a^2}{4}.$
- 4 petals: $r = a \cos(2\theta) \Rightarrow S = \frac{\pi a^2}{4}.$

Exercise

Consider the Archimedean spiral defined in polar coordinates by:

$$r = a\theta, \quad a > 0.$$

1. Compute the area swept by the radius vector as θ changes from 0 to a value θ .
2. Deduce the area corresponding to one full cycle ($0 \leq \theta \leq 2\pi$).

Solution

Recall that in polar coordinates, the area is given by:

$$S = \int_0^\theta \frac{r^2}{2} d\theta.$$

Here:

$$r = a\theta.$$

Thus:

$$S(\theta) = \int_0^\theta \frac{(a\theta)^2}{2} d\theta = \frac{a^2}{2} \int_0^\theta \theta^2 d\theta.$$

We obtain:

$$S(\theta) = \frac{a^2}{2} \cdot \frac{\theta^3}{3} = \frac{a^2 \theta^3}{6}.$$

For one full cycle ($\theta = 2\pi$):

$$S(2\pi) = \frac{a^2(2\pi)^3}{6} = \frac{8a^2\pi^3}{6} = \frac{4a^2\pi^3}{3}.$$

Exercise

Let a curve be defined in polar coordinates by

$$r = e^{k\theta}, \quad k > 0$$

(logarithmic spiral).

1. Compute the arc length of the spiral between $\theta = 0$ and $\theta = \Theta$.
2. Compute the area swept by the radius vector between $\theta = 0$ and $\theta = \Theta$.
3. Determine the radius of curvature $\rho(\theta)$.

Solution

Preliminary: Expression of the arc length element in polar coordinates

In the plane, the arc length element is given by:

$$dl = \sqrt{(dx)^2 + (dy)^2}.$$

In polar coordinates, we have:

$$x = r \cos \theta, \quad y = r \sin \theta, \quad r = r(\theta).$$

Differentiating:

$$dx = \left(\frac{dr}{d\theta} \cos \theta - r \sin \theta \right) d\theta, \quad dy = \left(\frac{dr}{d\theta} \sin \theta + r \cos \theta \right) d\theta.$$

Squaring and adding:

$$\begin{aligned} (dx)^2 + (dy)^2 &= \left[\left(\frac{dr}{d\theta} \right)^2 (\cos^2 \theta + \sin^2 \theta) \right. \\ &\quad \left. + r^2 (\sin^2 \theta + \cos^2 \theta) \right] d\theta^2 \\ &= \left[r^2 + \left(\frac{dr}{d\theta} \right)^2 \right] d\theta^2. \end{aligned}$$

Thus we obtain:

$$dl = \sqrt{r^2 + \left(\frac{dr}{d\theta} \right)^2} d\theta$$

1. Arc length

We have:

$$r = e^{k\theta}, \quad \frac{dr}{d\theta} = ke^{k\theta}.$$

Thus:

$$dl = e^{k\theta} \sqrt{1 + k^2} d\theta.$$

The length of the spiral between 0 and Θ is:

$$L = \int_0^\Theta e^{k\theta} \sqrt{1 + k^2} d\theta = \frac{\sqrt{1 + k^2}}{k} (e^{k\Theta} - 1).$$

$$L = \frac{\sqrt{1 + k^2}}{k} (e^{k\Theta} - 1)$$

2. Swept area

The area swept by the radius vector is given by:

$$A = \frac{1}{2} \int_0^\Theta r^2 d\theta = \frac{1}{2} \int_0^\Theta e^{2k\theta} d\theta.$$

We obtain:

$$A = \frac{1}{4k} (e^{2k\Theta} - 1).$$

$$A = \frac{1}{4k} (e^{2k\Theta} - 1)$$

3. Radius of curvature

The radius of curvature for a curve defined in polar coordinates is:

$$\rho = \frac{\left(r^2 + \left(\frac{dr}{d\theta} \right)^2 \right)^{3/2}}{\left| r^2 + 2 \left(\frac{dr}{d\theta} \right)^2 - r \frac{d^2r}{d\theta^2} \right|}.$$

We have:

$$\frac{dr}{d\theta} = k e^{k\theta}, \quad \frac{d^2r}{d\theta^2} = k^2 e^{k\theta}.$$

Thus:

$$\rho(\theta) = \sqrt{1 + k^2} e^{k\theta}.$$

$$\rho(\theta) = \sqrt{1 + k^2} e^{k\theta}$$

Differential Equations

Theorem

1. **(Local Cauchy–Lipschitz Theorem)** Consider the Cauchy problem

$$y'(t) = f(t, y(t)), \quad y(t_0) = y_0.$$

If f is continuous in t and Lipschitz in y in a neighbourhood of (t_0, y_0) , then this problem admits a **unique local solution**.

2. **(First-order linear ODE)** The equation

$$y'(t) + a(t)y(t) = b(t)$$

has the general solution

$$y(t) = \frac{1}{\mu(t)} \left(\int \mu(t)b(t) dt + C \right),$$

where $\mu(t) = \exp\left(\int a(t) dt\right)$ is the integrating factor.

3. **(Homogeneous first-order linear ODE)** For $y'(t) + a(t)y(t) = 0$, the solution space is one-dimensional.

4. **(Homogeneous second-order linear ODE)** The equation $y''(t) + a(t)y'(t) + b(t)y(t) = 0$ admits two linearly independent solutions y_1, y_2 . The general solution is $y(t) = C_1y_1(t) + C_2y_2(t)$.

5. **(Grönwall's inequality)** If u is non-negative, continuous and satisfies

$$u(t) \leq A + \int_{t_0}^t k(s)u(s) ds,$$

then

$$u(t) \leq A \exp\left(\int_{t_0}^t k(s) ds\right).$$

6. **(Global uniqueness)** If f is continuous and globally Lipschitz in y on $[t_0, T] \times \mathbb{R}$, then the Cauchy problem has a unique solution defined on the whole interval $[t_0, T]$.

7. **(Peano's theorem)** If f is continuous near (t_0, y_0) , then the Cauchy problem admits at least one local solution.

8. **(Superposition principle)** For the nonhomogeneous linear equation $y' + a(t)y = b(t)$, the general solution = particular solution + general solution of the homogeneous equation.

9. **(Constant-coefficient ODEs)** For $ay'' + by' + cy = 0$, the solutions are determined by the characteristic equation $ar^2 + br + c = 0$.

10. **(Variation of parameters)** Let y_1, y_2 be two linearly independent solutions of $y'' + a(t)y' + b(t)y = f(t)$ is

$$y_p(t) = C_1(t)y_1(t) + C_2(t)y_2(t),$$

where

$$\begin{cases} C_1'(t)y_1(t) + C_2'(t)y_2(t) = 0, \\ C_1'(t)y_1'(t) + C_2'(t)y_2'(t) = f(t). \end{cases}$$

11. **(Matrix Cauchy–Lipschitz for linear systems)** For $y'(t) = A(t)y(t) + b(t)$ with $A(t)$ continuous, the Cauchy problem has a unique global solution.

12. **(Linear stability)** Consider $y' = f(y)$ with $f \in C^1$. If all eigenvalues of the Jacobian $Df(y^*)$ at an equilibrium y^* have negative real parts, then y^* is asymptotically stable.

13. **(Lyapunov stability)** If an equilibrium admits a strictly decreasing Lyapunov function, then it is stable.

14. **(Maximal existence)** Any maximal solution of a Cauchy problem is defined on a maximal open interval containing t_0 .

15. **(Continuous dependence on initial data)** Solutions depend continuously on the initial conditions.

Exercise

Solve $ay' + by + c = 0$, $a \neq 0$.

Solution

1. **Standard form.** Divide by a :

$$y' + \frac{b}{a}y = -\frac{c}{a}.$$

2. **Case $b \neq 0$ (first-order linear with constant coefficients).** Rewrite as:

$$y' = -\frac{b}{a}y - \frac{c}{a}.$$

Set $\alpha = -\frac{b}{a}$ and $\beta = -\frac{c}{a}$. Then:

$$y' = \alpha y + \beta.$$

Separate variables:

$$\frac{dy}{\alpha y + \beta} = dx.$$

Integrate both sides:

$$\int \frac{dy}{\alpha y + \beta} = \int dx \implies \frac{1}{\alpha} \ln |\alpha y + \beta| = x + C_0.$$

Multiply by α :

$$\ln |\alpha y + \beta| = \alpha x + \alpha C_0.$$

Exponentiate:

$$\alpha y + \beta = C e^{\alpha x}, \quad C = \pm e^{\alpha C_0}.$$

Thus:

$$y = \frac{C}{\alpha} e^{\alpha x} - \frac{\beta}{\alpha}.$$

Substitute back $\alpha = -\frac{b}{a}$ and $\beta = -\frac{c}{a}$:

$$\frac{\beta}{\alpha} = \frac{-c/a}{-b/a} = \frac{c}{b}, \quad \frac{1}{\alpha} = -\frac{a}{b}.$$

Hence:

$$y(x) = -\frac{a}{b} C e^{-(b/a)x} - \frac{c}{b}.$$

Renaming $K = -\frac{a}{b} C$ (an arbitrary constant), we obtain:

$$y(x) = K e^{-(b/a)x} - \frac{c}{b}.$$

3. Case $b = 0$. The equation reduces to $ay' + c = 0$, i.e. $y' = -\frac{c}{a}$. Direct integration gives:

$$y(x) = -\frac{c}{a} x + K.$$

Therefore,

$$y(x) = \begin{cases} K e^{-(b/a)x} - \frac{c}{b}, & b \neq 0, \\ -\frac{c}{a} x + K, & b = 0. \end{cases}$$

Exercise

Solve the differential equation:

$$y' = y^2 - t,$$

where t is a constant.

Solution

1. Separate variables. Rewrite the equation as:

$$\frac{dy}{y^2 - t} = dx.$$

2. Partial fraction decomposition. We consider the case $t > 0$ (for $t < 0$, the solution involves arctangent; for $t = 0$, it is a simple rational integral). Assume $t > 0$ and set $a = \sqrt{t}$. Then:

$$\frac{1}{y^2 - a^2} = \frac{1}{2a} \left(\frac{1}{y - a} - \frac{1}{y + a} \right).$$

3. Integrate both sides.

$$\int \frac{dy}{y^2 - a^2} = \frac{1}{2a} \int \left(\frac{1}{y - a} - \frac{1}{y + a} \right) dy = \int dx.$$

Thus:

$$\frac{1}{2a} (\ln |y - a| - \ln |y + a|) = x + C.$$

4. Simplify.

$$\frac{1}{2a} \ln \left| \frac{y - a}{y + a} \right| = x + C \implies \ln \left| \frac{y - a}{y + a} \right| = 2ax + 2aC.$$

Exponentiate:

$$\frac{y - a}{y + a} = Ke^{2ax}, \quad \text{where } K = \pm e^{2aC}.$$

5. Solve for y .

$$y - a = Ke^{2ax}(y + a) \implies y - Ke^{2ax}y = a + Kae^{2ax}.$$

$$y(1 - Ke^{2ax}) = a(1 + Ke^{2ax}) \implies y(x) = a \frac{1 + Ke^{2ax}}{1 - Ke^{2ax}}.$$

6. Special cases.

- If $t < 0$, let $a = \sqrt{-t}$. Then $\frac{1}{y^2 + a^2}$ integrates to $\frac{1}{a} \arctan \frac{y}{a}$:

$$y(x) = a \tan(a(x + C)).$$

- If $t = 0$, separate variables: $\frac{dy}{y^2} = dx$ gives $y(x) = -\frac{1}{x + C}$.

Therefore, the general solution is:

$$y(x) = \begin{cases} \sqrt{t} \frac{1 + Ke^{2\sqrt{t}x}}{1 - Ke^{2\sqrt{t}x}}, & t > 0, \\ \sqrt{-t} \tan(\sqrt{-t}(x + C)), & t < 0, \\ -\frac{1}{x + C}, & t = 0. \end{cases}$$

Exercise

Solve the homogeneous linear differential equation:

$$ay'' + by' + cy = 0, \quad a \neq 0.$$

Solution

1. Operator notation. Let $D = \frac{d}{dx}$. Then $D^2 = \frac{d^2}{dx^2}$. The equation becomes:

$$(aD^2 + bD + c)y = 0.$$

2. Factorization of the characteristic polynomial. The characteristic equation is $ar^2 + br + c = 0$. Let $\Delta = b^2 - 4ac$ and denote the roots:

$$r_1 = \frac{-b - \sqrt{\Delta}}{2a}, \quad r_2 = \frac{-b + \sqrt{\Delta}}{2a}.$$

Then the polynomial can be factored as:

$$ar^2 + br + c = a(r - r_1)(r - r_2).$$

Hence the differential operator factorises:

$$aD^2 + bD + c = a(D - r_1)(D - r_2).$$

3. Solving using the factorised form. Consider the equation $(D - r_2)y = 0$ first. This gives $y' - r_2y = 0$, whose solution is $y = C_2e^{r_2x}$.

Now, to solve $(D - r_1)(D - r_2)y = 0$, set $u = (D - r_2)y$. Then $(D - r_1)u = 0$, so $u = C_1e^{r_1x}$.

Thus:

$$(D - r_2)y = C_1e^{r_1x} \implies y' - r_2y = C_1e^{r_1x}.$$

4. First-order linear ODE for y . Multiply by the integrating factor e^{-r_2x} :

$$\frac{d}{dx}(ye^{-r_2x}) = C_1e^{(r_1-r_2)x}.$$

Integrate:

$$ye^{-r_2x} = \begin{cases} C_1 \frac{e^{(r_1-r_2)x}}{r_1 - r_2} + C_2, & r_1 \neq r_2, \\ C_1x + C_2, & r_1 = r_2. \end{cases}$$

5. General solution.

- **Case $\Delta > 0$ (distinct real roots $r_1 \neq r_2$):**

$$y(x) = C_1e^{r_1x} + C_2e^{r_2x}.$$

- **Case $\Delta = 0$ (double root $r = r_1 = r_2$):**

$$y(x) = (C_1x + C_2)e^{rx}.$$

- **Case $\Delta < 0$ (complex conjugate roots $r = \alpha \pm i\beta$):** Using Euler's formula $e^{i\beta x} = \cos \beta x + i \sin \beta x$, the real general solution is:

$$y(x) = e^{\alpha x} (C_1 \cos(\beta x) + C_2 \sin(\beta x)),$$

$$\text{where } \alpha = -\frac{b}{2a} \text{ and } \beta = \frac{\sqrt{-\Delta}}{2a}.$$

6. Conclusion.

$$y(x) = \begin{cases} C_1 e^{r_1 x} + C_2 e^{r_2 x}, & \Delta > 0, \\ (C_1 x + C_2) e^{rx}, & \Delta = 0, \\ e^{\alpha x} (C_1 \cos(\beta x) + C_2 \sin(\beta x)), & \Delta < 0. \end{cases}$$

Exercise

Solve $y'' - 5y' + 6y - 12 = 0$.

Solution

1. Operator form. $(D^2 - 5D + 6)y = 12$.

2. Homogeneous solution. Characteristic: $r^2 - 5r + 6 = (r - 2)(r - 3) = 0$, roots $r = 2, 3$. Thus

$$y_h = C_1 e^{3x} + C_2 e^{2x}.$$

3. Particular solution. Since RHS is constant 12, try $y_p = A$:

$$6A = 12 \implies A = 2.$$

4. General solution.

$$y(x) = C_1 e^{3x} + C_2 e^{2x} + 2.$$

Exercise

Solve $y'' + 6y' + 9y + 18 = 0$.

Solution

1. Operator form. $(D^2 + 6D + 9)y = -18$.

2. Homogeneous solution. Characteristic: $r^2 + 6r + 9 = (r + 3)^2 = 0$, double root $r = -3$. Thus

$$y_h = (C_1 + C_2 x) e^{-3x}.$$

3. Particular solution. $y_p = A$ constant: $9A = -18 \Rightarrow A = -2$.

4. General solution.

$$y(x) = (C_1 + C_2 x) e^{-3x} - 2.$$

Exercise

Solve $y'' + 4y' + 13y - 26 = 0$.

Solution

1. Operator form. $(D^2 + 4D + 13)y = 26$.

2. Homogeneous solution. Characteristic: $r^2 + 4r + 13 = 0$, $\Delta = 16 - 52 = -36$, roots $r = -2 \pm 3i$. Thus

$$y_h = e^{-2x} (C_1 \cos 3x + C_2 \sin 3x).$$

3. Particular solution. $y_p = A$ constant: $13A = 26 \Rightarrow A = 2$.

4. General solution.

$$y(x) = e^{-2x}(C_1 \cos 3x + C_2 \sin 3x) + 2.$$

Exercise

Solve $y'' + 2y' - 8 = 0$.

Solution

1. Operator form. $(D^2 + 2D)y = 8$.

2. Homogeneous solution. Characteristic: $r^2 + 2r = r(r + 2) = 0$, roots $r = 0, -2$.
Thus

$$y_h = C_1 + C_2 e^{-2x}.$$

3. Particular solution. Since RHS is constant and $r = 0$ is a root, try $y_p = kx$:

$$y'_p = k, y''_p = 0 \Rightarrow 2k = 8 \Rightarrow k = 4.$$

Thus $y_p = 4x$.

4. General solution.

$$y(x) = C_1 + C_2 e^{-2x} + 4x.$$

Exercise

Solve $y'' - y' - 2y + 4 = 0, y(0) = 0, y'(0) = 1$.

Solution

1. Operator form. $(D^2 - D - 2)y = -4$.

2. Homogeneous solution. Characteristic: $r^2 - r - 2 = (r - 2)(r + 1) = 0$, roots $r = 2, -1$. Thus

$$y_h = C_1 e^{2x} + C_2 e^{-x}.$$

3. Particular solution. $y_p = A$ constant: $-2A = -4 \Rightarrow A = 2$. So

$$y(x) = C_1 e^{2x} + C_2 e^{-x} + 2.$$

4. Apply initial conditions.

$$y(0) = C_1 + C_2 + 2 = 0 \Rightarrow C_1 + C_2 = -2.$$

$$y'(x) = 2C_1 e^{2x} - C_2 e^{-x} \Rightarrow y'(0) = 2C_1 - C_2 = 1.$$

Solving: $3C_1 = -1 \Rightarrow C_1 = -\frac{1}{3}, C_2 = -2 + \frac{1}{3} = -\frac{5}{3}$.

5. Final solution.

$$y(x) = -\frac{1}{3}e^{2x} - \frac{5}{3}e^{-x} + 2.$$

Exercise

Solve $ay''' + by'' + cy' + dy = k$, $a \neq 0$.

Solution

1. Operator form. $(aD^3 + bD^2 + cD + d)y = k$.

2. Homogeneous solution. Characteristic polynomial: $a\lambda^3 + b\lambda^2 + c\lambda + d = 0$. The form of y_h depends on the roots:

- Three distinct real roots λ_i : $y_h = \sum C_i e^{\lambda_i x}$.
- One double root λ_1 , one simple root λ_2 : $y_h = (C_1 + C_2 x)e^{\lambda_1 x} + C_3 e^{\lambda_2 x}$.
- Triple root λ : $y_h = (C_1 + C_2 x + C_3 x^2)e^{\lambda x}$.
- One real root λ and a complex conjugate pair $\alpha \pm i\beta$: $y_h = C_1 e^{\lambda x} + e^{\alpha x}(C_2 \cos \beta x + C_3 \sin \beta x)$.

3. Particular solution. Since RHS is constant k , try $y_p = A$ constant:

$$dA = k \implies A = \frac{k}{d} \quad (d \neq 0).$$

If $d = 0$, increase the degree of the trial polynomial according to the multiplicity of $\lambda = 0$.

4. General solution. $y = y_h + y_p$.

Exercise

Solve $y''' - y = 0$.

Solution

1. Operator form. $(D^3 - 1)y = 0$.

2. Characteristic equation. $\lambda^3 - 1 = 0$. Roots:

$$\lambda_1 = 1, \quad \lambda_2 = e^{i\pi/3} = \frac{1}{2} + i\frac{\sqrt{3}}{2}, \quad \lambda_3 = e^{-i\pi/3} = \frac{1}{2} - i\frac{\sqrt{3}}{2}.$$

3. Real general solution. For the complex pair $\alpha \pm i\beta$ with $\alpha = 1/2$, $\beta = \sqrt{3}/2$:

$$y(x) = C_1 e^x + e^{x/2} \left(C_2 \cos \frac{\sqrt{3}}{2} x + C_3 \sin \frac{\sqrt{3}}{2} x \right).$$

$$y(x) = C_1 e^x + e^{x/2} \left(C_2 \cos \frac{\sqrt{3}}{2} x + C_3 \sin \frac{\sqrt{3}}{2} x \right).$$

Exercise

Solve $y''' - y + 1 = 0$.

Solution

1. **Operator form.** $(D^3 - 1)y = -1$.
2. **Homogeneous solution.** From Exercise 11.10:

$$y_h = C_1 e^x + e^{x/2} \left(C_2 \cos \frac{\sqrt{3}}{2} x + C_3 \sin \frac{\sqrt{3}}{2} x \right).$$

3. **Particular solution.** $y_p = A$ constant: $0 - A = -1 \Rightarrow A = 1$.
4. **General solution.**

$$y(x) = y_h + 1.$$

Exercise

Solve $y''' - 3y'' + 3y' - y = 0$.

Solution

1. **Operator form.** $(D^3 - 3D^2 + 3D - 1)y = 0$.
2. **Characteristic polynomial.** $\lambda^3 - 3\lambda^2 + 3\lambda - 1 = (\lambda - 1)^3 = 0$, triple root $\lambda = 1$.
3. **General solution.**

$$y(x) = (C_1 + C_2 x + C_3 x^2) e^x.$$

Exercise

Solve $y''' + y'' + y' + y = 4$.

Solution

1. **Operator form.** $(D^3 + D^2 + D + 1)y = 4$.
2. **Characteristic polynomial.** $\lambda^3 + \lambda^2 + \lambda + 1 = (\lambda + 1)(\lambda^2 + 1) = 0$, roots $\lambda = -1, \pm i$.

$$y_h = C_1 e^{-x} + C_2 \cos x + C_3 \sin x.$$

3. **Particular solution.** $y_p = A$ constant: $0 + 0 + 0 + A = 4 \Rightarrow A = 4$.
4. **General solution.**

$$y(x) = C_1 e^{-x} + C_2 \cos x + C_3 \sin x + 4.$$

Exercise

Solve $y''' - 2y'' - y' + 2y = 0$, $y(0) = 1$, $y'(0) = 0$, $y''(0) = 2$.

Solution

1. **Operator form.** $(D^3 - 2D^2 - D + 2)y = 0$.
2. **Characteristic polynomial.** $\lambda^3 - 2\lambda^2 - \lambda + 2 = (\lambda - 1)(\lambda - 2)(\lambda + 1) = 0$, roots $\lambda = 1, 2, -1$.

$$y_h = C_1 e^x + C_2 e^{2x} + C_3 e^{-x}.$$

3. Apply initial conditions.

$$\begin{aligned}y(0) &= C_1 + C_2 + C_3 = 1, \\y'(0) &= C_1 + 2C_2 - C_3 = 0, \\y''(0) &= C_1 + 4C_2 + C_3 = 2.\end{aligned}$$

Solving: Adding the first two: $2C_1 + 3C_2 = 1$. Adding the second and third: $2C_1 + 6C_2 = 2$. Subtracting: $3C_2 = 1 \Rightarrow C_2 = \frac{1}{3}$. Then $2C_1 + 1 = 1 \Rightarrow C_1 = 0$, and $C_3 = 1 - \frac{1}{3} = \frac{2}{3}$.

4. Final solution.

$$y(x) = \frac{1}{3}e^{2x} + \frac{2}{3}e^{-x}.$$

Exercise

Solve $xy'' + 2xy' + (x - 1)y = 0$.

Solution

1. Substitution. Let $y = e^{-x}u$. Then $y' = e^{-x}(u' - u)$, $y'' = e^{-x}(u'' - 2u' + u)$. Substitute:

$$x(u'' - 2u' + u) + 2x(u' - u) + (x - 1)u = 0.$$

Simplify: $xu'' - 2xu' + xu + 2xu' - 2xu + xu - u = xu'' - u = 0$.

Thus $xu'' - u = 0$ or $u'' - \frac{1}{x}u = 0$.

2. Frobenius method. The indicial equation is $r(r - 1) = 0$ with roots $r = 0, 1$. The solutions are $u_1 = x$ and $u_2 = x \ln x$ (up to constants).

3. General solution in u : $u(x) = C_1x + C_2x \ln x$.

4. Back to y : $y = e^{-x}u$, thus

$$y(x) = e^{-x}(C_1x + C_2x \ln x).$$

Exercise

Solve $x^2y'' + 3xy' + y = 0$.

Solution

1. Euler–Cauchy equation. Look for $y = x^m$. Substituting: $m(m - 1) + 3m + 1 = 0 \Rightarrow m^2 + 2m + 1 = (m + 1)^2 = 0$, double root $m = -1$.

2. General solution. For a double root,

$$y(x) = (C_1 + C_2 \ln x)x^{-1}.$$

Exercise

Solve $y'' + \frac{1}{x}y' + \left(1 - \frac{1}{4x^2}\right)y = 0$.

Solution

1. Bessel equation. Multiply by x^2 : $x^2 y'' + xy' + (x^2 - \frac{1}{4})y = 0$, so $\nu = \frac{1}{2}$.

2. Bessel functions:

$$J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \sin x, \quad Y_{1/2}(x) = -\sqrt{\frac{2}{\pi x}} \cos x.$$

3. General solution:

$$y(x) = \frac{1}{\sqrt{x}} (A \sin x + B \cos x).$$

Exercise

Solve $(1 - x^2)y'' - 2xy' + 2y = 0$.

Solution

1. Legendre equation. Here $n(n+1) = 2$, so $n = 1$ (taking the positive integer root).

2. First solution: $P_1(x) = x$.

3. Second solution:

$$Q_1(x) = \frac{x}{2} \ln\left(\frac{1+x}{1-x}\right) - 1.$$

4. General solution:

$$y(x) = C_1 x + C_2 \left[\frac{x}{2} \ln\left(\frac{1+x}{1-x}\right) - 1 \right].$$

Exercise

Solve $y'' + (\tan x)y' + (\cos^2 x)y = 0$.

Solution

1. Change of variable. Let $z = \sin x$, $\frac{dz}{dx} = \cos x$. Then

$$y' = \cos x \frac{dy}{dz}, \quad y'' = \cos^2 x \frac{d^2 y}{dz^2} - \sin x \frac{dy}{dz}.$$

2. Substitute: $\cos^2 x \ddot{y} - \sin x \dot{y} + \tan x (\cos x \dot{y}) + \cos^2 x y = 0$. Since $\tan x \cos x = \sin x$, the $\dot{y}$ terms cancel, leaving $\cos^2 x (\ddot{y} + y) = 0$. For $\cos x \neq 0$, $\ddot{y} + y = 0$.

3. Solution in z : $y(z) = A \cos z + B \sin z$.

4. Back to x : $z = \sin x$, so

$$y(x) = A \cos(\sin x) + B \sin(\sin x).$$

Exercise

Solve $y''' - y + 1 = 0$.

Solution

1. Operator form. $(D^3 - 1)y = -1$.

2. Homogeneous solution. Characteristic: $\lambda^3 - 1 = 0$, roots $1, \frac{1}{2} \pm i\frac{\sqrt{3}}{2}$. Thus

$$y_h = C_1 e^x + e^{x/2} \left(C_2 \cos \frac{\sqrt{3}}{2} x + C_3 \sin \frac{\sqrt{3}}{2} x \right).$$

3. Particular solution. Constant $y_p = A$ gives $-A = -1 \Rightarrow A = 1$.

4. General solution.

$$y(x) = y_h + 1.$$

Exercise

Solve $y''' - 3y'' + 3y' - y = 0$.

Solution

1. Operator form. $(D^3 - 3D^2 + 3D - 1)y = 0$.

2. Characteristic polynomial. $(\lambda - 1)^3 = 0$, triple root $\lambda = 1$.

3. General solution.

$$y(x) = (C_1 + C_2 x + C_3 x^2) e^x.$$

Exercise

Solve $y''' + y'' + y' + y = 4$.

Solution

1. Operator form. $(D^3 + D^2 + D + 1)y = 4$.

2. Characteristic polynomial. $\lambda^3 + \lambda^2 + \lambda + 1 = (\lambda + 1)(\lambda^2 + 1) = 0$, roots $\lambda = -1, \pm i$.

$$y_h = C_1 e^{-x} + C_2 \cos x + C_3 \sin x.$$

3. Particular solution. $y_p = A$ constant: $0 + 0 + 0 + A = 4 \Rightarrow A = 4$.

4. General solution.

$$y(x) = C_1 e^{-x} + C_2 \cos x + C_3 \sin x + 4.$$

Exercise

Solve $y''' - 2y'' - y' + 2y = 0$, $y(0) = 1$, $y'(0) = 0$, $y''(0) = 2$.

Solution

1. Operator form. $(D^3 - 2D^2 - D + 2)y = 0$.

2. Characteristic polynomial. $\lambda^3 - 2\lambda^2 - \lambda + 2 = (\lambda - 1)(\lambda - 2)(\lambda + 1) = 0$, roots $\lambda = 1, 2, -1$.

$$y(x) = C_1 e^x + C_2 e^{2x} + C_3 e^{-x}.$$

3. Apply conditions.

$$y(0) = C_1 + C_2 + C_3 = 1,$$

$$y'(0) = C_1 + 2C_2 - C_3 = 0,$$

$$y''(0) = C_1 + 4C_2 + C_3 = 2.$$

Solving: $C_1 = 0$, $C_2 = \frac{1}{3}$, $C_3 = \frac{2}{3}$.

4. Final solution.

$$y(x) = \frac{1}{3}e^{2x} + \frac{2}{3}e^{-x}.$$

Exercise

Solve $xy'' + 2xy' + (x - 1)y = 0$.

Solution

1. Substitution. Let $y = e^{-x}u$. Then

$$y' = e^{-x}(u' - u), \quad y'' = e^{-x}(u'' - 2u' + u).$$

Substitute and simplify: $xu'' - u = 0$ or $u'' - \frac{1}{x}u = 0$.

2. Frobenius. The indicial equation $r(r - 1) = 0$ gives $r = 0, 1$. Solutions: $u_1 = x$, $u_2 = x \ln x$.

3. Back to y :

$$y(x) = e^{-x}(C_1x + C_2x \ln x).$$

Exercise

Solve $x^2y'' + 3xy' + y = 0$.

Solution

1. Euler–Cauchy. Substitute $y = x^m$: $m(m - 1) + 3m + 1 = m^2 + 2m + 1 = (m + 1)^2 = 0$, double root $m = -1$.

2. General solution.

$$y(x) = (C_1 + C_2 \ln x)x^{-1}.$$

Exercise

Solve $y'' + \frac{1}{x}y' + \left(1 - \frac{1}{4x^2}\right)y = 0$.

Solution

1. Bessel form. Multiply by x^2 : $x^2y'' + xy' + (x^2 - \frac{1}{4})y = 0$, so $\nu = \frac{1}{2}$.

2. Bessel functions.

$$J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \sin x, \quad Y_{1/2}(x) = -\sqrt{\frac{2}{\pi x}} \cos x.$$

3. Elementary form.

$$y(x) = \frac{1}{\sqrt{x}} (A \sin x + B \cos x).$$

Exercise

Solve $(1 - x^2)y'' - 2xy' + 2y = 0$.

Solution

1. Legendre equation. $n(n+1) = 2 \Rightarrow n = 1$.

2. First solution: $P_1(x) = x$.

3. Second solution: $Q_1(x) = \frac{x}{2} \ln\left(\frac{1+x}{1-x}\right) - 1$.

4. General solution.

$$y(x) = C_1 x + C_2 \left[\frac{x}{2} \ln\left(\frac{1+x}{1-x}\right) - 1 \right].$$

Exercise

Solve $y'' + (\tan x)y' + (\cos^2 x)y = 0$.

Solution

1. Change $z = \sin x$. Then $\frac{d}{dx} = \cos x \frac{d}{dz}$, and

$$y'' = \cos^2 x \frac{d^2 y}{dz^2} - \sin x \frac{dy}{dz}.$$

2. Substitution. After simplification, the terms $\frac{dy}{dz}$ cancel, leaving $\frac{d^2 y}{dz^2} + y = 0$.

3. Solution in z : $y(z) = A \cos z + B \sin z$.

4. Back to x : $z = \sin x$, so

$$y(x) = A \cos(\sin x) + B \sin(\sin x).$$

Exercise

Study the motion of a simple pendulum of length L and mass m . Consider small and large oscillations.

Solution

1. Equation of motion. Projecting forces: $mL\ddot{\theta} = -mg \sin \theta$, so

$$\ddot{\theta} + \frac{g}{L} \sin \theta = 0.$$

2. Small oscillations. $\sin \theta \approx \theta$:

$$\ddot{\theta} + \frac{g}{L} \theta = 0 \implies \theta(t) = \theta_m \cos\left(\sqrt{\frac{g}{L}} t + \varphi\right), \quad T_0 = 2\pi \sqrt{\frac{L}{g}}.$$

3. Large oscillations. Exact energy conservation: multiply by $\dot{\theta}$ and integrate:

$$\frac{1}{2}\dot{\theta}^2 - \frac{g}{L}\cos\theta = \text{constant}.$$

With $\theta(0) = \theta_0$, $\dot{\theta}(0) = 0$, we get $\dot{\theta}^2 = \frac{2g}{L}(\cos\theta - \cos\theta_0)$. The period is

$$T = 4\sqrt{\frac{L}{g}} \int_0^{\pi/2} \frac{d\phi}{\sqrt{1 - k^2 \sin^2 \phi}}, \quad k = \sin \frac{\theta_0}{2}.$$

4. Conclusion.

$$\begin{cases} T_0 = 2\pi\sqrt{L/g}, & \text{small oscillations,} \\ T = 4\sqrt{L/g} K(\sin \frac{\theta_0}{2}), & \text{large oscillations,} \end{cases}$$

where $K(k)$ is the complete elliptic integral of the first kind.

Exercise

An RL series circuit (resistance R and inductance L) is supplied by a DC voltage source E applied at time $t = 0$.

Determine:

1. The differential equation for the current $i(t)$.
2. The expression for $i(t)$ and the time constant τ .
3. The voltages across R and L : $U_R(t)$ and $U_L(t)$.
4. Plot the curves and determine τ graphically.
5. Verify τ using tangents at selected points.

Solution

1. Differential equation

Kirchhoff's voltage law gives:

$$E - Ri(t) - L\frac{di}{dt} = 0 \quad \Rightarrow \quad \frac{di}{dt} + \frac{R}{L}i = \frac{E}{L}$$

Set $\alpha = \frac{R}{L}$ and $\beta = \frac{E}{L}$:

$$\frac{di}{dt} + \alpha i = \beta$$

2. Solution using the integrating factor

Multiply by $e^{\alpha t}$:

$$e^{\alpha t} \frac{di}{dt} + \alpha e^{\alpha t} i = \beta e^{\alpha t}$$

The left-hand side is the derivative of $i(t)e^{\alpha t}$:

$$\frac{d}{dt} [i(t)e^{\alpha t}] = \beta e^{\alpha t}$$

Integrate both sides:

$$i(t)e^{\alpha t} = \frac{\beta}{\alpha}e^{\alpha t} + C$$

where C is the integration constant.

Divide by $e^{\alpha t}$:

$$i(t) = \frac{\beta}{\alpha} + Ce^{-\alpha t}$$

Now $\frac{\beta}{\alpha} = \frac{E/L}{R/L} = \frac{E}{R}$. Thus:

$$i(t) = \frac{E}{R} + Ce^{-\frac{R}{L}t}$$

3. Initial condition

At $t = 0$, the inductor opposes sudden current changes, so $i(0) = 0$:

$$0 = \frac{E}{R} + C \Rightarrow C = -\frac{E}{R}$$

Hence:

$$i(t) = \frac{E}{R} \left(1 - e^{-\frac{R}{L}t}\right)$$

Define:

$$I_0 = \frac{E}{R} \quad \text{and} \quad \boxed{\tau = \frac{L}{R}}$$

Thus:

$$\boxed{i(t) = I_0 \left(1 - e^{-t/\tau}\right)}$$

4. Voltages across components

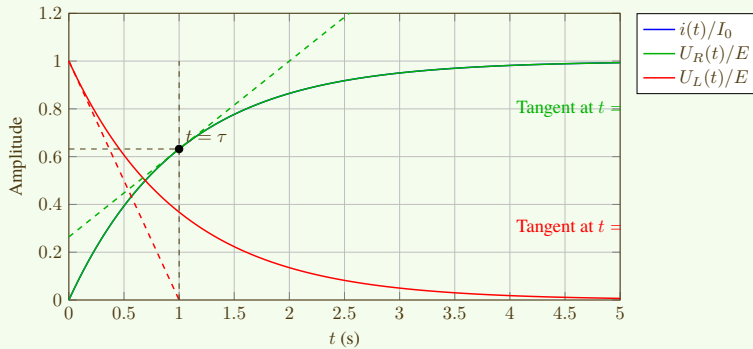
$$U_R(t) = Ri(t) = \boxed{E \left(1 - e^{-t/\tau}\right)}$$

$$U_L(t) = L \frac{di}{dt} = L \cdot \frac{d}{dt} \left[I_0 \left(1 - e^{-t/\tau}\right) \right] = L \cdot \frac{I_0}{\tau} e^{-t/\tau}$$

Since $\frac{I_0}{\tau} = \frac{E}{R} \cdot \frac{R}{L} = \frac{E}{L}$, we obtain:

$$\boxed{U_L(t) = E e^{-t/\tau}}$$

5. Graphical determination of τ



6. Methods for graphical determination of τ

- **Method 1:** Value at 63% of final: $i(\tau) = I_0(1 - e^{-1}) \approx 0.632I_0$
- **Method 2:** Tangent to $U_L(t)$ at $t = 0$:

$$\text{Slope} = \left. \frac{dU_L}{dt} \right|_{t=0} = -\frac{E}{\tau}$$

This tangent intersects the time axis at $t = \tau$.

- **Method 3:** Tangent to $U_R(t)$ at $t = \tau$:

$$\text{Slope} = \left. \frac{dU_R}{dt} \right|_{t=\tau} = \frac{E}{\tau} e^{-1} \approx 0.368 \frac{E}{\tau}$$

This tangent meets the asymptote $U_R = E$ at $t = 2\tau$.

7. Physical interpretation

- At $t = 0^+$: $i = 0$, $U_R = 0$, $U_L = E$ (inductor opposes current change)
- As $t \rightarrow \infty$: $i = I_0$, $U_R = E$, $U_L = 0$ (steady state)

$$\tau = \frac{L}{R} \quad \text{time constant of the RL circuit}$$

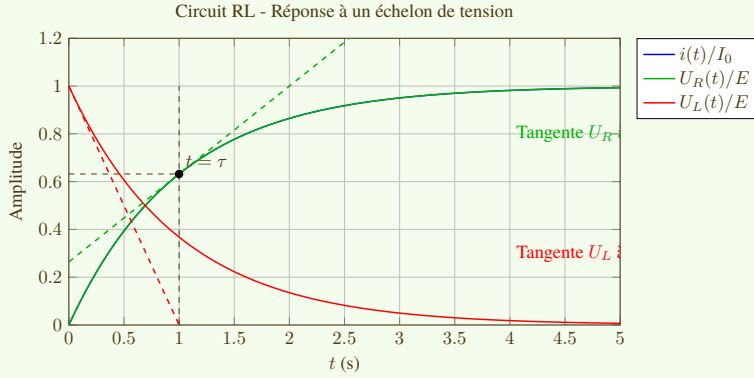
Exercise

An RL series circuit (resistance R and inductance L) is supplied by a DC voltage source E applied at $t = 0$.

Determine:

1. The differential equation for $i(t)$.
2. The expression for $i(t)$ and the time constant τ .
3. The voltages $U_R(t)$ and $U_L(t)$.
4. The energy stored in the inductor $W_L(t)$.
5. Plot the curves and determine τ graphically.
6. Verify τ using tangents.
7. Study the discharge phase of the inductor.

Solution



1. Differential equation and solution

Same as Exercise 11.21, we obtain:

$$i(t) = I_0 \left(1 - e^{-t/\tau} \right) \quad \text{with} \quad I_0 = \frac{E}{R}, \quad \tau = \frac{L}{R}$$

2. Voltages

$$U_R(t) = E \left(1 - e^{-t/\tau} \right), \quad U_L(t) = E e^{-t/\tau}$$

3. Energy stored in the inductor

$$W_L(t) = \frac{1}{2} L i^2(t) = \frac{1}{2} L I_0^2 \left(1 - e^{-t/\tau} \right)^2$$

At steady state ($t \rightarrow \infty$):

$$W_L(\infty) = \frac{1}{2} L I_0^2 = \frac{1}{2} L \left(\frac{E}{R} \right)^2$$

4. Discharge phase

When the circuit is suddenly opened after steady state, the inductor maintains the current momentarily:

$$L \frac{di}{dt} + Ri = 0 \quad \Rightarrow \quad i(t) = I_0 e^{-t/\tau}$$

$$U_L(t) = -R I_0 e^{-t/\tau} \quad (\text{large negative voltage!})$$

This negative voltage can cause sparks across the switch.

5. Limit values

- At $t = 0^+$ (charging): $i = 0$, $U_R = 0$, $U_L = E$, $W_L = 0$
- At $t \rightarrow \infty$ (charging): $i = I_0$, $U_R = E$, $U_L = 0$, $W_L = \frac{1}{2} L I_0^2$

$$\tau = \frac{L}{R}$$

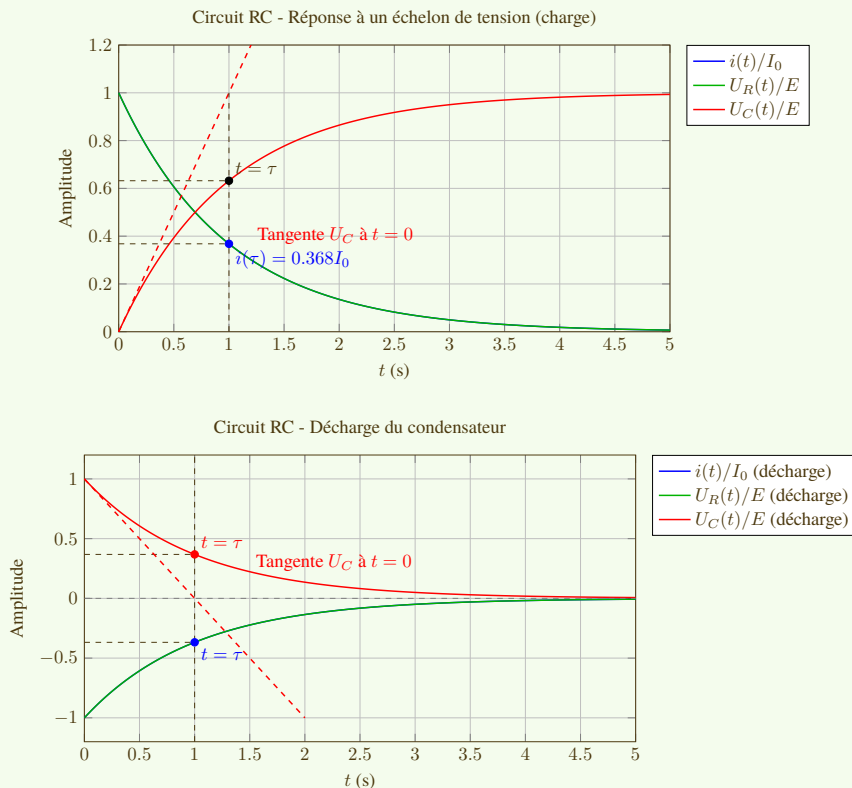
Exercise

An RC series circuit (resistance R and capacitance C) is supplied by a DC voltage source E applied at $t = 0$. The capacitor is initially uncharged.

Determine:

1. The differential equations for $q(t)$ and $i(t)$.
2. The expression for $i(t)$ and the time constant τ .
3. The voltages $U_R(t)$ and $U_C(t)$.
4. The energy stored in the capacitor $W_C(t)$.
5. Plot the curves and determine τ graphically.
6. Study the discharge phase of the capacitor.
7. Compare with the RL circuit behaviour.

Solution



1. Differential equation

Kirchhoff's voltage law:

$$E - Ri(t) - \frac{q(t)}{C} = 0$$

With $i(t) = \frac{dq}{dt}$, we obtain:

$$R\frac{dq}{dt} + \frac{q}{C} = E \quad \Rightarrow \quad \frac{dq}{dt} + \frac{1}{RC}q = \frac{E}{R}$$

Set $\alpha = \frac{1}{RC}$, $\beta = \frac{E}{R}$:

$$\frac{dq}{dt} + \alpha q = \beta$$

2. Solution for charge and current

Using the integrating factor $e^{\alpha t}$:

$$q(t) = \frac{\beta}{\alpha} + Ke^{-\alpha t} = EC + Ke^{-t/(RC)}$$

Initial condition $q(0) = 0$ gives $K = -EC$, so:

$$q(t) = EC \left(1 - e^{-t/(RC)}\right)$$

The current is:

$$i(t) = \frac{dq}{dt} = \frac{E}{R}e^{-t/(RC)}$$

Define $I_0 = \frac{E}{R}$ and $\tau = RC$:

$$\boxed{i(t) = I_0 e^{-t/\tau}},$$

$$\boxed{q(t) = EC \left(1 - e^{-t/\tau}\right)}$$

3. Voltages

$$U_R(t) = Ri(t) = \boxed{E e^{-t/\tau}}, \quad U_C(t) = \frac{q(t)}{C} = \boxed{E \left(1 - e^{-t/\tau}\right)}$$

4. Energy stored in the capacitor

$$W_C(t) = \frac{1}{2}CU_C^2(t) = \frac{1}{2}CE^2 \left(1 - e^{-t/\tau}\right)^2$$

At steady state: $W_C(\infty) = \frac{1}{2}CE^2$

5. Discharge phase

When the charged capacitor is connected to R alone (source removed) at $t = 0$:

$$Ri(t) + \frac{q(t)}{C} = 0, \quad i(t) = -\frac{dq}{dt}$$

$$\frac{dq}{dt} + \frac{1}{RC}q = 0 \quad \Rightarrow \quad q(t) = CE e^{-t/\tau}$$

$$i(t) = -\frac{E}{R}e^{-t/\tau}, \quad U_C(t) = E e^{-t/\tau}, \quad U_R(t) = -E e^{-t/\tau}$$

6. Comparison between RC and RL circuits

Aspect	RL Circuit	RC Circuit
Time constant	$\tau = L/R$	$\tau = RC$
Current (charge)	$I_0(1 - e^{-t/\tau})$	$I_0 e^{-t/\tau}$
Reactive voltage	$U_L = E e^{-t/\tau}$	$U_C = E(1 - e^{-t/\tau})$
Stored energy	$W_L = \frac{1}{2} L I_0^2 (1 - e^{-t/\tau})^2$	$W_C = \frac{1}{2} C E^2 (1 - e^{-t/\tau})^2$
At $t = 0^+$ (charge)	Inductor = open circuit	Capacitor = short circuit
At $t \rightarrow \infty$ (charge)	Inductor = wire	Capacitor = open circuit
Discharge current	$i(t) = I_0 e^{-t/\tau}$	$i(t) = -I_0 e^{-t/\tau}$
Discharge voltage	$U_L = -R I_0 e^{-t/\tau}$	$U_C = E e^{-t/\tau}$

7. Limit values (charging phase)

- At $t = 0^+$: $i = I_0$, $U_R = E$, $U_C = 0$, $W_C = 0$
- At $t \rightarrow \infty$: $i = 0$, $U_R = 0$, $U_C = E$, $W_C = \frac{1}{2} C E^2$

$$\tau = RC$$

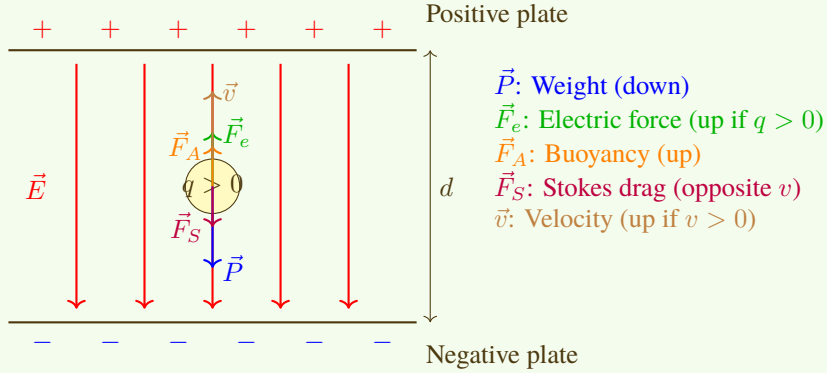
Exercise

In Millikan's experiment, a spherical oil drop of radius r is placed between two vertical parallel plates separated by a distance d with a uniform electric field $\vec{E}$. The drop is subject to:

1. Weight: $\vec{P} = m\vec{g}$, $m = \rho V$, $V = \frac{4}{3}\pi r^3$
2. Electric force: $\vec{F}_e = q\vec{E}$, $E = U/d$, $q = ne$
3. Buoyancy force: $\vec{F}_A = -m_0\vec{g}$, $m_0 = \rho_0 V$
4. Viscous drag (Stokes' law): $\vec{F}_S = -k\vec{v}$

Determine:

1. A diagram showing all forces.
2. The differential equation of motion.
3. The solution for $v(t)$.
4. The terminal velocity v_∞ and time constant τ .
5. The graph of $v(t)$.
6. The expression for q when the drop is stationary.
7. How Millikan determined the elementary charge e .
8. The case where the electric field is reversed.

Solution
1. Force diagram

2. Differential equation

Applying Newton's second law (upward positive):

$$m \frac{dv}{dt} = F_e + F_A - P - F_S$$

Substituting expressions:

$$\frac{4}{3}\pi r^3 \rho \frac{dv}{dt} = qE + \frac{4}{3}\pi r^3 \rho_0 g - \frac{4}{3}\pi r^3 \rho g - 6\pi \eta r v$$

Rearranging:

$$\frac{dv}{dt} + \frac{9\eta}{2\rho r^2} v = g \left(\frac{\rho_0}{\rho} - 1 \right) + \frac{3qE}{4\pi r^3 \rho}$$

Thus:

$$\boxed{\frac{dv}{dt} + \alpha v = \beta}$$

with $\alpha = \frac{9\eta}{2\rho r^2}$ and $\beta = g \left(\frac{\rho_0}{\rho} - 1 \right) + \frac{3qE}{4\pi r^3 \rho}$.

3. Solution

Using the integrating factor $e^{\alpha t}$:

$$v(t) = \frac{\beta}{\alpha} + \left(v_0 - \frac{\beta}{\alpha} \right) e^{-\alpha t}$$

4. Terminal velocity and time constant

$$v_\infty = \frac{\beta}{\alpha} = \frac{2r^2 g}{9\eta} (\rho_0 - \rho) + \frac{qE}{6\pi \eta r}$$

$$\boxed{\tau = \frac{1}{\alpha} = \frac{2\rho r^2}{9\eta}}$$

5. Stationary drop condition ($v_\infty = 0$)

$$\frac{2r^2g}{9\eta}(\rho_0 - \rho) + \frac{qE}{6\pi\eta r} = 0$$

Multiplying by $6\pi\eta r$:

$$\frac{4}{3}\pi r^3g(\rho_0 - \rho) + qE = 0$$

Thus:

$$q = \frac{4\pi r^3g}{3E}(\rho - \rho_0)$$

6. Determination of the elementary charge

Millikan observed that all measured charges were integer multiples of:

$$e = 1.602 \times 10^{-19} \text{ C} \quad \Rightarrow \quad \boxed{q = ne}$$

7. Reversed electric field

If the field is reversed ($E \rightarrow -E$), the electric force changes sign:

$$v'_\infty = \frac{2r^2g}{9\eta}(\rho_0 - \rho) - \frac{qE}{6\pi\eta r}$$

8. Physical interpretation

- $\tau \propto r^2$: larger drops have longer time constants
- $\tau \propto \rho$: denser drops respond more slowly
- The Stokes drag is valid for $Re = \frac{\rho v r}{\eta} \ll 1$ (laminar flow)

$$\boxed{\tau = \frac{2\rho r^2}{9\eta}},$$

$$\boxed{v_\infty = \frac{2r^2g}{9\eta}(\rho_0 - \rho) + \frac{qE}{6\pi\eta r}}$$

Numerical Series

Theorem

[Fundamental Numerical Series] **Definition of a series:** A numerical series is an infinite sum of the form:

$$\sum_{n=0}^{+\infty} u_n = u_0 + u_1 + u_2 + \cdots$$

where (u_n) is a sequence of real or complex numbers.

Partial sum: The partial sum of order N is:

$$S_N = \sum_{n=0}^N u_n$$

Convergence: The series converges if the sequence (S_N) converges to a finite limit S .

Necessary condition for convergence: If $\sum u_n$ converges, then $\lim_{n \rightarrow \infty} u_n = 0$.

Geometric series: For $a \neq 0$ and $q \in \mathbb{R}$:

$$\sum_{n=0}^{+\infty} aq^n \text{ converges } \iff |q| < 1$$

In this case: $\sum_{n=0}^{+\infty} aq^n = \frac{a}{1-q}$.

Riemann series: The series $\sum_{n=1}^{+\infty} \frac{1}{n^\alpha}$:

$$\begin{cases} \text{converges} & \text{if } \alpha > 1 \\ \text{diverges} & \text{if } \alpha \leq 1 \end{cases}$$

Harmonic series: Special case $\alpha = 1$:

$$\sum_{n=1}^{+\infty} \frac{1}{n} \text{ diverges}$$

Ratio test (d'Alembert): Let $\sum u_n$ with $u_n > 0$. If $\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \ell$:

$$\begin{cases} \ell < 1 & \text{converges} \\ \ell > 1 & \text{diverges} \\ \ell = 1 & \text{inconclusive} \end{cases}$$

Root test (Cauchy): Let $\sum u_n$ with $u_n \geq 0$. If $\lim_{n \rightarrow \infty} \sqrt[n]{u_n} = \ell$:

$$\begin{cases} \ell < 1 & \text{converges} \\ \ell > 1 & \text{diverges} \\ \ell = 1 & \text{inconclusive} \end{cases}$$

Alternating series test (Leibniz): Let $\sum (-1)^n a_n$ with $a_n \geq 0$. If:

1. (a_n) is decreasing
2. $\lim_{n \rightarrow \infty} a_n = 0$

then the series converges.

Absolute convergence: A series $\sum u_n$ is absolutely convergent if $\sum |u_n|$ converges.

Absolutely convergent $\Rightarrow$ Convergent

Alternating harmonic series:

$$\sum_{n=1}^{+\infty} \frac{(-1)^{n-1}}{n} = \ln 2 \quad (\text{converges but not absolutely})$$

Exponential series: For all $x \in \mathbb{C}$:

$$\sum_{n=0}^{+\infty} \frac{x^n}{n!} = e^x \quad (\text{absolutely convergent})$$

Integral test: Let $f : [1, +\infty[\rightarrow \mathbb{R}^+$ be continuous and decreasing. Then:

$$\sum_{n=1}^{+\infty} f(n) \text{ and } \int_1^{+\infty} f(x) dx \text{ have the same nature}$$

Telescoping series: If $u_n = v_n - v_{n+1}$, then:

$$\sum_{n=0}^N u_n = v_0 - v_{N+1}$$

and $\sum u_n$ converges iff (v_n) converges.

Important classical series:

- Gregory-Leibniz series: $\sum_{n=0}^{+\infty} \frac{(-1)^n}{2n+1} = \frac{\pi}{4}$

- Basel series: $\sum_{n=1}^{+\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$

- Differentiated geometric series: $\sum_{n=1}^{+\infty} nx^{n-1} = \frac{1}{(1-x)^2}$ for $|x| < 1$

Remainders of a convergent series: If $\sum u_n$ converges, then the remainder of order N :

$$R_N = \sum_{n=N+1}^{+\infty} u_n \rightarrow 0 \quad \text{as } N \rightarrow \infty$$

Remainder bound (alternating series): For an alternating series satisfying Leibniz's conditions:

$$|R_N| \leq a_{N+1}$$

where a_{N+1} is the absolute value of the first omitted term.

Exercise

Determine the nature of the series $\sum_{n=1}^{+\infty} \frac{1}{n(n+1)}$ and compute its sum.

Solution

1. Partial fraction decomposition

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$$

2. Partial sum

$$S_N = \sum_{n=1}^N \left(\frac{1}{n} - \frac{1}{n+1} \right)$$

This is a telescoping sum:

$$S_N = 1 - \frac{1}{N+1}$$

3. Limit

$$\lim_{N \rightarrow \infty} S_N = \lim_{N \rightarrow \infty} \left(1 - \frac{1}{N+1} \right) = 1$$

Conclusion

The series converges and:

$$\boxed{\sum_{n=1}^{+\infty} \frac{1}{n(n+1)} = 1}$$

Exercise

Study the convergence of $\sum_{n=1}^{+\infty} \frac{n^2}{2^n}$.

Solution

1. Apply the ratio test (d'Alembert)

Let $u_n = \frac{n^2}{2^n}$. Then:

$$\frac{u_{n+1}}{u_n} = \frac{(n+1)^2}{2^{n+1}} \cdot \frac{2^n}{n^2} = \frac{1}{2} \cdot \frac{(n+1)^2}{n^2}$$

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \frac{1}{2} \cdot 1 = \frac{1}{2} < 1$$

2. Conclusion

By the ratio test, the series converges.

Exercise

Study the convergence of $\sum_{n=1}^{+\infty} \frac{(-1)^n}{\sqrt{n}}$.

Solution

1. Verify Leibniz's test conditions

Let $a_n = \frac{1}{\sqrt{n}}$.

1. $a_{n+1} = \frac{1}{\sqrt{n+1}} \leq \frac{1}{\sqrt{n}} = a_n$: the sequence is decreasing

2. $\lim_{n \rightarrow \infty} a_n = 0$

2. Apply Leibniz's test

Both conditions are satisfied, so the alternating series converges.

3. Absolute convergence

The series $\sum_{n=1}^{+\infty} \left| \frac{(-1)^n}{\sqrt{n}} \right| = \sum_{n=1}^{+\infty} \frac{1}{\sqrt{n}}$ diverges (Riemann series with $\alpha = \frac{1}{2} \leq 1$).

Conclusion

The series converges (by Leibniz) but not absolutely.

Exercise

Show that the harmonic series $\sum_{n=1}^{+\infty} \frac{1}{n}$ diverges.

Solution

1. Classical grouping method

Group the terms as follows:

$$\begin{aligned} & 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4} \right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} \right) + \cdots \\ & > 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4} \right) + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} \right) + \cdots \\ & = 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \cdots \end{aligned}$$

This last sum tends to infinity.

2. Comparison with an integral

For $x \in [n, n+1]$, we have $\frac{1}{x} \leq \frac{1}{n}$, so:

$$\int_1^{N+1} \frac{1}{x} dx \leq \sum_{n=1}^N \frac{1}{n}$$

But $\int_1^{N+1} \frac{1}{x} dx = \ln(N+1) \rightarrow +\infty$.

Conclusion

The harmonic series diverges to $+\infty$.

Exercise

Determine the nature of $\sum_{n=1}^{+\infty} \frac{\sin n}{n^2}$.

Solution

1. Bound the general term

$$\left| \frac{\sin n}{n^2} \right| \leq \frac{1}{n^2}$$

2. Compare with a convergent series

The series $\sum_{n=1}^{+\infty} \frac{1}{n^2}$ converges (Riemann series with $\alpha = 2 > 1$).

3. Apply comparison test

By comparison, $\sum_{n=1}^{+\infty} \left| \frac{\sin n}{n^2} \right|$ converges.

Conclusion

The series converges absolutely, therefore it converges.

Exercise

Compute $\sum_{n=0}^{+\infty} \frac{1}{2^n}$.

Solution

1. Recognize a geometric series

This is a geometric series with $a = 1$ and $q = \frac{1}{2}$.

2. Convergence condition

$|q| = \frac{1}{2} < 1$, so the series converges.

3. Compute the sum

$$\sum_{n=0}^{+\infty} \frac{1}{2^n} = \frac{1}{1 - \frac{1}{2}} = 2$$

Conclusion

$$\sum_{n=0}^{+\infty} \frac{1}{2^n} = 2$$

Exercise

Show that for all $x \in]-1, 1[$:

$$\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}.$$

Solution

Let $S_n = 1 + x + x^2 + \cdots + x^n$ be the n -th partial sum.

1. Compute S_n

Multiply S_n by $1 - x$:

$$(1-x)S_n = (1-x)(1+x+x^2+\cdots+x^n).$$

Expanding:

$$(1-x)S_n = (1+x+x^2+\cdots+x^n) - (x+x^2+x^3+\cdots+x^{n+1}).$$

All terms cancel in pairs, leaving:

$$(1 - x)S_n = 1 - x^{n+1}.$$

Thus, for $x \neq 1$,

$$S_n = \frac{1 - x^{n+1}}{1 - x}.$$

2. Limit

If $|x| < 1$, then $x^{n+1} \rightarrow 0$ as $n \rightarrow \infty$. Hence:

$$\lim_{n \rightarrow \infty} S_n = \frac{1}{1 - x}.$$

By definition of the sum of a series:

$$\sum_{k=0}^{\infty} x^k = \frac{1}{1 - x}.$$

Conclusion:

$$\sum_{k=0}^{\infty} x^k = \frac{1}{1 - x}, \quad \text{for } |x| < 1.$$

Remark: The series diverges for $|x| \geq 1$.

Exercise

Show that for all $x \in]-1, 1[$:

$$\sum_{k=0}^{\infty} (-x)^k = \frac{1}{1 + x}.$$

Solution

Method 1: Direct application of the theorem

For $x \in]-1, 1[$, we have $|-x| = |x| < 1$. Applying the geometric series to $-x$ gives:

$$\sum_{k=0}^{\infty} (-x)^k = \frac{1}{1 - (-x)} = \frac{1}{1 + x}.$$

Method 2: Direct proof

Let

$$S_n = 1 - x + x^2 - x^3 + \cdots + (-x)^n.$$

Multiply by $1 + x$:

$$(1 + x)S_n = (1 + x)(1 - x + x^2 - x^3 + \cdots + (-x)^n).$$

Expanding and simplifying:

$$(1+x)S_n = 1 + (-1)^n x^{n+1}.$$

Thus, for $x \neq -1$,

$$S_n = \frac{1 + (-1)^n x^{n+1}}{1+x}.$$

If $|x| < 1$, then $x^{n+1} \rightarrow 0$, so:

$$\lim_{n \rightarrow \infty} S_n = \frac{1}{1+x}.$$

Conclusion:

$$\sum_{k=0}^{\infty} (-x)^k = \frac{1}{1+x}.$$

Exercise

For $|x| < 1$, show that:

$$\sum_{n=1}^{+\infty} nx^n = \frac{x}{(1-x)^2}.$$

Solution

Start from the geometric series:

$$\sum_{n=0}^{+\infty} x^n = \frac{1}{1-x}, \quad |x| < 1.$$

Differentiate term by term with respect to x :

$$\sum_{n=1}^{+\infty} nx^{n-1} = \frac{1}{(1-x)^2}.$$

Multiply both sides by x :

$$\sum_{n=1}^{+\infty} nx^n = \frac{x}{(1-x)^2}.$$

This completes the proof.

Exercise

For $|x| < 1$ and any integer $k \geq 1$, show that:

$$\sum_{n=k}^{+\infty} n(n-1) \cdots (n-k+1)x^n = \frac{k!x^k}{(1-x)^{k+1}}.$$

Solution

Consider the geometric series:

$$\sum_{n=0}^{+\infty} x^n = \frac{1}{1-x}, \quad |x| < 1.$$

Differentiate k times term by term with respect to x :

$$\sum_{n=k}^{+\infty} n(n-1) \cdots (n-k+1)x^{n-k} = \frac{k!}{(1-x)^{k+1}}.$$

Multiply both sides by x^k :

$$\sum_{n=k}^{+\infty} n(n-1) \cdots (n-k+1)x^n = \frac{k!x^k}{(1-x)^{k+1}}.$$

The formula is thus proved.

Exercise

Compute the following integral using a geometric series:

$$I = \int_0^1 \frac{1}{1+x^2} dx.$$

Solution

Recall that for $|t| < 1$,

$$\frac{1}{1-t} = \sum_{k=0}^{\infty} t^k.$$

1. Series expansion

Write:

$$\frac{1}{1+x^2} = \frac{1}{1-(-x^2)}.$$

Setting $t = -x^2$, for $x \in [0, 1[$:

$$\frac{1}{1+x^2} = \sum_{k=0}^{\infty} (-1)^k x^{2k}.$$

2. Termwise integration

The convergence is uniform on every segment $[0, r]$ with $0 < r < 1$, so we can integrate term by term:

$$I = \int_0^1 \sum_{k=0}^{\infty} (-1)^k x^{2k} dx = \sum_{k=0}^{\infty} (-1)^k \int_0^1 x^{2k} dx.$$

3. Elementary calculation

$$\int_0^1 x^{2k} dx = \frac{1}{2k+1}.$$

Thus:

$$I = \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1}.$$

4. Recognize the series

This is the Gregory-Leibniz series:

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} = \frac{\pi}{4}.$$

Conclusion

$$\int_0^1 \frac{1}{1+x^2} dx = \frac{\pi}{4}.$$

Exercise

Compute the following integral using a series expansion:

$$I = \int_0^{\infty} \ln(1 - e^{-x}) dx.$$

Solution

For $x > 0$, we have $0 < e^{-x} < 1$, so the series expansion is valid.

1. Series expansion of the logarithm

Recall that for $|u| < 1$,

$$\ln(1 - u) = - \sum_{n=1}^{\infty} \frac{u^n}{n}.$$

Setting $u = e^{-x}$, we obtain:

$$\ln(1 - e^{-x}) = - \sum_{n=1}^{\infty} \frac{e^{-nx}}{n}.$$

2. Termwise integration

Integrate over $[0, +\infty[$:

$$I = - \int_0^{\infty} \sum_{n=1}^{\infty} \frac{e^{-nx}}{n} dx.$$

The convergence is uniform on every interval $[a, +\infty[$ with $a > 0$, allowing termwise integration:

$$I = - \sum_{n=1}^{\infty} \frac{1}{n} \int_0^{\infty} e^{-nx} dx.$$

3. Elementary integral

For $n \geq 1$,

$$\int_0^{\infty} e^{-nx} dx = \frac{1}{n}.$$

Thus:

$$I = - \sum_{n=1}^{\infty} \frac{1}{n^2}.$$

4. Value of the series

The Basel series is well known:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Conclusion

$$\int_0^{\infty} \ln(1 - e^{-x}) dx = -\frac{\pi^2}{6}.$$

Remarks:

1. The result is negative because $\ln(1 - e^{-x}) < 0$ for all $x > 0$.
2. This integral appears in statistical physics (black body radiation, Bose gas).

Exercise

Compute the following integral using a series expansion:

$$I = \int_0^{\infty} \ln(1 + e^{-x}) dx.$$

Solution

1. Series expansion of the logarithm

Recall that for $|u| < 1$,

$$\ln(1 + u) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{u^n}{n}.$$

For $x > 0$, we have $0 < e^{-x} < 1$. Setting $u = e^{-x}$, we obtain:

$$\ln(1 + e^{-x}) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{e^{-nx}}{n}.$$

2. Termwise integration

$$I = \int_0^{\infty} \ln(1 + e^{-x}) dx = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} \int_0^{\infty} e^{-nx} dx.$$

We have:

$$\int_0^{\infty} e^{-nx} dx = \frac{1}{n}.$$

Thus:

$$I = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2}.$$

3. Value of the series

It is known that:

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2} = (1 - 2^{1-2}) \zeta(2) = \frac{1}{2} \cdot \frac{\pi^2}{6} = \frac{\pi^2}{12}.$$

Conclusion

$$\boxed{\int_0^{\infty} \ln(1 + e^{-x}) dx = \frac{\pi^2}{12}.$$

Remarks:

1. The integral is positive because $\ln(1 + e^{-x}) > 0$.
2. We note the relation:

$$\int_0^{\infty} \ln(1 - e^{-x}) dx = -2 \int_0^{\infty} \ln(1 + e^{-x}) dx.$$

3. These integrals appear in statistical physics (Fermi gas).

Exercise

Determine the nature of the series:

$$1. \sum_{n=1}^{+\infty} \frac{n^3}{3^n}$$

$$2. \sum_{n=2}^{+\infty} \frac{1}{n \ln n}$$

$$3. \sum_{n=1}^{+\infty} \frac{n!}{n^n}$$

Solution

1. Study of $\sum_{n=1}^{+\infty} \frac{n^3}{3^n}$

Method: Ratio test (d'Alembert)

Let $u_n = \frac{n^3}{3^n}$. Compute:

$$\frac{u_{n+1}}{u_n} = \frac{(n+1)^3}{3^{n+1}} \cdot \frac{3^n}{n^3} = \frac{1}{3} \cdot \left(\frac{n+1}{n}\right)^3.$$

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \frac{1}{3} \cdot 1 = \frac{1}{3} < 1.$$

Conclusion

By the ratio test, the series converges.

2. Study of $\sum_{n=2}^{+\infty} \frac{1}{n \ln n}$

Method: Integral test

Consider $f(x) = \frac{1}{x \ln x}$ for $x \geq 2$.

$$\int_2^{\infty} \frac{dx}{x \ln x} = [\ln(\ln x)]_2^{\infty} = +\infty.$$

Conclusion

By the integral test, the series diverges.

3. Study of $\sum_{n=1}^{+\infty} \frac{n!}{n^n}$

Method: Ratio test (d'Alembert)

Let $u_n = \frac{n!}{n^n}$. Compute:

$$\frac{u_{n+1}}{u_n} = \frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} = \frac{n+1}{n+1} \cdot \left(\frac{n}{n+1}\right)^n = \left(1 - \frac{1}{n+1}\right)^n.$$

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \frac{1}{e} < 1.$$

Conclusion

By the ratio test, the series converges.

Summary

$$\sum_{n=1}^{+\infty} \frac{n^3}{3^n} \text{ converges, } \sum_{n=2}^{+\infty} \frac{1}{n \ln n} \text{ diverges, } \sum_{n=1}^{+\infty} \frac{n!}{n^n} \text{ converges.}$$

Exercise

Show that $\sum_{n=1}^{+\infty} \frac{1}{n^2}$ converges and that its sum is $\frac{\pi^2}{6}$.

Solution

1. Convergence

The series $\sum_{n=1}^{+\infty} \frac{1}{n^2}$ is a Riemann series with $\alpha = 2 > 1$, hence it converges.

2. Sum calculation (classical method)

Approach using Fourier series

Consider the function $f(x) = x^2$ on $[-\pi, \pi]$.

Fourier coefficients

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 dx = \frac{2\pi^2}{3},$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \cos(nx) dx = \frac{4(-1)^n}{n^2}.$$

Fourier series

$$x^2 = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos(nx).$$

Evaluate at $x = \pi$

$$\pi^2 = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} (-1)^n = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{1}{n^2}.$$

Solve for the sum

$$4 \sum_{n=1}^{\infty} \frac{1}{n^2} = \pi^2 - \frac{\pi^2}{3} = \frac{2\pi^2}{3}.$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Conclusion

$$\boxed{\sum_{n=1}^{+\infty} \frac{1}{n^2} = \frac{\pi^2}{6}}.$$

Exercise

Study the convergence of $\sum_{n=1}^{+\infty} \frac{(-1)^n}{n^p}$ according to the value of $p > 0$.

Solution

1. For any $p > 0$

Alternating series test (Leibniz)

Let $a_n = \frac{1}{n^p}$.

1. $a_{n+1} = \frac{1}{(n+1)^p} \leq \frac{1}{n^p} = a_n$: the sequence is decreasing
2. $\lim_{n \rightarrow \infty} a_n = 0$

Conclusion for simple convergence

For all $p > 0$, the alternating series $\sum_{n=1}^{+\infty} \frac{(-1)^n}{n^p}$ converges (by Leibniz's test).

2. Absolute convergence

Absolute value series

$$\sum_{n=1}^{+\infty} \left| \frac{(-1)^n}{n^p} \right| = \sum_{n=1}^{+\infty} \frac{1}{n^p}.$$

Nature according to p

- If $p > 1$: the series converges absolutely
- If $0 < p \leq 1$: the series converges but not absolutely

Summary

$$\begin{cases} \text{Converges for all } p > 0 \\ \text{Absolutely convergent if } p > 1 \\ \text{Conditionally convergent if } 0 < p \leq 1 \end{cases}$$

Exercise

Compute $\sum_{n=1}^{+\infty} \frac{1}{n(n+2)}$.

Solution

1. Partial fraction decomposition

$$\frac{1}{n(n+2)} = \frac{1}{2} \left(\frac{1}{n} - \frac{1}{n+2} \right).$$

2. Partial sum

$$S_N = \frac{1}{2} \sum_{n=1}^N \left(\frac{1}{n} - \frac{1}{n+2} \right).$$

3. Telescoping

Expand the first few terms:

$$\begin{aligned} S_N &= \frac{1}{2} \left[\left(1 - \frac{1}{3} \right) + \left(\frac{1}{2} - \frac{1}{4} \right) + \left(\frac{1}{3} - \frac{1}{5} \right) + \cdots + \left(\frac{1}{N} - \frac{1}{N+2} \right) \right] \\ &= \frac{1}{2} \left(1 + \frac{1}{2} - \frac{1}{N+1} - \frac{1}{N+2} \right). \end{aligned}$$

4. Limit

$$\lim_{N \rightarrow \infty} S_N = \frac{1}{2} \left(1 + \frac{1}{2} \right) = \frac{3}{4}.$$

Conclusion

$$\sum_{n=1}^{+\infty} \frac{1}{n(n+2)} = \frac{3}{4}.$$

Exercise

Show that $\sum_{n=0}^{+\infty} \frac{x^n}{n!}$ converges for all $x \in \mathbb{R}$.

Solution

1. Apply the ratio test (d'Alembert)

Let $u_n = \frac{x^n}{n!}$ for $x \neq 0$ (if $x = 0$, the series converges trivially).

2. Compute the ratio

$$\frac{|u_{n+1}|}{|u_n|} = \frac{|x|^{n+1}}{(n+1)!} \cdot \frac{n!}{|x|^n} = \frac{|x|}{n+1}.$$

3. Limit

$$\lim_{n \rightarrow \infty} \frac{|u_{n+1}|}{|u_n|} = 0 \quad \text{for all } x \in \mathbb{R}.$$

4. Conclusion

Since the limit is $0 < 1$ for all $x \in \mathbb{R}$, the series converges absolutely.

5. Remark

This series defines the exponential function:

$$\sum_{n=0}^{+\infty} \frac{x^n}{n!} = e^x.$$

Conclusion

$$\sum_{n=0}^{+\infty} \frac{x^n}{n!} \text{ converges absolutely for all } x \in \mathbb{R}.$$

Exercise

Study $\sum_{n=1}^{+\infty} \frac{\cos n}{n}$.

Solution

1. Absolute convergence

Absolute value series

$$\sum_{n=1}^{+\infty} \left| \frac{\cos n}{n} \right|.$$

Comparison

Since $|\cos n| \geq \cos^2 n = \frac{1+\cos 2n}{2}$, we have:

$$\left| \frac{\cos n}{n} \right| \geq \frac{1}{2n} + \frac{\cos 2n}{2n}.$$

Divergence

The series $\sum \frac{1}{2n}$ diverges, so $\sum \left| \frac{\cos n}{n} \right|$ diverges. Thus the series does not converge absolutely.

2. Simple convergence

Abel's transformation

Write $\frac{\cos n}{n} = a_n b_n$ with $a_n = \cos n$ and $b_n = \frac{1}{n}$.

Partial sums of $\cos n$

$$\left| \sum_{k=1}^n \cos k \right| = \left| \frac{\sin(n + \frac{1}{2}) - \sin \frac{1}{2}}{2 \sin \frac{1}{2}} \right| \leq \frac{1}{\sin \frac{1}{2}}.$$

The sequence of partial sums of $\cos n$ is bounded.

Apply Abel's criterion

Since:

1. $(\frac{1}{n})$ is decreasing and tends to 0

2. The partial sums of $\cos n$ are bounded
the series converges.

Conclusion

$$\sum_{n=1}^{+\infty} \frac{\cos n}{n} \text{ converges but not absolutely.}$$

Exercise

Physical application: Total distance traveled by a bouncing ball with coefficient of restitution e .

Solution

1. Assumptions

- The ball is dropped from an initial height h_0
- Coefficient of restitution: $0 < e < 1$
- After each bounce, the height is multiplied by e^2

2. Successive heights

$$\begin{aligned} \text{Initial drop: } & h_0 \\ \text{First bounce: } & h_1 = e^2 h_0 \\ \text{Second bounce: } & h_2 = e^4 h_0 \\ & \vdots \\ n\text{-th bounce: } & h_n = e^{2n} h_0 \end{aligned}$$

3. Distance traveled

Each cycle (down + up):

$$\begin{aligned} \text{Initial drop (down): } & h_0 \\ \text{First cycle: } & h_0 + 2h_1 = h_0 + 2e^2 h_0 \\ \text{Second cycle: } & 2h_2 = 2e^4 h_0 \\ & \vdots \end{aligned}$$

4. Total series

$$D = h_0 + 2e^2 h_0 + 2e^4 h_0 + 2e^6 h_0 + \cdots$$

5. Compute the sum

$$D = h_0 + 2h_0 \sum_{n=1}^{\infty} e^{2n}.$$

6. Geometric series

$$\sum_{n=1}^{\infty} e^{2n} = \frac{e^2}{1 - e^2} \quad (\text{since } |e^2| < 1).$$

7. Total distance

$$D = h_0 + 2h_0 \cdot \frac{e^2}{1 - e^2} = h_0 \left(1 + \frac{2e^2}{1 - e^2} \right) = h_0 \frac{1 + e^2}{1 - e^2}.$$

Conclusion

$$D = h_0 \frac{1 + e^2}{1 - e^2}.$$

Special cases

- $e \rightarrow 0$: $D \rightarrow h_0$ (no bounce)
- $e \rightarrow 1$: $D \rightarrow \infty$ (infinite bounces with no energy loss)

Exercise

Compute $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n$ using series.

Solution

1. Logarithm of the expression

$$\ln \left(1 + \frac{1}{n} \right)^n = n \ln \left(1 + \frac{1}{n} \right).$$

2. Series expansion

For $|x| < 1$:

$$\ln(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

With $x = \frac{1}{n}$:

$$\ln \left(1 + \frac{1}{n} \right) = \frac{1}{n} - \frac{1}{2n^2} + \frac{1}{3n^3} - \frac{1}{4n^4} + \dots$$

3. Multiply by n

$$n \ln \left(1 + \frac{1}{n} \right) = 1 - \frac{1}{2n} + \frac{1}{3n^2} - \frac{1}{4n^3} + \cdots$$

4. Limit

$$\lim_{n \rightarrow \infty} n \ln \left(1 + \frac{1}{n} \right) = 1.$$

5. Exponentiation

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n = e^{\lim_{n \rightarrow \infty} n \ln(1 + \frac{1}{n})} = e^1.$$

Conclusion

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n = e.$$

Remark

This limit is the classical definition of the number e .

Special Functions

Theorem

[Special Functions - Extended Version]

Fundamental Functions

1. **Euler's Gamma function (extended definition).** The Gamma function is defined for all $x > 0$ by:

$$\Gamma(x) = \int_0^{+\infty} t^{x-1} e^{-t} dt.$$

It satisfies the following properties:

- Functional relation: $\Gamma(x+1) = x \Gamma(x)$
- Special values: $\Gamma(n) = (n-1)!$, $\Gamma(\frac{1}{2}) = \sqrt{\pi}$
- Reflection formula: $\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)}$, $z \notin \mathbb{Z}$

2. **Stirling's formula (complete version).** As $x \rightarrow +\infty$, we have the asymptotic approximation:

$$\Gamma(x) \sim \sqrt{2\pi} x^{x-\frac{1}{2}} e^{-x} \left(1 + \frac{1}{12x} + \frac{1}{288x^2} - \frac{139}{51840x^3} + O\left(\frac{1}{x^4}\right) \right)$$

3. **Euler's Beta function and properties.** For $x > 0$ and $y > 0$, the Beta function is defined by:

$$B(x, y) = \int_0^1 t^{x-1} (1-t)^{y-1} dt,$$

and satisfies the relations:

$$B(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)},$$

$$B(x, y) = B(y, x),$$

$$B(x, y) = 2 \int_0^{\pi/2} \sin^{2x-1} \theta \cos^{2y-1} \theta d\theta.$$

4. **Digamma function and derivatives.** The digamma function is defined by:

$$\psi(x) = \frac{\Gamma'(x)}{\Gamma(x)}, \quad x > 0.$$

Important properties:

- $\psi(x+1) = \psi(x) + \frac{1}{x}$
- $\psi(1) = -\gamma, \psi\left(\frac{1}{2}\right) = -\gamma - 2\ln 2$
- $\gamma \approx 0.5772156649$ (Euler-Mascheroni constant)

5. **Bessel functions (complete types).** For $\nu \in \mathbb{R}$, the Bessel functions are defined by:

- **First kind:** $J_\nu(x) = \sum_{k=0}^{+\infty} \frac{(-1)^k}{k! \Gamma(k + \nu + 1)} \left(\frac{x}{2}\right)^{2k+\nu}$
- **Second kind (Neumann):** $Y_\nu(x) = \frac{J_\nu(x) \cos(\nu\pi) - J_{-\nu}(x)}{\sin(\nu\pi)}, \nu \notin \mathbb{Z}$
- **Modified functions:** $I_\nu(x) = i^{-\nu} J_\nu(ix), K_\nu(x) = \frac{\pi}{2} \frac{I_{-\nu}(x) - I_\nu(x)}{\sin(\nu\pi)}$

They are solutions of the equations:

$$x^2 y'' + xy' + (x^2 - \nu^2)y = 0 \quad (\text{standard Bessel})$$

$$x^2 y'' + xy' - (x^2 + \nu^2)y = 0 \quad (\text{modified Bessel})$$

6. **Airy functions (integral definitions).** The Airy functions Ai and Bi are solutions of:

$$y'' - xy = 0.$$

With the integral representations:

$$\text{Ai}(x) = \frac{1}{\pi} \int_0^\infty \cos\left(\frac{t^3}{3} + xt\right) dt,$$

$$\text{Bi}(x) = \frac{1}{\pi} \int_0^\infty \left[\exp\left(-\frac{t^3}{3} + xt\right) + \sin\left(\frac{t^3}{3} + xt\right) \right] dt.$$

7. **Legendre polynomials (complete properties).** The Legendre polynomials (P_n) are defined by:

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} [(x^2 - 1)^n], \quad n \in \mathbb{N}.$$

Properties:

- Differential equation: $(1 - x^2)y'' - 2xy' + n(n+1)y = 0$
- Orthogonality: $\int_{-1}^1 P_m(x)P_n(x)dx = \frac{2}{2n+1}\delta_{mn}$
- Generating function: $\frac{1}{\sqrt{1-2xt+t^2}} = \sum_{n=0}^{\infty} P_n(x)t^n, |t| < 1$

8. **Hermite polynomials (extended properties).** The Hermite polynomials (H_n) are defined by:

$$H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} (e^{-x^2}).$$

Properties:

- Differential equation: $H_n'' - 2xH_n' + 2nH_n = 0$
- Recurrence relation: $H_{n+1}(x) = 2xH_n(x) - 2nH_{n-1}(x)$
- Orthogonality: $\int_{-\infty}^{\infty} H_m(x)H_n(x)e^{-x^2}dx = 2^n n! \sqrt{\pi} \delta_{mn}$
- Generating function: $e^{2xt-t^2} = \sum_{n=0}^{\infty} \frac{H_n(x)}{n!} t^n$

9. **Laguerre polynomials (generalized version).** The generalized Laguerre polynomials are defined by:

$$L_n^{(\alpha)}(x) = \frac{x^{-\alpha} e^x}{n!} \frac{d^n}{dx^n} (e^{-x} x^{n+\alpha}).$$

For $\alpha = 0$, we denote $L_n(x) = L_n^{(0)}(x)$.

10. **Error function (advanced properties).** The error function is defined by:

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt.$$

Properties:

- $\operatorname{erf}(-x) = -\operatorname{erf}(x)$
- $\operatorname{erf}(0) = 0$, $\lim_{x \rightarrow \infty} \operatorname{erf}(x) = 1$
- Complementary function: $\operatorname{erfc}(x) = 1 - \operatorname{erf}(x)$
- Asymptotic approximation: $\operatorname{erf}(x) \sim 1 - \frac{e^{-x^2}}{x\sqrt{\pi}} \left(1 - \frac{1}{2x^2} + \frac{3}{4x^4} - \cdots \right)$, $x \rightarrow \infty$

11. **Lambert W function (detailed properties).** The Lambert W function is defined as the inverse of:

$$x \mapsto xe^x,$$

i.e., $W(x)e^{W(x)} = x$. Properties:

- Derivative: $\frac{d}{dx} W(x) = \frac{W(x)}{x(1+W(x))}$, $x \neq 0, -\frac{1}{e}$
- Taylor series: $W(x) = \sum_{n=1}^{\infty} \frac{(-n)^{n-1}}{n!} x^n$, $|x| < \frac{1}{e}$
- Special values: $W(0) = 0$, $W(e) = 1$, $W(-e^{-1}) = -1$

12. **Important orthogonality relations.**

- **Hermite:** $\int_{-\infty}^{\infty} H_m(x)H_n(x)e^{-x^2}dx = 2^n n! \sqrt{\pi} \delta_{mn}$
- **Legendre:** $\int_{-1}^1 P_m(x)P_n(x)dx = \frac{2}{2n+1} \delta_{mn}$
- **Laguerre:** $\int_0^{\infty} L_m^{(\alpha)}(x)L_n^{(\alpha)}(x)x^{\alpha}e^{-x}dx = \frac{\Gamma(n+\alpha+1)}{n!} \delta_{mn}$

13. Main applications of special functions.

- **Physics:** Wave propagation, quantum mechanics, electrostatics
- **Probability:** Normal, Gamma, Beta, Chi-square distributions
- **Applied mathematics:** PDE solving, perturbation theory
- **Engineering:** Wave transmission, signal processing, heat transfer

14. Main generating functions.

- **Hermite:** $e^{2xt-t^2} = \sum_{n=0}^{\infty} \frac{H_n(x)}{n!} t^n$
- **Legendre:** $\frac{1}{\sqrt{1-2xt+t^2}} = \sum_{n=0}^{\infty} P_n(x)t^n, |t| < 1$
- **Laguerre:** $\frac{e^{-xt/(1-t)}}{(1-t)^{\alpha+1}} = \sum_{n=0}^{\infty} L_n^{(\alpha)}(x)t^n, |t| < 1$

Exercise

Show that for all $x > 0$:

$$\Gamma(x+1) = x\Gamma(x).$$

Solution

1. Par définition : $\Gamma(x+1) = \int_0^{+\infty} t^x e^{-t} dt.$
2. Intégrons par parties : posons $u = t^x, dv = e^{-t} dt.$ Alors $du = xt^{x-1} dt, v = -e^{-t}.$
3. $\int_0^{+\infty} t^x e^{-t} dt = [-t^x e^{-t}]_0^{+\infty} - \int_0^{+\infty} (-e^{-t}) xt^{x-1} dt.$
4. Le terme intégré s'annule car $\lim_{t \rightarrow +\infty} t^x e^{-t} = 0$ et $\lim_{t \rightarrow 0} t^x e^{-t} = 0.$
5. Donc $\Gamma(x+1) = x \int_0^{+\infty} t^{x-1} e^{-t} dt = x\Gamma(x).$

$$\boxed{\Gamma(x+1) = x\Gamma(x)}.$$

Exercise

Using the relation $\Gamma(x+1) = x\Gamma(x)$, show that for every integer $n \geq 1$:

$$\Gamma(n) = (n-1)!.$$

Solution

1. Par récurrence sur n .
2. Initialisation : $n = 1$. $\Gamma(1) = \int_0^\infty e^{-t} dt = 1 = 0!$.
3. Hérédité : supposons $\Gamma(n) = (n-1)!$. Alors $\Gamma(n+1) = n\Gamma(n) = n \cdot (n-1)! = n!$.
4. Donc la propriété est vraie pour tout $n \in \mathbb{N}^*$.

$$\Gamma(n) = (n-1)!.$$

Exercise

Compute:

$$\Gamma(1), \quad \Gamma(2), \quad \text{and} \quad \left(\frac{1}{2}\right)!.$$

Solution

1. $\Gamma(1) = \int_0^\infty e^{-t} dt = 1$.
2. $\Gamma(2) = 1 \cdot \Gamma(1) = 1$.
3. $\left(\frac{1}{2}\right)! = \Gamma\left(\frac{3}{2}\right) = \frac{1}{2}\Gamma\left(\frac{1}{2}\right)$.
4. Or $\Gamma\left(\frac{1}{2}\right) = \int_0^\infty t^{-1/2} e^{-t} dt$. Posons $t = u^2$, $dt = 2udu$: $\Gamma\left(\frac{1}{2}\right) = \int_0^\infty u^{-1} e^{-u^2} \cdot 2udu = 2 \int_0^\infty e^{-u^2} du = \sqrt{\pi}$.
5. Donc $\left(\frac{1}{2}\right)! = \frac{1}{2}\sqrt{\pi}$.

$$\Gamma(1) = 1, \quad \Gamma(2) = 1, \quad \left(\frac{1}{2}\right)! = \frac{\sqrt{\pi}}{2}.$$

Exercise

Show that for $0 < x < 1$:

$$\Gamma(x) = \frac{\Gamma(x+1)}{x}.$$

Solution

1. De la relation fonctionnelle $\Gamma(x+1) = x\Gamma(x)$, valable pour tout $x > 0$.
2. On en déduit immédiatement $\Gamma(x) = \frac{\Gamma(x+1)}{x}$.

$$\Gamma(x) = \frac{\Gamma(x+1)}{x}.$$

Exercise

By iterating the previous relation, show that for $\alpha \in \mathbb{N}$ and $x > 0$ with $x + \alpha > 0$:

$$\Gamma(x) = \frac{\Gamma(x+\alpha)}{x(x+1)(x+2)\cdots(x+\alpha-1)}.$$

Solution

1. Itérons la relation $\Gamma(x) = \frac{\Gamma(x+1)}{x}$:
2. $\Gamma(x) = \frac{\Gamma(x+1)}{x} = \frac{\Gamma(x+2)}{x(x+1)} = \frac{\Gamma(x+3)}{x(x+1)(x+2)} = \dots$
3. Après α itérations :

$$\Gamma(x) = \frac{\Gamma(x+\alpha)}{x(x+1)(x+2)\cdots(x+\alpha-1)}.$$

$$\Gamma(x) = \frac{\Gamma(x+\alpha)}{x(x+1)\cdots(x+\alpha-1)}.$$

Exercise

Conclude the domain of definition of the gamma function.

Solution

1. L'intégrale $\Gamma(x) = \int_0^{+\infty} t^{x-1} e^{-t} dt$ converge pour $x > 0$.
2. La formule de l'exercice 13.5 permet de prolonger $\Gamma(x)$ aux valeurs négatives non entières.
3. Les pôles apparaissent quand le dénominateur s'annule : $x = 0, -1, -2, \dots$
4. Donc le domaine maximal (par prolongement analytique) est :

$$\mathbb{R} \setminus \{0, -1, -2, -3, \dots\}.$$

Exercise

Compute:

$$\Gamma(0^+), \quad \Gamma((-1)^-), \quad \Gamma\left(\frac{1}{2}\right), \quad \Gamma\left(-\frac{1}{2}\right).$$

Solution

1. $\Gamma(0^+) = \lim_{x \rightarrow 0^+} \frac{\Gamma(x+1)}{x} = \lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty$.
2. $\Gamma((-1)^-) = \lim_{x \rightarrow -1^-} \frac{\Gamma(x+1)}{x} = \frac{\Gamma(0)}{-1} = -\Gamma(0)$. Comme $\Gamma(0) = +\infty$, le signe négatif donne $+\infty$.
3. $\Gamma(1/2) = \sqrt{\pi}$ (démontré en 13.3).
4. $\Gamma(-1/2) = \frac{\Gamma(1/2)}{-1/2} = -2\sqrt{\pi}$.

$$\Gamma(0^+) = +\infty, \quad \Gamma((-1)^-) = +\infty, \quad \Gamma(1/2) = \sqrt{\pi}, \quad \Gamma(-1/2) = -2\sqrt{\pi}.$$

Exercise

Let $n \in \mathbb{N}$. Show that:

$$\Gamma\left(n + \frac{1}{2}\right) = \frac{(2n)!}{2^{2n} n!} \sqrt{\pi}.$$

Solution

1. Par récurrence sur n en utilisant $\Gamma(x+1) = x\Gamma(x)$.
2. Initialisation $n = 0$: $\Gamma(1/2) = \sqrt{\pi} = \frac{0!}{2^0 \cdot 0!} \sqrt{\pi}$.
3. Hérédité : supposons $\Gamma(n-1/2) = \frac{(2n-2)!}{2^{2n-2}(n-1)!} \sqrt{\pi}$.
4. $\Gamma(n+1/2) = (n-1/2)\Gamma(n-1/2) = \frac{2n-1}{2} \cdot \frac{(2n-2)!}{2^{2n-2}(n-1)!} \sqrt{\pi}$.
5. $(2n-1)(2n-2)! = (2n-1)! \text{ et } \frac{1}{(n-1)!} = \frac{n}{n!}$.
6. Donc $\Gamma(n+1/2) = \frac{(2n-1)! n}{2^{2n-1} n!} \sqrt{\pi} = \frac{(2n)!}{2^{2n} n!} \sqrt{\pi}$.

$$\Gamma\left(n + \frac{1}{2}\right) = \frac{(2n)!}{2^{2n} n!} \sqrt{\pi}.$$

Exercise

Prove Euler's reflection formula:

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)}, \quad z \notin \mathbb{Z}.$$

Solution

1. Pour $0 < z < 1$, considérons $B(z, 1-z) = \int_0^1 t^{z-1}(1-t)^{-z} dt = \frac{\Gamma(z)\Gamma(1-z)}{\Gamma(1)} = \Gamma(z)\Gamma(1-z)$.
2. Effectuons le changement $t = \frac{u}{1+u}$, $dt = \frac{du}{(1+u)^2}$.
3. Alors $t^{z-1}(1-t)^{-z} = \left(\frac{u}{1+u}\right)^{z-1} \left(\frac{1}{1+u}\right)^{-z} = u^{z-1}(1+u)^{-1}$.
4. Donc $B(z, 1-z) = \int_0^{+\infty} \frac{u^{z-1}}{1+u} du$.
5. Cette intégrale se calcule par résidus : $\int_0^{+\infty} \frac{u^{z-1}}{1+u} du = \frac{\pi}{\sin(\pi z)}$.
6. Par prolongement analytique, la formule est valable pour tout $z \notin \mathbb{Z}$.

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)}.$$

Exercise

Let $a, b > 0$.

1. Show that $B(a, b) = 2 \int_0^{\pi/2} \sin^{2a-1} \theta \cos^{2b-1} \theta d\theta$.
2. Deduce $\int_0^{\pi/2} \sin^m \theta \cos^n \theta d\theta = \frac{1}{2} B\left(\frac{m+1}{2}, \frac{n+1}{2}\right)$ for $m, n > -1$.
3. Compute $\int_0^{\pi/2} \sin^4 \theta \cos^6 \theta d\theta$.

Solution

1. $B(a, b) = \int_0^1 t^{a-1} (1-t)^{b-1} dt$. Posons $t = \sin^2 \theta$, $dt = 2 \sin \theta \cos \theta d\theta$:

$$B(a, b) = \int_0^{\pi/2} \sin^{2a-2} \theta \cos^{2b-2} \theta \cdot 2 \sin \theta \cos \theta d\theta = 2 \int_0^{\pi/2} \sin^{2a-1} \theta \cos^{2b-1} \theta d\theta.$$

2. Posons $m = 2a - 1$, $n = 2b - 1$, soit $a = \frac{m+1}{2}$, $b = \frac{n+1}{2}$:

$$\int_0^{\pi/2} \sin^m \theta \cos^n \theta d\theta = \frac{1}{2} B\left(\frac{m+1}{2}, \frac{n+1}{2}\right).$$

3. Pour $m = 4$, $n = 6$: $I = \frac{1}{2} B\left(\frac{5}{2}, \frac{7}{2}\right) = \frac{1}{2} \cdot \frac{\Gamma(5/2)\Gamma(7/2)}{\Gamma(6)}$.

$$\Gamma(5/2) = \frac{3}{4}\sqrt{\pi}, \Gamma(7/2) = \frac{15}{8}\sqrt{\pi}, \Gamma(6) = 120.$$

$$\text{Donc } I = \frac{1}{2} \cdot \frac{\frac{45}{32}\pi}{120} = \frac{3\pi}{512}.$$

$$I = \frac{3\pi}{512}.$$

Exercise

For $\nu > -1$, the Bessel functions satisfy:

$$1. \frac{d}{dx} [x^\nu J_\nu(x)] = x^\nu J_{\nu-1}(x).$$

$$2. \frac{d}{dx} [x^{-\nu} J_\nu(x)] = -x^{-\nu} J_{\nu+1}(x).$$

$$3. \text{Deduce: } J_{\nu-1}(x) + J_{\nu+1}(x) = \frac{2\nu}{x} J_\nu(x) \text{ and } J_{\nu-1}(x) - J_{\nu+1}(x) = 2J'_\nu(x).$$

Solution

$$1. J_\nu(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k + \nu + 1)} \left(\frac{x}{2}\right)^{2k+\nu}.$$

$$2. x^\nu J_\nu(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k + \nu + 1)} \frac{x^{2k+2\nu}}{2^{2k+\nu}}.$$

$$3. \text{Dérivons : } \frac{d}{dx} [x^\nu J_\nu(x)] = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k + \nu + 1)} \frac{(2k + 2\nu)x^{2k+2\nu-1}}{2^{2k+\nu}} = x^\nu J_{\nu-1}(x).$$

$$4. x^{-\nu} J_\nu(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k + \nu + 1)} \frac{x^{2k}}{2^{2k+\nu}}.$$

$$5. \text{Dérivons : } \frac{d}{dx} [x^{-\nu} J_\nu(x)] = -x^{-\nu} J_{\nu+1}(x).$$

$$6. \text{En développant : } \nu x^{\nu-1} J_\nu + x^\nu J'_\nu = x^\nu J_{\nu-1} - \nu x^{-1} J_\nu + J'_\nu = J_{\nu-1}.$$

$$7. -\nu x^{-\nu-1} J_\nu + x^{-\nu} J'_\nu = -x^{-\nu} J_{\nu+1} - \nu x^{-1} J_\nu + J'_\nu = -J_{\nu+1}.$$

$$8. \text{Addition : } 2J'_\nu = J_{\nu-1} - J_{\nu+1}.$$

$$9. \text{Soustraction : } 2\nu x^{-1} J_\nu = J_{\nu-1} + J_{\nu+1}.$$

$$J_{\nu-1}(x) + J_{\nu+1}(x) = \frac{2\nu}{x} J_\nu(x), \quad J_{\nu-1}(x) - J_{\nu+1}(x) = 2J'_\nu(x).$$

Exercise

For Legendre polynomials:

1. Verify $P_0(x) = 1$, $P_1(x) = x$, $P_2(x) = \frac{1}{2}(3x^2 - 1)$.
2. Show $\int_{-1}^1 P_0 P_1 dx = 0$.
3. Show orthogonality $\int_{-1}^1 P_n P_m dx = 0$ for $n \neq m$.
4. Compute $\|P_2\|^2$.

Solution

1. $P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n$ donne $P_0 = 1$, $P_1 = x$, $P_2 = \frac{1}{2}(3x^2 - 1)$.
2. $\int_{-1}^1 x dx = 0$ (fonction impaire).
3. P_n et P_m vérifient $(1 - x^2)P_n'' - 2xP_n' + n(n+1)P_n = 0$. Multiplions par P_m , soustrayons, intégrons par parties : $[n(n+1) - m(m+1)] \int_{-1}^1 P_n P_m dx = 0$. Pour $n \neq m$, on a $\int_{-1}^1 P_n P_m dx = 0$.
4. $\int_{-1}^1 [P_2(x)]^2 dx = \int_{-1}^1 \frac{1}{4} (3x^2 - 1)^2 dx = \frac{1}{4} \int_{-1}^1 (9x^4 - 6x^2 + 1) dx = \frac{1}{4} \left(\frac{18}{5} - 4 + 2 \right) = \frac{2}{5}$.

$$\|P_2\|^2 = \frac{2}{5}.$$

Exercise

For Hermite polynomials:

1. Compute H_0 , H_1 , H_2 , H_3 .
2. Verify H_2 satisfies $y'' - 2xy' + 4y = 0$.
3. Show $\int_{-\infty}^{\infty} H_0 H_1 e^{-x^2} dx = 0$.
4. Show $H'_n(x) = 2nH_{n-1}(x)$ using the generating function.

Solution

1. $H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2}$ donne $H_0 = 1$, $H_1 = 2x$, $H_2 = 4x^2 - 2$, $H_3 = 8x^3 - 12x$.
2. $H_2'' - 2xH_2' + 4H_2 = 8 - 2x(8x) + 4(4x^2 - 2) = 0$.
3. $\int_{-\infty}^{\infty} 2xe^{-x^2} dx = 0$ (fonction impaire).
4. Fonction génératrice : $e^{2xt-t^2} = \sum \frac{H_n(x)}{n!} t^n$. Dérivons par rapport à x : $2te^{2xt-t^2} = \sum \frac{H'_n(x)}{n!} t^n$. Mais $2te^{2xt-t^2} = 2 \sum \frac{H_n(x)}{n!} t^{n+1} = 2 \sum \frac{H_{n-1}(x)}{(n-1)!} t^n$. Identification : $\frac{H'_n(x)}{n!} = 2 \frac{H_{n-1}(x)}{(n-1)!}$ $H'_n(x) = 2nH_{n-1}(x)$.

$$H'_n(x) = 2nH_{n-1}(x).$$

Exercise

1. Show $\Gamma(x) = 2 \int_0^\infty t^{2x-1} e^{-t^2} dt$.
2. Deduce $\int_0^\infty e^{-t^2} dt = \frac{\sqrt{\pi}}{2}$.
3. Compute $\Gamma(3/4)\Gamma(1/4)$.
4. Show $\Gamma(x)\Gamma(1-x) = \frac{\pi}{\sin(\pi x)}$ for $0 < x < 1$.

Solution

1. $\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt$. Posons $t = u^2$, $dt = 2udu$: $\Gamma(x) = \int_0^\infty u^{2x-2} e^{-u^2} 2udu = 2 \int_0^\infty u^{2x-1} e^{-u^2} du$.
2. Pour $x = 1/2$: $\Gamma(1/2) = \sqrt{\pi} = 2 \int_0^\infty e^{-u^2} du$ $\int_0^\infty e^{-t^2} dt = \sqrt{\pi}/2$.
3. Par la formule de réflexion : $\Gamma(1/4)\Gamma(3/4) = \frac{\pi}{\sin(\pi/4)} = \pi\sqrt{2}$.
4. Voir exercice 13.9.

$$\int_0^\infty e^{-t^2} dt = \frac{\sqrt{\pi}}{2}, \quad \Gamma(1/4)\Gamma(3/4) = \pi\sqrt{2}.$$

Exercise

Hypergeometric functions:

1. Show $\frac{1}{1-z} = {}_2F_1(1, 1; 1; z)$.
2. Show $\operatorname{erf}(x) = \frac{2x}{\sqrt{\pi}} {}_1F_1(\frac{1}{2}; \frac{3}{2}; -x^2)$.
3. Verify $P_n(x) = {}_2F_1(-n, n+1; 1; \frac{1-x}{2})$ for $n = 0, 1, 2$.
4. Show $J_\nu(x) = \frac{(x/2)^\nu}{\Gamma(\nu+1)} {}_0F_1(\nu+1; -\frac{x^2}{4})$.

Solution

1. ${}_2F_1(1, 1; 1; z) = \sum_{n=0}^\infty \frac{(1)_n(1)_n}{(1)_n} \frac{z^n}{n!} = \sum_{n=0}^\infty z^n = \frac{1}{1-z}$.
2. ${}_1F_1(\frac{1}{2}; \frac{3}{2}; -x^2) = \sum_{n=0}^\infty \frac{(1/2)_n}{(3/2)_n} \frac{(-x^2)^n}{n!} = \sum_{n=0}^\infty \frac{(-1)^n x^{2n}}{(2n+1)n!}$. Donc $\frac{2x}{\sqrt{\pi}} {}_1F_1 = \frac{2}{\sqrt{\pi}} \sum \frac{(-1)^n x^{2n+1}}{(2n+1)n!} = \operatorname{erf}(x)$.
3. Pour $n = 0$: ${}_2F_1(0, 1; 1; z) = 1 = P_0$. $n = 1$: ${}_2F_1(-1, 2; 1; z) = 1 - 2z$, avec $z = (1-x)/2$ donne $x = P_1$. $n = 2$: ${}_2F_1(-2, 3; 1; z) = 1 - 6z + 6z^2$ donne $\frac{1}{2}(3x^2 - 1) = P_2$.
4. ${}_0F_1(\nu+1; z) = \sum_{k=0}^\infty \frac{z^k}{(\nu+1)_k k!}$. Donc $\frac{(x/2)^\nu}{\Gamma(\nu+1)} {}_0F_1(\nu+1; -\frac{x^2}{4}) = \sum_{k=0}^\infty \frac{(-1)^k}{k! \Gamma(k+\nu+1)} \left(\frac{x}{2}\right)^{2k+\nu} = J_\nu(x)$.

Exercise

Bessel integrals:

1. Show $J_n(x) = \frac{1}{\pi} \int_0^\pi \cos(x \sin \theta - n\theta) d\theta$ for integer n .

2. Conclude $|J_n(x)| \leq 1$.
3. Compute $\int_0^\infty e^{-ax} J_0(bx) dx$ for $a, b > 0$.
4. Orthogonality: $\int_0^\infty J_\nu(ax) J_\nu(bx) x dx = \frac{1}{a} \delta(a - b)$.

Solution

1. $e^{ix \sin \theta} = \sum_{n=-\infty}^{\infty} J_n(x) e^{in\theta}$. Multiplions par $e^{-in\theta}$ et intégrons sur $[0, 2\pi]$: $\frac{1}{2\pi} \int_0^{2\pi} e^{i(x \sin \theta - n\theta)} d\theta = J_n(x)$. En prenant la partie réelle : $J_n(x) = \frac{1}{\pi} \int_0^\pi \cos(x \sin \theta - n\theta) d\theta$.
2. $|J_n(x)| \leq \frac{1}{\pi} \int_0^\pi 1 \cdot d\theta = 1$.
3. $\int_0^\infty e^{-ax} J_0(bx) dx = \sum_{k=0}^{\infty} \frac{(-1)^k}{(k!)^2} \left(\frac{b}{2}\right)^{2k} \frac{(2k)!}{a^{2k+1}} = \frac{1}{a} \sum_{k=0}^{\infty} \binom{2k}{k} \left(-\frac{b^2}{4a^2}\right)^k = \frac{1}{a} \cdot \frac{1}{\sqrt{1+b^2/a^2}} = \frac{1}{\sqrt{a^2+b^2}}$.
4. Résultat admis (relation d'orthogonalité des fonctions de Bessel).

$$\int_0^\infty e^{-ax} J_0(bx) dx = \frac{1}{\sqrt{a^2 + b^2}}.$$

Exercise

Airy functions:

1. Show $\text{Ai}(x) = \frac{1}{\pi} \int_0^\infty \cos\left(\frac{t^3}{3} + xt\right) dt$.
2. Show $\text{Ai}(x)$ satisfies $y'' - xy = 0$.
3. Asymptotics: $\text{Ai}(x) \sim \frac{1}{2\sqrt{\pi}} x^{-1/4} e^{-(2/3)x^{3/2}}$ as $x \rightarrow \infty$.
4. Compute $\text{Ai}(0)$ and $\text{Ai}'(0)$.

Solution

1. $\text{Ai}(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i(t^3/3 + xt)} dt = \frac{1}{\pi} \int_0^\infty \cos\left(\frac{t^3}{3} + xt\right) dt$.
2. Dérivons sous le signe intégral : $\text{Ai}'(x) = -\frac{1}{\pi} \int_0^\infty t \sin(\cdot) dt$, $\text{Ai}''(x) = -\frac{1}{\pi} \int_0^\infty t^2 \cos(\cdot) dt$. Intégrons $\text{Ai}''(x)$ par parties : $\text{Ai}''(x) - x\text{Ai}(x) = 0$.
3. Par la méthode du col, pour $x \rightarrow +\infty$: $\text{Ai}(x) \sim \frac{1}{2\sqrt{\pi}} x^{-1/4} e^{-\frac{2}{3}x^{3/2}}$.
4. $\text{Ai}(0) = \frac{\Gamma(1/3)}{2\pi 3^{1/6}}$, $\text{Ai}'(0) = -\frac{\Gamma(2/3)}{2\pi 3^{1/6}}$.

$$\text{Ai}(0) = \frac{\Gamma(1/3)}{2\pi 3^{1/6}}, \quad \text{Ai}'(0) = -\frac{\Gamma(2/3)}{2\pi 3^{1/6}}.$$

Exercise

Spherical Bessel functions: $j_n(x) = \sqrt{\frac{\pi}{2x}} J_{n+1/2}(x)$.

1. Compute j_0, j_1, j_2 .
2. Show j_n satisfies $x^2 y'' + 2xy' + [x^2 - n(n+1)]y = 0$.

3. Show $j_n(x) = (-x)^n \left(\frac{1}{x} \frac{d}{dx} \right)^n \frac{\sin x}{x}$.

4. Compute $\int_0^\infty j_0(ax)j_0(bx)x^2 dx$.

Solution

1. $J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \sin x$ $j_0(x) = \frac{\sin x}{x}$. $J_{3/2}(x) = \sqrt{\frac{2}{\pi x}} \left(\frac{\sin x}{x} - \cos x \right)$ $j_1(x) = \frac{\sin x}{x^2} - \frac{\cos x}{x}$. $J_{5/2}(x) = \sqrt{\frac{2}{\pi x}} \left[\left(\frac{3}{x^2} - 1 \right) \sin x - \frac{3}{x} \cos x \right]$ $j_2(x) = \left(\frac{3}{x^2} - 1 \right) \frac{\sin x}{x} - \frac{3 \cos x}{x^2}$.

2. En substituant $y = x^{-1/2} J_{n+1/2}(x)$ dans l'équation de Bessel, on obtient l'équation donnée.

3. Par récurrence : $j_0(x) = \frac{\sin x}{x}$. Supposons vraie à l'ordre n , alors $j_{n+1}(x) = -x \frac{d}{dx} \left(\frac{j_n(x)}{x} \right)$ donne la formule.

4. $\int_0^\infty j_0(ax)j_0(bx)x^2 dx = \int_0^\infty \frac{\sin(ax)}{ax} \frac{\sin(bx)}{bx} x^2 dx = \frac{1}{ab} \int_0^\infty \sin(ax) \sin(bx) dx = \frac{\pi}{2ab} \delta(a-b)$.

$$\int_0^\infty j_0(ax)j_0(bx)x^2 dx = \frac{\pi}{2ab} \delta(a-b).$$

Exercise

Hypergeometric functions:

1. Show $\sum_{n=0}^\infty \frac{(1)_n(1)_n(1)_n}{(2)_n(2)_n} \frac{z^n}{n!} = \frac{1}{z} [\text{Li}_2(z) - z]$.

2. Show $P_n^{(\alpha, \beta)}(x) = \frac{(\alpha+1)_n}{n!} {}_2F_1(-n, n+\alpha+\beta+1; \alpha+1; \frac{1-x}{2})$.

3. Show ${}_1F_2(1; \frac{3}{2}, 2; -\frac{x^2}{4}) = \frac{4}{x^2} \left(1 - \frac{\sin x}{x} \right)$.

Solution

1. $(1)_n = n!$, $(2)_n = (n+1)!$, donc $\frac{(1)_n(1)_n(1)_n}{(2)_n(2)_n} \frac{1}{n!} = \frac{1}{(n+1)^2}$. La série vaut $\sum_{n=0}^\infty \frac{z^n}{(n+1)^2} = \frac{1}{z} \sum_{m=1}^\infty \frac{z^m}{m^2} = \frac{1}{z} \text{Li}_2(z)$.

2. Formule standard des polynômes de Jacobi.

3. ${}_1F_2(1; \frac{3}{2}, 2; -\frac{x^2}{4}) = \sum_{n=0}^\infty \frac{(1)_n}{(\frac{3}{2})_n(2)_n} \frac{(-x^2/4)^n}{n!} = \sum_{n=0}^\infty \frac{(-1)^n x^{2n}}{(2n+1)(2n+2)}$. On vérifie que $\frac{4}{x^2} \left(1 - \frac{\sin x}{x} \right) = \sum_{n=0}^\infty \frac{(-1)^n x^{2n}}{(2n+1)(2n+2)}$.

$${}_1F_2 \left(1; \frac{3}{2}, 2; -\frac{x^2}{4} \right) = \frac{4}{x^2} \left(1 - \frac{\sin x}{x} \right).$$

Exercise

Mathieu functions: $y'' + (a - 2q \cos 2x)y = 0$.

1. For $q = 0$, solutions are $\cos(\sqrt{a}x)$, $\sin(\sqrt{a}x)$.

2. Substituting $y = \sum c_n e^{inx}$ gives recurrence $(a - n^2)c_n - q(c_{n-2} + c_{n+2}) = 0$.

3. For each q , there exists a discrete set of a giving periodic solutions.

4. Modified Mathieu functions: $y'' - (a - 2q \cosh 2x)y = 0$ obtained by $x \rightarrow ix$.

Solution

1. Pour $q = 0$: $y'' + ay = 0$, équation harmonique.
2. $y = \sum c_n e^{inx}$ donne $\sum [-n^2 c_n + a c_n - q(c_{n-2} + c_{n+2})] e^{inx} = 0$ $(a - n^2)c_n - q(c_{n-2} + c_{n+2}) = 0$.
3. La récurrence définit un problème aux valeurs propres pour a .
4. En posant $x = ix$, $\cos 2x \rightarrow \cosh 2x$, d'où l'équation modifiée.

$$(a - n^2)c_n - q(c_{n-2} + c_{n+2}) = 0.$$

Exercise

Elliptic integrals:

1. Show $K(0) = \pi/2$ where $K(k) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1-k^2 \sin^2 \theta}}$.
2. Expand $K(k)$ to order k^4 .
3. Jacobi elliptic function: $u = \int_0^{\text{sn}(u,k)} \frac{dt}{\sqrt{(1-t^2)(1-k^2 t^2)}}$.
4. Show $\left(\frac{d}{du} \text{sn}\right)^2 = (1 - \text{sn}^2)(1 - k^2 \text{sn}^2)$.

Solution

1. $K(0) = \int_0^{\pi/2} d\theta = \frac{\pi}{2}$.
2. $(1 - k^2 \sin^2 \theta)^{-1/2} = 1 + \frac{1}{2}k^2 \sin^2 \theta + \frac{3}{8}k^4 \sin^4 \theta + \dots$ $\int_0^{\pi/2} \sin^2 \theta d\theta = \frac{\pi}{4}$, $\int_0^{\pi/2} \sin^4 \theta d\theta = \frac{3\pi}{16}$. Donc $K(k) = \frac{\pi}{2} \left(1 + \frac{k^2}{4} + \frac{9k^4}{64} + \dots\right)$.
3. Définition de la fonction elliptique de Jacobi $\text{sn}(u, k)$.
4. Dérivons la définition : $\frac{du}{dy} = \frac{1}{\sqrt{(1-y^2)(1-k^2 y^2)}}$, avec $y = \text{sn}(u, k)$. Donc $\left(\frac{dy}{du}\right)^2 = (1 - y^2)(1 - k^2 y^2)$.

$$K(k) = \frac{\pi}{2} \left(1 + \frac{k^2}{4} + \frac{9k^4}{64} + \dots\right).$$

Exercise

Riemann zeta and polylogarithms:

1. Show $\zeta(2) = \pi^2/6$.
2. Show $\text{Li}_s(1) = \zeta(s)$ for $\Re(s) > 1$.
3. Prove Euler's identity for the dilogarithm: $\text{Li}_2(z) + \text{Li}_2(1-z) = \frac{\pi^2}{6} - \ln z \ln(1-z)$.
4. Compute $\int_0^1 \frac{\ln(1-x)}{x} dx$.

Solution

1. $\zeta(2) = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$ (problème de Bâle).
2. $\text{Li}_s(1) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \zeta(s)$ pour $\Re(s) > 1$.
3. Dérivons $F(z) = \text{Li}_2(z) + \text{Li}_2(1-z) : F'(z) = -\frac{\ln(1-z)}{z} + \frac{\ln z}{1-z}$. Dérivons $G(z) = \frac{\pi^2}{6} - \ln z \ln(1-z) : G'(z) = -\frac{\ln(1-z)}{z} + \frac{\ln z}{1-z}$. Donc $F(z) = G(z) + C$. En $z = 1/2$, on trouve $C = 0$.
4. $\int_0^1 \frac{\ln(1-x)}{x} dx = -\sum_{n=1}^{\infty} \frac{1}{n^2} = -\zeta(2) = -\frac{\pi^2}{6}$.

$$\boxed{\int_0^1 \frac{\ln(1-x)}{x} dx = -\frac{\pi^2}{6}}.$$

Appendix

Appendix

Definition and elementary properties

The **Zeraoulia function**, denoted by $\mathcal{Z}(x)$, was introduced by Zeraoulia Rafik and Mabrouk Mohamed Cherif in their Master 2 thesis (2021). It is defined for real $x \geq 0$ by

$$\mathcal{Z}(x) = \int_0^x (e^{-t^2})^{\text{erf}(t)} dt = \int_0^x e^{-t^2 \text{erf}(t)} dt. \quad (14.1)$$

The integrand involves the Gaussian e^{-t^2} raised to the power of the error function $\text{erf}(t)$. The function $\mathcal{Z}$ is smooth, strictly increasing, concave–convex and tends very rapidly to a finite limit.

For parameters $\lambda > 0$, $\mu \in \mathbb{R}$, we define

$$C_{\lambda,\mu} = \int_0^\infty e^{-\xi^2(\lambda + \mu \text{erf}(\xi))} d\xi,$$

$$\mathcal{Z}_{\lambda,\mu}(x) = \frac{1}{C_{\lambda,\mu}} \int_0^x e^{-\xi^2(\lambda + \mu \text{erf}(\xi))} d\xi \quad (x \geq 0).$$

$\mathcal{Z}_{\lambda,\mu}(x)$ is a cumulative distribution function (CDF) on $[0, \infty)$; its probability density is

$$f_{\lambda,\mu}(x) = \frac{1}{C_{\lambda,\mu}} e^{-x^2(\lambda + \mu \text{erf}(x))}.$$

For $\mu = 0$, after an appropriate normalization, we recover the error function: $\mathcal{Z}_{\lambda,0}(x) = \text{erf}(\sqrt{\lambda}x)$.

Numerical value and OEIS

The improper integral converges extremely rapidly. With more than 100 decimal places,

$$\begin{aligned} \mathcal{Z}(\infty) &= \int_0^\infty e^{-x^2 \text{erf}(x)} dx \\ &= 0.972106992769178593151077875442391175554272 \dots \end{aligned}$$

This constant has been assigned the number **A328968** in the On-Line Encyclopedia of Integer Sequences (OEIS).

Series expansion and non-elementary representability

Expanding the exponential and the error function yields the Taylor expansion near 0:

$$\mathcal{Z}(x) = x - \frac{x^4}{2\sqrt{\pi}} + \frac{x^6}{9\sqrt{\pi}} + \frac{2x^7}{7\pi} - \frac{x^8}{40\sqrt{\pi}} - \frac{4x^9}{27\pi} + \frac{(\pi - 28)x^{10}}{210\pi^{3/2}} + O(x^{11}).$$

Explicit approximations

Several accurate approximations have been established.

a) Approximation of $\mathcal{Z}(\infty)$

$$\mathcal{Z}(\infty) \approx \frac{\pi^{1/6} \Gamma(1/3)}{2^{1/3}} = 0.97216864\dots$$

(absolute error $\approx 6.16 \times 10^{-5}$).

b) Padé approximant of the integrand

Using the approximation $\operatorname{erf}(x)^2 \approx 1 - \exp(-(1 + \pi)^{2/3} \log^2(2) x^2)$, we obtain a series representation involving exponential integrals. A finer Padé approximant for $\log(1 - \operatorname{erf}^2 x)$ yields a maximum error of 0.00035.

c) Chebyshev expansion

On $[0, 1.5]$, an 11th-degree Chebyshev expansion gives

$$\mathcal{Z}(x) \approx 1.00003x - 0.00104x^2 + \dots + 0.01379x^{11}.$$

Connections with other functions

$\mathcal{Z}(x)$ cannot be expressed in terms of usual elementary functions. This follows from Liouville's theorem on elementary primitives: the differential equation $1 = h'(x) + h(x)(-x^2 \operatorname{erf}(x))$ admits no rational solution $h(x)$. Let $\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-z^2/2} dz$ be the CDF of the standard normal distribution. Then

$$\mathcal{Z}(x) = \int_0^x e^{t^2(1-2\Phi(t\sqrt{2}))} dt.$$

The function $y(x) = \mathcal{Z}_{\lambda, \mu}(x)$ satisfies the second-order nonlinear differential equation

$$\sqrt{\pi} x y''(x) = 2e^{-x^2} y'(x) \left(\sqrt{\pi} e^{x^2} \ln(y'(x)) - \mu x^3 \right).$$

For $\mu = 0$, we obtain

$$x y'' = 2y' \ln(y'),$$

whose general solution involves erfi .

Advection equation and wave propagation

Consider $f(x, t) = e^{-t^2 \operatorname{erf}(x)}$ (this is the density associated with $\mathcal{Z}(x)$ with $\lambda = 0, \mu = 1$ after normalization). Then

$$\frac{\partial f}{\partial t} + a(x, t) \frac{\partial f}{\partial x} = 0, \quad a(x, t) = \frac{\operatorname{erf}(-x) \sqrt{\pi}}{2t e^{-x^2}}.$$

Thus f is a solution of the linear advection equation. For $x > 0$, $a(x, t) < 0$ (downwind scheme), which implies instability of standard centered schemes while upwind schemes stabilize the numerical resolution by introducing artificial diffusion.

Membership in Hardy spaces

The inverse function $\mathcal{Z}^{-1} : (-1, 1) \rightarrow \mathbb{R}$ extends analytically to the unit disk and has two logarithmic singularities at ± 1 . Consequently, $\mathcal{Z}^{-1}$ belongs to the Hardy spaces H^p for all $p > 0$ (in particular H^2), and its Taylor coefficients are square-summable.

Applications in probability and physics

1. Approximation of the error function

Solving $\text{erf}(1) = \mathcal{Z}_{\lambda, \mu}(1)$ and $\text{erf}'(1) = \mathcal{Z}'_{\lambda, \mu}(1)$ yields $\lambda \approx 0.16716$, $\mu \approx 0.84497$; the function $\mathcal{Z}_{\lambda, \mu}$ then coincides with $\text{erf}(x)$ to within 10^{-3} on $[0, 2]$.

2. Maxwell–Boltzmann cumulative distribution function

For a scale parameter $a > 0$, the Maxwell–Boltzmann CDF can be approximated by a combination of $\mathcal{Z}_{\lambda, \mu}$ and a Gaussian term (see thesis, Fig. 9.2).

3. New symmetric probability distribution

With parameters $\sigma \in \mathbb{R}$, $\mu \in \mathbb{R}$ define

$$F(z, \mu, \sigma) = 1.010575 \frac{\sqrt[4]{2\pi}}{\sqrt[4]{1 + \mu^2/4}} (z - \sigma)^2 \exp\left(-\frac{(z - \sigma)^2 \sqrt{2\pi}}{\sqrt{1 + \mu^2/4}}\right) \times \text{erf}\left(\frac{(z - \sigma)^2 \sqrt{2\pi}}{\sqrt{1 + \mu^2/4}}\right).$$

This density is symmetric about σ , its skewness is zero; the variance and kurtosis are computed numerically. Normality tests (Kolmogorov–Smirnov, etc.) show that this distribution is not Gaussian.

4. Connection with the plasma dispersion function and relativistic quantum mechanics

Expressing $\text{erf}(x^2)$ using the Faddeeva function $w(z)$ (or the plasma dispersion function $Z(z) = i\sqrt{\pi}w(z)$), we obtain

$$\exp(-x^2 \text{erf}(x^2)) = \exp\left(i \frac{x^2}{\sqrt{\pi}} (e^{-x^4} Z(-ix^2) - i\sqrt{\pi})\right).$$

Choosing $x = k(z - \sigma)$ with $k = \sqrt[4]{2\pi/(1 + \mu^2/4)}$ and interpreting $z = \sqrt{E}$ (energy), $\sigma = t$ (time) and $\mu = 2p/(m_0 c)$ (momentum, rest mass, speed of light), we obtain a wave function for a massive charged particle:

$$\psi(X, t) = kE_0 \exp(iCX - i\sqrt{\pi}E_0), \quad C = \frac{\sqrt{2}E_0}{eE_0^2}, \quad X = Z(-ix^2).$$

5. Representation of the golden ratio

There exists a constant $t_0 \approx 2.74953926380898 \dots$ such that

$$\int_0^{t_0} (\exp(\sqrt{x} - x^2))^{\text{erf}(x)} dx = \frac{1 + \sqrt{5}}{2},$$

the golden ratio ϕ , with an accuracy better than 10^{-50} .

Other properties and perspectives

- The function is monotonic and satisfies $\mathcal{Z}(0) = 0$, $\mathcal{Z}(\infty) < \infty$.
- The radius of convergence of its series expansion is finite (approximately 1.6).
- Future work aims to connect $\mathcal{Z}(x)$ to the Riemann zeta function and the Riemann hypothesis via random matrix theory (GUE) and chaotic operators.

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